



Welcome to Expedition

Expedition is simply the best tactical and navigation software available.

Expedition has been in development since the mid 1990s by veteran Volvo Ocean Race navigator and Whitbread winner, Nick White.

Expedition has been used in multiple Volvo Ocean, America's Cup and Grand Prix events and is the most advanced and usable software available - for racing around the world or your local harbour.

Expedition is continually refined and developed with input from a core group of world-renowned navigators.

Prerequisites

For Windows 11 or 10 with at least 16GB memory. Windows 11 with minimum 16GB memory and Intel graphics is recommended.

Expedition is a highly multi-threaded application. CPUs with more cores will increase performance, especially for optimal routing.

Features

Charting

Chart-plotter that seamlessly selects, mosaics and rotates charts.

S-57 ENC's

S-63 ENC's

BSB RNC's

C-Map X

C-Map 4D

Navionics

Various legacy formats

Advanced weather display and tools

Sophisticated display options

Grib 1 and 2

NetCDF

Intelligent merging and display of multiple grib file datasets

Integrated WRF models

Integrated GRIB server

Saildocs, Squid and other weather sources

SailFlow weather observations

Rainviewer weather radar

Weather observations

Tides & currents

Satellite images

Weather charts

Simply the best weather routing available

- Multiple routing algorithms
- Display multiple optimum routes
- Optimal fleet routing
- Ensemble multiple weather model routing
- Optimal route sensitivity
- Wave corrections and avoidance
- Routing sensitivity
- Reverse isochrones for graphical interpretation of routing
- Asymmetric polars

Start display and advanced buoy racing functions

- Windward-leeward course support
- Time to marks, laylines, next leg etc.
- Layline bounds
- What if? functionality
- America's cup race management
- Sail charts
- Handicap support
- Start line functions
- Time to ends & line
- Time to burn
- Line bias
- Hold wind and current
- Display grid
- ... and much more

Instrument connectivity with all popular instrument systems

AIS, DSC and AIS-SART receivers

A+T

B&G Hercules, Hydra & GoFree H390 to H5000

Bravo Systems

Cosworth

Digital Yacht

DMK Yacht

Garmin USB

Iridium GO!

Koden radar

KVH Quadro

Laser rangefinders

Navico radar

Nexus NX2 FDX

NKE

NMEA 0183

NMEA 2000

Ockam

Optimizer

Sailmon

Pixel sur Mer Exocet

Tacktick

Ventus

Vesper AIS

VSPARS

Yacht Devices

Various compasses, MOB and other sensors

Networking support

Also

Stripchart to plot and analyse data

Sail chart

YB Tracking

AIS target tracking

Race tracking functions to track and analyse competitors

Display marks, courses, laylines, track, AIS targets, radar, tracks, race schedule information etc. on the chart

Polar functions to output, analyse and create or modify a yacht's performance polars

Logging functions to record race data that can be replayed or analysed later to assist with instrument calibration or polar modifications

GPX import and export

Number boxes to display any system number

GPS window

Alarms

Full screen mode

... and much more!



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User agrees to comply with all laws, regulations, and requirements of the Country in which you are using the Products and acknowledges and agrees that:

- i. Nautical navigation is an inherently dangerous act and that this product is only a navigational aid and neither the sole nor primary method of navigation.
- ii. No national hydrographic office or official agency of any country has verified the data or information in this product, including but not limited to nautical charts or other images in digital or physical form and no such office will accept responsibility or liability for the accuracy of any reproduction or any modification made to the data or information produced by the product.

- iii. No national hydrographic office or agency warrants that this product satisfies national or international regulations regarding the use of the appropriate products for navigation.
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Installing Expedition

Expedition can be downloaded from the Expedition web site at www.ExpeditionMarine.com.

Run the setup .exe file to start the installation process.

It is good practice to keep a copy of the installation file on your computer.

After the installation is complete, you can launch Expedition by



double clicking the Expedition icon installed onto your desktop.

When Expedition launches for this first time, it will ask for a licence key which will be unique for your computer. Email the site code the program provides to Expedition or one of the resellers listed. The authorisation window contains functionality to automatically create this email for you. If you already own or have paid for a licence to use Expedition, a key will be emailed to you. If not, you can arrange for payment or a temporary licence at this time.

Licence keys (licence.xml) should be copied to the Expedition data folder. There are linked to the data folder on the [Application](#) button menu and also the licence window.

Installations and updates may be obtained from www.ExpeditionMarine.com. Periodic updates are made available on the web site.

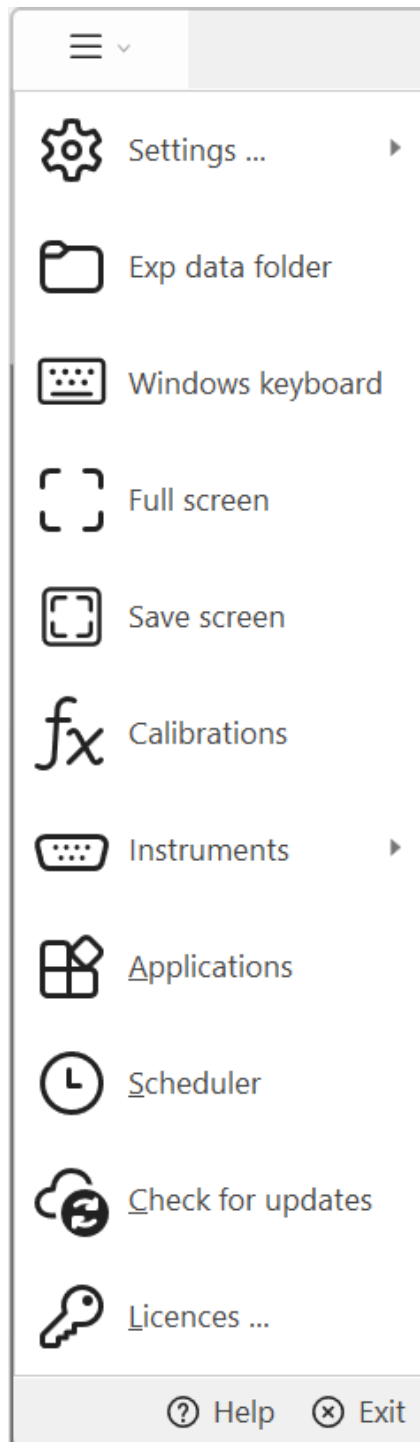


Expedition interface

Expedition uses Microsoft's Office Fluent User Interface which will be familiar to anyone used to products such as Microsoft Office.

Application button and system menu

Some quick, easily accessed functions.



Settings (Ctrl-S)

Accesses Expedition's various [settings](#).

Save settings

Saves various settings in Exp without the need to exit.

Export settings

Exports all Expedition settings saved in the registry to a file for backup etc. purposes. Note that this also exports display states etc.

The data exported is from:

HKEY_CURRENT_USER\SOFTWARE\Expedition

Save user configuration

Load user configuration

These options allow some settings such as boat length, polars, sailchart and advanced optimal routing settings to be saved to an xml file.

This is to ease switching settings if sailing on different boats with the same computer or doing on-shore planning for different boats.

The files are saved in the config_user folder in the Expedition data folder.

Expedition data folder

Opens the App Data folder used to store Expedition data files, such as log, archive and configuration files.

Normally this is <c:\programdata\expedition\>

Archive folder

Several critical files are archived to the archive folder whenever Expedition is started. These are moved to the recycle bin after

30 days.

It is recommended that the archive folder is saved regularly in a secure location (e.g. a USB memory stick).

The files archived include:

- marks database
- database in use
- race tracking
- polar files
- config folder and files
- instruments folder and files
- states folder and files
- stripchart folder and files

Errors folder

Some Expedition errors are logged to the errors folder. These are moved to the recycle bin after 60 days.

Windows keyboard

Launches Windows' on-screen keyboard. This can be useful for touch screen displays.

Also see tablet mode in [system settings](#).

Full screen

Shows a full screen, chart only display.

Save display

Exports the chart display to a file.

Calibration

Interface to calibration functions used by Expedition. For instrument data, calibration of the instrument system is preferred.

Instruments (Ctrl-I)

Displays the [instrument](#) pages.

Also has various menu options:

GPS

Select the [GPS](#) source to use.

Reconnect

Reconnects all network connections. This is the same as on the individual [connection](#) pages. Useful for dropped wireless connections. Also on the [QAT](#).

Send reconnect

Tells any other computers connected with the Expedition instrument protocol to reconnect their network instrument connections.

Number of network connections

Expedition can have several network connections for receiving instrument data over. Most users will only require 1 (the default value). For practicality, the number is limited to 8.

Marks to instruments

Sends mark and route information to instruments where applicable.

Instrument specific options

Shortcuts to instrument system settings. Depending on the instrument types selected, some of these options may appear.

Start radar

Start radar transmission

Stop radar

Stop radar transmission

AIS

Applications

[Stripchart](#)

[Garmin2Exp](#)

[Cosworth2Exp](#)

[Navico radar](#)

[Dfw2Exp](#)

[Expedition System Restore](#)

Scheduler

User controlled timer for various functions.

Position reports

Monitor folder for Expedition/Volvo/MaxSea position reports

Monitors a selected folder for Expedition, Volvo or MaxSea type [position report](#) files.

The first line must be EXPEDITION or POSRELEASE.

Replaces the old Volvo SatC application, originally designed to detect [position reports](#) received by an Inmarsat C terminal in the Volvo Ocean race.

Wx to grib

Creates grib file data from wind data from all boat, weather boat or AIS data.

Check for updates

Checks www.ExpeditionMarine.com to see if there is a later release available.

Also see the [system settings](#).

Licences

Opens the licence window.

If Expedition is licensed, the text will say 'Licence OK'.

If the computer is not connected to the internet, please send the site code (eg 1234-5678) to info@expeditionmarine.com or your reseller. Please do not send a photo or screen image.

Data folder

Opens the data folder as above to copy a key file to.

Get key

Gets a key from the Expedition internet server if one has already been issued for the computer.

Request key

Sends a licence key request email for the computer.

Help

Opens Expedition's help documentation.

Exit

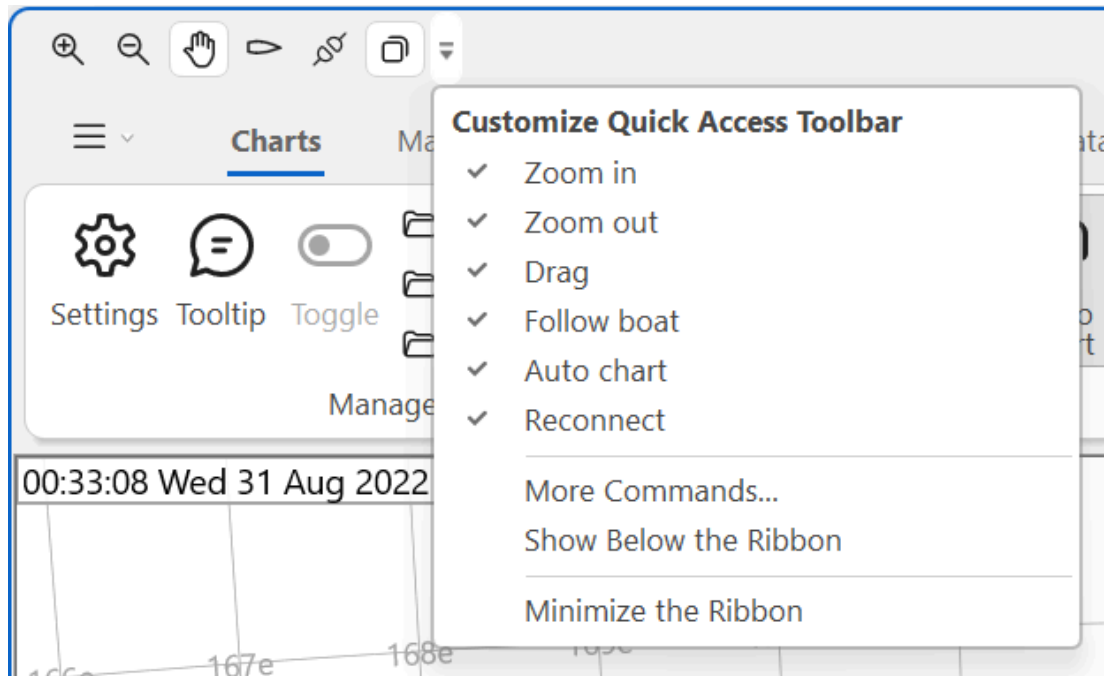
Closes Expedition.

Quick access toolbar

The Quick access toolbar (QAT) is just to the right of the Application button.

The QAT can be customised to show any button or ribbon pane by clicking on the preset shortcut or 'More commands' on the pop-up menu.

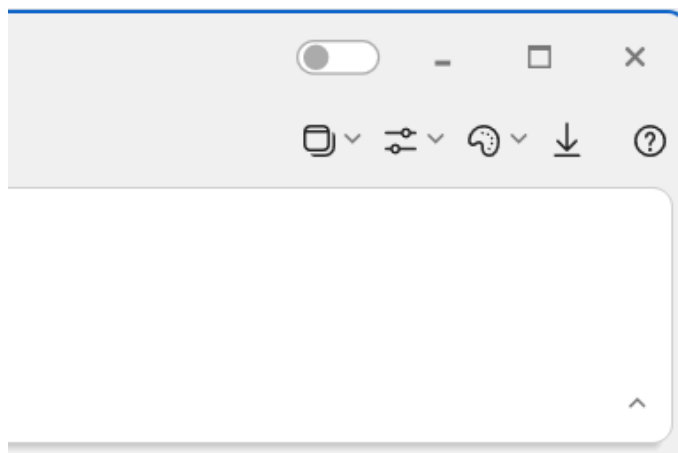
Selecting 'Minimise the ribbon' hides the ribbon. It will appear when selecting a ribbon menu item.



Simple-classic switch

Toggles the Simple ribbon setting, a simple and small ribbon with a caret button at the bottom right of the ribbon.

This option is also available in Expedition's [System](#) settings.



Download

Toggles the [download manager](#) window.

Data downloads can be cancelled or resumed here.

Windows

These menu items toggle Expedition's [windows](#) on and off.

Lock panes

Locks [number boxes](#) and most panes.

States

The position of all [windows](#), contents of the [number boxes](#) and QAT etc. can be saved as 'Display states'. This enables easy switching between configurations. The active display state will be displayed at the top of the main Expedition window.

Display states available

Default display state

Pre-start display state

Race display state

Inshore display state

Offshore display state

Cruising display state

Save

Saves any changes to the display state currently being used.

Save as ...

Allows the current display state to be copied and saved as another.

Delete all

Deletes the saved positions of windows and toolbar configurations.

Theme

The visual appearance of Exp can be modified to suit the user's preferences. This is purely cosmetic and options include:

Colourful

Dark grey

White

Black

Expedition can also modify the display colours for day, dusk or night conditions.

Note that C-Map 4D does not have a dusk palette.

Day

Dusk

Night



Expedition windows

The Expedition interface allows a lot of customisation as different users will have differing requirements that may vary from time to time.

Available from the button near the top-right of Expedition.

Windows available

AIS & DSC

Displays a list of AIS and DSC targets. The data displayed can be selected on the [AIS & DSC settings](#) page.

Double-clicking on the target list toggles the toolbar on and off. The AIS toolbar allows the AIS targets on the chart to be toggled on and off and has a direct link to the AIS settings.

The targets can be sorted by any of the columns items in the target list by clicking on the column header.

Route

A list of the legs of the route with wind angle, range, bearing etc. data.

GPS

Shows the GPS SV constellation, with GPS mode, satellite signal to noise ratios etc.

Number boxes

There are two [number box](#) windows available. Clicking the window toggles the toolbar on and off.

Number list

List of numbers, similar to the number box windows. Double-clicking on the list toggles the toolbar on and off.

Radar

Displays radar echoes in a simple PPI display. Double-clicking on the radar window toggles the toolbar on and off.

Sail chart

See the [sail chart](#) help. Double-clicking on the sail chart toggles the toolbar on and off.

Sail graphic

A graphical representation of the boat, mark, wind etc. See the sail graphic help.

What-if?

The [What-if?](#) window.

Wind

Plots of TWD and TWS.

May be damped. Use the toolbar to set the damping.

Can also display TWD & TWS from the loaded grib file(s). This is for the current position rather than previous boat positions.

Lock panes

From the Window menu. Locks most panes and also [number boxes](#).

Docking

Windows can be dragged around the screen and left floating or docked to each other or the side of the main Expedition window. As a window is dragged, small controls will appear. Just drag the window to these controls to dock the window to an existing window or other windows. These windows can even be docked as tabs over each other.

The configuration of these windows, Quick Access Toolbar and number window contents are saved as [display states](#).

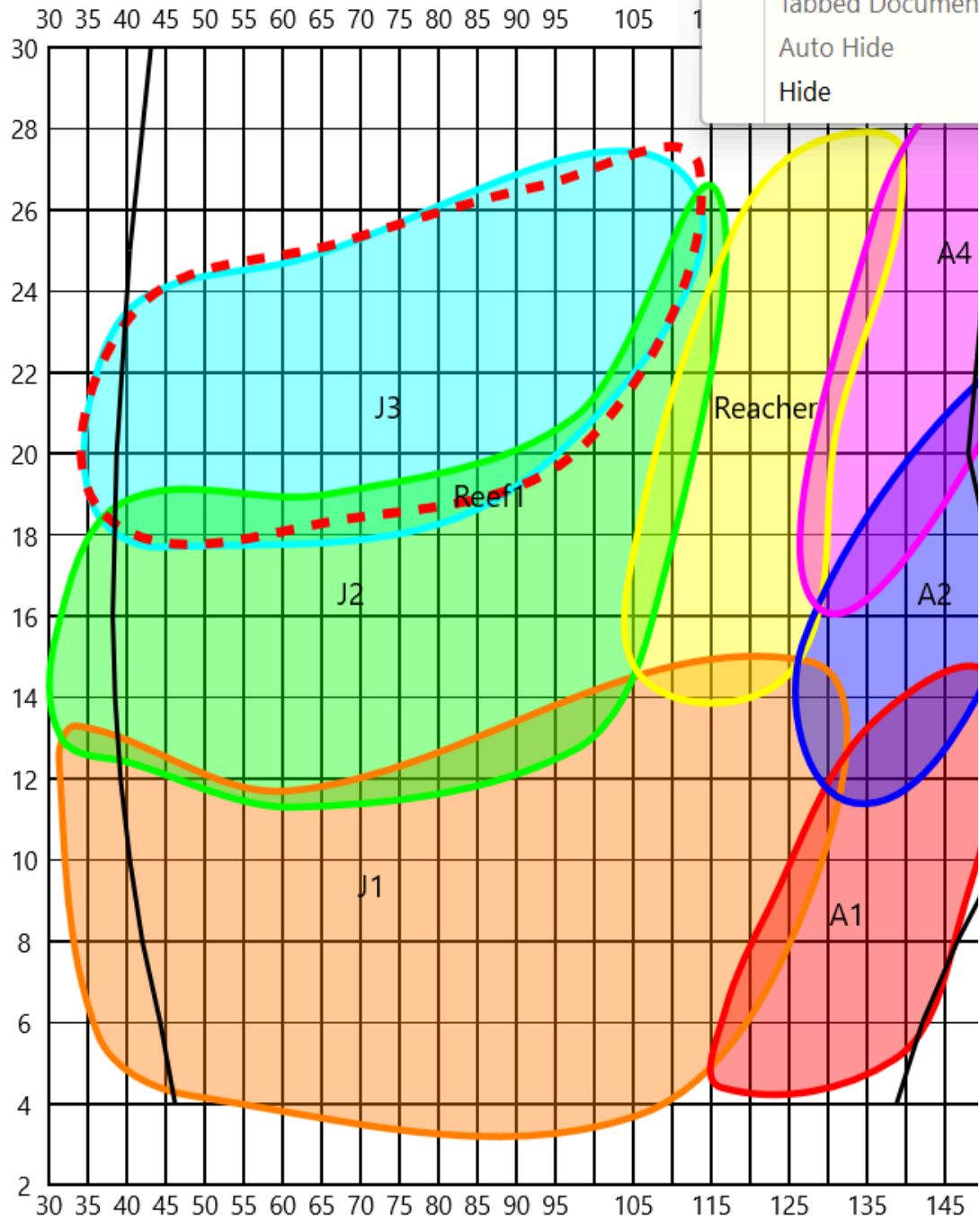
Auto-hide

Expedition windows can be made to auto-hide by clicking on the pin icon on the window header or by right-clicking on the window header and selecting Auto Hide from the pop-up menu. In this case, they appear as small tags on the side of the Exp window and only appear when the mouse pointer is moved over the tag.

C:\ProgramData\Expedition\sails\KNDSailChart.xml



- ☒ Floating
- Docking
- Tabbed Document
- Auto Hide
- Hide





Number boxes

There are three number box windows available in Expedition.

Number box toolbar

To save screen space, the toolbar for the Number-box windows is hidden by default. This can be toggled on and off just by clicking anywhere in the number-box window.

The toolbar is toggled on and off in the number list window by double-clicking on the list.

Adding numbers

The contents of these windows can be edited by clicking on the toolbar button in each.

The largest box/window on the right of edit interface contains all the numbers or channels that Expedition can display. Simply left mouse click on the number you want to have displayed in this large centre box. This selected number will immediately appear in the previously empty left hand box with your boat number (i.e. 0: Depth). All numbers displayed in the left hand box will be displayed on the number box or list window being edited.

Some of the numbers are for specialized uses, so not all of these numbers will be available from your particular instrument system. However, the numbers that are coming from your instruments (i.e. Latitude, Longitude, boat speed, depth etc.) will be available for display on your main chart window.

To remove a number from the display list, simply move your mouse to the left hand box (where the data to display is listed), click on the desired number and then click on the Remove button.

Notes

The individual numbers in the number box windows can also be edited by right-clicking on them.

Number boxes



TWD °M	Depth
337	17.5
TWS	HDG °M
4.1	046
BSP	PolBsp%
5.24	104.0
SOG	COG °M
4.86	064
Tm on P	Tm on S
03h 11m	41m 40s
MkBrg °M	MkRng
007	12.62
MkTwa	NxtMkTwa
-34	-116
Set °M	Drift
264	0.1
05-Dec-2021	Mark Tm
14:32:45	03h 52m

The displayed numbers can be damped or streaming straight from your instrument system.

Up to ten [alternating](#) channels (i.e. alternating/flashing two different numbers) to be displayed and/or exported into the external channels of your instrument system for display on deck.

See [Appendix A](#) for a list and explanation of Expedition's numbers.

Boat

This drop down window is in the lower left corner of the page. Expedition has the ability and power to display data from up to sixteen boats. This feature is useful for an America's Cup level weather program or a two (or more) boat testing program. Most users will only have data coming in from their own instruments and therefore should have Boat 0 selected.

Unless you are running a fancy multiple boat system, your boat is Boat 0 and it should always be selected.

Damping

Damped vars will be *italicised*.

The system damped value or a longer period may be selected. The amount of system damping to be applied for particular function/number is set on the [User settings](#) tab.

Note that some vars cannot be damped. For example it makes no sense to damp time to layline.

Boat list

As mentioned above, most users will leave this at '0', but if you do have multiple sources of information, first select the desired *boat number* in the drop down window before you select the number or channel you want displayed.

Dragging a number box

Number boxes can be dragged around the number box window by left-clicking on them and dragging.

A number box may be deleted by dragging off the number box window.

Right click to change an individual number box

Any individual number box settings, including background colour, can be changed by right-clicking on it.

Lock

Locks number boxes and most panes.

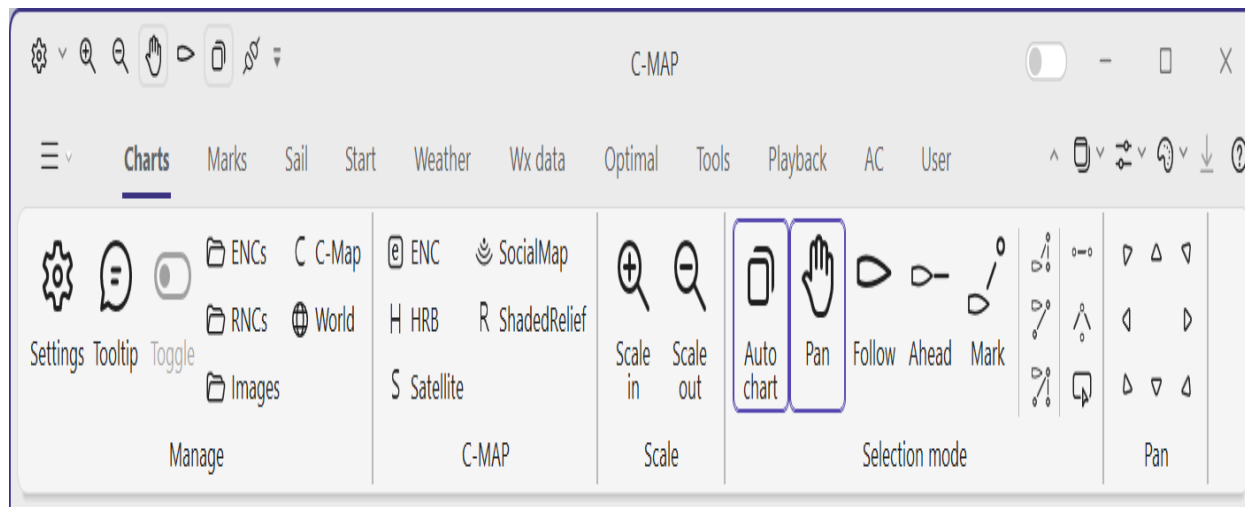
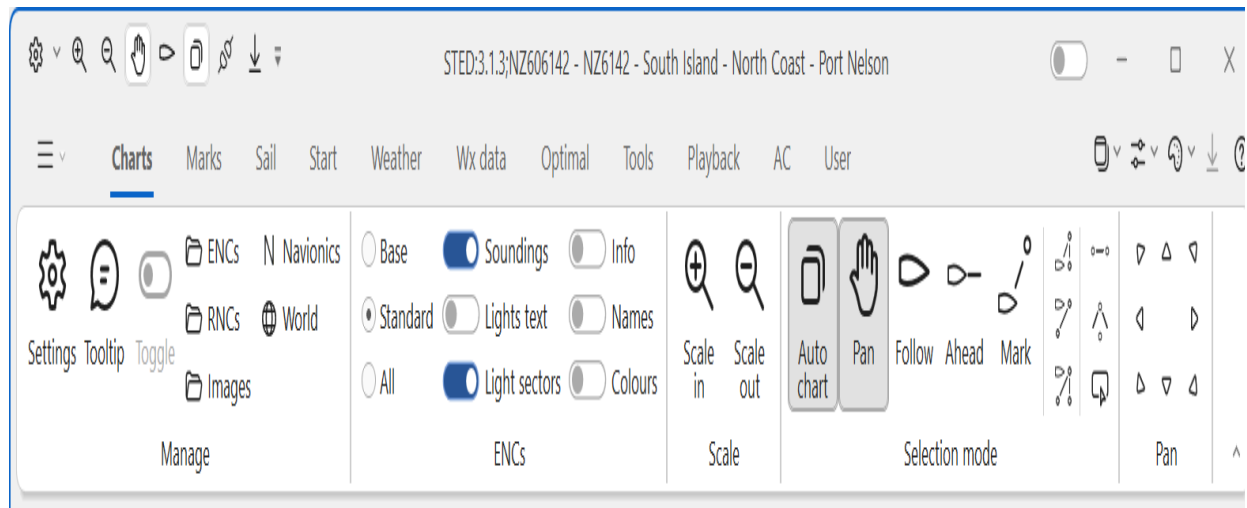
Note about true wind angle in Expedition

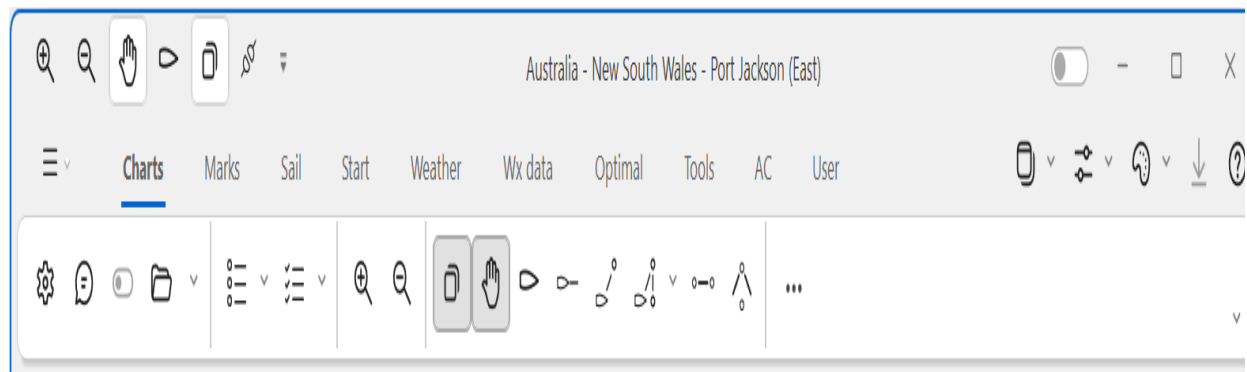
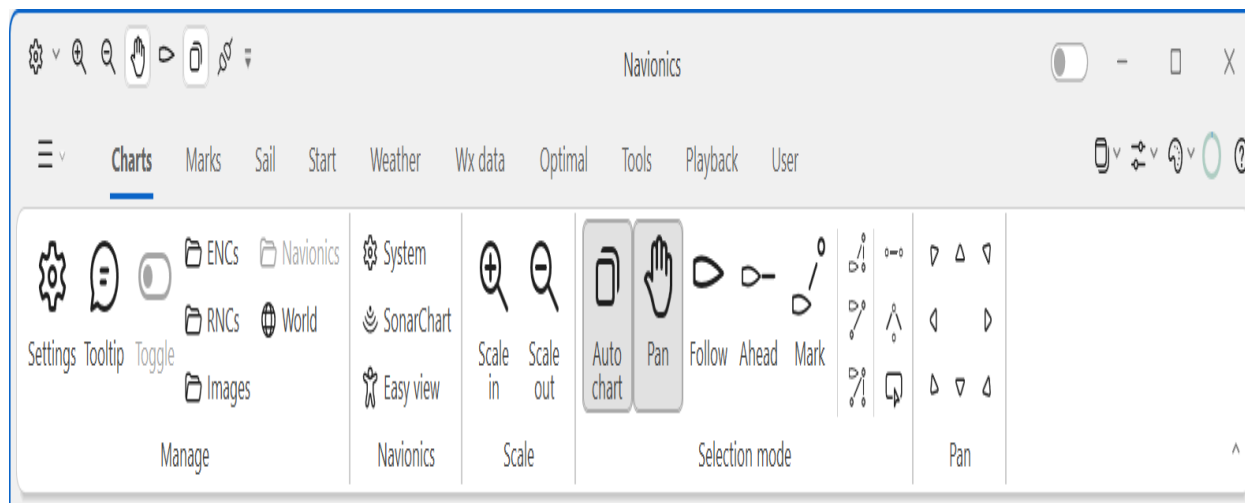
Please note that True wind angle displayed in Expedition and StripChart includes the effects of leeway wherever possible.

Polar data from designers generally present TWA in this manner (TWA to the boat's centreline plus leeway equals TWA track) as do Ockam and modern B&G systems. For systems that do not add in the effect of leeway into TWA, leeway is added to the TWA received from the instrument system wherever possible. There are several benefits to this. For example, calculations of VMG have to include leeway and it makes the optimal routing function easier to use. This is why you may notice that Twa as reported by Expedition may be a few degrees wider than Twa reported by the instruments.



Charts ribbon





Right-click menu options

On the **pop-up** menu, use Open best chart to select the best chart available at the mouse pointer or use Open chart to select any of the available charts at the mouse pointer.

Manage

Settings

Chart [settings](#) is used to manage how charts are displayed in Expedition.

Toggle

Toggles the display between the active chart and an image.

ENCs

ENC chart [management](#) is used to manage and select ENCs.

RNCs

RNC chart [management](#) is used to manage and select raster charts such as BSB, Maptech and NV Charts.

Images

Image [management](#) is used to manage and select raster images such as Squid satellite images and .bmp, .jpeg etc. images.

World chart

Selects the default world chart.

Navionics

Selects [Navionics](#) charting.

C-Map

Selects C-Map charting.

ENCs

Only shown if using ENCs.

These are also available in ENC [settings](#).

Base

Standard

All

Toggle between levels of chart display information.

Soundings

Adds soundings and depth contours to the Standard ENC display mode.

Lights text

Toggles lights text.

Light sectors

Toggles light sectors.

Info

Shows chart information objects on the chart.

Names

More names etc. text.

Colours

ENCs default to many structures being solid black.

If they have a colour attribute, this option displays them with that colour.

Mainly affects beacons, buoys and top marks.

Navionics

Only shown if using [Navionics](#) charting.

System

Navionics settings and cartridge management functions.

SonarChart™

Toggles Navionics SonarChart™ display.

Easy view

The Easy View feature allows users to magnify icons and texts on the map to provide better reading.

When Easy View is turned ON, the chart texts and objects on the screen will be larger.

C-MAP

Only shown if using C-Map charting.

C-MAP

Selects C-MAP charting is not in C-MAP mode.

If in C-MAP mode, shows C-MAP chart areas available and provides access to C-MAP chart management tools. For [MAX](#), this is a shortcut to NT Selector. For [4D](#), this provides access to My Maps and the C-MAP Web Store.

For C-Map 4D, this will also indicate whether the OpenGL to Expedition mechanism is accelerated.

Advanced settings

More C-MAP options.

Safe route check

See the safe route check [help](#).

C-MAP 4D

Raster charts

Toggles raster chart mode.

Satellite

Toggles satellite image mode.

Scale

Scale in or out. If Auto-chart is selected, this will automatically select the next scale chart.

These functions can also be performed using the + and - keyboard keys, the right click drop down menu or the mouse scroll button.

Scaling in can also be accomplished in the Zoom+centre mode by holding the left mouse button down while dragging the mouse pointer. This scales in one level.

Selection mode

Controls how Expedition selects and centres charts. Note that the chart cannot be manually panned if one of the automatic chart display functions such as 'Follow boat' are selected.

Auto chart (also known as Open Best Chart)

This function only applies to scaling in and out and in the Follow boat and Centre ahead of boat modes.

With Auto chart mode selected, the next higher or lower scale chart will be selected when scaling in or out. If this mode is not selected, scaling in or out will display the current chart at different scales. See the help on chart enhancement.

This is ignored when panning.

It is recommended that this mode is selected for most use.

Navionics note

Auto-chart is not available with Navionics charting with Follow boat mode selected.

C-Map note

This mode also influences C-MAP MAX chart behaviour. With it unselected, you can scale in and out on the same chart level.

Pan

In this mode, click on the chart and drag in the direction to pan. This technique will be more familiar to Google Earth and Photoshop users and can be easier on touch screens.

Follow (boat)

Automatically centres the chart on the boat.

Ahead

Selects the best chart to display the boat and predictor line.

If the chart pans around too much because of frequent changes of heading, de-select this feature.

(Boat to) mark

Selects the best chart to display the boat and the active mark of the route.

Centre on active leg

Selects the best chart to display the boat and the active leg of the route.

Centre on boat, active mark and next mark

Selects the best chart to display the boat, the active mark and the following mark.

Centre on active route

Selects the best chart to display the boat and the active route.

Zoom + centre

The chart can be re-re-entered by clicking the left mouse button. This option is preferred by some users as it can be more efficient. This mode is can also be used to scale in.

Start display

Race display

This is an un-cluttered display mode with no charting and scaled to fit the start line or course to the screen.

After the start, it will display the active mark and the boat.

Pan

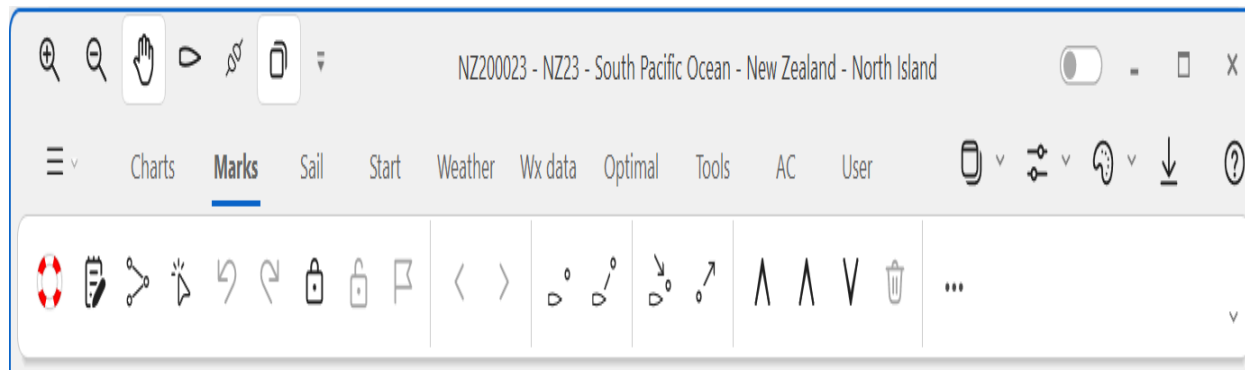
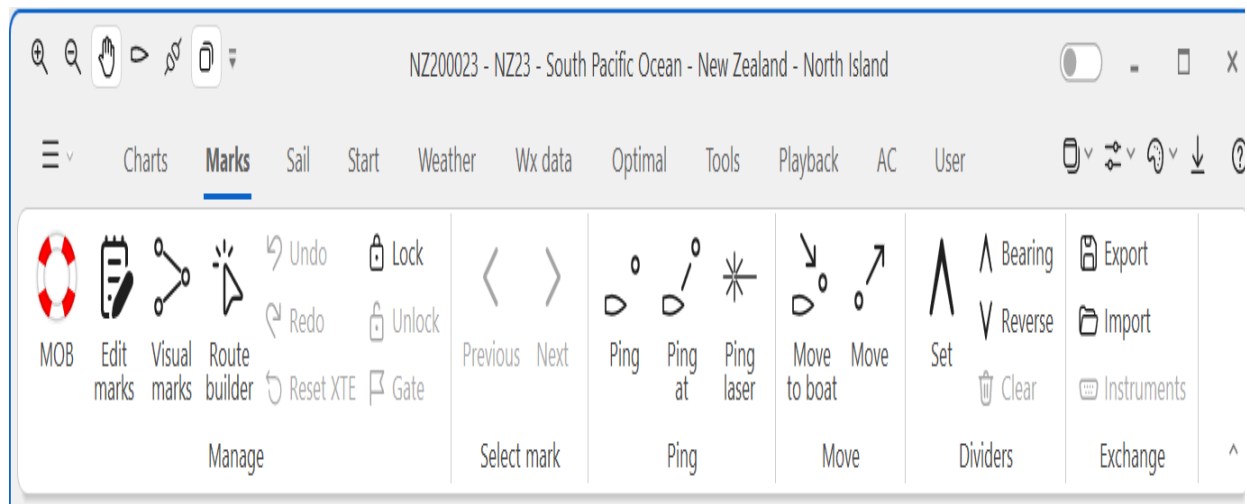
Pans the chart up, down, left or right.

Also ctrl-up, down, left and right.

See also [auto-pan](#).



Marks ribbon



Manage

Man over-board (F2)

See the [Man overboard](#) help.

Edit marks (Ctrl-M)

[Mark management.](#)

Visual marks

Sets the [Visual marks](#) route creation mode. This is a quick and easy way to graphically set a new route.

Route builder

Enables easy creation of a route from marks displayed on the chart as 'always drawn'.

To use, select this mode and then select each mark in turn. Options will include leaving each added mark to port or starboard or to create a new route.

When finished, deselect the Course builder mode.

Undo

Reverses the last mark or route modification.

Redo

Redo a mark or route modification that was undone.

Reset XTE

The Reset XTE option resets the preceding position for the XTE calculation to the boat position when using a boat position rather than the previous mark for the XTE calculation.

See also the XTE calculated between marks option in [System settings](#).

Lock

Locks all marks in the active route.

Unlock

Unlocks all marks in the active route.

Gate

Toggles between the active mark and a gate mark for calculated numbers (laylines, mark bearings, distances and times etc.).

Select mark

Next mark (right key)

Select the next mark as the active mark The right arrow is a keyboard shortcut to previous mark.

Previous mark (left key)

Select the previous mark as the active mark The left arrow is a keyboard shortcut to previous mark.

Left and right keyboard arrows are short cuts to these functions.

Ping

Ping

Set a mark. Defaults to boat position.

Ping at boat

Set a mark at the current boat position - either at the bow or GPS based on the Set start at bow (else at GPS) function in [Racing](#) settings.

This is helpful if you want to ping a mark and then later right click or use mark management and add it to a route. For example, I often use this function to ping gate marks instead of using the mark gate function.

Ping at

Creates a mark at a given range and bearing from the boat. Option of setting the created mark to be the active

mark/route or as an always drawn mark to be added to the route (e.g. with the right click insert mark function).

Ping laser

Creates a mark at the last laser position.

Move

Move to boat

If it isn't locked, moves the active mark to the boat's position - either the bow or GPS based on the Set start at bow (else at GPS) function in [Racing](#) settings.

Move

Moves the active mark.

Dividers

Set dividers

The dividers tool allows the user to draw and calculate the range and bearing between points on the main chart window. It is very useful in race planning or coastal navigation. The number of dividers (actually a range and bearing between two points) displayed is unlimited and they

can be all cleared at once using the Clear button or the [right-click pop-up](#) menu.

Bearing

Set dividers given range and bearing from the boat. This tool can be used with running bearings on a mark to refine it's position.

A W/L race is a good example of the use of this tool.

The windward mark can be set from the range and bearing from the race committee using the [Create WL course](#) function. However this is only approximate and once rounded, the mark can be dragged to its correct position.

Before arriving at the mark, its position can be refined using this tool from different places on the race course - the actual mark position is the intersection of the divider lines drawn using this tool.

Reverse

Reverses bearings displayed on dividers.

Clear

Clears all dividers.

Exchange

Instruments

Sends mark and route information to instruments (NMEA 0183, NMEA 2000, Nexus FDX etc.).

Note that PGN 129285, 130067 and/or 130074 must be enabled to send for [NMEA 2000](#). Different manufacturers may support different PGNs for this. For example, some Navico plotters support 130074.

Also available from the [application menu](#).

Export

Export marks and routes to various file formats.

GPX

Exports Expedition's mark and route database to a GPX eXchange format (*.gpx) file for use in a range of other mapping and GPS products.

CSV

Exports marks to a CSV (*.csv) file.

The format is

mark name, latitude, longitude, mark group name,
range circle, comment

Export

Import marks and routes from various file formats.

GPX

Imports mark and route information from a GPX file.

Expedition

Imports marks from an Expedition marks.dat file.

Dfw

Imports marks from a Deckman marks .d file.

CSV

Imports marks into Expedition from a CSV (*.csv) file.

The expected format is

mark name, latitude, longitude, mark group name,
range circle, comment

The mark group name, range circle and comment are optional.

For example:

North Head,50 42.69n,01 35.52w
North Ryde Middle,50 46.61n,01 14.31w
North Sturbridge,50 45.33n,01 08.23w
Paul Jackson,50 44.33n,01 28.19w
Peel Bank,50 45.49n,01 13.34w
Peters & May,50 46.13n,01 22.19w
Prince Consort,50 46.42n,01 17.59w
Quinnell,50 47.07n,01 19.88w

The latitude and longitude could also be in decimal degrees. For example:

My mark,50.1234,-1.0234

Cowes

Imports a race course into Expedition from a text (*.txt) file in the format used by Cowes Weeks SMS course messages.

The format is

IRC 3; Saturday; 8Y(RS); 8S(RS); 44(RP); 31(RP);
3H(RP); FINISH at 80

Class;Day of week;
markref(rounding);markref(rounding);...markref(rounding);Finish at markref

First field: name of class

Second field: Day of week

Third to n-1th field: mark of the course

Nth field: finish mark in the format shown

Class and day are usually defined by the organisers.

Each mark is shown with the standard 2 character reference (per the Cowes Week SIs, which include some Cowes Week specific marks) together with (in brackets) a rounding/passing designator as follows:

The markref must be two characters and the corresponding mark in Exp's database should be the same two characters or start with the same characters and a space. For example

4Y NRYDM

Rounding may be

(RP) - round to port

(RS) - round to starboard

(PP) - pass to port

(PS) - pass to starboard

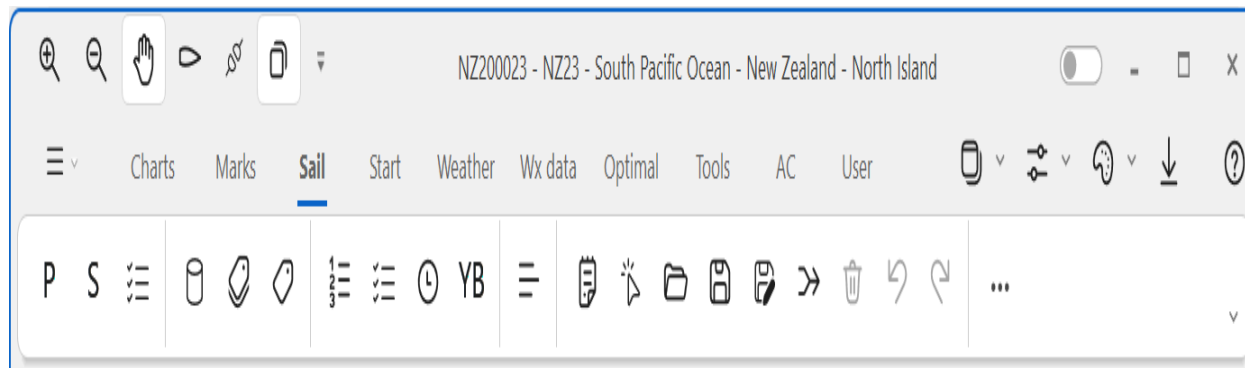
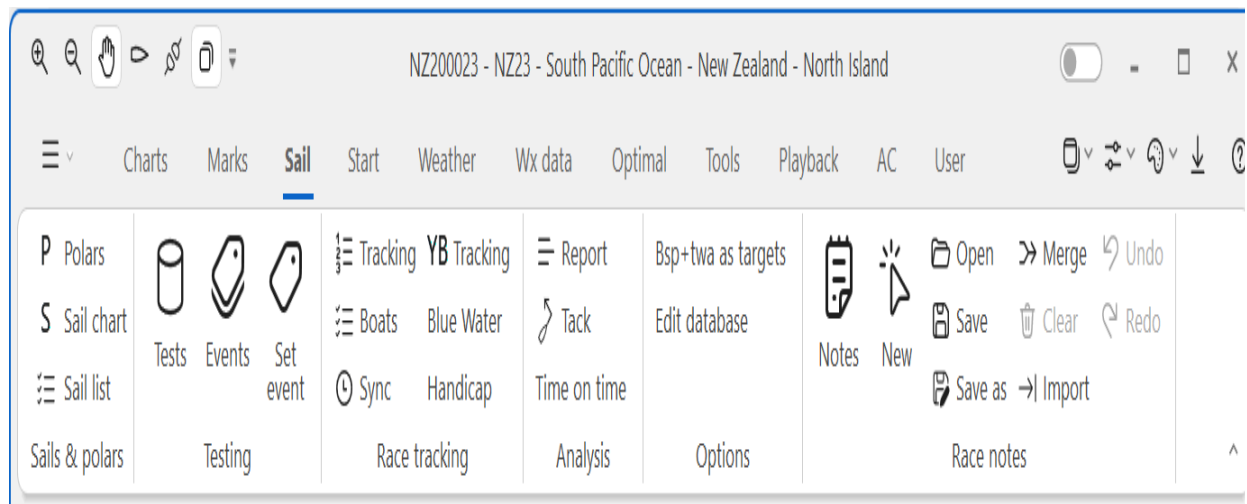
(GT) - gate mark (sets as sail to mark as we don't know the gate details)

Expedition also supports the 2013 roundings:

(LP) - leave to Port

(LS) - leave to Starboard

The starting line/CV is not defined in the SMS message file but defined manually using the Start Panel.



Polars

Polars (Ctrl-P)

See [Polars](#).

Sail chart

Displays the active [Sail chart](#).

Sail list

Allows the [Sail list](#) used in Exp to be edited.

Testing

Tests (Ctrl-T)

Displays test data stored in the [database](#) for selection to plot or exporting from Expedition.

Also available with the Ctrl-T keyboard short-cut or from a polar edit window.

See the [sail test analysis](#) help and *Saving tests* in the [Stripchart](#) help.

Event list

Expedition's event database.

Set event (Ctrl-E)

Set an event. Note that only sails selected in [Sail settings](#) are displayed in the drop lists.

Racing tracking

Race tracking

Accesses Expedition's [race tracking](#) functions.

Yellow Brick

Downloads and imports race positions from [YB Tracking](#).

Note the YB Tracking button has a one-click sub-item that can be added to the [Quick Access Toolbar](#). This will use the last race name and other settings used for YB Tracking downloads.

Custom

For other race position file downloads.

Expedition will automatically recognise Expedition, MaxSea and Volvo position report formats.

URL (scheme, host, path)

For example

`ftp://ftp.me.com`

`ftp://ftp.me.com/dir`

<https://yb.tl>

<https://yb.tl/Links/Fastnet2023>

Username and password

If required for ftp.

Files

The file(s) you wish to download.

For example, using the Fastnet example from
<https://yb.tl/Links/Fastnet2023>

fastnet2023-expedition.txt

fastnet2023-boatids.txt The file name may include a wildcard for ftp. This currently only downloads the newest file found, but may be extended for multiple files when needed. For example:

vendeeglobe_posreport_*.txt

Blue Water

Downloads and imports race positions from [Blue Water Tracks](#).

Vessel Finder

Downloads and imports AIS positions from [Vessel Finder](#) as race positions using the vesselslist method.

Sync time

Sets the display time to the time of the latest position report for the reference boat.

Boats

Edits boats used in [race tracking](#) and the [handicap](#) calculator.

Handicap

[Handicap](#) calculator.

Analysis

Report

Recent sailing performance, implemented for a Volvo project.

Tack analysis

For analysing tack performance / cost. Also populates the 'Tack/gybe loss time' and 'Tack/gybe loss metres' channels.

Miscellaneous

Inst targets

Instructs Exp to use damped bsp and twa instrument values instead of targets to calculate laylines.

MMAG

Interface to the Marine Mammal Advisory Group (MMAG)
[Marine Strike Log](#).

To use this feature, a user account is needed.

Get strike and observation data

Click the MMAG button.

The reports will be displayed on the if selected in
Display [settings](#).

Moving the cursor over a strike icon will show a tooltip
with details of the strike or observation.

Settings

From the sub-menu on the right of the MMAG button.

User name and password.

Days is the period for which to retrieve strike data.

Create

From the sub-menu on the right of the MMAG button.

Allows the user to send a strike or observation report to MMAG.

Edit database

Race notes

Race notes are annotations on the chart. They can be used for race planning and can be used to set areas to avoid or to reduce speed when using route [optimisation](#).

The right-click pop-up [menu](#) can also be used to edit a race note.

There is also a number channel 'GPS time to race note', which is the time to reach the nearest race note area boundary based on Cog and Sog.

Notes

Lists all race notes.

Add

Creates a new race note. Select this, then click on the chart to create a new race note area.

Open

Loads a race notes file.

Merge

Merges data from another race notes file.

Save as

Saves a race notes file.

Clear

Clears all loaded race notes.

Import

Imports race note areas from a text file. The csv format is:

Area name 1

lat,lon

lat,lon

lat,lon

lat,lon

Area2

lat,lon

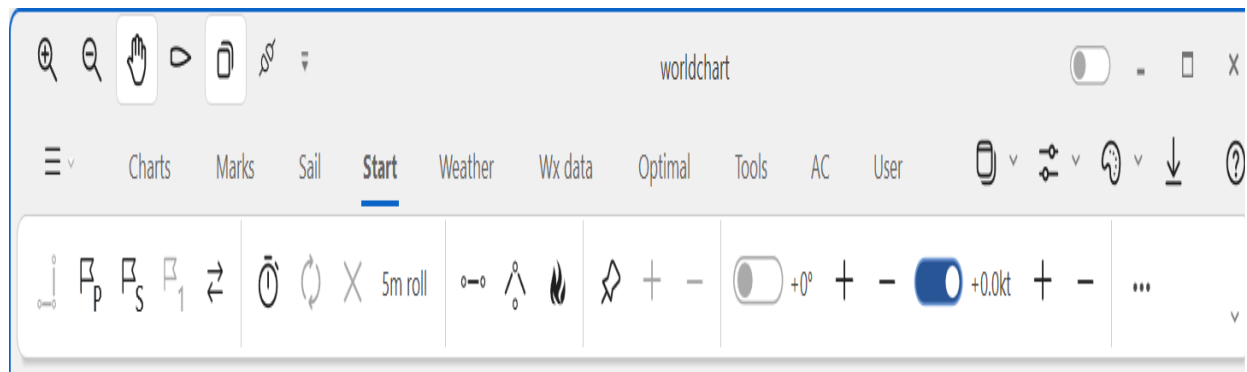
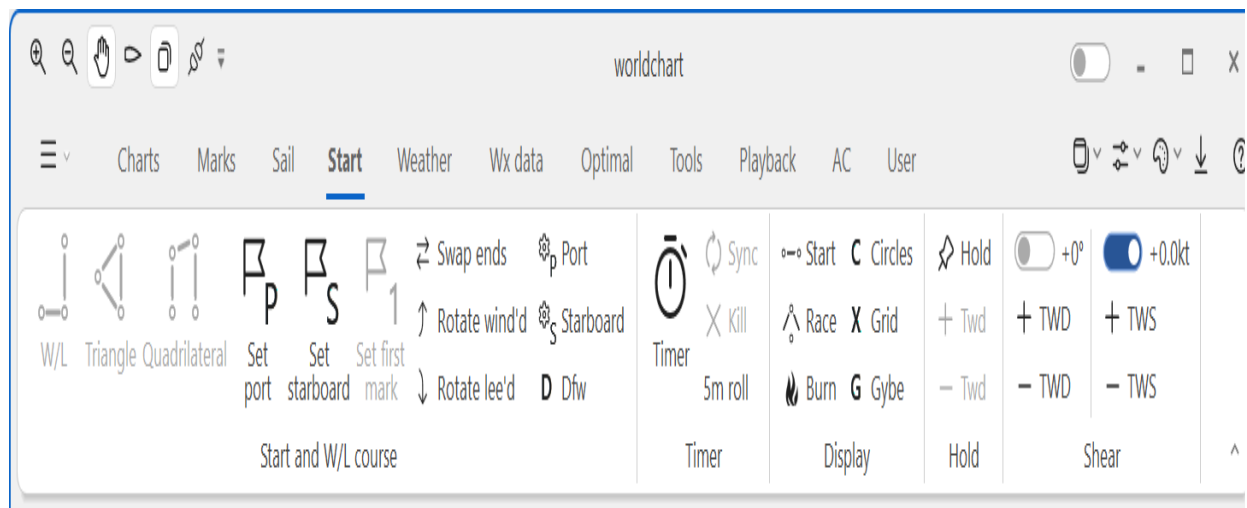
lat,lon

lat,lon

lat,lon

...

Start ribbon



These tools can be useful at the start of any race and for windward/leeward style buoy races.

Start and W/L course

W/L

This is the quickest way to configure a standard windward/leeward course. It easily allows you to configure a course where the leeward mark (or gate) is upwind of the starting line.

Triangle

Olympic triangle course wizard.

Quadrilateral

Quadrilateral course wizard.

Set port

Sets the port end of the starting line to the position of the boat.

Set starboard

Sets the starboard end of the starting line to the position of the boat.

Select whether to set start marks at the bow or the GPS on the [user settings](#) page. If pinging the mark at the bow, this requires the distance between the GPS and the bow to also be entered correctly.

The position used will be the last position received when the OK button is clicked.

Set first mark

Adds a mark at a range and bearing from the starboard end of the line to the beginning of the active course.

Swap ends

If you pinged them wrong?

Rotate wind'd

Rotates W/L marks to allow for a course change partway through the race.

Rotate lee'd

Rotates W/L marks to allow for a course change partway through the race.

If the race committee announces a change in the axis of the windward mark when you are rounding the first leeward mark simply press rotate windward mark and enter the new axis/heading in the dialogue box that pops open and then click OK to accept your change. Keep in mind that with these course change buttons, the course rotates around the

leeward mark (or the windward mark), not the starting line (unless you configured the course so the leeward mark is at the starting line).

Port options

Starboard Options

Provides options for each end.

Laser port

Sets the end of the starting line to the position that most recently *pinged* with the Laser gun.

Nudge upwind

Nudges the end of the start line upwind or downwind in metres or system distance units. Positive values nudge the mark upwind, negative values nudge it downwind.

From starboard

Set the end of the start line given a range & bearing from the starboard end. Range is in metres or system distance units. Bearing is in degrees magnetic or true, depending on the system settings.

From port

Set the end of the start line given a range & bearing from the port end. Range is in metres or system distance units. Bearing is in degrees magnetic or true, depending on the system settings.

Timer

Timer

The start time controls are used to control Expedition's starting line functionality by setting the time to go before the start. Take note that the pre-start countdown starts when you select OK on the pop-up window that appears.

Start time

Sets the start time to any time.

Time to gun

Sets the time to the gun to any number of minutes and seconds.

Sync start time

Rounds the count-down timer (time to gun) to the nearest minute.

Kill

Kills the start countdown timer.

Roll

Resets the countdown timer at the gun. Defaults to 5 minutes, but may be set in [Racing & Start settings](#).

Display

Start display

Shows the boat and start line.

Race display

This is an un-cluttered display mode with no charting and scaled to fit the start line or course to the screen.

After the start, it will display the active mark and the boat.

See also the [chart](#) panel help.

Burn

Switches from time to go to the line, line ends etc. to time to burn to the line etc.

Circles

Toggle the [start range circles](#) on and off.

Grid

Toggle the [start range grid](#) on and off.

Gybe

Shows gybes to the line. See also the start [settings](#).

Time ratios

Displays start times as a ratio of time to gun.

For example, time to line is displayed as $\frac{\text{time to gun}}{\text{time to line}}$.

Hold

Hold wind and/or current. Often, especially in the pre-start, it may be desirable to fix the wind and/or current to set values. The hold button offers a quick way to hold the twd, tws, set or drift to user-defined values rather than from the instruments.

Exp can release these settings at the gun - See Release hold wind and current at start in [Start settings](#).

Shear

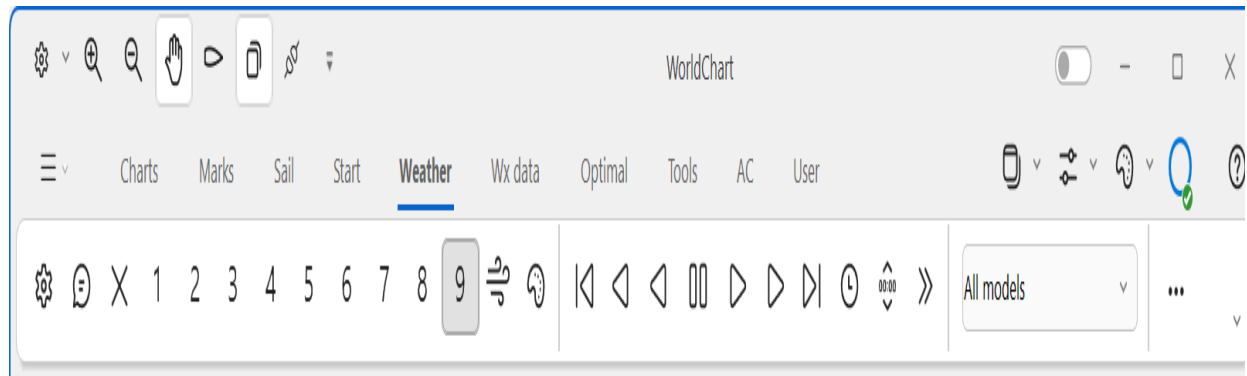
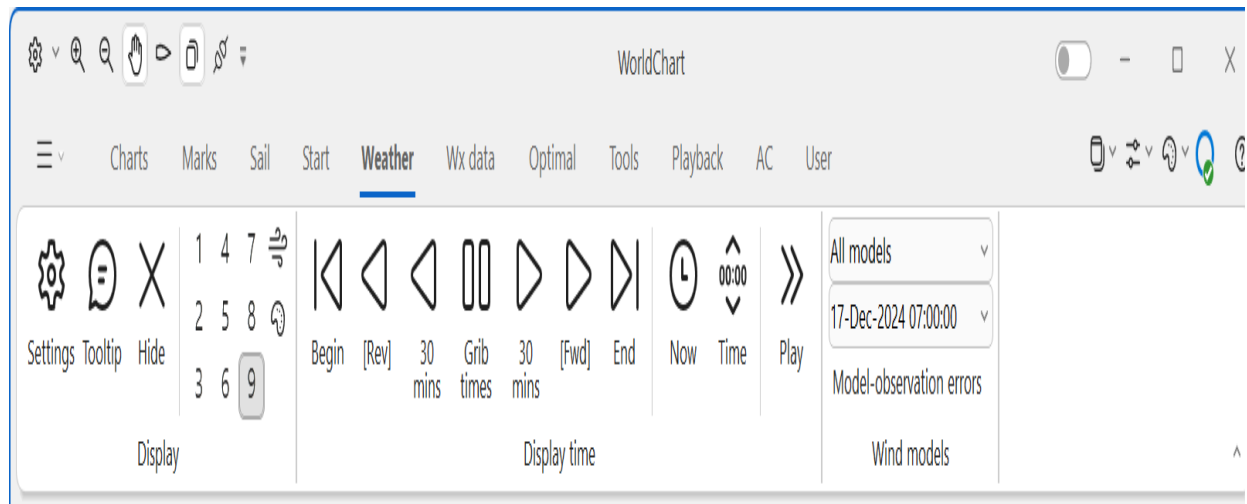
Twist

Adds to TWD for laylines etc.

Gradient

Adds to TWS for laylines etc.

Weather ribbon



Display

Settings (Ctrl-W)

Shortcut to the [weather settings](#) page, also available from the keyboard as *Ctrl-W*.

Tooltip

Toggles the weather tooltip on and off. The weather tooltip can be used to examine forecast weather and tide conditions anywhere on the screen.

See also the weather table from the [pop-up](#) menu.

1 2 3 4 5 6 7 8 9

Weather display states. Allows easy switching between different weather display settings.

Wind flow

Animated wind flow. The colour can be set in Weather [settings](#).

Ensemble settings

Set colour for each weather model. Also sets the default colour for [ensemble optimal routing](#) .

Hide

A convenient one-click control to globally turn weather display on or off. This is very useful if you want to quickly 'de-clutter' the screen.

Display time

The keyboard up and down arrows are shortcuts to the forward and reverse buttons. The keyboard *space bar* toggles between stepping by times in the grib file or the user defined interval (See *xx minutes* below).

The time is displayed in the upper left corner of the chart window.

Begin

Sets the display time at the first time in the grib file(s).

Reverse

Steps the GRIB weather display backward in time. The keyboard down arrow is a shortcut to this function.

xx minutes

Steps the grib display forward or backward in time.

Grib times / xx min

Expedition may interpolate between the steps of the GRIB file or time steps set on the [weather settings](#) page, depending on the selection of the Grib/xx min button.

Forward

Steps the GRIB weather display forward in time. The keyboard up arrow is a shortcut to this function.

End

Sets the display time at the last time in the grib file(s).

Now

Synchronizes the weather display time with that of the computer clock. It will always display the weather for now.

Time

Sets the display time to any user-selectable time.

Play

Animates the weather display.

Wind models

Quickly switch the display between models or ensemble runs loaded.

For known models, the name of the model will be displayed.

If the model is not known, the National Centre and codes for the sub-centre and process will be displayed.

Occasionally the weather model will be depicted as a code.

Most private weather modellers do not have a National Centre code, so invariably they use 255.

Expedition shows the code to discern between models.

However several use the same codes, even within their own products. This means ensemble routing and weather may be unpredictable for such products.

For example, the Squid GCWF model will appear as 255 0 96.

Times

Quickly switch between time steps in the grib file(s).

Model-observation errors

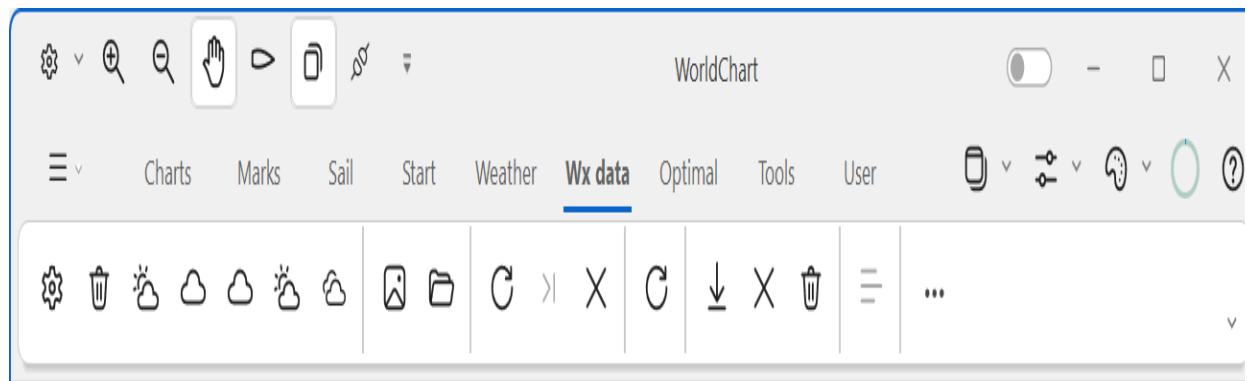
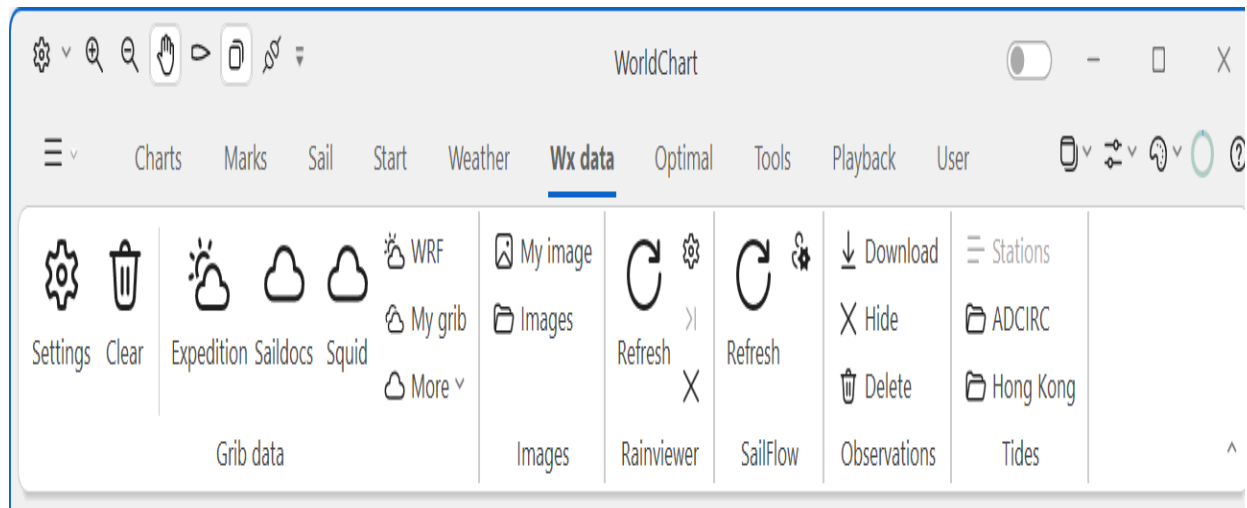
Compares loaded weather models with available wind observations, ASCAT grib data and damped instrument values.

Smaller RMS errors are good.

The idea is it may be of some use to quickly see which model is closest to the current situation.



Weather data ribbon



Settings (Ctrl-W)

Shortcut to the [weather settings](#) page, also available from the keyboard as Ctrl-W.

Clear

Clears all GRIB and NetCDF data, as in the Clear All button in [Weather data](#) settings.

Grib data

Most downloads messages and progress are shown on the weather ribbon.

Downloads may be cancelled at any time by selecting the Cancel button in the Download section of the ribbon.

Expedition

Interface to the Expedition GRIB [service](#).

Saildocs

Interface for the [Saildocs](#) grib service. Select this, then left-click on the chart and drag the mouse pointer to select an area.

Great Circle - Squid

Interface to the [Great Circle - Squid](#) grib service.

WRF

Expedition WRF [models](#).

WRF model data may also be downloaded by right-clicking on the chart and selecting Expedition WRF from the [pop-up](#) menu.

My grib

User defineable grib file downloader. The files are listed in \config\MyGrib.xml in the Exp program files folder.

MyGrib data may also be downloaded by right-clicking on the chart and selecting the MyGrib from the [pop-up](#) menu.

It may be better to use a dedicated file downloader over satellite connections, depending on circumstances.

Tidetechn

Opens the [Tidetechn](#) interface to tidal and ocean current data from Tidetechn. Select this, then left-click on the chart and drag the mouse pointer to select an area.

Tidetechn data may also be downloaded by right-clicking on the chart and select the Tidetechn from the [pop-up](#) menu.

Images

Images

Image [management](#) is used to manage and select raster images such as from My Image and Squid satellite.

Also see the [Chart ribbon](#).

My image

Downloads and automatically geo-references [images](#), including satellite images and weather charts, from the web.

Rainviewer

[Rainviewer](#) is a very nice service that provides radar images for many locations.

Downloads may be cancelled at any time by selecting the Cancel button in the Download section of the ribbon.

Rainviewer is a free service from Ukraine, so we encourage people to support them with [Patreon](#).

Refresh

Downloads radar images for the current chart area from [Rainviewer](#).

Settings

Select the radar colour palette to download.

Adjust the transparency of the radar images displayed on the chart.

Select the number of hours to retain data.

Download latest image is selected by default. If deselected, Expedition will download available images.

Fast downloads will download lower resolution data in, resulting in smaller downloads.

Latest

Sets the Expedition display time to the last Rainviewer images at the chart scale/level.

Hide

Toggles Rainviewer imagery on or off.

SailFlow

Downloads may be cancelled at any time by selecting the Cancel button in the Download section of the ribbon.

Refresh

Downloads weather observations for the current chart area from [SailFlow](#).

Settings

The SailFlow username and password.

Observations

See also the Weather Observations options in [display settings](#).

Note that the colour of the observations can be determined by the weather 10m wind [settings](#) or the colour specified in display settings.

Download

Downloads observation data from the web from the BOM, NOAA's [NDBC](#) web site, ship and buoy observations as well as 3 hourly land synoptic reports.

Hide

Toggles weather observations display.

Delete

Deletes downloaded observation data to the recycle bin.

Tides

Stations

Access a list of tidal stations currently loaded into Exp. Select any individual station and clicking the *Predictions* button will display a window showing tide heights (or stream) information for that station.

If you don't find a desired location in the Tides section, look in the Secondary Tides section.

ADCIRC

Imports [ADCIRC](#) current model output in fort.64 format. Requires the associated .grd mesh grid file and a user input model run time.

Hong Kong

Imports [Hong Kong Hydrographic Office](#) current model [predictions](#).

The format is

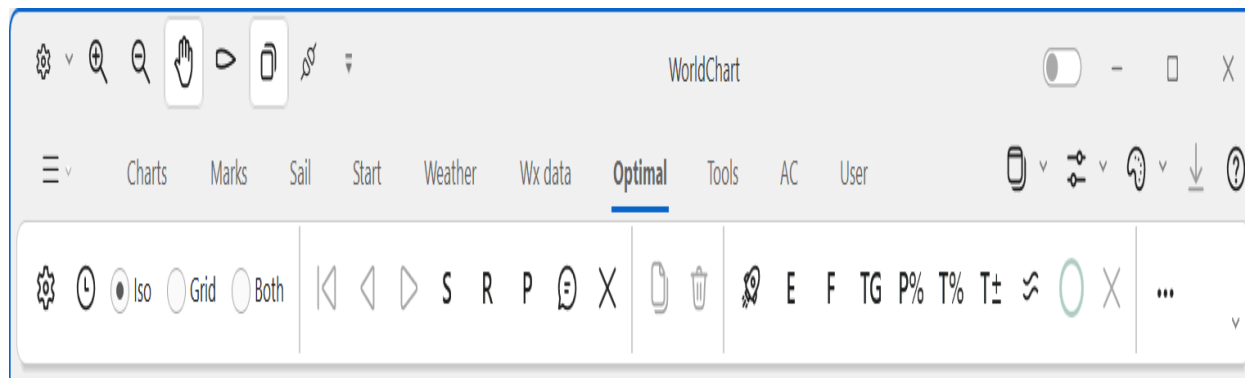
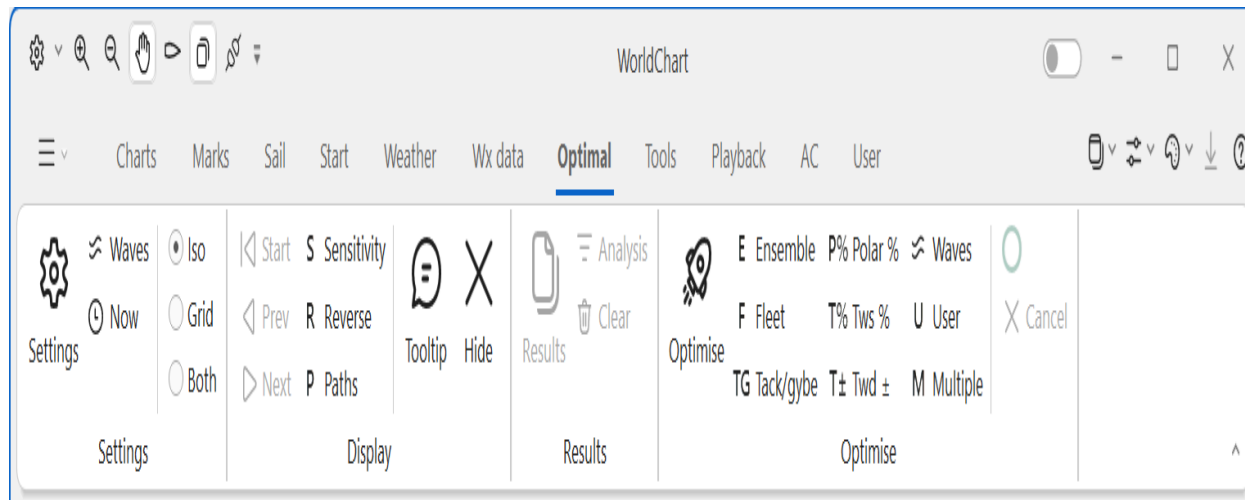
Date,Time,Knots,Degree,Latitude(WGS84),Longitude(WGS84)

2016-07-

19,00:00,0.10,59,22.4618675408,114.3535103735

Times are local (Hong Kong), so select this when prompted.

Optimal ribbon



Settings

Settings

This is a shortcut to the [optimal](#) settings page

Waves

Correction factors (percentages) for waves. Wave heights are in metres. Angles are to the boat. See the optimal routing settings [help](#).

Now

Set manual start time to now, as in [settings](#).

Isochronal or grid based routing

Expedition has two route optimisation algorithms.

The isochronal algorithm should yield a slightly 'better' result as it is a better way of performing the simulation.

The grid algorithm was originally developed for routing studies where it was important to compare similar simulations.

They should yield very similar results, but a user may prefer one or the other or use both for comparison of results.

See the optimal routing [topic](#) and optimal routing settings [help](#).

Display

Sensitivity

Toggles sensitivity display.

Reverse isochrones

Toggles reverse isochrones display.

Paths

Toggles paths display.

Begin

Sets the display time to the start of the optimal routing. Also on the [weather display](#) ribbon.

Previous

Sets the display time to the previous step in the optimal routing.

Also Shift-Dn.

Next

Sets the display time to the next step in the optimal routing.

Also Shift-Up.

Results

Results

Displays [tables](#) or meteograms of the optimised routes.

Clear

Clears all optimised routing.

Optimise

Produces a summary of the optimal routes. This may be exported to a .csv file that may be opened by Microsoft Excel.

Tooltip

Enables a tooltip to show time, speed and weather data over the analysed area. This is for the most recent solution and should be coloured as for the optimised solution.

DNM% is the distance from the next mark as a percentage of the closest point to the mark on an isochrone.

Optimise

Optimise

Computes a single [optimal](#) route. This processing is performed in the background. The results will be displayed on the chart when complete.

Ensemble

Automatically performs route optimisation for each model or ensemble run loaded.

Colours are set by the definitions for the various [weather models](#).

Fleet

Performs route optimisation for boats in the fleet that are drawn on the chart at the last [race tracking](#) time for the reference boat.

Each boat needs to have a polar specified on the [Boats](#) interface.

Note that [All optimal paths](#) needs to be selected to draw the other boats on the chart.

Tack / gybe

Performs route optimisation with the 'Prefer tacks and gybes at routing steps' option off and on.

May be useful for sensitivity decisions.

Polar %

Runs a series of route optimisations at 90, 95, 100, 105 and 110% of the navigation polar.

These are in addition to any global polar scaling.

Tws %

Runs a series of route optimisations at 90, 95, 100, 105, 110, 115 and 120% of the grib wind speeds.

These are in addition to any global wind scaling.

Twd +/-

Runs a series of route optimisations at -10, -5, 0, +5 and +10 TWD offsets.

These are in addition to any global wind offsets.

User wind

Perform a route optimisation with a user defined wind.
Useful for exploring current effects.

Instrument

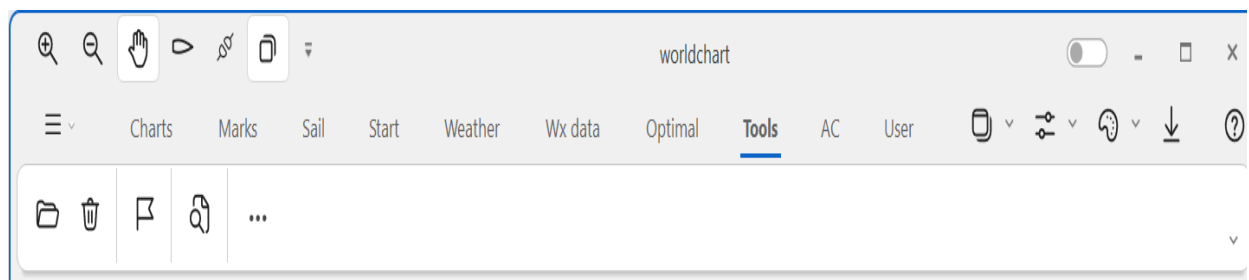
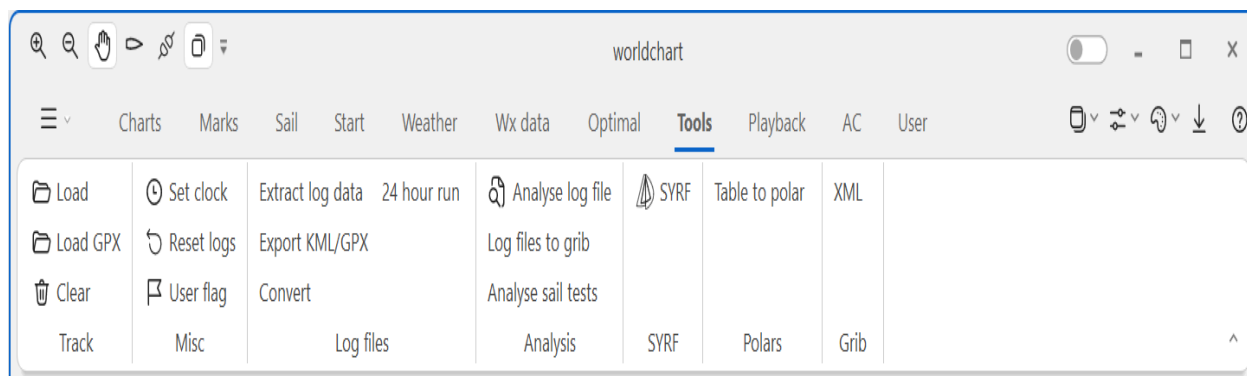
Optimise route with instrument wind.

Multiple

Compute multiple optimal routes with staggered start times.



ribbon



Track

Load

Loads saved boat positions (track)

and currents from a log file. See the help on the [display](#) page for more information.

Load GPX

Loads track data from a GPX file.

Clear

Clears the boat

[track](#).

Misc

Set clock from GPS

Updates the computer clock to the
time from the GPS.

This may not be successful if you
do not have the required privileges on your computer. Expedition must have the
[SE_SYSTEMTIME_NAME](#) privilege. This privilege is disabled by default.

One easy solution is to run Exp
as administrator - right click Expedition.exe and select Run as administrator from
the pop-up menu.

Reset logs

Resets Log Bsp and Log Sog to zero.

User flag

Toggles the ExSysBoolUserFlag system
channel for third party use. May be retrieved or cleared with
[ExpDLL](#). See GetSysBool() in ExpDll.h.

Log files

Extract log data

Extract selected data from a log
file. This can be useful for import into Excel etc.

All data extracts all data into
the log file.

The Euro option replaces decimal
points with commas and commas with semi-colons.

Data may also be restricted to
the times between Stripchart wands, which may also be set from the right-click pop-up
menu. This is useful to extract a section of a race.

Export KML/GPX

Exports any selected log file to
a file in KML for GPX formats. The KML output can be read by
[Google Earth](#).

24 hour run

Finds the maximum 24 hour run from
a series of log files.

As each log file covers 24 hours,
at least 2 consecutive files are needed.

Convert

Converts Yacht Devices Recorder
and NKE or NMEA 0183 format log file to Exp log file.

The Yacht Devices NMEA 2000 recorder
requires PGNs 126992 or 129029 for this conversion.

Analysis

Analyse log file

Automatically extracts

[test](#) style data means from a log file.

Log files to grib

Create grib files of sea temperature,
depth or current set and drift from your log files.

Analyse sail tests

See the

[sail test analysis](#) help.

Polars

Table to polar

Can create an Expedition format

from a regular polar table. This can be useful to estimate the targets.

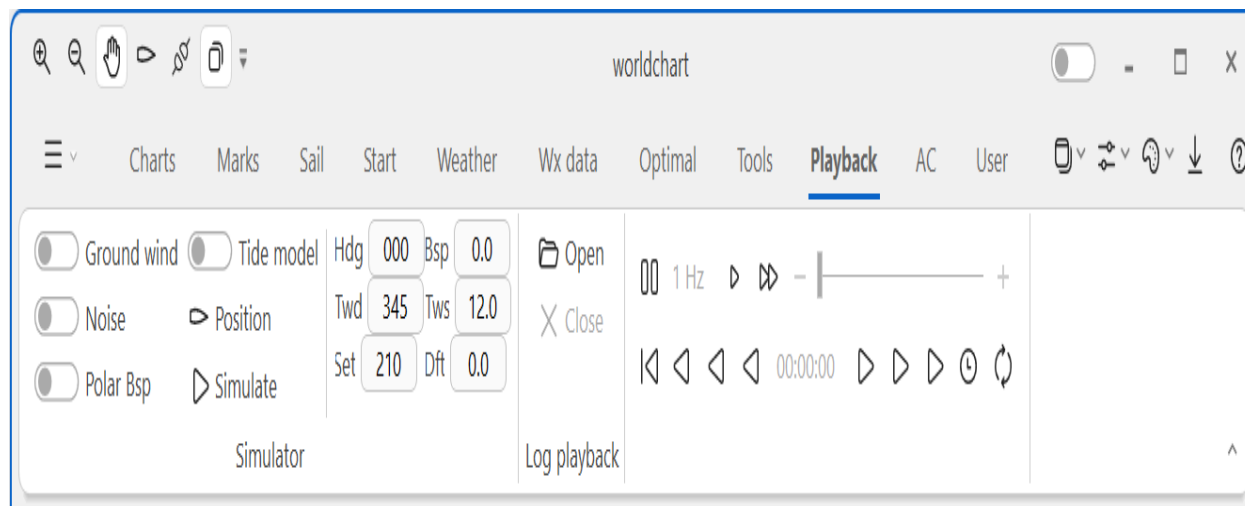
The format is a (normally) tab

delimited text file with two increasing across and two increasing down.

Example

	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180
4	2.51	3.56	4.37	4.91	5.24	5.38	5.34	5.12	5.04	4.97	4.62	4.10	3.65	3.19	2.73	2.27
6	3.77	5.15	6.11	6.68	6.99	7.11	7.07	6.98	7.09	6.98	6.56	5.92	5.20	4.58	4.01	3.43
8	4.73	6.27	7.17	7.62	7.85	7.94	7.92	8.03	8.13	8.06	7.77	7.27	6.60	5.90	5.24	4.57
10	5.38	6.84	7.58	8.01	8.33	8.50	8.52	8.43	8.60	8.71	8.52	8.12	7.60	6.96	6.32	5.67
12	5.73	7.11	7.78	8.22	8.57	8.87	9.02	8.96	8.96	9.16	9.18	8.83	8.29	7.71	7.16	6.67
14	5.92	7.24	7.92	8.38	8.76	9.10	9.43	9.52	9.31	9.59	9.84	9.50	8.89	8.35	7.85	7.50
16	5.99	7.32	8.01	8.50	8.91	9.31	9.71	10.02	9.91	10.0	10.49	10.20	9.63	8.96	8.45	8.14
20	5.88	7.38	8.12	8.66	9.15	9.66	10.18	10.78	11.25	10.89	11.67	11.91	11.65	10.62	9.86	9.29
25	5.33	7.27	8.13	8.75	9.34	9.96	10.70	11.57	12.49	13.15	13.32	14.23	14.92	13.59	12.32	11.05
30	3.95	6.75	8.01	8.72	9.39	10.13	11.08	12.20	13.49	14.98	15.18	16.83	18.39	17.32	15.39	13.46

Playback ribbon



Simulator

Simulate

Starts or stops the simulator.

Position

Manually set the boat position.

Noise

Applies pseudo-random noise to the simulated vars.

Polar Bsp

Uses polar bsp as bsp from the simulator TWA and TWS.

TWD, TWS, HDG & BSP

User defined vars for the simulator. Bsp is ignored if Polar Bsp is selected.

Log Playback

Open

Opens a log file.

Close

Closes all open log files.

Sync

Synchronises the display time with the time in the log file.

There are two groups of controls for managing the playback:

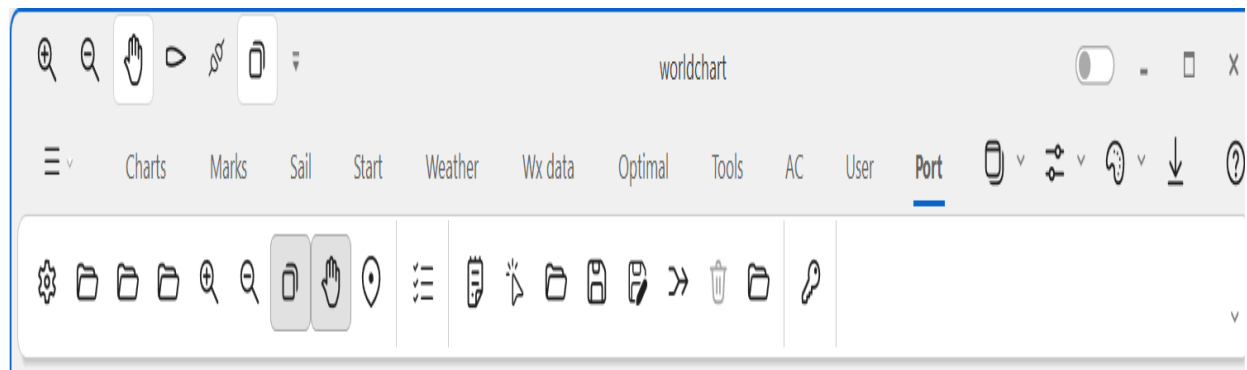
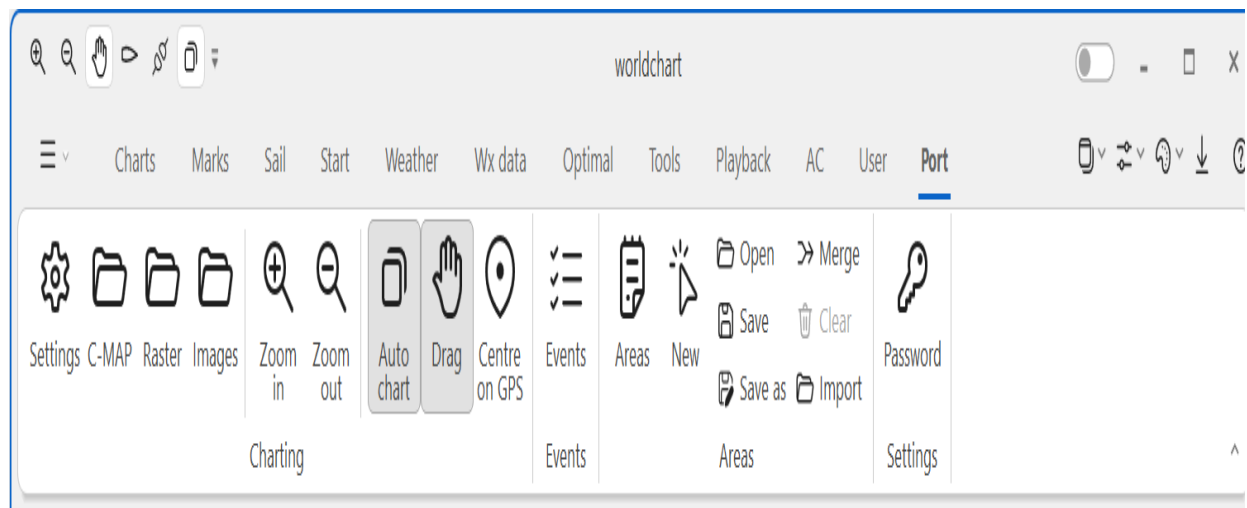
Speed

Pause,
Playback at 1Hz,
Playback at 2, 3, 4 -10Hz.

Position in log file

Rewind to beginning,
Step back,
Step forward,
Go to a user selected time in the log file.

Port ribbon



Expedition's Port mode shows the Port panel and hides many features not used in this configuration.

Charting

Settings

Chart [settings](#) is used to manage how charts are displayed in Expedition.

C-MAP

Shows C-MAP chart areas available, selects C-MAP charting mode and provides access to C-MAP chart management tools.

There may be a pop-up menu on the bottom of the C-MAP button for further functionality.

Raster

Chart [management](#) is used to manage and select raster charts such as BSB, Maptech and NV Charts.

Scale in and out

Scale in or out. If Auto-chart is selected, this will automatically select the next scale chart.

These functions can also be performed using the + and - keyboard keys, the right click drop down menu or the mouse scroll button.

Auto chart

This function only applies to the Centre on GPS mode and scaling in and out.

With *Auto chart* selected button selected, the next higher or lower scale chart will be selected when you scale in or out. If you have this button de-selected, scaling in or out will display the current chart at different scales.

Note: It is recommended that this button is selected for most use.

Drag

In this mode, click on the chart and drag in the direction to pan. This technique will be more familiar to Google Earth and Photoshop users and can be easier on touch screens.

Centre on GPS

Automatically centres the chart on the boat.

Events

Event list

Expedition's event database.

Areas

Areas are annotations on the chart.

The right-click pop-up [menu](#) can also be used to edit, delete or add points to a race note.

Notes

Lists all race notes.

Add

Creates a new race note. Select this, then click on the chart to create a new race note area.

Open

Loads a race notes file.

Merge

Merges data from another race notes file.

Save as

Saves a race notes file.

Clear

Clears all loaded race notes.

Settings

Configuration

Manages the Vesper AIS connection. See also [AIS](#) help. The instrument connection may also be accessed from the [Application](#) button.



Pop-up menu

Right mouse-clicking anywhere on the screen will display the pop-up menu, which may have any of the following menu items. You will notice that the mouse pointer changes when near a mark - when the mouse pointer is near enough a mark, many of the following functions will be available from the pop-up menu, else they are hidden.

Open chart

Allows any available chart at the mouse pointer position to be opened. Hovering the mouse pointer over this menu item causes a sub-menu to be displayed (normally to the right) with all the available charts for the mouse pointer latitude/longitude. The C-MAP charting system will be at the top of the list. Below C-MAP will be a list of charts sorted by increasing scale.

If you wish to exclusively use [C-MAP](#) charts, open the C-MAP charting system (either using this method, by selecting the check box under [chart management](#) or by scaling in or out to a C-MAP chart, then de-select the [Auto-chart](#) button.

Open image

Allow any available image at the mouse pointer position to be opened.

Open best chart

Opens best chart at the selected location.

Update image time

Set the validity time for the displayed [image](#).

Query C-MAP object

Displays [C-MAP](#) objects information.

C-MAP mark: xxxx (e.g.. Lighthouse)

Creates an always drawn mark based on the selected C-MAP object.

Astro

Sun rise and set etc.

Add *mark*

Allows the selected mark to be inserted at any point in the active route. Hovering the mouse pointer over this menu item causes a sub-menu to be displayed (normally to the right). Use this sub-menu to insert the selected mark at the beginning of the active route or after any desired mark in the route.

Add *mark* as gate to

Set active mark and route

Creates a new route and sets a new mark at the pointer.

Ping mark

Pings a mark, not part of the active route.

Insert mark

Insert a mark in the active route. The inserted mark will always be placed in the nearest great circle leg of the route to the pointer. See [right click to build a route](#).

Add new first mark

Adds a mark to the beginning of the active route.

Add new end mark

Adds a mark to the end of the active route.

Add mark at range and bearing

Adds a mark at a specified range and bearing from the selected mark.

Edit *mark*

Displays a dialog to edit the selected mark.

Remove *mark* from *route*

Removes the selected mark from the active route without deleting it.

Lock *mark*

Locks the selected mark.

Unlock *mark*

Unlocks the selected mark.

Delete *mark*

Deletes the selected mark.

Delete divider

Deletes the selected divider.

Set *mark* as port end

Sets the selected mark to be the port end of the start line.

Set *mark* as starboard end

Sets the selected mark to be the starboard end of the start line.

Port start

Sets the port end of the start line at the pointer.

Starboard start

Sets the starboard end of the start line at the pointer.

Route to mouse pointer

Computes an optimal route from the boat to the mouse pointer.

Route via mouse pointer

View optimal route

Displays details for selected optimised route.

Meteogram

Displays graphs or a table of weather data from the grib file or files loaded valid for the location selected.

Most colours and line widths are controlled as for [weather display](#) settings.

SkewT

Draws a SkewT diagram at the selected location. See the [weather](#) settings help.

Tides

Displays [tidal](#) information for the tide station selected.

MyGrib

Opens the [MyGrib](#) interface to download weather and oceanographic data.

MyImage

Opens the [MyImage](#) interface to download weather and oceanographic data.

Insert point into race note area

Inserts a point into a race note polygon.

New race note area

Creates a new race note area with the first point at this location.

Edit race note

Edit the race note attributes - colour, transparency, text.

Note that if editing the notes (as shown in the tooltip), text can be shown on a new line by using the Ctrl+Enter keys. This can make the tooltips much more readable.

Delete race note point

Deletes the nearest race note polygon point.

Chart details

Displays various details about the chart such as type, manufacturer, datum, projection.

Open best chart

Opens the best chart at the pointer.

Stripchart wand

If displaying a track, creates a wand on Stripchart at the time corresponding to the point on the track selected.

Query AIS target

Shows AIS target details.

Map AIS target to boat

Map a target to an Expedition boat (1 to 7). This can be useful if tracking a competitor. The cog, sog, range and bearing can then be displayed in number boxes, sent to instrument displays etc.



Status Bar

The status bar has no controls and is purely for information. Most of the displayed numbers relate to the mouse pointer or chart.

Use the [Display](#) settings page to select which of the items listed below to display on the status bar.

From left to right:

Chart scale and scale factor

The published scale of the chart. In most cases, you will be using charts at real size, but you may wish to over-zoom or under-zoom them. In which case, the scale factor will be displayed to the left of the colon.

Mouse position

Latitude and longitude.

Magnetic variation

The magnetic variation at the mouse pointer.

Mouse range and bearing

Range and bearing to the mouse pointer.

Time to mouse

The time to the mouse pointer position at the present GPS cog and sog. The polar time is also given in parentheses. These will not be displayed if you have no position information.

Polar time to mouse

Based on polars.

Chart datum

The published datum of the chart. If the datum of the chart cannot be determined or allowed for, UNKNOWN DATUM will be displayed. In this case, exercise extra prudence when navigating with the chart.

Position fix warning

If no position information is available, NO POSITION FIX will be displayed.

Log warning

This will only be displayed if you not logging data.

Active mark

The active mark name.

Chart depth units

The chart depth units for raster images. For vector charts, the depth units selected under [system](#) settings.



Navigation and yacht racing

Navigation.



On deck displays

The most common Expedition configuration, on a race boat at least, is a dedicated computer in the nav station and a roaming tablet that is linked to this computer.

While it is possible to run Expedition on a tablet, there are several advantages to having a fixed computer in the navstation, with everything displayed on the main computer being reflected on the on-deck display.

- A physical connection to the instruments, so lessens the risk of lost connections
- There is no chance of dropping it over-board or otherwise damaging it
- A permanent power supply

Note that a good, fast wireless connection is essential.

Tablets

Various options , see

www.expeditionmarine.com/support/tablets

Tip

Turning off unnecessary animations in Windows can help performance. To do this, go to Windows' Control panel, then 'Ease of access', then 'Make the computer easier to see' and select 'Turn off all unnecessary animations (when possible)'.

Software

There are several solutions to use a remote tablet with the navigation computer.

Remote desktop is generally faster than VNC. This is because the remote desktop system sends instructions on how to draw the screen, whereas VNC systems generally send a picture of the desktop from the server.

Microsoft Remote Desktop

Included with Windows and also freely available for [Android](#) and [iPad](#).

Remote desktop tends to be faster and more secure than VNC.

VNC

VNC tends to be slow and less secure than remote desktop, but can be used by more users.

There are several options.

[RealVNC](#) Android, Apple and Windows Client Software are free, but server software is not.

Splashtop

[Splashtop](#). Easy interface, but not as good as it once was.

IP Code

The IP code rates the degree of protection against dust, water etc.

The first digit indicates the level of protection that the enclosure provides against access to hazardous parts (e.g., electrical conductors, moving parts) and the ingress of solid foreign objects.

5 - Dust protected, mostly dust protected

6 - Dust tight, no ingress of dust

The second digit indicates the level of protection that the enclosure provides against harmful ingress of water.

4 - Splashing of water

5 - Water jets

6 - Powerful water jets

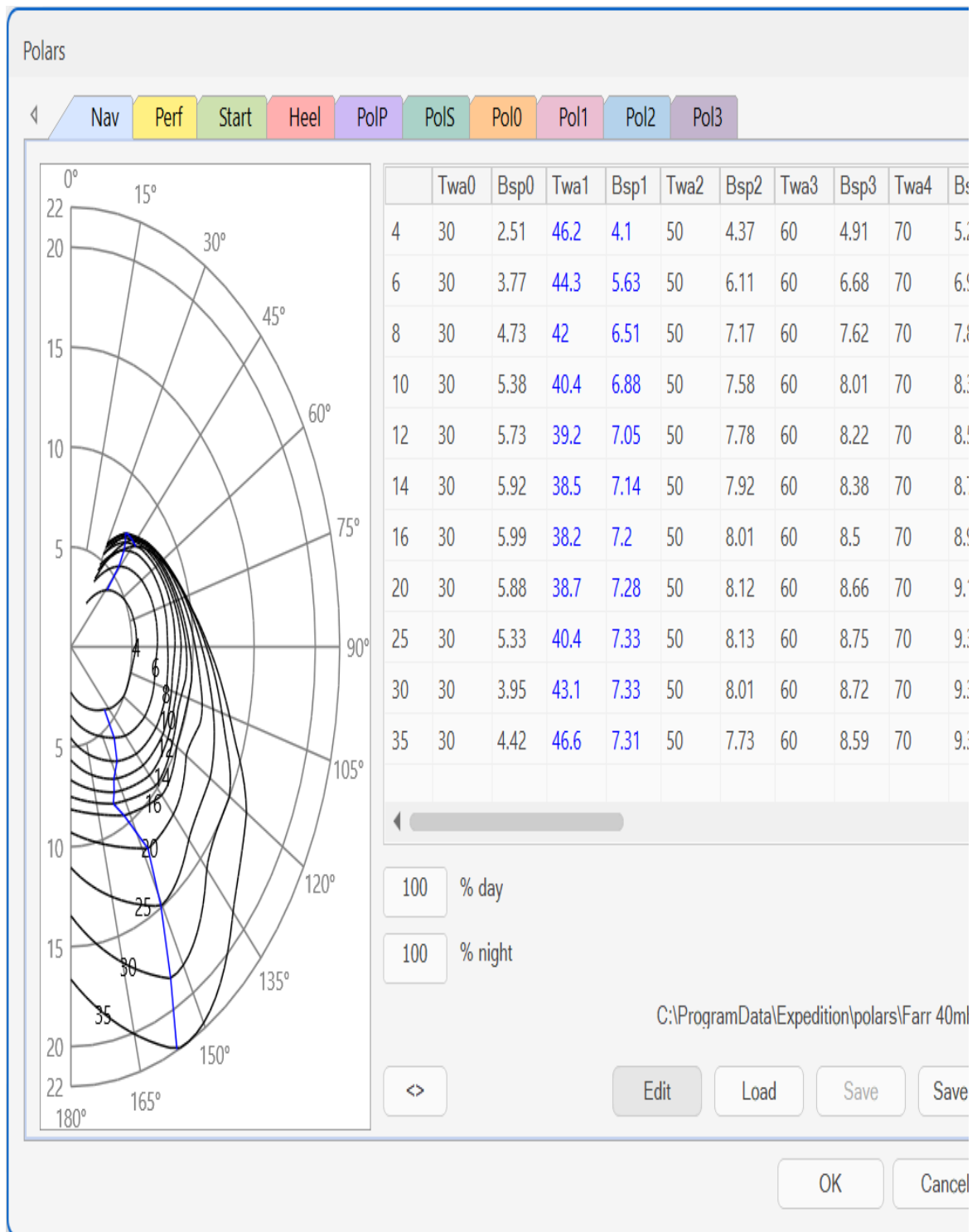
7 - Immersion up to 1m for 30 minutes

8 - Immersion below 1m for indefinite period



Targets and polars

Your **polars** are an extremely valuable input into the navigation process - for tactics and performance analysis. At their simplest, polars are a table of boat-speed for varying true wind speed and angles as seen in the table of the right of the picture below. This performance data is normally referred to as *polars* because it is often presented graphically in a polar form - as seen on the left of the picture.



Polars may be obtained from many sources - your boat designer, class association, your IMS certificate or US Sailing to name just a few. Expedition has some powerful tools to edit and manipulate polars.

Targets

Since a yacht can't sail directly upwind or downwind, but has to sail at an angle to the wind, we aren't making ground to weather at the boat-speed. Rather, progress to weather is at *Velocity made good (Vmg)* - that is, the component of speed either upwind or downwind.

Targets are the *true wind angle* and *true wind speed* at which the boat should be sailed to maximise VMG either upwind or downwind. These are indicated by the blue radial lines on the left of the picture above. So, in an ideal world, you will sail best upwind or downwind sailing at your target angle and speed.

Polar performance

If the boat isn't sailing upwind or downwind, but is reaching, then the target numbers aren't a lot of use. In this case, we use the predicted speed from the polar to tell us how well we are doing. Expedition has numbers such as polar bsp and polar bsp %. For example if the polar bsp% is 96%, then the boat is only doing 96% of the speed the polar suggests it should be able to do. Obviously, the use of polars is a very powerful means of monitoring how well you are sailing and is a very important input into any tactical decision.

Obviously, your polar numbers will vary slightly from day to day with changes in shear and air density. 96% of polar speed might be all that is possible some days. For this reason, Expedition uses a concept of wind weight. If the air feels heavier, you may want to use a wind weight of 1.1 for example.



Calibration

The data received from the various sensors by the instrument system will be of varying quality and accuracy. Obviously, if you have inaccurate data, then any calculations made by the instrument system and any decisions based on them will reflect those errors. So, we need to calibrate the instruments, bearing in mind that updating these calibrations is an on-going task.

The basic calibrations are performed first. Most of these will also be available on lower end instrument systems.

Speed

The basic instrument calibration done in all systems is to calibrate the speed. This will be explained in the manual.

Compass

Calibrating a compass involves two steps. The first is to swing it. This involves motoring the boat through several steady circles so the compass can learn about any magnetic effects in it or the boat and correct for them. The compass can also be corrected to line up with the boat - normally just linear addition or

subtraction from the heading so when the boat is point due north, the compass reads 0°.

Apparent wind

The apparent wind speed is normally calibrated by the factory and isn't changed by the user. However, the apparent wind angle will need to be calibrated as the wind sensor may not be aligned with the boat. Your instrument manual will explain how to do this.

Generally, all the other sensors (depth, temperature etc.) can also be calibrated.

Unfortunately, this is only the first step in the calibration process. There are other influences on the wind sensors that can (and do) vary from day to day such as upwash, heel and wind shear. So, the next step is to calibrate the outputs. The ability to do this is what differentiates the various instrument systems available. To correct for all the errors in Awa and Aws would be impossible if we tried to correct them all individually, but we can make some very easy approximations and correct the outputs for all the combined sources of error. Experience has shown this approach works very well.

Speed

Paddle-wheels are notoriously non-linear, especially as the boundary layer breaks down with increasing speed. Because of

this, sonic speed sensors are replacing paddle-wheels in many installations. Higher end systems often use a table of calibration values, effectively giving different calibration values at different speeds.

True wind angle

As the air flows around and over the sails, it is distorted. We call this upwash. What this means is that the apparent wind angle and speed measured at the mast-head will vary with wind speed, angle to the wind, the wind shear and even the sails being used.

Furthermore, as the boat heels, the wind angle measured by the wind sensor will vary and there are also effects due to twist in the mast.

Finally, wind speed and direction will vary at different heights about the water. Generally the wind speed will be more higher up and twisted, but by how much will vary from day to day.

The easiest way of calibrating the true wind angle this is to compare true wind direction when sailing at the same angle on port and starboard. Twa is treated as negative by the instruments internally, so if the wind direction is higher when on port tack than starboard, we add half the difference between the wind directions to the true wind angle. Conversely, if the wind direction is higher on starboard, we subtract half the difference

from the true wind angle.

In this way, we build up a TWA calibration table based on TWA and TWS that collectively corrects for upwash and twist.

True wind speed

In general, the wind sensor will read more wind downwind than upwind because of upwash - the wind accelerates over the sails. This effect is more pronounced downwind. B&G instruments have a simple, but effective solution in which the difference is entered in a calibration table and subtracted downwind. Less is subtracted as TWA decreases. Expedition and Ockam T1 systems can take this one step further and offer different calibration values at different wind angles as well as wind speeds.

Apparent wind angle and speed

Effectively what the instrument system or Expedition is doing here is correcting the true wind values for errors in the apparent wind inputs. Expedition then takes this one step further and recalculates the apparent wind speed and angle. You can think of these as calibrated or corrected apparent winds.



Inside the instrument system

Obviously, the raw information from the instruments is of limited use for many navigation problems. Fortunately, the instrument system (and computer) can make use of the raw data to provide a range of very useful information.

True wind angle and speed

At a basic level, the instrument system receives apparent wind angle and speed and speed.

Using some basic mathematics, it then calculates *true wind speed* and *true wind angle*. Tws is the speed of the wind over the water (remember aws is the speed of the wind over the boat) and Twa is the angle of the wind to the boat's heading if the boat wasn't moving. These are some of the most useful numbers you can get from the instrument system. Your sail chart and boat [polars](#) will be in terms of these two numbers.

Leeway

As the boat sails through the water, it also slides sideways a little. If the boat has a heel sensor, then an approximate leeway can be calculated. Expedition, B&G and Ockam systems all do

this in a bid to increase the accuracy of the wind triangle calculations.

The commonly used equation is $\text{Leeway} = -1 \cdot k \cdot \text{heel} / \text{bsp}^2$

So, leeway increases as you heel more, but decreases as you go faster.

Note that heel should be -ve on port, yielding a +ve leeway value.

Expedition and some instrument systems include leeway in Twa (so twa is relative to the boat's track through the water instead of its heading). This has important consequences for calculating laylines and is generally preferable.

True wind direction

If the boat has a compass, the instrument system can then add heading into the calculations and calculate the *true wind direction*. Twd is the direction, from north, the wind is coming from over the water.

Current set and drift

The GPS gives the boat's route over the ground (Cog) and speed over the ground (Sog). Since boat speed is relative to the water and we know the heading of the boat from the compass, then the instrument system can calculate the speed and

direction of the water. The is the current *set* and *drift*. For obscure reasons, the current set is the direction the current is going to - this is the opposite sense to the way the true wind direction is described.

Laylines

In combination with your targets and [polars](#), Expedition and high end instrument systems can use all this information to calculate laylines, polar speeds, time to laylines and many other numbers.

These calculations can be significantly enhanced by [calibrating](#) both the sensors and the derived numbers.



The last couple of decades have seen the most profound changes to navigation paradigms in centuries with the advent of satellite navigation systems - the latest of these being the US GPS system.

Prior to this, navigation primarily involved position fixing - a time consuming and, by today's standards, relatively inexact art. With the advent of GPS, position information is available to anyone possessing the barest of skills and experience with astounding accuracy and timeliness.

This and the increasing array on instrumentation have changed the role of the racing navigator from being a pure navigator to a tactician and manager of information.

The GPS, instrument system and boat computer are often connected using the RS232 and NMEA electrical standard, although there are several proprietary communication protocols.

Instrument systems

The instrument system supplies the much of the information available to the navigator. Systems by Ockam and B&G are the main ones chosen by racing navigators, but there are other manufacturers of instrumentation.

The five fundamental sensors connected to a yacht system are wind, depth, compass, speed and GPS. There are many other sensors available that provide extra information and/or add to the accuracy of the information from these main inputs. For example, barometer, heel, trim, sea temperature and load cells.

GPS

The GPS provides position (latitude and longitude), route and speed over the ground (Cog and Sog) as well as being an accurate time source.

Wind

The raw information received from the wind sensor are the apparent wind angle (awa) and apparent wind speed (aws). These are obviously relative to the boat, so will vary depending on boat speed and heading.

Speed

This is historically measured by a paddlewheel, but in high-end applications is slowly being replaced by sonic-speed devices, which provide better linearity and repeatability.

Heading

This is the direction your boat is heading and is often measured by a fluxgate or gyro compass.

Depth

The depth sensor is a fundamental safety and navigation instrument, but is an input that can be treated as separate from the rest of the system as far as the racing navigator is concerned.

Heel

A heel sensor (inclinometer) measures the heel of the boat to port or starboard. It can be a useful measure of how powered up the boat is and is often used in calculations of leeway.

Trim

A trim sensor (inclinometer) measures the fore and aft trim of the boat.

Load cell

These measure loads, for example forestay load/tension and can be useful tuning aids.

Sea temperature

The sea temperature sensor (an electronic thermometer) is often packaged with the speed sensor. It can be very useful to know the sea temperature when racing through the Gulf Stream for example.

Barometer and air temperature

These can be useful for monitoring the weather.



Expedition charting

Expedition features a highly efficient and class leading charting system that seamlessly rotates, mosaics and moves between charts. Expedition supports a range of chart formats from different manufacturers.

Electronic charts are available in several formats. Expedition can use ENC's, RNC's, C-Map, Navionics and other raster charts.

See the Expedition [charts](#) web page for chart sources.

Unlike RNC's, objects on ENC, C-MAP and Navionics charts can be interrogated using the right-click [pop-up](#) menu.

ENCs

[S-57 and S-63 ENC's](#)

Vector charts are databases of all objects in a chart and their attributes - you can think of them as an intelligent chart. Vector charts are normally more efficient, require less storage and often allow better clarity. However, their main advantage is that the user or software can choose to display objects on the chart or not.

ENC charts use the WGS-84 datum and are displayed on a Mercator projection.

Some countries supply these free of charge, for example NZ and the USA. Others charge for them - Canada, Hong Kong and the South China etc.

Official S-63 ENCs are also available through global providers, such as ChartWorld and the Admiralty Vector Chart Service (AVCS).

RNCs and raster charts

RNCs and raster charts are originals or scans of paper charts that contain the geo-referencing information necessary to convert between pixels in the image and position. Most raster charts also contain other information such as the date of update, datum and depth datum.

Many navigators prefer raster charts over vector charts because they are more familiar.

Raster charts are most commonly available as varieties of the BSB format.

RNCs are generally being replaced by ENCs.

RNC and other raster charts do not always use the WGS-84 datum and a Mercator projection. Where possible, all raster charts are corrected to WGS84.

RNCs from various manufacturers have near global coverage.

Note that not all raster charts are official charts and may be sold or supplied as an Aid to Navigation.

C-Map

C-MAP X and 4D have near global coverage.

C-MAP charts are similar to ENC's in appearance and are based on ENC data, but are not official charts and are sold as an Aid to Navigation.

C-Map charts use the WGS-84 datum and are displayed on a Mercator projection.

Navionics

Navionics charts have near global coverage.

Navionics charts are similar to ENC's in appearance and are based on ENC data, but are not official charts and are sold as an Aid to Navigation.

Navionics charts use the WGS-84 datum and are displayed on a Mercator projection.



Chart settings

Accessed from the Chart tab on the settings sheet, see the [chart](#) panel. Ctrl+C.

Important warning

Most charts are referenced to the WGS84 datum. Non WGS84 charts are automatically corrected by Expedition to the WGS84 datum where possible. This is to make everything consistent within the system and with the position provided by your GPS. Most charts contain enough information to facilitate this, but if the words unknown datum or unknown ellipsoid appear in the status bar at the bottom of Expedition's main chart window, then no conversion is possible. In this case, extra care should be exercised if using Expedition as an aid to navigation.

There are small errors involved in datum conversions, potentially up to tens of metres. Some projections (for example ED50 around the Mediterranean) may have errors larger than this for various reasons.

A chart's datum can also be viewed in the [chart management](#) dialog or using the [right-click pop-up](#) menu. There are inherent errors in any datum conversion, so the user should be especially careful with non WGS84 charts. For optimum accuracy on a

given chart, the user may want to apply an offset to the GPS position. Most GPSs have a protocol to enter a user determined correction to be applied to the position. This sort of correction should only be done if you are confident of its accuracy and you must remember to remove/change it if you switch to a different chart. Luckily this sort of manipulation is rarely necessary.

Because of this, it is normally recommended that your GPS is configured to output position based on the WGS84 datum unless you are an expert user and your chart datum is unknown to Expedition.

General

Expedition has no printing features due to licensing restrictions with the chart manufacturers.

Chart orientation

Expedition can rotate raster charts to suit individual needs, but most navigators accustomed to paper charts will prefer the chart up display setting.

The chart up mode is faster when displaying raster (e.g. Bsb) images.

Chart up

The default and most common setting. Display charts oriented as they are drawn (most charts are drawn with true north up).

True north up

Display charts so that true north is always to the top of the screen.

Course up

Useful for windward/leeward courses where tactical viewing is more desirable than navigational viewing.

Heading up

Orients the chart to the current heading.

Start up

Like Course up, this setting displays charts with the starting line oriented horizontally across the screen.

Projection

Expedition can use different projections for the simple world vector chart. Currently the options are Mercator or orthographic.

Mercator

The standard map projection for nautical navigation as rhumb lines or loxodromes are straight lines that preserve angles with the meridians.

The latitude scale is distorted towards the poles, thus distorting areas.

ENCs, RNCs, C-Map charting and most paper charts use a Mercator projection.

Orthographic

Preserves neither area nor angles, but great circles near the centre latitude and longitude will appear nearly straight.

Use ENCs and/or RNCs

Restrict Expedition to automatically using the selected chart types.

This could be useful if using ENCs, but want to temporarily use or refer to an RNC.

Palette

By default, Expedition displays uses a standard S-52 colour palette. However, a grey scale colour scheme may also be

used.

Note that the dusk palette can be used with ENC's in night mode, see below.

ENCs

Shallow, Safety and Deep depths

Adjusts colours of depth areas, soundings and the safety contour.

Safety depth

This is an important setting.

Affects the bold displayed safety contour and colour of soundings.

The safety depth will also affect some symbols - for example wrecks and other obstructions.

Display class

Also selectable from the Chart ribbon [panel](#).

See IMO Resolution A.817(19), appendix 2.

If in any doubt, use the All display option.

Display base

Not intended to be sufficient for safe navigation.

Permanently retained on the ECDIS display, consisting of:

1. Coastline (high water),
2. Own ship's safety contour, to be selected by the mariner,
3. Indication of isolated underwater dangers of depths less than the safety contour which lie within the safe waters defined by the safety contour,
4. Indication of isolated dangers which lie within the safe water defined by the safety contour such as bridges, overhead wires, etc., and including buoys and beacons whether or not these are being used as aids to navigation,
5. Traffic routing systems,
6. Scale, range, orientation and display-mode,
7. Units of depth and height.

Standard display

To be displayed when the chart is first displayed by ECDIS. Note this does not include soundings and depth contours, which are shown in the All display option or can be selected in Standard by Exp.

1. Display base,
2. Drying line,
3. Indication of fixed and floating aids to navigation,
4. Boundaries of fairways, channels, etc.,
5. Visual and radar conspicuous features,
6. Prohibited and restricted areas,
7. Chart scale boundaries,
8. Indication of cautionary notes.

All

Standard display and additional information.

1. Spot soundings,
2. Submarine cables and pipelines,
3. Ferry routes,
4. Details of all isolated dangers,

5. Details of aids to navigation,
6. Contents of cautionary notes,
7. ENC edition date,
8. Geodetic datum,
9. Magnetic variation,
10. Graticule,
11. Place names.

Chart display settings

General

Chart bounds

Draws available chart bounds.

Image bounds

Draws image bounds.

Quilt charts

Will attempt to fill in areas of the screen where the active chart has no coverage. Will try to use the next

scale out chart.

One possible use of disabling quilting with RNCs is it will show detail around the edges of the chart.

For very slow computers, it could be useful to disable quilting.

Fill world chart land

Fill world chart sea

Default is to not shade the sea on the simple word chart to improve clarity.

World chart names

Place names.

ENCs

Soundings and depth contours

These are not displayed in the Standard display. This causes them to be shown in Standard display mode.

Anti-clutter

Hides chart data depending on the display scale.

Land features

Not implemented.

Two colour depth shading

Else uses 4 colours for depth shading based on Shallow, Safety and Deep depth settings.

Simplified boundaries

Draws boundaries with simplified symbology. Default on.

Simplified symbols

Draws symbols with simplified symbology. Default is off for paper symbols.

Colour

ENCs default to many structures being solid black.

If they have a colour attribute, this option displays them with that colour.

Mainly affects beacons, buoys and top marks.

Use dusk for night palette

Often the ENC night palette may be too dark. This allows the ENC dusk palette to be used instead.

Information objects

Information about the chart, quality etc. - M_NSYS, M_QUAL, CATZOC etc.

ENC Lights

Light sectors

Displays light sectors. Default is on.

The light sectors will be replaced by a simple flare symbol.

This is a non-standard option for small screens where the user may want to reduce clutter during daylight hours.

Lights text

Light descriptions.

Full light sector lines

Draw radial lines depicting light sector bounds to the nominal range of the light.

If deselected to reduce screen clutter, will draw shorter radial lines.

Small light sectors

ENCs are normally drawn with light sectors at 20 or 25mm radius.

This is often too large for users and may clutter the chart. Use this option to reduce the size of the drawn light sectors.

ENC Text

Large text

Draws soundings and other text with a larger font.

Important text

Important text, including bridge, cable and pipeline clearances, recommended tracks etc.

Lights

Lights.

All other

Any other text.

ENCs

ENCs are available in S-57 and [S-63](#) encrypted formats.

Expedition mostly follows the IHO S-52 Presentation Library edition 4.0, with some customisations to suit the needs of sailing navigators. For example, the options to make light sectors smaller or shown as flares to reduce clutter.

David Burch's 'Introduction to Electronic Chart Navigation' is recommended reading on ENCs. It is available from www.starpath.com.

Management

ENC chart management is accessed from the ENC button on the [chart panel](#).

Also see the [chart panel](#) and [chart settings](#) help.

Performance note

ENCs are slower to load and render than RNCs. A fast computer helps. If desired, deselecting quilting in [chart display settings](#) will improve performance.

Open

Opens the selected chart cell in the main chart window.

Clear DB

Clears the chart cell list. Does not delete cells from your system.

Find chart

Type any part of a chart name and the Exp will filter the chart list only show charts with names containing that text.

Search

This will search your computer for charts and build a database of all the ENC's in the selected folder. Expedition will ask you where to search - you can search the whole computer or just update specific folders in your chart database to save time. After the search is complete, you will be able to see and access your updated chart list.

Selecting any chart in the list will display its attributes right of the dialog.

The active chart's details (chart name, datum, etc.) may be queried by right mouse clicking anywhere on the chart and selecting Chart details from the [popup](#) menu.

Once selected, a chart will remain displayed in the main chart window until you either scale in, scale out, re-centre or pan the chart window off the present chart.

S-63 ENC's

Many ENC's are supplied in an encrypted form for licensing reasons. S-63 is the encrypted distribution format for S-57 ENC's.

The cell licences are for use on a single system and are valid for a set period. This ensures the charts cells are kept up to date. Expired cells may still be displayed if not updated, but Expedition will display a warning.

The encrypted ENC management functions are accessed from the ENC Permits button on the ENC management window.

User Permit

We supply User Permits with the Expedition on request.

The User Permit will also be displayed on the ENC Permits window.

Please keep a copy of the User Permit for your records and for obtaining ENCs from ENC providers.

User Permits supplied for Expedition are unique to the Expedition licence and will not work with other systems.

ENC Permits		
User permit: xxxxxx		
Cell	Data Ser...	Expiry date
NZ		
NZ14600E	NZ	30/11/2022
NZ14600W	NZ	30/11/2022
NZ200021	NZ	30/11/2022
NZ200023	NZ	30/11/2022
NZ200024	NZ	30/11/2022
NZ200025	NZ	30/11/2022
NZ200026	NZ	30/11/2022
NZ200028	NZ	30/11/2022
NZ200029	NZ	30/11/2022
NZ20022E	NZ	30/11/2022
NZ20022W	NZ	30/11/2022
NZ20027E	NZ	30/11/2022
NZ20027W	NZ	30/11/2022
NZ20031E	NZ	30/11/2022
NZ20031W	NZ	30/11/2022
NZ300041	NZ	30/11/2022
NZ300042	NZ	30/11/2022
NZ300043	NZ	30/11/2022
NZ300045	NZ	30/11/2022
NZ300046	NZ	30/11/2022
NZ300051	NZ	30/11/2022
NZ300053	NZ	30/11/2022
NZ300054	NZ	30/11/2022
NZ300055	NZ	30/11/2022

Purchasing ENC licences

Your ENC provider will need the 28 character User Permit when purchasing ENC licences.

The provider will supply a Permit (PERMIT.TXT) and the ENC data, commonly online.

Chartworld note

Do not enter anything for Hardware ID.

Installing ENCs

ENC data

Copy the supplied ENC data to your computer. The directory structure must be exactly as supplied.

A typical location would be

C:\ProgramData\Expedition\S63\

It is good practice to keep different chart sets separate. For example, AusENCs could be stored in the folders

C:\ProgramData\Expedition\S63\AusENC\enc-base\

C:\ProgramData\Expedition\S63\AusENC\enc-update\

Certificates

Expedition is distributed with the default IHO.PUB certificate. Some servers may use and distribute different certificates.

In this case, copy the certificate (e.g. UKHO.PUB) to the Expedition S63 data folder, normally

C:\ProgramData\Expedition\S63\

Cell permits

The ENC distributor will supply permits to use the ENC cells.

To install the permits, click the Import Permits button on the ENC Permits window and search for the supplied PERMIT.TXT file.

Old or unwanted permits may be removed by selecting them from the list and clicking the Remove Permits or Remove All buttons.

Please keep back-up copies of the Cell permit files.

Install charts

After importing the cell permits, click the Search button from the main ENC chart management window (see above) and select the base data folder the ENC data was copied to.

Then repeat the procedure for any update data folders.

SSE-21 Decryption failed, no valid cell permit found warning

Expedition may give this warning if cells have been updated in a subsequent update. This error will be resolved on importing the update(s).

The base and update data will have the same directory structure, similar to

```
[ENC_ROOT]
[EXTRAS]
[INFO]
IHO.CRT
SERIAL.ENC
```

ENC Updates

Periodically, or as advised by your supplier, copy the chart updates and permits to your computer.

Both the permits and chart updates need to be imported into Exp. The permits must be imported first.

1. Import the updated PERMITS.TXT file supplied as above.
2. Update the ENC database by selecting the [search](#) button.

AusENC

At 01 October 2017, AusENC data is available at

<http://www.hydro.gov.au/software/products/products.jsp>

See [AusENC Fact sheets](#)

Using AusENC with Expedition requires

1. AusENC Base data
2. AusENC Update data
3. AusENC cell permits

The Base data is released on a six monthly basis in January and July. The update data is released every two weeks thereafter.

The updates are available as a cumulative file or as fortnightly updates. Applying the cumulative file data is simpler and recommended to reduce the risk of missing updated cell data.

At 01 October 2017, the Base data file is 185MB and the cumulative update file 24MB. This is one reason for not re-issuing the complete data set every two weeks.

The zip download options are easier for most users than the iso downloads.

Installing AusENCs

1. Unzip and copy the AusENC Base data to the desired location, such as
C:\ProgramData\Expedition\S63\AusENC\enc-base\
2. Unzip and copy the Cumulative AusENC to the desired folder.
C:\ProgramData\Expedition\S63\AusENC\enc-update\
3. Install the permits (PERMITS.TXT) as [above](#).
4. Tell Expedition to import the base data, then the update data, see [search](#) above.

Updating AusENCs

If significant time has elapsed between the expiry of your old AusENC service and the start of the new service, we recommend that you remove all previous data and corresponding permits from Expedition prior to installing the new service.

1. Unzip and copy the Cumulative AusENC to the folder used to store the ENC update data as above.
2. Install the permits (PERMITS.TXT) as [above](#).
3. Tell Expedition to re-import the chart cells again after the update, see [search](#) above.



RNC charts

Raster chart management is accessed from the RNC button on the [chart panel](#).

Also see the [chart panel](#) and [chart settings](#) help.

Open

Opens the selected chart in the main chart window.

Clear DB

Clears the RNC chart list. Does not delete charts from your system.

Find chart

Type any part of a chart name and the Exp will filter the chart list only show charts with names containing that text.

Search

This will search your computer for charts and build a database of all the raster charts in the selected folder. Expedition will ask

you where to search - you can search the whole computer or just update specific folders in your chart database to save time. After the search is complete, you will be able to see and access your updated chart list.

Selecting any chart in the list will display its attributes right of the dialog.

The active chart's details (chart name, datum, etc.) may be queried by right mouse clicking anywhere on the chart and selecting Chart details from the [popup](#) menu.

Once selected, a chart will remain displayed in the main chart window until you either scale in, scale out, re-centre or pan the chart window off the present chart.

Note that Expedition does not support notice to mariner updates.



Custom images

Accessed from the images button on the [chart panel](#).

This opens the raster image management window. This is very similar to the [raster chart management](#) window.

Also see [My Image](#) on the [weather data](#) panel.

The custom image support is **DEFINITELY** not for navigation.

Set display time

Set's Expedition weather display time to the validity time of the image. This can be set from the right-click [pop-up menu](#) or when the image is geo-referenced.

Squid

Quick import of Squid satellite images.

Open

Opens the selected image in the main chart window.

Clear DB

Clears the image list. Does not delete images from your system.

Find chart

Type any part of a chart name and the Exp will filter the chart list only show charts with names containing that text.

Search

This will search your computer for charts and build a database of all the raster images in the selected folder. Expedition will ask you where to search - you can search the whole computer or just update specific folders in your image database to save time. After the search is complete, you will be able to see your updated image list and access them as described below.

After running search and building the data base, your images are arranged alphabetically in a tree arrangement based on their format.

On selecting any image in the image management window its attributes will appear on the right of the dialog.

You can import individual images using the Import button (see below) or import several images at once using the Search function.

You can also examine the present image's details (name, etc.) by right mouse clicking anywhere on the chart and selecting

Image details from the popup menu.

Once selected, an image will remain displayed in the main chart window until you either scale in, scale out or re-centre (pan) the chart window off the present image.

Import

Allows import of individual images (saves a full search) in BMP, GIF, JPEG, PNG or TIFF formats. The image to be imported has to be north up, a Mercator or equidistant lat/lon projection. As part of the import process, the image needs to be geo-referenced.

Expedition requires at least two geo-referencing points that should be as far apart as possible on the chart for best results - for example in the top left and bottom right or top right and bottom left corners.

It also allows the import of Squid .sat satellite images, which already contain the geo-referencing.

For example, you can import images of weather faxes, sea temperature charts, satellite images, 'QuikScat' wind satellite charts and ocean current charts to plot your position on or overlay grib weather.

Note: Do not use these custom images files for navigation. The ability to display these is only provided as an aid to strategy and decision making.



Expedition supports [Navionics](#) charting.

Features

<https://www.navionics.com/charts/features/navionics>

Navionics charting is available as Navionics+ and Platinum+.

Navionics+

Platinum+ adds relief, satellite and aerial imagery. Expedition does not currently support Platinum+ 3D chart view.

Daily updates.

SonarChart™ bathymetry

Understand depths and bottom detail with innovative 0.5 m bathymetry charts created using Navionics proprietary systems that process sonar data contributed by boaters with existing content.

Community Edits

Useful local knowledge from fellow boaters in this crowd sourced content layer packed with points of interest, navigation aids and variety of edits made by users of our Navionics Boating app.

Notes

Navionics charting for Garmin plotters is not supported as it is a different product.

Online web app

Navionics has a useful [web app](https://webapp.navionics.com) to explore cartography.

<https://webapp.navionics.com>

Using Navionics charting in Expedition

To switch to Navionics mode from other charting, select the Navionics button on the [Charts](#) ribbon. Expedition will then switch to display Navionics charting.

World chart

A Navionics world chart is distributed with Expedition.

A larger world chart containing satellite imagery is available, currently WorldMap_5.0.

If desired, the normal location to copy it to is

C:\ProgramData\Expedition\navionics\

Do not otherwise edit or delete files in the Navionics folder.

Chart Installer

Navionics uses the Chart Installer app to activate and update charts.

Simply insert the Navionics SD card into an SD reader and run Chart Installer to activate your cartridge.

Available at

www.navionics.com/my-card/activate-update

Tutorial

<https://www.navionics.com/support/chart-installer-tutorial/activate>

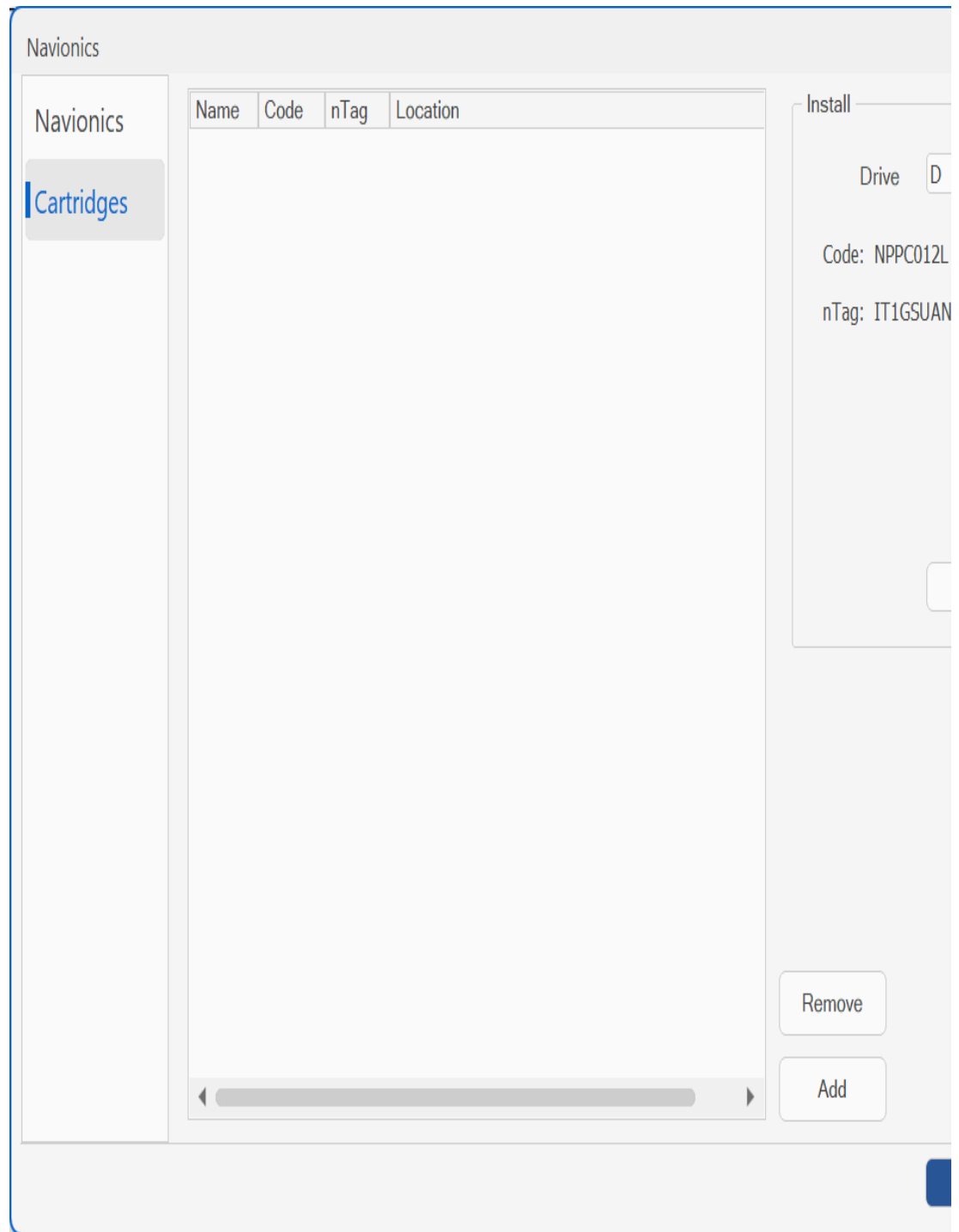
Installing Navionics charts for use with Expedition

Expedition can use charts directly from the SD card, but we recommend copying to the local disk for performance. Keep the SD card in a safe place!

If not in Navionics charting mode, select the Navionics button on the Charting ribbon.

Select the Settings button.

Select the Cartridges tab



Expedition will automatically detect available removable drives.

To copy or update an existing cartridge already installed, select the appropriate drive and select the Install button. Note that copying the cartridge may take several minutes.

Cartridges may also be manually added or removed with the Add and Remove buttons.

Notes

As with any SD card, it is recommended to Eject it in Windows before removing from the SD card reader.

Internet connection required

Expedition, with an activated and installed Navionics chart, must be started at least once with an internet connection to register your PC with Navionics.

There is a limit (5) to the number of computers you can copy cartridges to on a licence.

Updates

The Navionics Chart Installer is used to update the cartridge. Updates will be available most days. After updating the cartridge, you need to install it to your local disk again.

Notes

Over-zooming charts will result in a hatched pattern on the chart. Unfortunately, there is no way to limit zooming in Navionics, so the Auto-chart button is disabled when in Follow boat mode.

Querying chart objects

Use the pop-up menu to query or set marks at Navionics chart objects.

Settings

Language

Available languages.

Safety depth

Safety depth allows the shaded display of depth areas using different shades of blue to distinguish between shallow and deep water.

Shallow depth

With Shallow Area, users can control a pattern made of red dots that will be drawn over the map to highlight areas that are not safe for navigation.

Layers

Mask

Transparency

Set transparency to 100% or select none to show no layers. Not currently available.

All chart details

Display mandatory and all optional layers, else only mandatory layers.

Base data only

Hides most items on the chart. Only useful if using the chart solely as a weather backdrop etc.

Chart bounds

Chart boundaries.

It is recommended this setting is enabled as it provides important visual indications of smaller scale charts.

Dynamic icons

Note that dynamic tide and current icons are displayed for the current clock time, not the Expedition display time.

Easy view

The Easy View feature allows users to magnify icons and texts on the map to provide better reading.

When Easy View is turned ON, the chart texts and objects on the screen will be larger.

International presentation mode

Else US.

Lat/lon grid

Light sectors

Multimedia

Nature of seabed

Seabed Areas display bottom composition details and are recognizable thanks to a specific pattern applied to the map.

Orientation

Set on the [chart settings](#) page.

System requirements

OpenGL

Navionics may use OpenGL to render the charts. OpenGL enables hardware acceleration of 3D graphics.

Drivers

Support is required from the graphics drivers for optimum performance. Graphics drivers that come with Microsoft Windows or that are downloaded from Windows Update often do not support OpenGL.

Especially with a bare install of Windows, additional OpenGL drivers may need to be installed. Common graphics driver links may be found at:

<https://www.expeditionmarine.com/resources.html>



C-Map X Charting

ExpeditionX supports [C-MAP X](#) charting.

Also see the [chart panel](#) help.

Features

[C-MAP X](#) charting is available in two variants - Discover X and Reveal X.

Discover X

- C-Map vector charting
- ENC cartography option
- High resolution bathymetry
- Tides and currents

Reveal X includes all the Discover X data and adds

- Satellite overlay
- Shaded relief
- Social Map bathymetry
- Social Map vegetation
- Social Map bottom hardness.

System requirements

OpenGL

C-MAP X charting uses OpenGL. OpenGL enables hardware acceleration of 3D graphics.

Drivers

Support is required from the graphics drivers for optimum performance. Graphics drivers that come with Microsoft Windows or that are downloaded from Windows Update may not support OpenGL.

Especially with a bare install of Windows, additional OpenGL drivers may need to be installed. Links to common graphics drivers may be found at:

<https://www.expeditionmarine.com/resources.html>

Notes

Expedition does not support C-Map X charts on SD cards.

Some users have found C-Map X charting performance is better if Expedition X is run As Administrator.

X-Chart Manager

Chart management, downloads and purchases are managed through C-Map's X-Chart Manager app (XCM), available at

www.c-map.com/update-my-chart

Expedition and X-Chart Manager XCM can not be used at the same time.

Using X-Chart Manager

Please read the XCM user guides before proceeding as they cover setting up an account etc. in more detail than the below.

User Guide available with XCM (select the 'i' button, then the 'User Guide' button).

There is also a [How to use X-Chart Manager](#) support topic.

Log in

Log in with your Navico (C-Map, B&G, Simrad, Lowrance) credentials.

If you do not have an account, XCM will let you set up a new account.

SD Cards

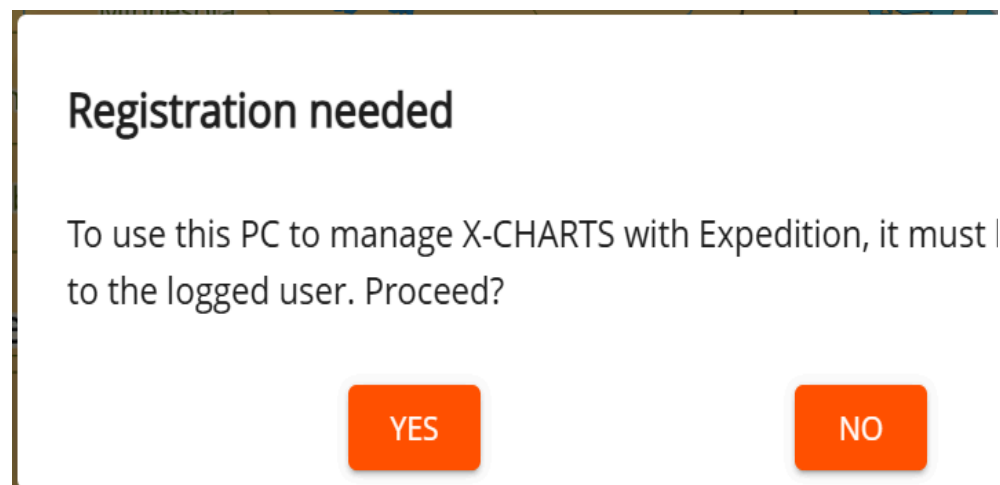
Expedition does not support C-Map X Charts on SD card.

Ignore the 'No SD cards mounted' window.

Registration

ExpeditionX requires the device to be registered with X-Chart Manager and will warn if not registered.

ExpeditionX needs to be run and the C-Map option to be selected before XCM for registration to proceed.



Purchasing charts

Click the Chart Settings hamburger button at the top left. This will present an option to purchase new charts.

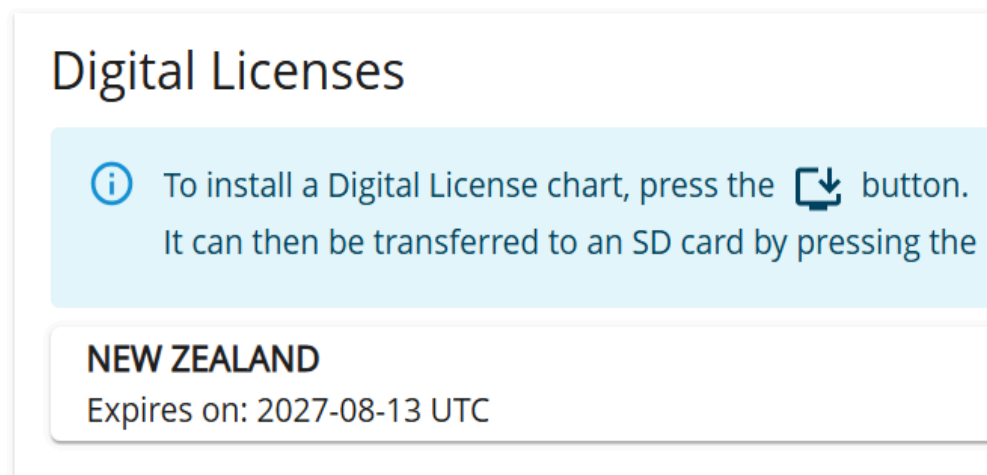
Selecting the 'Purchase New Charts' button will show a C-Map Store window.

When purchasing charts, select the Digital Licence option, not the MicroSD card option.

Digital licences

One difference from SD cards for plotters is that the Digital Licence needs to be installed on the computer.

Click the install button.



Downloading charts

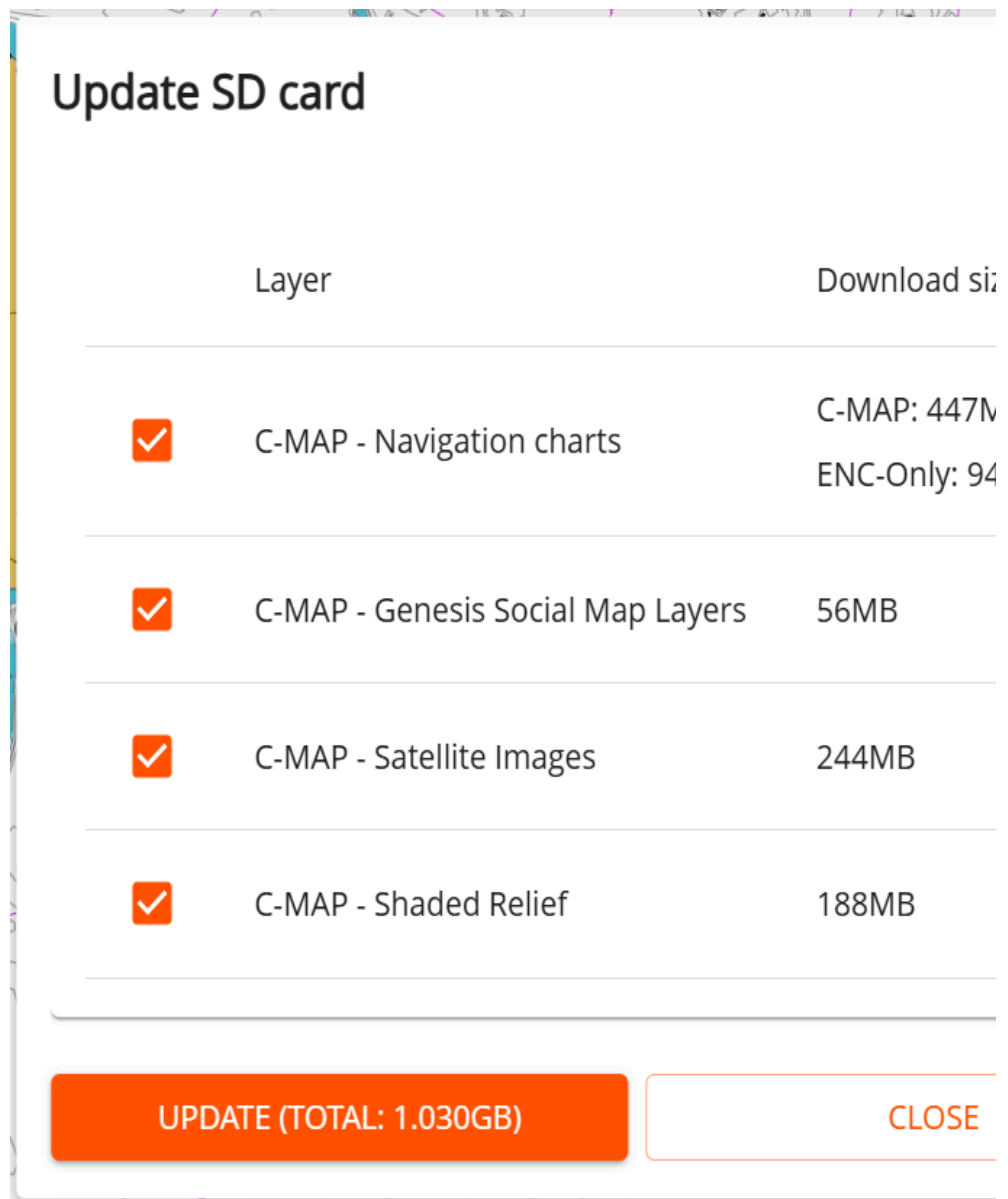
To download chart data, select the 'Add a new area' button.

Then use the mouse button to select an area. Use the left mouse button to add points and the right button to delete the last point added.

When satisfied, name the area and select the 'Create Area' button.

Several areas can be defined.

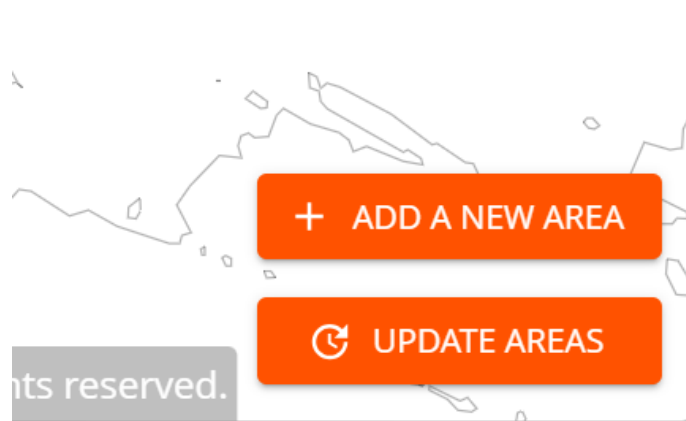
Finally, select the 'Update Areas' button. This will give options for data to download. Ignore the SD card title and click the Update button.



Updating charts

C-Map X Charts are regularly updated.

To update downloaded chart areas, select the 'Update Areas' button.



Settings

Logging

Enables C-Map X logging for C-map debugging. Off by default.

Require login at next start-up

Expedition remembers the last C-Map X user details to save logging in every session.

If these change or there is a problem with X-Chart manager, select this one-time setting to instruct Expedition to prompt to login it at the next start-up.

The Expedition System Restore [app](#) can also be used to more completely reset Expedition's C-Map settings.

Display settings

Also see the [chart settings](#) and [chart panel](#) help topics.

Language

Available C-MAP languages.

Orientation

Set on the [chart settings](#) page.

Satellite mode

Satellite opacity

Coverage mode and opacity when displaying the satellite overlay.

Shaded relief opacity

Opacity when displaying the shaded relief overlay.

Safety depth

Shallow depth

Chart bounds

Chart boundaries.

It is recommended this setting is enabled as it provides important visual indications of smaller scale charts.

Depth contour labels

Label depth contour lines.

ENC cartography only

ENC cartography without added C-Map data and overlays.

See also the E button on the [chart panel](#).

High resolution bathymetry

See also the H button on the [chart panel](#).

International presentation mode

International or US symbols and palette.

The US option uses simplified symbols, similar to the Expedition S-52 ENC Simplified symbols option.

National text

Select the text to be displayed, using the S-57 convention of defining an international text for features and a national text for the same features (when available).

Raster charts

Display raster charting if available.

Limited coverage.

Satellite overlay

Display satellite image overlay if available. See also the **S** button on the [chart panel](#).

Shaded relief

Display shaded relief if available. See also the **R** button on the [chart panel](#).

Shallow depth alert

Highlight dangerous depth areas having depth lower than the specified safety depth value.

User areas

Enables/disables the display of SKU coverage and user defined areas to show where data is available.

Genesis social map

Community source mapping.

Bathymetry

See also the Social map button on the [chart panel](#).

Bottom hardness

Vegetation

C-MAP objects

Use the Query C-MAP object menu item on the right-click [pop-up menu](#) to interrogate any C-MAP object.

Select any item in the tree control on the left of the C-MAP object window to display its attributes in the right hand window.

Expedition4D supports the [C-MAP 4D](#) charting.

Also see the [chart panel](#) help.

Features

C-MAP 4D gives access to coastal, inter-coastal and waterway maps, detailed oceanic and enhanced GPS maritime navigation, precise fishing bathymetry and digital raster maps.

As well as the base data, C-MAP 4D charting is available in two variants:

- MAX - Charts, nav aids, photographs, tides and currents. features etc.
- 4D - All the Max data plus 4D charting, satellite and raster chart overlays as well as VAD data.

Important warning

C-MAP charts, are referenced to the WGS84 datum.

Because of this, it is normally recommended that your GPS is configured to output position based on the WGS84 datum.

System requirements

C-MAP 4D is limited to 2732x2048 pixels, but higher DPI displays will work with 4D in Expedition.

Maps Manager

Expedition4D requires Maps Manager to be installed. It may be downloaded from the Expedition [web site](#).

Web browser

Map management and purchases are managed through a web browser interface.

If there are any issues, especially with purchases, ensure the default web browser is up to date.

OpenGL

C-MAP 4D uses OpenGL to render the charts. OpenGL enables hardware acceleration of 3D graphics.

In addition, using the data in Exp can be accelerated. This is indicated in the first line of the C-MAP chart summary, available from the C-MAP Charts button on [Chart](#) ribbon panel. This will indicate whether the OpenGL to Expedition transfer is accelerated.

Drivers

Support is required from the graphics drivers for optimum performance. Graphics drivers that come with Microsoft Windows or that are downloaded from Windows Update often do not support OpenGL.

Especially with a bare install of Windows, additional OpenGL drivers may need to be installed. Common graphics driver links may be found at:

<https://www.expeditionmarine.com/resources.html>

C-MAP Chart Management

The C-Map Web store and My Maps chart management functions are accessed by clicking the C-Map button on the Charts [panel](#).

Chart purchase is managed by the Web Store. Users who want to buy charts from the Web Store need to create an account to become a C-MAP customer first.

The Web Store allows charts to be purchased online. The charts may be purchased in as either MAX or full 4D.

My Maps controls downloading of purchased charts.

C-MAP Web Store

This will open the Web Store, see image below. To buy a chart

1. Click on Maps Catalogue
2. Select the cartridge price class type - Local or Wide
3. Select the desired chart by using the map on the right, the search box or list of chart areas.
The paper charts in the selected electronic chart area will be listed at the bottom of the page.
4. Click Show HOs and Show VADs to show lists of Hydrographic Offices and available value Added Data.
5. Select the cartridge type
 - 4D MAX (contains only vector chart data)
 - Full 4D (contains all data - vector chart, satellite, 3D, raster chart etc.)
6. Click the Buy button
7. In some countries (currently EMEA areas), it may be possible to purchase charts using a special SD Card, called a Transport Card. This allows you to use the charts without downloading them from the Internet. Once you have received the Transport Card, insert it into your PC. The Maps Manager recognises the inserted SD Card and uploads the SD content into the PC. The Transport Card is a special SD Card which is used ONLY for downloading charts - Expedition cannot read charts directly from the Transport Card.
8. My Maps can be used to download and update chart areas.

C-MAP 4D

Select price class:

☐ Local

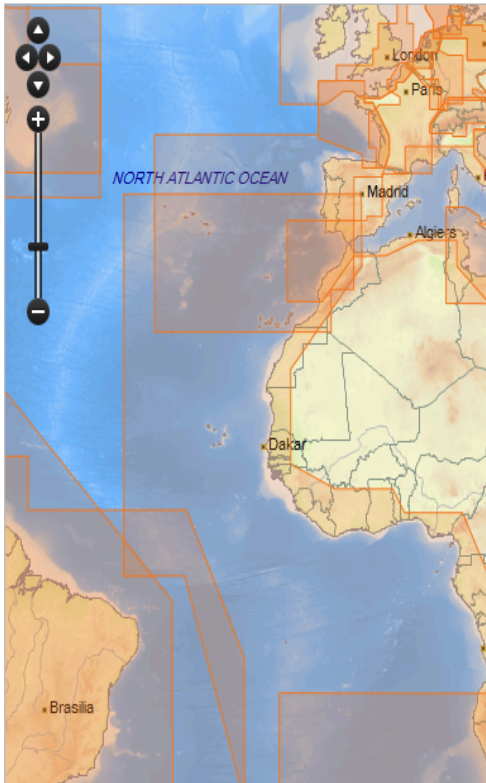
☒ Wide

Search:

Search

Code Description

AF-D003.38	ANTARCTICA
AF-D209.38	SOUTH - EAST AFRICA
AF-D210.38	NORTH - WEST AFRICA
AN-D013.38	KAMCHATKA PENINSULA AND KURIL ISLANDS
AN-D050.38	ASIA NORTH CONTINENTAL
AN-D204.38	JAPAN, AND NORTH AND SOUTH KOREA
AS-D050.38	ASIA SOUTH CONTINENTAL
AS-D205.38	PHILIPPINES, PAPUA NEW GUINEA, E. INDONESIA
AS-D214.38	CHINA, TAIWAN AND VIETNAM
AU-D010.38	VICTOR HARBOR TO WELLESLEY ISLANDS
AU-D011.38	CERVANTES TO SEAL ROCKS



My Maps

Once the chart has been purchased, it is linked with the C-MAP account registered on the computer and has to be downloaded.

Click My Maps in Expedition as shown above. A table of charts associated with the current registered account will be shown.

These will be either:

1. On the Cloud - purchased, but not yet downloaded,
2. On the device - downloaded and available for Expedition.

To download a chart listed as On the Cloud, just click the Download button. Download progress can also be monitored here.

Be aware some of these chart files are large (several GB), so allow sufficient time for download.

The charts may also be disabled or removed using Your Maps. Note that if removed, they must be downloaded again. This can be a problem if a chart that was purchased some years previously is no longer available for download.

Display settings

Also see the [chart settings](#) and [chart panel](#) help topics.

Language

Available C-MAP languages.

Palette

Available colour palettes. The NOAA colour palette is default.

Raster mode

Raster Opacity

Select raster charts and/or satellite images.

Raster charts have limited coverage.

The opacity (or transparency) of the raster charts can also be specified.

Depth shading

Available colouring shading options. The Safety depth is used with the Safe depth colouring option.

Base data only

Hides most items on the chart. Only useful if using the C-MAP chart solely as a weather backdrop etc.

Chart bounds

Chart boundaries.

It is recommended this setting is enabled as it provides important visual indications of smaller scale charts.

Large text

Large icons

Draws the chart with larger icons and/or text for improved readability.

Lat/lon grid

Orientation

Set on the [chart settings](#) page.

Satellite overlay

Tides & currents

Tide-ways, water turbulence and tide height. In the case of C-MAP MAX charting, also displays tidal streams.

Continuous animation with C-MAP tides and currents showing can be slow.

C-MAP tidal streams are for display purposes only. See the tide settings help for Expedition's tides and tidal streams.

VAD (value added data) bounds

Great Barrier Reef, photos etc. Also tide stations in C-MAP MAX.

Multimedia

C-MAP objects

Use the Query C-MAP object menu item on the right-click [pop-up menu](#) to interrogate any C-MAP object.

Select any item in the tree control on the left of the C-MAP object window to display its attributes in the right hand window.

In the case of a tide station, times of high and low tides as well as sunrise and sunset will be displayed.

In the case of a C-MAP image, a picture will be displayed. Note that to suit smaller screens, expanding the dialog will show more detail in the displayed window. This will also be retained in the docking [C-MAP pane](#).

Errors

WWD303 More free Base maps must not be updated once installed or the C-Map 4D system may stop working and give this error.

Core::initialize ERROR!!!

Error when initialize CM library (WWB Cartridge NOT found)

If this does occur, one solution is

1. Stop the C-Map Manager Service (see below)
2. Within the folder C:\ProgramData\C-MAP\MapsManager delete the files:

WWD303.conf
WWD303.conf.bak
bkmngmt.config
3. Restart the C-Map Manager Service (see below)
4. Restart Expedition

To stop and restart the C-Map Manager Service

1. Click on the Windows start button on Windows' taskbar and type services
2. An option for the Services desktop app will appear. Select it.

3. Scroll down and select 'C-Map Maps Manager'
4. Select 'Stop the service'
5. To restart the service, select 'Restart the service'

This procedure may also be useful for other C-Map 4D errors after downloading chart areas.



C-Map MAX

[C-MAP](#) MAX/PC charting is no longer supported.

Reference guides

C-MAP provides two quick reference charts that can be downloaded from the [Expedition](#) or C-MAP web sites:

C-MAP chart legend: A visual reference and description of objects on C-MAP NT+ electronic charts.

C-MAP NT Cartography Reference Guide: Detailed information about C-MAP charts, the different formats and features.

C-MAP NT/MAX PC Selector

MAX charting requires C-Map's [NT/MAX PC Selector](#) for installation and licence management.

C-MAP MAX charts are normally distributed in iso format.

Please see the help [below](#).

C-Map MAX and Expedition

See the [chart](#) panel help.

Open Best chart

With Auto chart mode selected, the next higher or lower scale chart will be selected when you scale in or out. If this mode is not selected, scaling in or out will display the current chart at different scales.

This mode also influences C-MAP MAX chart behaviour. With it unselected, you can scale in and out on the same chart level.

C-MAP MAX scales between levels that contain chart data for different scales as displayed in the [status bar](#) and may be over-zoomed as you scale in.

For example, Level B/0 means level B data that will contain data between 1:500,000 and 1:1,500,000. Level B/1 is level B data scaled in one step. The display scale is also shown in the status bar.

Expedition should step over any levels that are empty as the display is scaled in and out.

These are represented below.

LEVEL_W 26,400,000

LEVEL_X 13,200,000

LEVEL_Y	8,800,000
LEVEL_Z	6,600,000
LEVEL_A	1,500,000 - 5,000,000
LEVEL_B	500,000 - 1,500,000
LEVEL_C	150,000 - 500,000
LEVEL_D	50,000 - 150,000
LEVEL_E	15,000 - 50,000
LEVEL_F	5,000 - 15,000
LEVEL_G	1,500 - 5,000

See the chart panel [help](#).

Palettes

By default, Expedition displays C-MAP charts using the default C-MAP colour palette. However, NOAA, Admiralty or grey scale palettes may be used. See the C-MAP chart settings [help](#).

Display settings

For other settings, see the C-Map 4D help [topic](#).

Colour schemes

Reference depth colouring

Colours depth areas less than the value specified in Reference depth (m).

This over-rides and disables Colour sea areas, Dynamic depth colouring and Invert depth colouring.

Dynamic depth colouring

Dynamic depth colouring.

Invert depth colouring

Inverts dynamic depth colouring.

Enhanced anti-clutter

Uses the C-MAP enhanced de-clutter algorithm for cleaner charts.

Enhanced coastal land colouring

Low lying areas can be hard to see on some chart plotters. This accentuates them.

Land features

Can be used to declutter land areas.

C-MAP objects

Use the Query C-MAP object menu item on the right-click [pop-up menu](#) to interrogate any C-MAP object.

Select any item in the tree control on the left of the C-MAP object window to display its attributes in the right hand window.

In the case of a tide station, times of high and low tides as well as sunrise and sunset will be displayed.

In the case of a C-MAP image, a picture will be displayed. Note that to suit smaller screens, expanding the dialog will show more detail in the displayed window.

C-MAP NT/MAX PC Selector

MAX charting requires C-Map's [NT/MAX PC Selector](#) for installation and licence management.

Please see the help included with NT/MAX Selector for detailed information.

NT/MAX PC Selector acts as an electronic chart catalogue to choose the charts needed. It generates licence code requests and manages chart licences and installation.

NT/MAX PC Selector can be launched from the C-Map button on the [charts](#) panel.

When using NT/MAX PC Selector to request a chart licence, make sure NT Selector shows Expedition as the client as illustrated below. If you have run Expedition before, it should be shown. If not, start Expedition and then start NT Selector.

If using a Eutron or Sentinal dongle, make sure the Eutron or Sentinal dongle is shown in the status-bar. If using a dongle and the status-bar shows C-MAP HID (as below), the dongle drivers probably need to be installed.



The user has two options when purchasing and loading these charts - they can be linked to the PC or to a security dongle. One advantage of the dongle is that the chart can be used on any computer the dongle is connected to.

The dongle security key solution makes migrating C-MAP charts to a new computer easier.

Important notes on C-MAP MAX NT Selector

The C-MAP destination folder must not be in Windows' Program Files. This is mainly a problem with old releases.

A common location is C:\ProgramData\C-MAP NT PC Selector\Charts\

If there are problems with the charts not being recognised, also check the folder security settings (right-click on the folder, select Properties etc.). These should be set for Full control, Modify and Read & Execute.

Using the NT/MAX backup and restore procedures are highly recommended. See the NT Selector help.

Legacy C-Map chart sources

Expedition can also utilise C-MAP NT+ charts from the following legacy sources:

C-Card

Requires a C-Card Reader. The required drivers should be provided with the cartridge reader, but may be downloaded from the [Expedition](#) or C-MAP web sites, and should be installed according to the on screen instructions when you first connecting the USB lead from the reader to your computer.

PCMCIA

The required drivers should be provided with the card, but may be downloaded from the [Expedition](#) or C-MAP web sites, and should be installed according to the on screen instructions when you first install the card in your computer.



Marks and routes

Marks.



overboard

The man overboard button on the [marks](#) panel sets a locked mark called MOB at the current boat position.

Also available from the and by using the F-2 keyboard shortcut.

The MOB attribute for a mark may also be set by the Ventus smart-phone app or an Exp MOB button for example.

Important note

Computers can be less reliable than a purpose built GPS, so always use your boat's GPS system as your primary MOB position fix system. Use Expedition as a backup - displaying the track on the screen

is a very good way visual of seeing your way back. Expedition not designed or intended

as a primary life safety control.



Visual marks

Also known as easy or quick marks. Accessed from the [Marks panel](#).

This is a quick and easy way to graphically set a new route. As per its name, this is often the easiest way to set up a new route.

First, scale out to a chart scale where you can see your entire route area. Now select Visual marks. A window will pop up where you can name the route or simply accept the name Easy by clicking OK. Now you can move the mouse pointer onto the main chart window. Wherever you left click, a mark will be laid and a route will be created following these marks in the order in which they were created.

Remember to deselect Visual marks when you're finished creating the route or else you will create an additional mark every time you use the left click button.

Marks can be easily added by using the insert mark item on the [pop-up](#) menu while you are creating a visual route.

Loop

If a route is set to be able to *loop*, moving to the next mark from the last mark sets the first mark as the active mark. Similarly, moving to the previous mark from the first mark selects the last mark as the first mark.

Fine tuning the position of a mark

After setting up an Quick Route you can then scale in to a particular mark and fine tune it.

Drag

You can drag a mark to a desired location by left mouse clicking it (an arrow will appear on the mark when it is drag-able) and whilst keeping the left click button depressed, move the mark where you want it.

Edit on chart

By right mouse clicking a mark in the main chart window you open a drop down window. If you select Edit mark, a Mark properties window will open where you can enter a precise latitude and longitude and perform a number of other functions including locking the mark so that it is not drag-able.

Edit in Mark management

You can get to the same Mark properties window by highlighting/selecting (with a left mouse click) any mark in the Mark management window and clicking the Edit button.

The ability to drag a mark is highly useful in buoy racing where the race committee does not always place the marks exactly where intended.

Mark Management and Visual Marks

If you accepted the name Easy when creating the Visual route your new marks are stored in the Easy folder in Mark management. They will be overwritten next time you use Visual marks. You could rename the mark and route folders using mark management, but it is easier to choose a unique route name when you first click this button.



Right mouse click to build a route

You can build a route by right clicking a point and selecting Set Active Mark and Route from the [pop-up menu](#). Subsequent marks in the route can be added by right clicking and selecting Insert Mark for example. The pop-up menu can also be used to add marks at the start and end of a route or to insert an already existing mark into any point of the route. See the [pop-up menu](#)

help for further assistance.

As with Easy Marks the marks you create in this manner can be found in the marks section of [mark management](#).

The [pop-up menu](#) tools can be very helpful when fine tuning or refining a route.

The inserted mark will always be placed in the nearest great circle leg of the route to the pointer.



Mark management

Accessed from the [marks](#) panel.

The mark management window can be used to modify existing routes and create new routes from existing marks that are in your mark folders.

To build a new route, first create an empty route by clicking the New button to the left of the routes section. The new route can be renamed by clicking the selected route.

Marks can be added to the selected route by selecting a mark and clicking the button. If the route is selected, the mark will be added to the end of the route. If a mark in the route is selected, the mark will be added after that mark.

To remove a mark from a route folder simply highlight it and click the right mark button.

Set a route to be the active route by selecting the it and then clicking Select.

Tips

Marks can be dragged between mark folders.

Mark and route names and their attributes may be edited by selecting the label in the tree control.

Marks

New

Creates a new mark or mark folder.

Edit

Edits a selected mark.

Delete

Deletes a selected mark or mark folder.

Lock

Locks a mark or all marks in a folder.

Unlock

Unlocks a mark or all marks in a folder.

MOB

The Man Over Board attribute for the mark. This may be set by the Ventus smart-phone app or an Exp MOB button for example.

Always draw

Sets a mark or all marks in a folder to be always drawn

Don't draw

Sets a mark or all marks in a folder to be not always drawn.

Port end

Sets the currently selected mark to be the port end of the start line.

Starboard end

Sets the currently selected mark to be the starboard end of the start line.

GPX

Exports the selected mark folder to GPX.

Table

Create a .csv format table of ranges and bearings between the marks in the selected mark group.

Routes

Leave to port, Leave to starboard or Sail to mark

It is also used when computing laylines - Exp adds or subtracts half a boat [width](#) when leaving a mark to port or starboard.

Loop

Toggles the route loop attribute on and off.

New

Creates a new route.

Delete

Deletes a route if selected or removes a mark from a route if selected.

Move up

Moves the selected mark of the route up one.

Move down

Moves the selected mark of the route down one.

Set active

Sets the selected route as the active route.

Clear active

Clears any selected, active route.

Details

Displays details about the currently selected/active route.

Can also edit a route's loop property. If a route is set to be able to loop, moving to the next mark from the last mark sets the first mark as the active mark. Similarly, moving to the previous mark from the first mark selects the last mark as the first mark.

Reverse

Reverses the order of marks in the selected route.

Reverse

Creates a copy of the selected route.

GPX

Exports the selected route to GPX.



Optimal routing

Route optimisation.



Weather settings

Available from the weather [display](#) panel.

The weather settings page is used to manage how weather data is displayed in Expedition and also to scale, colour etc. tidal and track currents.

Displaying and working with GRIB data in the main chart window

See the [weather display panel](#) and the [weather data settings](#) help.

Display control

The controls on the left of the weather settings page are used to select what information from the grib data is drawn on the chart and how. Just change the options for each grib parameter (wind, mean sea level pressure, current etc.).

The Defaults button has some useful pre-sets to populate these.

Options include:

Contours

For example, contouring mean sea level pressure will give you a weather map similar to the ones we are all used to. Contouring wind speed gives a visual representation of areas of light or strong wind and wind speed gradients. For example, it may be helpful to display wind speed contours whilst stepping forward in time and studying an optimal route, as this will depict the hot and cold spots of wind velocity more obviously than standard wind barbs.

Shading

Fills the chart with transparent shading, varying in intensity between the shade min and shade max values. Colours used in the shading can be set using the colour and colour mode properties.

Fade colours

Use brighter colours at larger values. This can be useful to visually show areas of light wind and may be combined with either the colour or spectrum options. The fade effect will be between the contour min and contour max values.

Colour

Colour contour lines, arrows or barbs in a single specified colour.

Colour mode

Select whether to draw contour lines, arrows, barbs or shading in the colour selected or in a spectrum based on red to blue, a grey scale or the beaufort scale.

Arrows

Select whether to draw vectors with arrows or barbs.

Line width

Select width of contours or barbs.

Contour step

How often to draw the contours. For example, selecting 4 for pressure will draw pressure lines every 4mb.

Contour high-light

Draws the selected level in bold. This may be useful for emphasising an area of strong or light winds, temperature or a jet stream for example.

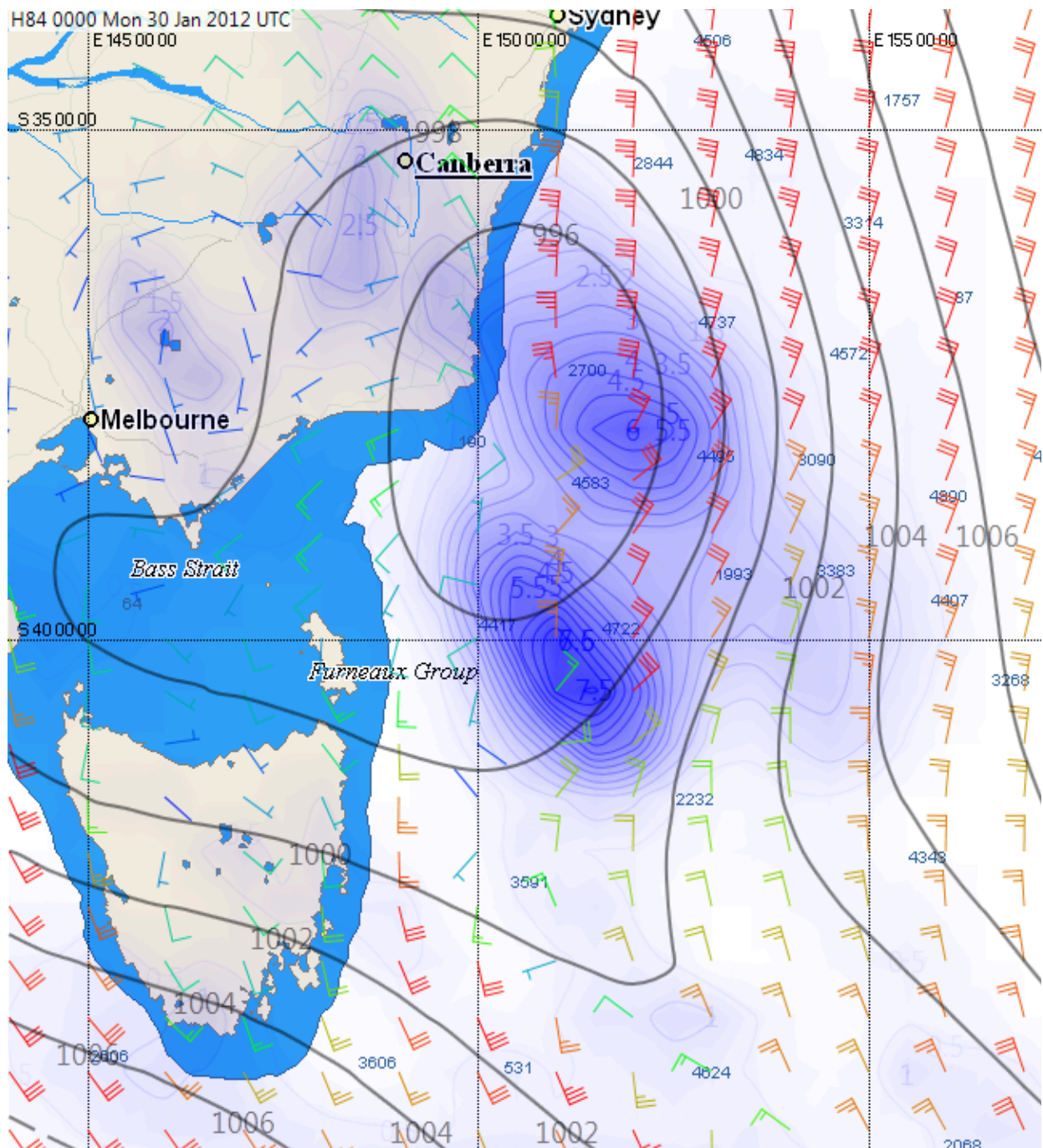
In the example below

Rain is drawn with shading and contouring selected. The contour interval is 0.5mm/hr.

Winds are drawn using barbs and a spectrum of colours. This can give a good visual reference. Wind speed is often contoured.

Surface pressure is drawn using 2 pixel width contours every 2mb.

The time is displayed in UTC at the top-left of the screen. Local time could be selected on the [System settings](#) page.



Display

Animation interval (minutes)

Set at 60 minutes by default, but the user can change the animation interval or time step when moving forward or backward in time whilst studying a GRIB or stepping through an optimal route.

Use the forward or reverse buttons on the [weather display panel](#) to step forward or backward in time. (You can also use the keyboard up arrow or down arrow to move forward/backward in time.) Time is displayed in the upper left corner of the main chart window and is in either UTC or local computer time depending on how you have configured Expedition in the *Local time* check box on the [system settings](#) page.

Always show currents and SST

Often, ocean current, temperature or height anomaly grib data will be an historic analysis, so won't normally be displayed at the same time as weather forecasts. Select this option to always display this data when showing weather forecasts.

This option is especially useful for races such as the Bermuda and Sydney to Hobart races where currents are very important.

Mode

Standard

The standard mode. Will select the best data or the model selected on the weather settings ribbon.

Ensemble

Will display all ensemble and individual model data loaded.

Ensemble statistics

Displays mean and standard deviation ensemble data.

Data

Scale currents

Expedition can apply a multiplication factor (in percent) to the currents in a GRIB file. (The default is 100%). This might be desirable if you believe the GRIB file is incorrectly predicting current speed. This factor has no effect on the current speed and direction reported by the instruments (or the *What If?* currents set by the user. These instrument or *What If?* currents drive the laylines and times whilst sailing. The GRIB file currents (which can be scaled here) are used for route optimizations.

Rotate 10m wind

Allows the user to rotate the GRIB winds by a fixed number of degrees. You might want to do this if you determine the

GRIB file is systematically incorrect in direction by a few degrees. Normally you leave this at 0, but if you do want to tweak the GRIB, a positive number rotates the winds in a clockwise direction and a negative number rotates in counter-clockwise in the specified number of degrees.

Scale 10m wind

Similarly, Expedition can scale the GRIB file's wind speed. Why scale the GRIB winds? Because wind speed usually increases with height and you are going to use these GRIB files for either route optimization or for study before a race. If it is the latter, you want the wind speed to be scaled to your masthead/instrument height so that the predictions match your expectations. If you are doing route optimisation, you want the wind speed to be scaled to the wind speed height that the [polar](#) is based upon. Although most yacht designers' polar files are scaled to a 10 metre height wind, most sailors want the polars and targets scaled to masthead instrument height.

Converting polar files to masthead height is simply done in Expedition as we will learn in the section on [polars](#). Wind speed generally increases with height. There is a simple formula that can be used to estimate how more wind is at masthead height than it is at 10 metres (which is a common height that GRIB files are scaled to).

$$Tws_{wand} = Tws_{10m} \cdot (h / 10)^a$$

where h is masthead/instrument height in metres and a is a constant (Hellmann exponent) which is normally in the range 0.11 to 0.14 at sea.

Example: If your mast/instrument height is 20 metres and your polars are scaled to this height then you can use the formula above to figure out what the percentage scaling you should apply to a 10 metre height GRIB file.

The answer is:

$$x = (20/10)^{0.12} = 1.09 \sim 109\%.$$

So you should enter 109 into the Scale winds % edit box.

User per-model scaling

Expedition allows the 10m win to be scaled independently for different wind speeds for different models.

To enable, select this checkbox. Click the Models button to edit.

The scalings are saved in the WxModels.xml file.



Weather data settings

Available from the Weather [data](#) panel.

The weather data settings page is used to manage weather data in Expedition.

Displaying and working with GRIB data in the main chart window

See the [weather display panel](#) and the [weather settings](#) help.

Loading GRIB files into Expedition

The main window on the weather page shows the Grib files (if any) that are currently loaded into Expedition. There is no practical limit to the number of Grib files that can be loaded at one time. For example, Expedition can use both current and wind files in route optimisation and can merge several Grib files that cover different times or areas.

Only grib files that are checked are actually used by Expedition.

Grib files may be loaded into Expedition several ways:

Select and drag

Drag them to the list on the left of the weather settings page.

Grib files can also be added by dragging them onto the main chart window if they have .grb, .grb1, .grb2 or .bz2 suffixes. This automatically loads the grib file into Expedition, but does not remove grib files from the grib file list. It is advisable to remove any unnecessary grib files from the list. Bz2 files will be automatically decompressed and the original .bz2 file moved to Windows' recycle bin.

Add file

The Add file button on the weather settings page opens a Windows' window that can be used to select grib files to add to the list.

If the Grib files do not have a .grb, .grb1, .grb2 or .bz2 suffix, they will not be automatically visible in the Windows open file dialog. Select All files at the bottom of the open file dialog and they will appear.

Note that more than one file may be selected.

Grib folder

As for Add file, but defaults to the Expedition grib download folder.

Squid

As for Add file, but defaults to the [Squid](#) grib download folder. Note this is only useful if using Squid to download Squid grib data. If using the Squid download integrated into Exp, these will be saved into Expedition's grib folder and automatically loaded.

If using Exp's integrated weather services (such as [Expedition](#), [Saildocs](#), [Squid](#) or [Tidetech](#)) to obtain grib files, these will be automatically loaded into Expedition.

To remove Grib and NetCDF file(s) from the list

Select the file or files to be removed, then click on the Clear All or Clear Selected buttons.

SkewT

Enable if using the pop-up [SkewT](#) diagram.

This is normally left unchecked as the extra data may make route optimisation etc. slower.



Available from the Saildocs button on the [Weather data panel](#).

[Saildocs](#) has an email or direct web delivery service for the delivery of grib weather data.

For more information, send email to the auto-respond addresses info@saildocs.com and gribnews@saildocs.com.

Requesting smaller areas with less fields, times and resolution will result in smaller grib files. This can be important with a slow or expensive email connection, especially at sea.

To use, click on the Saildocs button, then select an area of the chart to obtain weather for.

Check VT

Allows a check on the analysis time of the grib file to be downloaded. Useful to check if the grib file has been updated.

Get Grib

Obtains a grib file directly from the web, rather than via email, and automatically load the received grib file into Expedition.

Clear grib file list

Automatically removes previous grib files from the file list. If not selected, it is advisable to [remove](#) any unnecessary grib files from the list.

Email request

Creates a Saildocs email query.

Expedition automates the creation of the email message required to receive a Saildocs grib file, but you can edit the Expedition-created email before sending.

Days

Number of days to subscribe for.

Subscribe

Selecting subscribe will change the email to request grib files for desired number of days.

Resolution

Select the desired resolution of your grib file. Higher resolution grib files will contain more detail, but a 1° grib file may be close to 4 times as large as a 2° grib file.

Note that higher resolution options may not be available for all models. For example, COAMPS data is available at 0.2°, RTOFS is available at 0.05° and GFS is available at 0.5° resolutions.

Model

Various models and data sources are available from Saildocs.

GFS

The default global model from NOAA/NCEP. Data is generally available at 0.25° resolution every 3 hours to 120 hours, then at 0.5° resolution to 240 hours then at 1.0° resolution to 384 hours, .

Updated every 6 hours and is generally available by about 5 hours after the main synoptic hours.

ECMWF

ECMWF data at 0.25° resolution every 3 hours to 144 hours, then 6 hourly to 240 hours.

Updated twice daily on Saildocs.

ICON

DWD ICON data at 0.25° resolution every 3 hours to 240 hours.

NAVGEM

A US Navy model. Data is available at 1° resolution every 3 hours out to 24, then every 6 hours to 96, then every 12 hours to 144.

Updated every 6 hours and is generally available by 6 hours after the main synoptic hours.

HRRR

NOAA's High-Resolution Rapid Refresh model for the US.

Updated hourly.

NAM

NOAA's NAM model.

NDFD

The National Digital Forecast Database is a US National Weather Service (NWS) project to make forecaster-generated weather forecast data available in gridded format for the NWS areas of responsibility. These data files are

prepared along with the forecast charts (radio fax charts) and text bulletins.

NDFD is a developing project and available data will be expanded. NDFD is still in "experimental" status which means that the data may not be available or correct.

RTOFS (Real Time Ocean Forecast System)

An ocean current forecast model from NOAA's RTOFS system.

Forecast times

Select the forecast times to request.

Clear

Clears all selected times.

Default

Selects default times to request.

Parameters

Select desired fields. Defaults are PRMSL and WIND.

Note that the parameters selection has no effect if no forecast times are selected.



Great Circle - Squid

Available from the Squid button on the [Weather data panel](#).

Web

www.squid-sailing.com

To use, click on the Squid button, then select an area of the chart to obtain weather for.

Models/areas

Grib data available for the area selected

Parameters

Select desired fields. If none of these are selected, the default is 10u, 10v and prmsl.

Note that to download wind, both u and v components (e.g. 10u & 10v) must be selected.

Forecast days & time-step

Select the forecast times to request.

Clear old grib files

Automatically removes previous grib files from the file list. If not selected, it is advisable to [remove](#) any unnecessary grib files from the list.

JPEG Compression

JPEG compression results in smaller GRIB files. They will be slower to load into Expedition though.

Update models list

Refreshes the list of available products available for the user's subscription.

Email

Sends a grib email request to Squid.

Download

Obtains a grib file directly from Squid.

Squid X app

Great Circle also provides an app that can be used to download grib and satellite data.

This also allows Iridium Go.

It can be a good idea to set the GRIB Export directory to the Expedition Grib data folder, normally
C:\ProgramData\Expedition\grib.



Expedition GRIB server

Available from the Expedition and button on the [Weather data panel](#)

Data from other sources, provided to complement [Saildocs](#) and [Great Circle - Squid](#).

To use, click on the Expedition grib button, then select an area of the chart.

Notes on resolution

From experience, $1/5^\circ$ - $1/4^\circ$ is a good compromise for global model data.

- $1/2^\circ$ is usually too coarse, but may be an option for slow or expensive connections.
- $1/8^\circ$ - $1/10^\circ$ data may be more useful, but file sizes will be larger.
- Doubling the resolution will increase file size by about a factor of 4.
- Higher resolution data will also take longer for the server to extract and package.

GFS	0.11° and 0.25°
UM	0.1°
ECMWF	0.2°
ICON	0.1°
GDPS	0.15°
Arpège	0.1°-0.25°
NBM	0.09°
Mercator	1/12°

The regional weather and oceanographic models are all higher resolution (Access, Arome, Harmonie, HRRR, RAP, ICON-EU/DE, UM-UKV, WRF, OFS, Copernicus etc.).

For slow connections, and depending on the area and number of time steps desired, it may be worth downloading fewer time steps or over-riding the standard internet time-outs in Expedition's [system settings](#).

Times

Select the forecast times to request.

Clear

Clears selected times.

All

Selects all displayed times.

Defaults

Selects default times to request.

Model

Various models and data sources are available from Expedition.

For details, see <https://www.expeditionmarine.com/weather.html>

Clear old grib files

If this is selected, previous Expedition grib files loaded will be removed from the file list.

Previous run

Gets the previous model run. This may be useful when the latest run is a shorter range forecast.

Halve resolution

Sub-samples the grib data at half resolution. This will reduce file size by up to 75% and may be useful for low speed connections.

bzip2 compression

Compresses the grib file to be downloaded.

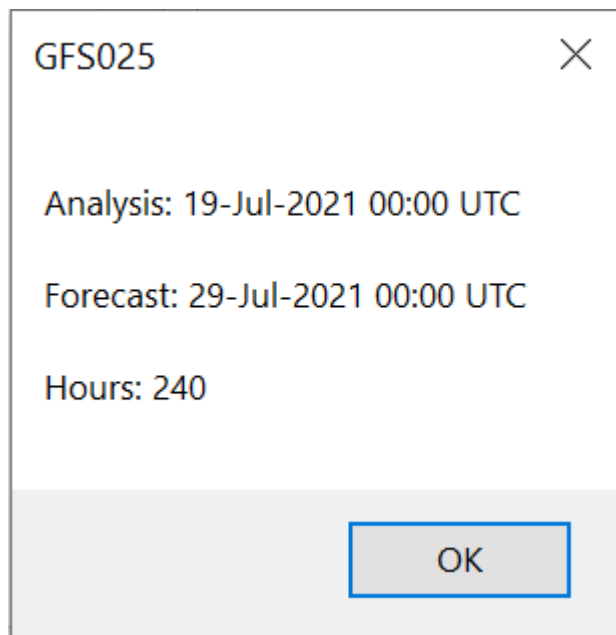
It is slower for the server to create, so unless file size is an issue, performance will be better if disabled.

Run time(s)

Shows the analysis time and last forecast time of the grib data available.

This is useful to check if the data has been updated.

Note that for most models, the data will gradually become available and the forecast time and hours is the maximum forecast currently available.



Download

Downloads the grib file.

Note that failed downloads due to lost connections etc. may be resumed for downloads from the Expedition grib server, but not for the other grib services.



Expedition WRF server

Available from the WRF button on the [Weather data panel](#) or from the right-click [pop-up menu](#).

Expedition WRF

Expedition high resolution WRF models, mostly at $1/12^\circ$, $1/36^\circ$ and $1/108^\circ$ resolutions. Most areas are updated 6 hourly, but some may be updated every 3 hours.

Selecting the WRF button on the [Weather data panel](#) will show all grib files available.

Selecting Expedition WRF from the right-click [pop-up menu](#) will show only grib files for that location.

Clear old grib files

If this is selected, previous Expedition WRF grib files loaded will be removed from the file list.

File info

Shows the file size, analysis time and creation time of the grib file to be downloaded.

Useful to check if the grib file has been updated.

Data options

There are 3 data options

All data - currently MSLP, rain, 10m wind and gusts, 20m wind, 2m temperature, CAPE and total cloud cover.

Base data - 10m wind, MSLP and rain

Wind data - 10m wind only

Download

Downloads the grib file.

Note that failed downloads due to lost connections etc. may be resumed for downloads from the Expedition grib server.



A collection of third party Grib and NetCDF files available on the internet.

Available from the MyGrib button on the [Weather data panel](#) or from the right-click [pop-up menu](#).

The MyGrib button on the [Weather data panel](#) will show all grib files.

Selecting MyGrib from the right-click [pop-up menu](#) will only show grib files for that location.

Clear old grib files

If this is selected, previous MyGrib files loaded will be removed from the Expedition file list.

File info

Shows the analysis time, issue time and size of the grib file to be downloaded. Analysis time is not available for some data.

Useful to check if the grib file has been updated.

Download

Downloads the grib file.



Available from the SailFlow Refresh button on the [Weather data panel](#).

Expedition can display weather observations from SailFlow. A SailFlow Plus, Pro or Gold membership is required.

Web

www.sailflow.com

Use

Click on the SailFlow Refresh button. The latest weather observations for the chart area will be downloaded.

The SailFlow user-name and password can be entered on the Weather data panel.

See also the Weather observation options in [Display Settings](#).



SHOM Tidal currents

Expedition includes [SHOM](#) tidal current models (Modèles de courant de marée en 2D).

Areas

557, Courants de maree dans le Pas de Calais

561, Courants de maree de la Baie de Seine, de Cherbourg a Fecamp

562, Courants de maree du Golfe Normand-Breton

563, Courants de maree Bretagne-Nord, des Heaux-de-Brehat a la Pointe de Pontusval

564, Courants de maree de la Manche de Dunkerque a Brest

558, Courants de maree de la Cote Sud de Bretagne, d' 'Audierne au Croisic

559, Courants de maree de la Cote ouest de France, de Saint-Nazaire a Royan

560, Courants de maree de la Cote ouest de Bretagne, de Goulven a Penmarc'h

565, Courants de maree dans le Golfe de Gascogne



Expedition integrates [Tidetech](#) sea temperature, ocean and tidal currents. A subscription to Tidetech is required for this functionality.

Select Tidetech from Weather data [ribbon](#) and then select an area of the chart.

TideTech

Meteorology

Currents

English Channel Currents (20.10mb)

Global Combined Currents (1557.32mb)

Global Ocean Currents (70.09mb)

Northwest European Shelf Currents (39.59mb)

Solent Currents (25.79mb)

Waves

Observations

API key:

☐ Clear Grib file list

[Tidetech.org](https://tidetech.org)

Update list

Cancel

Download

Enter your Tidetech API-Key, available from your Tidetech web dashboard.

The Update list button updates the list of products associated with your subscription.

Note that the list of products are only those available at the location selected. Areas available can be drawn on the chart, see [display settings](#).

To download a Tidetech grib file, just select the item in the list and then the download button.



Weather and tides

GRIB files

A Grib file is a digital weather or current file. These are available from a variety of sources - see www.ExpeditionMarine.com for examples.

Expedition can seamlessly merge and use multiple Grib files, automatically using the best available data in the selected Grib files for its calculations.

There is some variation in the types of Grib files that are available, but Expedition can read the most common types.

See the [weather settings](#) help for further help on using grib files.

NetCDF files

Expedition supports some NetCDF data. This can be expanded as support for new datasets is requested.

Tides and currents

See the [tide settings](#) help.



Optimal routing settings

Available from the [Optimal](#) panel.

See below for [advanced](#) settings.

See also the optimal routing [help](#).

Configuration

Start time

The time and date to start the routing from.

Now

Sets the time to now. The time will either be local or UTC depending on how you've configured Expedition using the local time check box on the [system settings](#) page.

Also see Optimise from current position and time below.

Resolution settings

Resolution

Determines the resolution and isochrone interval to use.

High

Will generally yield a 'better' solution than medium, but will take longer.

Medium

Default, suitable for most uses.

Custom

Allows the user to specify the grid and isochrone resolutions.

It is normal to select Medium or High resolution, which direct Expedition to automatically choose the routing resolution based on the grib time steps and grid resolution and leg length.

If not using auto-resolution, the grid resolution or isochrone time step can be defined. The smaller the resolution or time step, the longer the calculation will take.

See notes below.

For the isochronal algorithm, the time step is controlled by the isochrone resolution.

For the grid based algorithm, the resolution is controlled by the grid resolution.

Initial scan angle

Initial search angle for isochronal route optimisation.

Expedition will initially search half this value each side of the route, but larger values will also increase the search area. Note that the Draw paths option is useful for visualising the search area.

Default is 200°. Larger values may be necessary at times, but will take longer to run.

Minimum isochrone resolution

Limits the minimum automatically determined isochrone time steps when using high and medium resolutions as allowing too low a resolution may yield worse results.

Also helps performance.

Custom isochrone resolution

Used in custom resolution (see above).

Not generally recommended.

Custom grid resolution

Used in custom resolution (see above).

Smaller values will generally yield more accurate solutions for the grid routing algorithm, but will be slower.

Optimised routes to keep

Optimal routes in excess of this will be discarded.

Drawing options

Route colour

Options for colouring the unique optimal route.

Shade

Shades the optimal routing area by the selected weather parameter.

Isochrone drawing interval

The time step to draw isochrones at. May be approximate with isochronal solutions.

Shade time sensitivity

Shades an area where the routing is within the specified number of minutes of the optimal path.

Note that the 'Calculate reverse isochrones and sensitivity' option must be selected for this.

Configuration options

Optimise from current position and time

Route from your current position and time. This is the normal mode during a race. For pre-race planning, deselect this and manually set the start time.

Optimise first leg only

If a multiple leg route, only optimise the first leg of the route or the leg to the active mark. Often this is all you are interested in and it can speed the calculation.

Isochronal routing may only optimise the first leg.

Optimise along great circle

Used for planning. Sails along a great circle route between marks. When current are included, will approximate a great circle.

Only valid for the grid based routing algorithm.

Prefer tacks and gybes at routing steps

May reduce usefulness of the output and may not be available for the grid based algorithm.

Not normally used or recommended, especially for the grid algorithm.

Note that is possible to encounter a scenario where the isochronal routing can't reach the finish with this option selected.

Adjust polar for air density

Includes an effect for air density. Requires pressure, temperature and preferably dew point or relative humidity. Will use dry air density if dew point and relative humidity are not available.

The reference for the polar is a standard atmosphere (1013.25hPa and 15C) and a relative humidity of 80%.

Correct polar for waves if available

Uses grib data and wave correction parameters to modify the optimal route for wave effects.

See [waves](#).

Sail polars

Advanced users only.

Calculate reverse isochrones and sensitivity

Enabled by default as are very useful, but may not be needed for (e.g.) routing studies.

See comments on display options below.

Use port and starboard asymmetric polars

Allows separate [polars](#) to be used instead of the nav polar.

The most likely use of this is if a foil is damaged on one side of a foiling boat.

Motor if speed below minimum sailing speed

As set on the advanced optimal routing settings page.

Land options

Avoid land

Instructs the route optimisation to avoid land.

This option uses the simple world chart installed with Expedition. This is an approximation of the real coastline.

It does slow the route optimisation, so may not be needed offshore or for routing studies.

Note that [race notes](#) can also be used to mark custom areas to avoid, such as shipping channels and ice exclusion zones.

Avoid ENC land and safety depth

If using ENC charting, the routing can attempt to avoid land areas as well as depth areas and contours shallower than the [safety depth](#) in ENCs currently being used.

Probably only useful for very fine scale routing. Not normally needed or used and will slow the route optimisation, especially for the grid algorithm.

Misc options

Load previous optimised routes at startup

Saves the route optimisations for when Expedition is restarted.

Export optimal route to marks database

Outputs the computed optimal path to the mark database.

This feature is rarely used, but it can be very helpful for archiving optimal routes for comparing the results using different GRIB files or polars.

Data options

Use tidal streams

With this feature selected, tidal streams from the tidal databases (see [tide settings](#)) will be used in the route optimisation. This will override any current GRIB files so be very careful when selecting this feature.

The use of tidal streams slows route optimisation, so if you are using a custom grib file for currents, routing away from any tidal stream data or across ocean an ocean, it may pay to turn this off.

Tidal stream data is used in preference to grib current data where both are present. So for example in a Newport to Bermuda race, Expedition will use tidal stream predictions

to start with then start using your grib data containing Gulf Stream information when you get away from land.

You need to select the tidal stream source to use on the tide settings page - NOAA in the US, Exp diamonds in most other places or Winning Tides in the Solent.

Use instrument wind

Starts the optimal routing calculation using the damped instrument wind instead of the wind from the Grib file. The utility of this feature is still under review and it isn't often used. The default is to leave this check box empty.

In practice, this option has negligible benefit.

Use weather boats

Performs the routing using weather boat data across the race track instead of forecast data.

Extend wind forecast in time

If your wind Grib file only covers the next 3 days, but it will take 3 days to get to your destination and you leave this box empty, the optimal routing calculation cannot complete. The best solution is to get a GRIB file that covers the final day, but that is not always possible. When you can't get the full time scale GRIB then you can force Expedition to return

an optimal route solution by checking this box. The program then assumes the last wind field in the GRIB continues on forever into the future with no change.

If the router can only complete part of the route before the weather GRIB comes to an end, the resultant Optimum route on the screen will converge back to the great circle route from the moment the GRIB ends.

Extend current forecast in time

This check box is even more useful as ocean current GRIB files are often only valid for a single time step and it is a reasonable assumption to consider that the flow pattern will remain fairly static over a few days.

Display options

Optimal route

Draws the optimal route calculated. You pretty much always want this feature enabled.

Important note

Dashed lines indicate tacking or gybing for that part of the optimal route. Sometimes this might be one tack or

gybe, but in other cases it might mean staying in a lane of stronger wind for example.

These areas will have TWA values in parentheses in the optimal route [summary](#), for example (-12).

Isochrones

Isochrones are curves that indicate where a boat could sail to in a certain amount of time. Isochrones are calculated when Expedition calculates an optimal route and are displayed on the main chart window if this box is selected. This is a useful tool to leave enabled. Depending on the length of the route, you will vary the time step for the drawn isochrones. For a 2000 mile race every 12 or 24 hours will usually suffice. For a 100 mile race, you might want them drawn every 2 hours or so.

Reverse Isochrones

When selected, reverse isochrones are also drawn. This is a very useful tool for studying the sensitivity of a calculated optimal route. Reverse Isochrones use fancy math to work the optimal route backwards through the GRIB fields from finish to start. When displayed concurrently with forward isochrones you can more easily see the moments in a race when a particular side of the route is especially good or bad.

Reverse isochrones are lines of points equidistant in time from the finish whereas isochrones are lines of points equidistant in time from the start. So, if another boat is on the same reverse isochrone as you, they should finish at the same time. Thus, the reverse isochrones can also be used as a way of seeing who is ahead.

Note that the 'Calculate reverse isochrones and sensitivity' option must be selected for this.

Paths

Draws all the possible paths (not just the fastest one) successful paths obtained whilst computing the optimal path.

Paths can be useful when studying routes or to help understand options for other boats around you.

It can also be useful for visualising the route optimisation search area.

Winds on optimal route

Draws expected wind barbs along the route. This is quite helpful in pre race planning when showing other crew members roughly what wind speeds and angles to expect but it adds to the screen clutter if you are also displaying a GRIB file's wind barbs.

Winds everywhere

Shows wind arrows everywhere Exp has explored for the route optimisation.

Currents on optimal route

As for winds.

Highlight previous optimal routes

Draws all previous optimal paths in bold (previous runs, fleet optimal routes etc.).

Shade time sensitivity

Shade time sensitivity for previous routes

These shade areas where the routing is within the specified number of minutes of the optimal path.

Note that the 'Calculate reverse isochrones and sensitivity' option must be selected for these.

Advanced settings

Avoid

Max gust

Maximum upwind TWS

Minimum upwind TWS

Maximum downwind TWS

Minimum downwind TWS

Forces the routing to avoid areas with wind speeds outside these limits.

Penalties

Adds user-defined seconds to each tack or gybe.

Not used for reverse isochrones.

Wind

Time shift

Apply a shift in minutes to the wind if a change is happening more or less quickly than the grib files suggest.

Use negative values if conditions are in advance of the forecast, positive values for the opposite.

For example, at 18z

With a value of 60, Expedition will use 17z grib data.

With a value of -60, Expedition will use 19z grib data.

Scale wind

These are in addition to the scale and rotate 10m wind options under [weather](#) settings.

One use is if the weather is different from forecast, but you expect it to become more in line with the grib forecast in the future.

Reduce scaling and rotation over <selected> minutes

Check this box and enter a value in minutes to linearly reduce the values entered in rotate and scale winds above. The amount of rotation will reduce linearly from the maximum value at the current time to zero after the desired time period.

Waves

Wave heights are in metres.

Avoid significant waves over <x> feet

Forces the routing to avoid areas with significant wave heights over the set value.

Swell and wind waves

Percentage values should be the maximum effect at the reference height.

The wind waves option is not normally selected as wind wave effects are normally as are in the polar.

See also the optimal routing waves [topic](#).

Crossed sea

See also the optimal routing waves [topic](#).



Optimal routing

See also the [Optimal routing](#) ribbon and [Optimal routing](#) settings help topics.

Expedition easily routes over complex routes with multiple GRIB fields containing both wind and current information.

Expedition has features such as reverse isochrones and sensitivity shading that help illustrate the sensitivity of the calculated optimal routing solution. It has features designed to help both the shore side study of an upcoming race (or a race underway that you are watching) and also the navigator/tactician in the middle of the race.

Before using, make sure you understand the [optimal routing](#) settings, [marks and routes](#) and [weather](#).

Garbage in = garbage out!

Obviously the final results are only as good as the polars and weather forecast.

Even with the best polars and weather forecast, the prudent navigator will study the solution carefully and make decisions based upon practical experience or intuition.

Isochronal and grid based routing algorithms

Expedition has two route optimisation algorithms that were designed for different purposes. They should yield similar results, but a user may prefer one or the other or use both for comparison of results.

The isochronal algorithm pushes an isochrone out from the start point, whereas the grid based algorithm uses an adaptive grid over the route area.

The isochronal algorithm is generally the default choice. The grid algorithm is useful for routing studies and as a comparison.

Isochronal

The isochronal algorithm is generally a slightly superior methodology, especially in areas of light wind, where the current has a significant effect or around obstructions such as land or race notes.

To an extent, the isochronal algorithm is allowed to sail around corners.

Note that it is possible to encounter a scenario where the routing can't reach the finish with the [Prefer tacks and gybes](#) at routing steps option selected.

Grid

The grid algorithm was originally developed for routing studies where it was important to compare similar simulations. As such, it is less commonly used.

The route along the great circle route option is only available for the grid based algorithm.

Note that the grid algorithm is not designed to route around corners. One solution is to use multiple marks for this.

Marks

Fewer marks in a course generally yield better results.

Leaving marks to port or starboard is not currently enabled for isochronal routing.

Avoiding land

The Avoid land [settings](#) option. Avoiding land may make the route optimisation slower, so may not be needed offshore or for routing studies.

Race notes

[Race notes](#) can also be used to mark custom areas to avoid, such as shipping channels and ice exclusion zones.

Race notes can also be used to reduce the boat speed. An example would be due to Sargasso weed.

Before performing a route optimisation

Make sure

All the settings in [optimal routing](#) settings are correctly configured.

Wind forecast GRIB data is loaded that covers the area and time span needed to sail the route.

A polar is loaded, see the [polars](#) help.

There is an active route, see [mark management](#).

If using the [route to mouse pointer](#) function, an active route is not required, but Exp does need a GPS position.

It is also important that the correct destination mark is selected as active.

A simple optimal route is calculated by Expedition with a click on the Optimise button on the [Optimal](#) panel.

A progress control and cancel button are displayed while Expedition is computing the optimised route. The optimisation may be cancelled at any point by clicking on the cancel button.

After the calculation complete, the optimal path and other features such as isochrones that you selected for display on the [optimal routing](#) tab will be displayed on the chart.

Notes

View the routing details from the Results button on the [optimal](#) panel after running an optimal route.

The [Draw paths](#) option can be useful for visualising the optimal routing solution.

Reverse Isochrones and route sensitivity

When selected in [optimal routing](#) settings, reverse isochrones are also calculated.

This is a very useful tool for studying the sensitivity of a calculated optimal route.

Reverse isochrones are lines of points equidistant in time from the finish whereas isochrones are lines of points equidistant in time from the start. So, another boat is on the same reverse isochrone should finish at the same time. Thus, the reverse isochrones can be used as a way of seeing who is ahead.

When displayed concurrently with forward isochrones you can more easily see the moments in a race when a particular side of the route is especially good or bad.

Their main use is in analysing the optimal route. For example, the router will reward a half degree right shift with an optimal route that goes all the way to the right hand side of the when going upwind, when in fact it doesn't really matter where you go (and a prudent tactician might be more inclined to play the shifts going up the middle of the route).

How to ...

To get an idea of how critical the optimal route is, look at how parallel the forward and reverse isochrones are.

When stepping forward along the optimal path using the forward or reverse buttons on the [weather display](#) panel, the forward and reverse isochrones for the display time will be highlighted.

If the forward and reverse isochrones are close together and parallel over a large distance along their length, then the optimal route isn't very critical, but if they are only close together over a small distance, then the optimal route is much more critical. In this instance a prudent navigator may need to have good reason to deviate much from this critical point.

Another example is where the forward and reverse isochrones come close together in two different places, but in one you are 10 seconds ahead. The computer is going to suggest a track through the area that puts you 10 seconds

ahead, whereas both routes are probably relevant. For example, the 10 second slower option might be a safer or more reliable option.

Sensitivity

Calculating the reverse isochrones allows a routing sensitivity envelope to be calculated.

Whereas the reverse isochrones are at a particular point in time, shading the route sensitivity will give an envelope across the entire route.

The size of the sensitivity to be shaded can be selected in [optimal routing](#) display settings.

Waves

Wave height, direction and period/wavelength can be used to correct the optimal route for wave effects.

See the [optimal settings](#) topic.

Requires the boat length to be correctly entered in [settings](#).

Swells

For swells, Expedition uses, in order of priority

1. Swell direction, period and height.

2. Primary wave direction and period and significant wave height.
3. Mean wave direction and period and significant wave height.

Wind waves

Not normally selected as wind wave effects are normally included in your polar.

Crossed sea

Mostly only used for multihulls.

For multihulls, a critical consideration is if there are two wave trains that cross at a significant angle.

That causes them to appear to put up columns of water. There are interesting articles, images and videos of crossed seas on the internet.

Expedition will apply the specified percentage to the polar if the following conditions are met:

1. Both swell and wind wave height are greater than the specified minimum.
2. The combined swell and wave height is greater than the specified value.

3. The angle between the wind waves and swell are within the defined angle from 90 degrees.

If primary waves are used for swell, then the secondary waves will be used instead of wind waves.

The results table

Available from the [Optimal ribbon panel](#).

Displays a summary or meteogram of each route optimisation.

This table displays many details about each step of the optimised route. The number of entries (rows) in this table is a function of the length of the route and the resolution that you configured in the [optimal routing settings](#) tab.

The table can be used when studying predicted wind speeds and angles in order to plan sail selection.

The optimal route can be exported to a .gpx or .csv files. It can also be exported to Expedition's mark database.

Important note

If any TWA value is in parentheses, for example (-12), then Exp has tacked or gybed for that part of the optimal route. Sometimes this might be one tack or gybe, but in other cases it might mean staying in a lane of stronger wind for example.

These will also be displayed as dashed lines on the chart.

Common errors

GRIB problem

The GRIB file does not cover the correct time span or geographic range. Check by displaying 'Wind' in the GRIB file and animate it to confirm its time span and geographic range (See the help on [weather](#) for more information).

Route problem

Something is wrong with your route. Maybe the wrong route is selected or the wrong destination mark is selected as active.

Polar problem

Something is wrong with your polar file or you have no polar file loaded. See the help on [polars](#) for more information.

Optimal Routing setup problem

Something is wrong in the [optimal routing](#) settings.

The most common errors are

1. Incorrect start time. The start time and finish time should fall within the time span covered by the GRIB.
2. The resolution is set too high, resulting in the route optimisation taking too long.



Racing tools

Tools for racing.



Create a WL course

Available from the Create W/L course button on the [Start](#) panel.

This function orients your race course around an existing starting line or committee boat.

Unless in the [chart-less start display](#) mode, select the start line checkbox on the [display](#) settings page to show the start line.

For any derived start numbers such as Time to line, you need to have a start polar loaded.

Note that if you have already set the ends of the start line, they won't be reset by this operation.

After setting up and loading a windward leeward course you can fine tune the position of the ends of the line with the Set port or Set starboard buttons. Fine tuning the starting line in this manner (after you have set up a course) will have no effect on the position of the other marks.

Use the Set dividers given range and bearing tool for tips on fine-tuning the position of the top mark.

Start to windward mark

The distance in nautical miles to the top mark from the start line.

Start to leeward mark

The distance in nautical miles between the start line and the leeward mark. You could then sail up to the gate/leeward mark and drag this buoy to its exact position.

To place the leeward mark below the start line, enter a negative number.

Course axis

Allows the course axis to be manually set in degrees magnetic. If the race committee has done a good job this will be aligned with the True Wind Direction. The Twd button updates the course axis with the current damped true wind direction from Exp.

Leave marks to port

Sets all created marks to be left to port.

[Rate of turn](#) can also be used to more accurately calculate time and distance to laylines.

Gate at leeward mark

Select whether or not to have a gate to the left (looking upwind) of the leeward mark and its width. When the leeward mark is the active mark, laylines will also be drawn from this mark.

Finish

Can set a finish mark different from the pin end.

Leeward mark

Sets the position of the leeward mark above either the committee boat, the middle of the line or above the pin end.



Boat editor

The [database](#) used by Exp can contain a list of all or some boats in any race.

These are used by [race tracking](#) and the [handicap](#) calculator.

Options

Plot

Use this boat on the chart etc.

Boat name

Colour

Colour to plot boat and track on the chart.

Reference

Is it the reference boat? Normally this is you.

Handicap

Handicap value.

Start time

Note that the handicap calculator has an option to use the time from the start of the race if the [start timer](#) is being used.

Polar

Boat polar used for [fleet](#) route optimisation.

Boat editor							
Draw	Cla...	Colour	Refere...	Handic...	Start time	Pol %	Polar
☒ 2 UNLIMITED				1.0000		100	
☒ ABBOTTS VAMP				1.0000		100	
☒ ABOUT TIME				1.0000		100	
☒ ABRACADABRA				1.0000		100	
☒ ACTIVE AGAIN				1.0000		100	
☒ ADMIRAL				1.0000		100	
☒ AIRSTREAM				1.0000		100	
☒ ALIVE				1.0000		100	
☒ ALLEGRO				1.0000		100	
☒ ANGER MANAGEMENT FLG				1.0000		100	
☒ ANOTHER PAINKILLER				1.0000		100	
☒ ARCADIA				1.0000		100	
☒ ARCH RIVAL				1.0000		100	
☒ AUDACIOUS				1.0000		100	
☒ AUDERE				1.0000		100	
☒ Activator				1.0000		100	
☒ Am Meer				1.0000		100	
☒ Anarchy				1.0000		100	
☒ Arbitare				1.0000		100	
☒ Atamai				1.0000		100	



Handicap calculator

The handicap calculator window is used to compare mark rounding and finish times with other [boats](#) using time correction factor (TCF) handicaps.

The elapsed, corrected and delta times are based on the time from each boat's gun time and the event.

Delta is just the differences in the corrected times.

The Owed column is just a running time owed since the gun. Negative numbers mean you owe time, positive numbers mean you are owed time.

Refresh

Refreshes the time owed column.

To set a boat event

Just click on the boat name in the list. For example at mark roundings or the finish.

Set all start times to gun

Sets the start time for all boats to the start gun time. On races with staggered starts, it may be necessary to set the start time in the edit boats window.

Set all boats

Sets the event time for all boats to the current time.

Edit boats

Edit individual boat handicaps and start times in the [boats](#) window. Also select boats to plot and display in the handicap and [race tracking](#) windows.

Handicap calculator - click on any boat to set...

Boat	TCF	Mark/finish time	Elapsed	Corrected	Delta	
Akela	1.0000	23/09/2014 15:08:13	17s	17s	0.4s	
Alchemy	1.0000					
Alfa Romeo	1.0000					
Amante	1.0000					
Cazador	1.0000					
Flash	1.0000					
Holua	1.0000					
Magnitude 80	1.0000					
Medicine Man	1.0000					
Mirage	1.0000					
OEX	1.0000					
Pegasus 50	1.0000	23/09/2014 15:08:13	17s	17s	0.4s	
Pyewacket	1.0000	23/09/2014 15:08:14	18s	18s	-0.6s	
Samba Pa Ti	1.0000	23/09/2014 15:08:13	17s	17s		
Valkyrie	1.0000					
Westerly	1.0000					

Set to gun Update owe Edit boats OK



Available from the [sail](#) panel or the Ctrl+P keyboard shortcut.

There is support for performance, navigation, start and heel polars in Expedition and each has a dedicated tab.

For details on the file format, see the bottom of this page.

In addition to the pre-defined polars listed below, there are several custom polars. These can be used for parameters such as keel angle. In this case, they would generate polar keel and polar keel angle %.

Performance polar

Expedition uses the Performance polar when calculating performance numbers - such as Polar Bsp and Polar Bsp%.

Navigation polar

This is the main polar used by Expedition. The Nav polar is used in optimal routing and all navigation calculations.

Being able to modify just the navigation polar to suit different conditions means the performance polar can be left to provide

the same reference from day to day.

Of course, you may just wish to use the same polar file for the navigation and performance numbers.

Start polar

The start polar drives Expedition's starting line program (which features such valuable tools as time to burn) so if you plan on using that feature, you will want to enter a starting polar. If you are always doing upwind starts, then you can simply use the same text file as the performance polar. If you are doing downwind starts, you might want to modify this file and slow down the polar points at the angles where the polar assumes you have a spinnaker set.

Heel polar

If entered, it allows you to use Expedition's Target heel number which has proved quite useful.

Port polar

Starboard polar

Can be used in [optimal routing](#) with the [Use port and starboard asymmetric polars](#) option selected.

Database tab

Use the Database page to add, delete or edit items in the database, which is selected from the [user settings](#) page. These items are used to identify your boat's configuration (i.e. helmsman etc.) when a test is saved in Strip Chart. The boats list used in [race tracking](#) may also be edited here, but is more easily edited from the Boats item on the [sail panel](#).

You can also edit Expedition's database (*.mdb) by opening it in a database program like Microsoft Access.

Sails tab

Edit the sails database attributes

Whether on-board. This is to reduce the number of sail options displayed when setting [events](#) or [tests](#),

Name,

Colour (as plotted in the edit polars window described above),

Sail type,

Any notes or comments.

Polar file details

When Expedition was installed it added some sample polars in the Expedition polars folder.

Expedition polars are just text files, so can also be edited in any text editor such as Microsoft Notepad. Users often manipulate the data using Microsoft Excel - just make sure you save them in a *.txt file format. Take note of the format of these polars as any new ones that you create for your boat must be in a similar format.

The Expedition polar format is very versatile. It does however have a few restrictions:

- The first column always contains the Tws values.
- Other columns are in pairs of Twa and Bsp (or heel etc.).
- Twa should increase across a row.
- Tws should increase downwards.
- The number of polar points does not need to be the same for each row.

Normally we have a polar point above upwind target and the last point at 180 twa for each row.

For performance, navigation and start polars, Exp automatically determines which point is the target (maximum upwind or downwind vmg).

Comments may be used in the polar file by adding a '!' character at the start of each comment line. Note that these comments

aren't saved by Exp, so are only useful if you only edit your polars manually.

There is no limit to the number of rows (True Wind Speed) or columns (True Wind Angle) that you can have in your polar, but you may find it easier to work with a simple polar like the sample ones.

Polar percentage (%)

This allows the polar to be scaled by any value. The default value is 100.

Polar percentage night (%)

This allows the polar to be scaled by any value between civil dusk and dawn. The default value is 100.

This could be useful for route optimisation if the performance is expected to be slower at night.

Load

This allows you to load a polar file (in *.txt format) from anywhere on your computer.

Save and Save As

These buttons allow you to save changes/edits that you have made to a polar file. It is highly recommended that you use Save As so you do not overwrite a valuable polar file.

Comments may be used in the polar file by adding a '!' character at the start of each comment line. Note that these comments aren't saved by Exp, so are only useful if you only edit your polars manually.

Converting from 10m to masthead height

Many boat designer's polars are given for 10m winds, so the polars may need scaling. See Scale winds in the [weather settings](#) help for a formula to assist with this.

Edit button

Displays an edit polar dialog, which allows the polar to be graphically edited by simply dragging polar points.

Important note:

Targets are defined as the point on a polar curve with the maximum VMG upwind or downwind. Editing a target point can cause it to have a lower VMG value than an adjacent point, which would then become the new target and be the displayed in the up-wind and down-wind curves (below).

For this reason, not having polar points too close to the targets can make editing a lot easier.

For example, if the target angle is 46, consider using TWA values of 0, 46, 60 ... instead of 0, 46, 50, 60, ...

Editing polars

Different views of the polar and target data are available:

Polar

Display polar curves on the familiar polar diagram. Select a polar curve for any wind-speed using the wind-speed drop-list on the top right. The polar curves at wind speeds above and below the selected polar curve will also be displayed to aid with editing.

TwS drop-list

This box is only accessible when you are displaying the individual polar curves. Select the wind speed for the polar curve to be displayed.

Upwind

Display upwind target bsp, twa and vmg. Select whether to plot tests against bsp, twa or vmg from the drop-list.

Downwind

Display downwind target bsp, twa and vmg. Select whether to plot tests against bsp, twa or vmg from the drop-list.

Log file

Extracts data from a log file to over-lay on the polar, similarly to tests.

Tests

Displays test data stored in the [database](#) for selection to plot on the polar edit window or exporting from Expedition. Tests to be displayed on the polar can be selected.

To edit the target upwind/downwind polar points, select the upwind or downwind buttons for display. To edit reaching polar points, select the polar button. If you select the latter, then the drop down window on the upper right shows which wind speed polar curve is being displayed and allows you to move through the

wind speed matrix to get to the curve or curves you want to edit.

For Upwind, Downwind and Polar radial editing, the Black line is boat speed, the Blue line is TWA, and the Grey line is the calculated VMG.

Tests saved from [Stripchart](#) will be displayed here. Using this feature you can drag a polar point to the middle of a cluster of tests and have a high degree of confidence that you have improved your polar. There are several controls to modify how these tests are displays:

Normalised

The tests displayed will not be for exactly the wind speed selected in most cases. This attempts to scale the test bsp for to the selected tws.

Colour tests by sail

By default, tests are displayed in red if the test tws less than that of the polar curve being edited and in blue if greater. Selecting this option colours the tests by the colour associated with the sail entered for that test.

If editing upwind or downwind targets, tests are displayed in green if on starboard ($twa > 0$) or in red if on port ($twa < 0$). If the Vmg of the test is less than that of the polar curve, the test point will be hollow, else it will be drawn as a solid point.



Race tracking

Available from the Race tracking button on the [Sail](#) panel.

Race tracking was initially designed for and used very successfully in the 2001-2 Volvo race. It enables you to track longer races that have some sort of position reporting on a regular basis.

See also System files [settings](#).

Interpreting Race Tracking

Once you have entered two position reports, the power of Expedition's race tracking tool becomes more apparent. When you have the race tracking dialog open (race tracking from the [Sail panel](#), you are presented with a lot of information about the race.

Most of the column headings should be obvious.

DMG is distance made good towards the next (selected) mark.

Cor DMG is the distance made good on a corrected time basis. This is a very valuable piece of information, as it may highlight some important corrected time information (e.g. Some smaller/slower boats are making big corrected time

gains on a certain part of the route). As mentioned above some functions like Gain, BRG (bearing) and RNG (range) will be calculated relative to the reference boat.

COG & SOG are derived from successive positions, whereas iCOG and iSOG are instrument values if supplied.

DFS is distance from the previous mark.

It is important that the correct route and waypoint are selected as active for the Distance to Finish and other calculations to be correct.

Scheduler

Expedition can automatically monitor a folder for new position report files in Expedition, Volvo or MaxSea format.

See the [scheduler](#) help.

Displaying the track of other boats on the main chart window

There are check boxes to draw Race tracking boat positions and tracks on the [display settings](#) tab. If selected, the tracks and/or positions of the fleet will be drawn on the chart. The tracks can also be drawn in bold. The boat positions will be drawn for the current display time (upper left of main chart window). There are 7 different pen colours available. So if, for example, you have 8 boats with position report entries, boat 17 will be drawn in the

same colour and style as boat 1. However, by the time you have 7 boats drawn, your screen may be too cluttered anyway so consider only selecting the important boats (class leaders, your division etc.) to plot.

If no boat is designated as the reference boat, position reports will not be plotted on the chart.

Tracking window

Race tracking item on the [Sail](#) panel.

After setting up the fleet, use this tool to enter position reports and/or operate Expedition's [race tracking](#) features.

Boats

See the [Boats](#) help.

Use this to add or boat names, handicaps or start times and whether to plot in the database.

One boat, typically your boat, should be selected as the reference boat. Positions, based on distance to the finish (active mark), as well as ranges and bearings to the other boats are relative to the position of the reference boat. Each boat can have a handicap rating (either a time correction factor (TCF) or a time on distance factor).

Sync time

Sets Expedition's screen display time to the latest race tracking time.

Schedule time (local) window

Displays the time (in local, computer Windows time zone) of the most recent position report. You can display old position reports/tracking using the drop down box in this window.

Export

Exports tracking to a text file. This is useful for printing out and taking on deck.

Manual

Manual entry of position reports. Select the boat, position and time of the report.

For entering positions, use a format as follows: 12 34.5n
123 12.4w (you can use as many decimal places as you like).

Delete

Deletes any selected/highlighted schedule report - good if you make an error.

View all

Displays all position report entries. This is good for finding an erroneous entry.

Clear all

Clears all position reports.

Import

Import positions from a file. Several formats are supported, selected from the drop-list.

Any position report file can also be imported by dragging it onto the race tracking window.

Expedition

This format has been designed as a generic race position report that is efficient to receive at sea. It requires two files - one for the boat names and one for the actual position reports. Using the boat name file means the boat names don't need to be downloaded

with every race position report. For example, it could be downloaded before a race.

The Expedition format is used by [YB Tracking](#) for example.

BoatId.csv

This file contains all the boat names possible in a position report. It can be edited with Microsoft Excel or a text editor and the format is:

```
id,name[,class,handicap]
```

```
1,Samba Pa Ti  
2,Morning Light  
3,Pendragon  
4,...
```

scheds_yymmddhhmm.csv

scheds_yymmdd_hhmm.csv

This file contains the actual position report.

The file can have any name, but if it is in the scheds_yymmddhhmm.csv format, then the time of the position reports need not be entered for every boat in the file. This is also useful for differentiating position report files.

Times are in UTC.

The minutes part of the file name is optional. For example, a position report for 1200UTC on the 15th of March 2008 would be.

Scheds_0803151200.csv

or

Scheds_08031512.csv

If used with the [scheduler](#), the first line should be

EXPEDITION

Each line of the position report file represents a position report for an individual boat and can contain various parameters. The format is:

id,latitude,longitude[,YYMMDDhhmm[ss],cog,sog,twd,tws]

At the very minimum, each line needs to contain a boat id, latitude and longitude. In this case, it is assumed the time will be obtained from the file time.

1,-10.1000,-125.9667

2,-10.1500,-124.2500

3,-10.6833,-124.9333,0803151230,2

Adding a UTC time field after longitude over-rides any time value obtained from the file name. The time field could be used for all reports or as it the example above, for boat 3 (Pendragon) who has reported 30 minutes later. The seconds part of the time field is optional.

Basic

Imports from a basic text file. The format for the text file required is:

Boat name,UTC date and time, LL LL.LL N,LLL
LL.LL W

A degree symbol may be used after the degrees value.

The data & time should be in the format used by your computer.

For example, any of the following work:

Boat,26/11/2007 07:02:00,29 58.40 N,021 26.24
W

Boat,26/11/2007 07:02:00,29 58.40°N,021°26.24
W

Boat,26/11/2007 07:02:00 UTC,29 58.40 N,021
26.24 W

Volvo Ocean Race

If used with the [scheduler](#), the first line should be

POSRELEASE

Imports position reports from a text file in the format used in Volvo Ocean races since 2001/2. The format for the text file required is:

```
Rank;Boat
Name;Latitude;Longitude;DateTimeGroup;WindDirection;WindSpeed
1;TYCO;49,16.74S;017,30.72E;0600;05Aug01;170;18
2;NEWS;49,16.74S;017,30.72E;0600;05Aug01;170;18
3;ILLB;49,16.74S;017,30.72E;0600;05Aug01;170;18
```

Times in this file should be in UTC.

GPX

Imports [GPX](#) <trk> and <trkpt> tracks.

YB Tracking XML

Imports [YB Tracking](#) xml data.

Vessel Finder

Imports [Vessel Finder](#) json data.

MaxSea

If used with the [scheduler](#), the first line should be

POSREPORT

Imports position reports from MaxSea text files. The format for the text file required is:

```
rank;boat id;lat;lon;date time;dtf;log;dmg;point-to-  
point  
1;1;48.2967N;5.0785W;11/28/04  
08:10:44;23967.3;0;0;0  
2;2;48.1660N;5.2808W;11/28/04  
09:06:00;23949.3;18.8;18.8;0.0  
3;3;48.1076N;5.3560W;11/28/04  
09:28:00;23941.8;26.5;26.5;0.0  
4;6;47.5552N;5.5308W;11/28/04  
10:18:00;23922.9;45.7;45.7;0.0  
5;5;47.3576N;6.2252W;11/28/04  
11:40:00;23896.4;73.7;73.7;0.0  
6;4;47.1724N;6.4916W;11/28/04  
13:10:00;23872.1;99.5;99.5;0.0
```

All information to the right of time is ignored.

Positions are degrees and decimal minutes-dd.mmmmm.

Times in this file should be in UTC.

Mapping identifiers to boats

In addition, a file to map boat identifiers and names is required in the same folder as the file(s) to be imported.

Its format is:

```
identifiant nom
1 VM MATERIAUX
2 VIRBAC-PAPREC
```

Possible names for this file include

```
identifiant.csv
identifiant.txt
Id_Maxsea.txt
```

Expedition also supports the YB Tracking file name convention

```
nb2022-maxsea.csv
nb2022-identifiant.csv
```

Another file naming convention supported is

theoceanraceeurope_FIRST_POSREPORT.txt

theoceanraceeurope_BOATS_IDS.txt

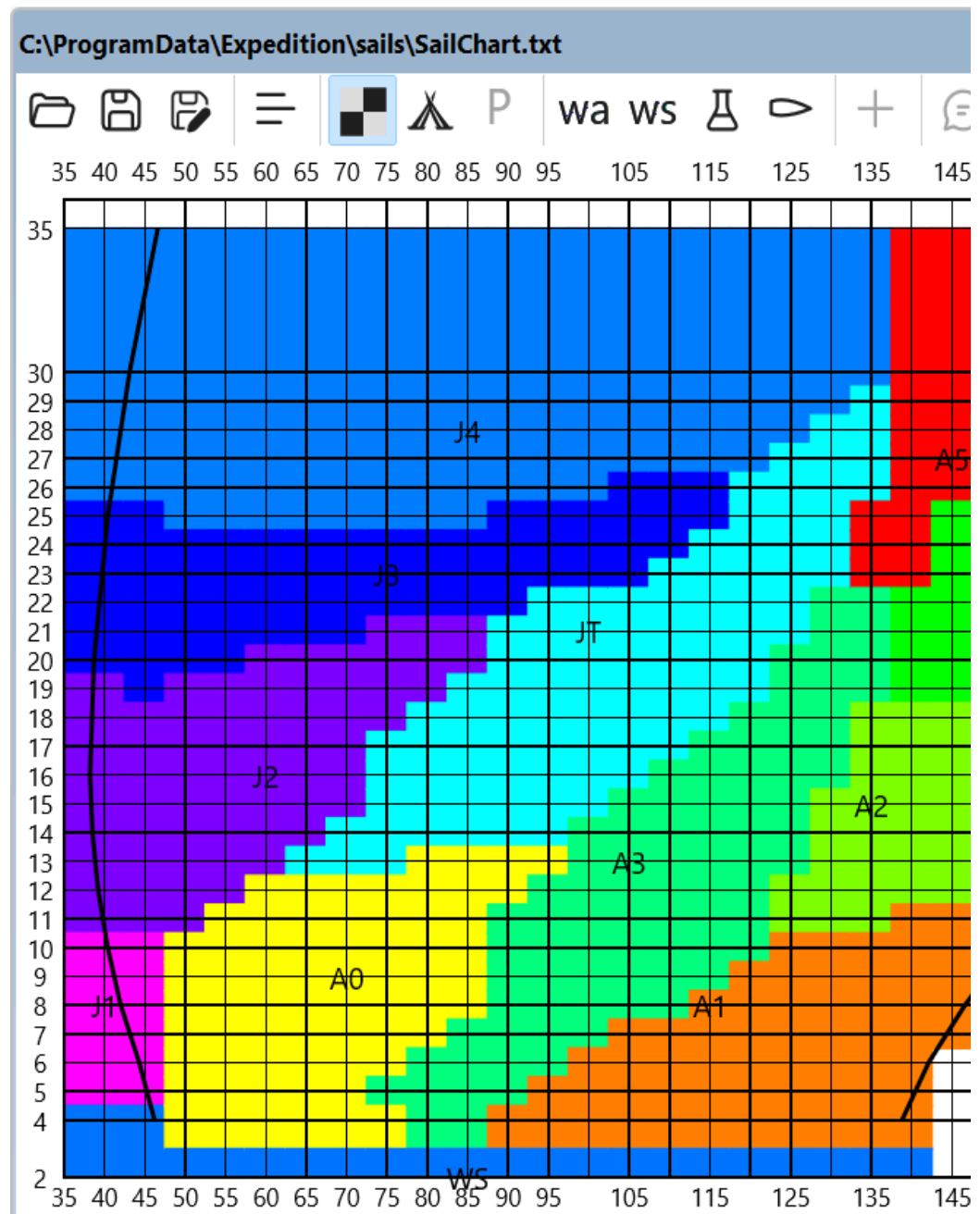


Sail chart

See the [Sail](#) panel help.

Sail charts

Basic sail chart



The basic sail chart loaded into Exp can be used for planning

The Route pane will show the sail indicated by the sail chart for each leg of the route.

The [optimal routing](#) summary will list the sail indicated by the sail chart for each section of the optimised route.

The current TWA/TWS point is drawn on the sail chart as a circle. Also, TWA to the active mark and next leg are drawn on the sail chart as 'M' and 'N'.

Editing

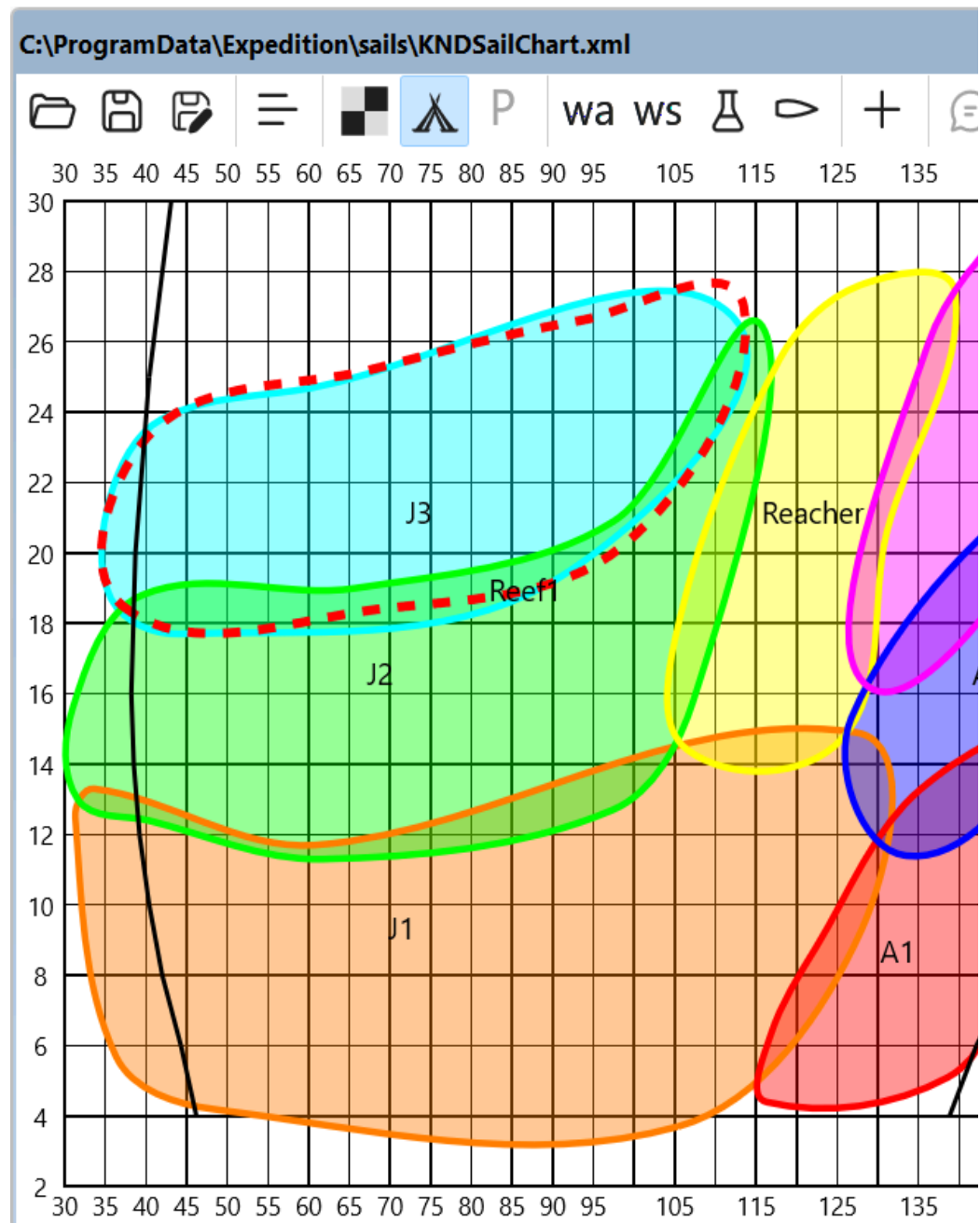
The format of the sail chart file is a simple text table, with TWA across, TWS down and the name of each sail separated by tabs. See the sample file included with Expedition. The file is also easily editable in Excel. Use the Load and save buttons to select different sail charts.

To edit the sail chart, simply right-click on any cell in the chart window and select the new sail from the pop-up menu.

Sail colours can be changed by clicking on the Sails button or editing the sail list (as above).

Sailing Performance sail chart

Note that this window should be visible for the sails to be used by route [optimisation](#), the route [pane](#) and to populate the sail, mark sail and next mark sail [channels](#).



Sailing Performance sail chart objects also have a sail, reef or other attribute. Note that Sailing Performance's Crossover Chart app must be v6.2.2 or later to support this.

Options include

Sail
Reef

Other (used for ballast, dagger boards etc.)

Editing

Limited editing of the sail chart is possible by right-clicking on any sail in the chart window and selecting the appropriate option.

Sail polars

Not currently rendered. For advanced users, also see the [sail test analysis](#) help.

Toolbar

Open

Selects a basic or Sailing Performance sail chart to display.

Save & Save As

Saves changes in the basic sail chart.

Sail list

Edit the sail list database

Use

Whether on-board. This is to reduce the number of sail options displayed when editing the sail chart or setting events or [tests](#),

Name,

Colour

Sail type

Select "Other" to not include in the optimal routing [filter](#) summary.

Any notes or comments

Basic

Displays the basic sail chart.

Sailing Performance

Displays the Sailing Performance sail chart.

Sail polars

wa & ws

Draw apparent wind curves.

Trails

Draws mark trails on the sail chart This can be useful for confidence.

Tooltip

Sail polars tooltip.

Sail polars settings and export

For advanced users.



Sail graphic

A graphical representation of the boat, sails, mark, wind etc.

See the [Expedition windows](#) help.



Sail tests

Saving sail tests

Tests (periods of good, consistent data) are saved using the [Stripchart](#) program. Note that saving sail changes as [Events](#) in Expedition or Stripchart can make post race creation of tests a lot easier.

Saving tests with Stripchart

See also *Saving tests* and *Events* in the [Stripchart](#) help.

Once you have set the wands at desired points in time, you can save the averages of the time span in Expedition's test database. A typical time span for a test would be in the 2 to 5 minute range.

Simply use the Save test to database button to save the averages within the *wanded* time span as a discrete test. A window will open where you can enter or change other test details such as sails or sea state as well as comments before finally saving.

Items such as sails available in the test database can be added to the test (see [polars](#)).

Once a test has been saved, that section of time will be shaded grey in Stripchart so you know not to save another test during that time span.

These saved tests are then available for viewing in the [tests list](#) and [analysis](#). The tests will also appear as data points in the [polar edit window](#) and can be displayed on the [sail chart](#).

Events

Events can be useful to record sail changes or items of interest and can be used to automatically populate sail tests.

There are several ways of saving an event:

- Set event on Expedition's [sail ribbon panel](#)
- The Event button on the [Stripchart](#) toolbar
- Right-click at a point in time in the [Stripchart](#) and select *Set event* from the right click pop-up menu.

Editing the polar

The test points will be displayed on the [polar edit](#) window. Note that log file data can also be drawn on the edit polar window, which can be an easier option than manually saving tests.

Editing the sail chart

The sails for each sail chart can be displayed on the [sail chart](#).



Sail test analysis

Available from the Analyse sail tests item on the [Tools](#) panel.

Currently disabled.

Tests

With time, the number of tests stored in the database can become very large (hundreds or even thousands of tests) and may be hard to manually draw conclusions from. Also, test data varies from day-to-day with changing conditions.

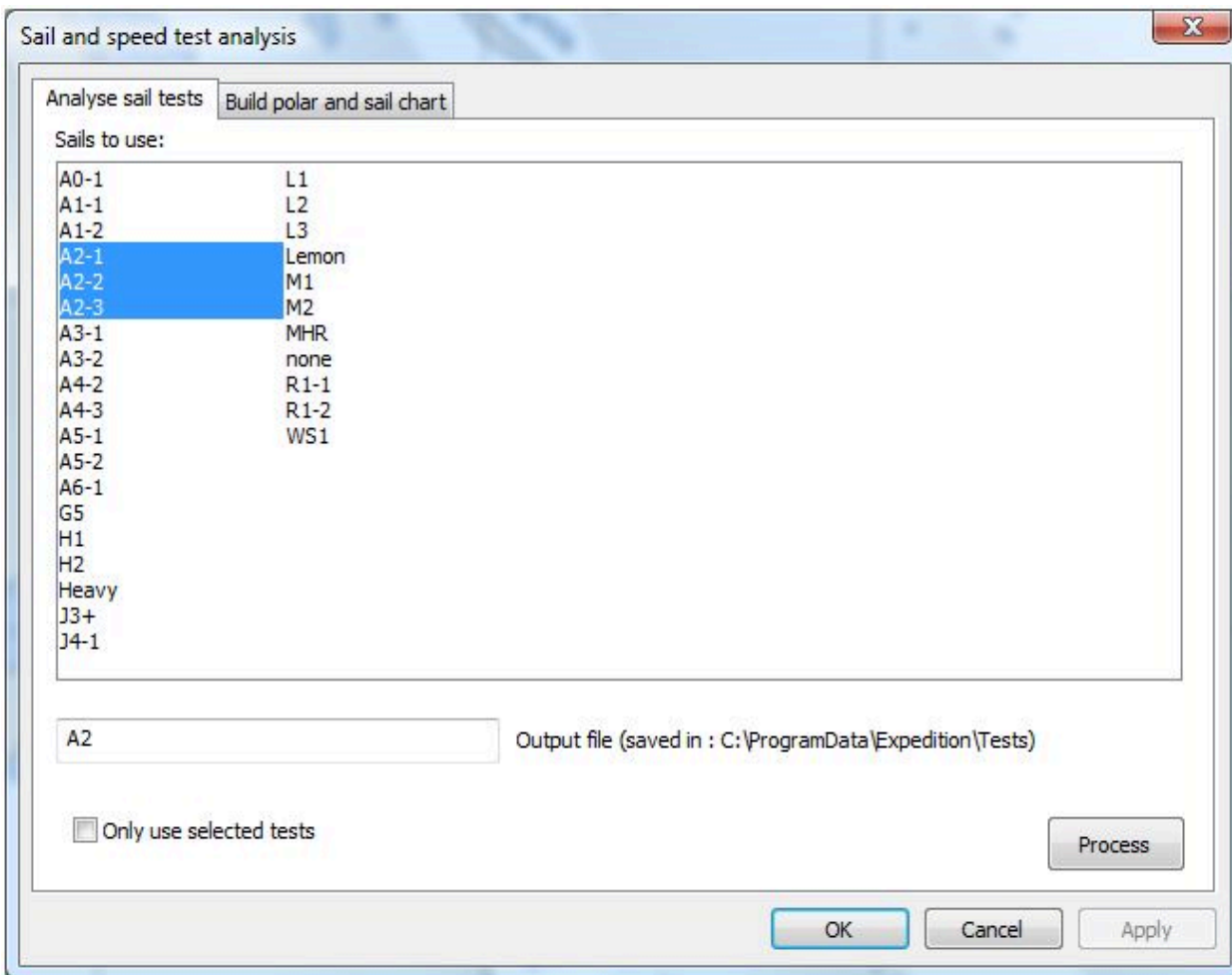
So, Exp has automated functions to automatically draw some statistical conclusions as to what the polar and sail chart should look like based on the saved tests.

This is a very powerful feature, but does require a lot of work to get the most of it and is really only suitable to expert users with large test databases.

Analyse sail tests

The first tab page in Sail and speed test analysis allows test data to be analysed for individual or subsets of sails. For example, all

sails A2-1, A2-2 and A2-3 can be selected as the test data for these analysed. For example:



It is a good idea to give the output file a short, descriptive name such as A2 in the example above as this name will be used in the final sail chart.

Also, this function requires the boat's performance [polar](#) to be loaded in Exp.

To actually perform the analysis, just click on the Process button. The output is a polar patch file for the sail or sails selected, which

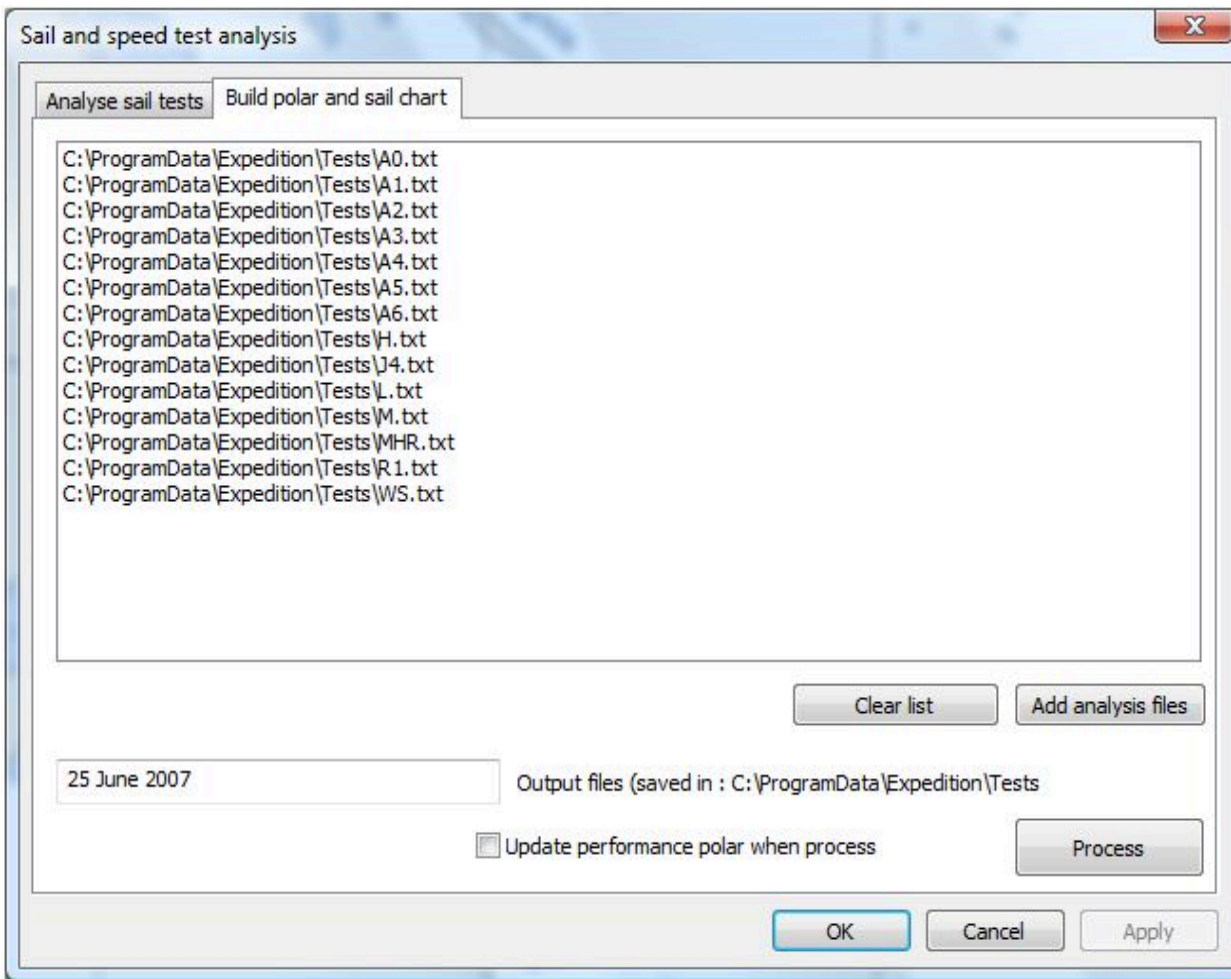
will be saved in the Tests folder in Expedition's data folder. There is a shortcut to the data folder on the [Application](#) button menu.

Repeat the analysis process for all sail types to end up with a collection of polars for each sail type.

On the test page accessed from the Tests item on the [Sail](#) panel, individual or all tests can be selected to be drawn on the [edit polars](#) window. Checking Only use selected tests directs Expedition to only use these selected tests in the analysis.

Build polar and sail chart

The second tab page in Sail and speed test analysis allows selected analysis files from Analyse sail tests to be combined to give a sail chart and polar. Select sail analyses to be used by clicking on Add analysis files. Enter an output file name, then click on the Process button as in the example below.



In this case, the output files will be labelled with the date.
Selecting Update performance polar when process directs Expedition to update the performance polar with the analysed polar speeds.

In this example, the output files will be

25 June 2007_polar.txt - the analysed polar file

25 June 2007_Sails.txt - the analysed sail chart

These will be saved in the Tests folder in Expedition's data folder.
There is a shortcut to the data folder on the [Application](#) button.

Note that in most cases, the test data will not be complete for every twa and tws value, so the output polar and sail chart files will probably have missing data. As more tests are included, the output will become more complete.



YB Tracking

Available from the YB Tracking button on the [Sail](#) panel.

Web

www.ybtracking.com

About

YB Tracking is the global standard in fleet tracking for yacht races.

YB Tracking provides position reports as small text for download at sea. This makes tracking your competition easy with Expedition's [race tracking](#) feature.

For each race, the organisers may publish a link to the position report files. For example, for the 2014 Rolex Middle Sea race, the link is

<http://yb.tl/links/rmsr2014>

Integrated download and import

The YB Tracking button on the [sail panel](#) will download the positions, import them into Exp's database and update the display with one click.

Race name

You must enter the supplied the race name. For example, this is rmsr2014 for the 2014 Rolex Middle Sea race. The race id is listed on the file links web page, in this case <http://yb.tl/links/rmsr2014>.

Update Boat IDs

Used with the Expedition format.

Downloads the boatids.txt file. This may only need to be downloaded once for each race and is normally not downloaded to save data.

Update handicaps

Updates the handicaps for each boat if in the file.

Update display time

Sets the Expedition display time to the time in the downloaded position report.

Archive position file

Saves a copy of the position file.

Format

The Expedition format is a small, efficient download.

The JSON options are used to download the most recent position or all positions since a specified time. The JSON options are much larger downloads, but also include GPS SOG and COG.

Airmail

Instructs Airmail to request position report files from YB Tracking.

Email

Creates an email request to YB Tracking in the default email client.

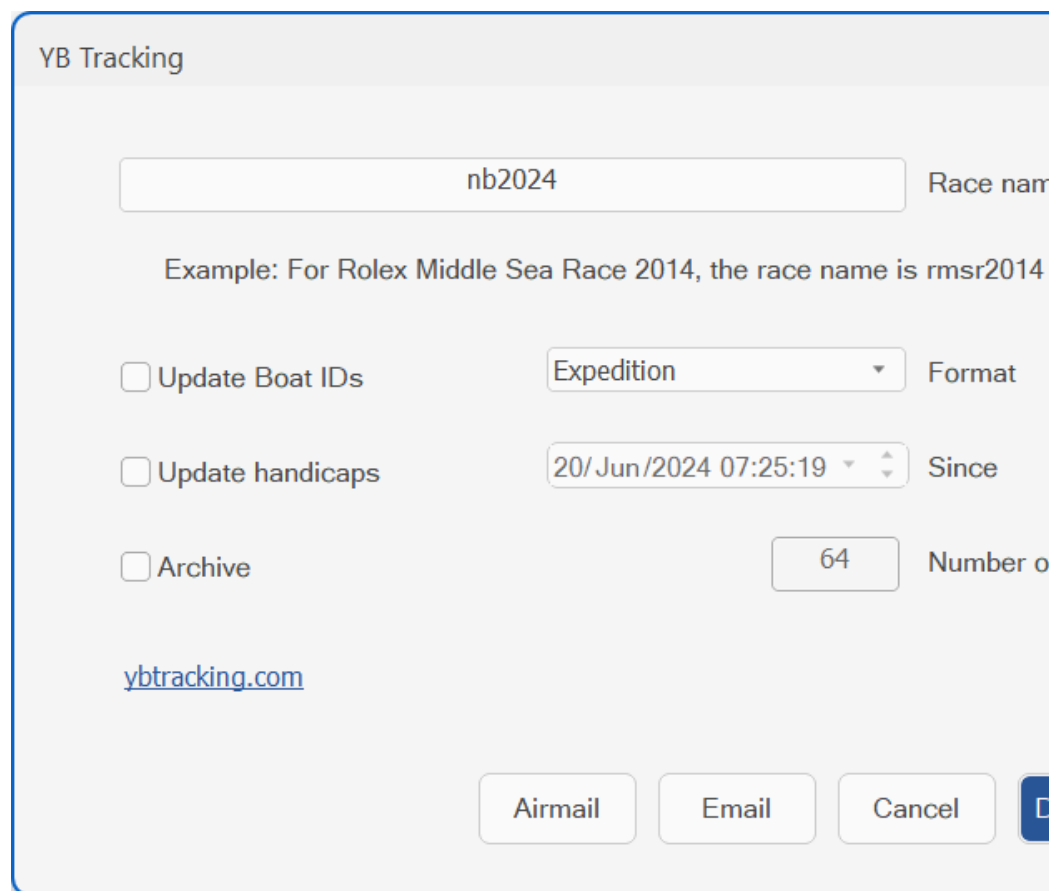
If you would like to subscribe to regular reports, finish the subject with 0.5, 1, 3, 4, 6, 12 or 24 hours.

For example:

subject: expedition 6

When you wish to stop your subscription, send an e-mail to:

nb2024@race.yb.tl with STOP as the subject.



The screenshot shows a web form titled "YB Tracking". It contains several input fields and checkboxes. The "Race name" field has "nb2024" entered. Below it is an example text: "Example: For Rolex Middle Sea Race 2014, the race name is rmsr2014". There are three checkboxes: "Update Boat IDs", "Update handicaps", and "Archive". To the right of "Update Boat IDs" is a dropdown menu showing "Expedition" with a label "Format". To the right of "Update handicaps" is a date/time dropdown showing "20/ Jun /2024 07:25:19" with a label "Since". To the right of "Archive" is a text input field containing "64" with a label "Number o". At the bottom left is a link "ybtracking.com". At the bottom right are three buttons: "Airmail", "Email", and "Cancel".

YB Tracking

Race name

Example: For Rolex Middle Sea Race 2014, the race name is rmsr2014

☐ Update Boat IDs Format

☐ Update handicaps Since

☐ Archive Number o

ybtracking.com

Manual download and import

Downloaded files may also be manually imported with Expedition's [race tracking](#) import interface.

The YB Tracking web page example above provides links to positions in various formats. To manually import YB Tracking data, you need the two files:

<http://yb.tl/rmsr2014-expedition.txt> <http://yb.tl/rmsr2014-boatids.txt>



What if?

Can be toggled on and off from the [Window](#) menu.

What-if? is a user definable wind and current value for use in certain calculations. This is a powerful tool for tacticians and navigators.

For example, a navigator might be expecting a 10° wind shift to the left (from 270 instead of 280) and want to see where the laylines would be with the new wind direction in order to avoid sailing extra distance.

If the What-if? true wind speed is zero, Exp will use the instrument tws value.

Note that What-if? is slightly different in Exp than in Deckman, which uses ground wind as an input whereas Exp uses true wind direction and speed.



Expedition has some very powerful start functions and displays, but these do require some advanced user input to use them to their full potential.

Race display

This is an un-cluttered display mode with no charting and scaled to fit the start line or course to the screen.

After the start, it will display the active mark and the boat. See the [start](#) and [chart](#) panels help.

Both ends of the line as well as boat position are required for the start screen to be displayed.

See [race](#) settings for more options, including the option to specify the display state used before and after the start.

Events

Events are automatically set in the database when pinging start marks and at the gun.

In the prestart

The start line is displayed with the port end to the left of the screen and the starboard end to the right.

The current heading is drawn as a thin blue line.

The GPS derived course of the ground is drawn as a solid blue line.

The boat track will be drawn if selected on the [display](#) settings page. This can be very useful to reposition start marks if you sail around them.

Laylines are drawn from each end of the line.

The time to each end is drawn at that end and at the top corner. The time to burn to the corner is immediately below it.

A bias line is drawn above the start line, indicating the favoured end. In the example below, the starboard end is favoured. The bias angle, length and line square wind are also displayed at the bottom right.

If wind or current is held, they will be displayed at the bottom of the screen. See the [start panel](#) help.

Turns to beat (or run) to the line are drawn as thin black lines. Note that some turns aren't drawn in order to keep the display uncluttered.

Many of these items stop being displayed 1 minute after the start to reduce screen clutter. At this point, only part of the line may be drawn.

Expedition draws numbers on the start line for times to the ends, the line etc. Clicking on the time to burn button on the [start](#) panel makes these numbers *time to burn* numbers instead of actual times.

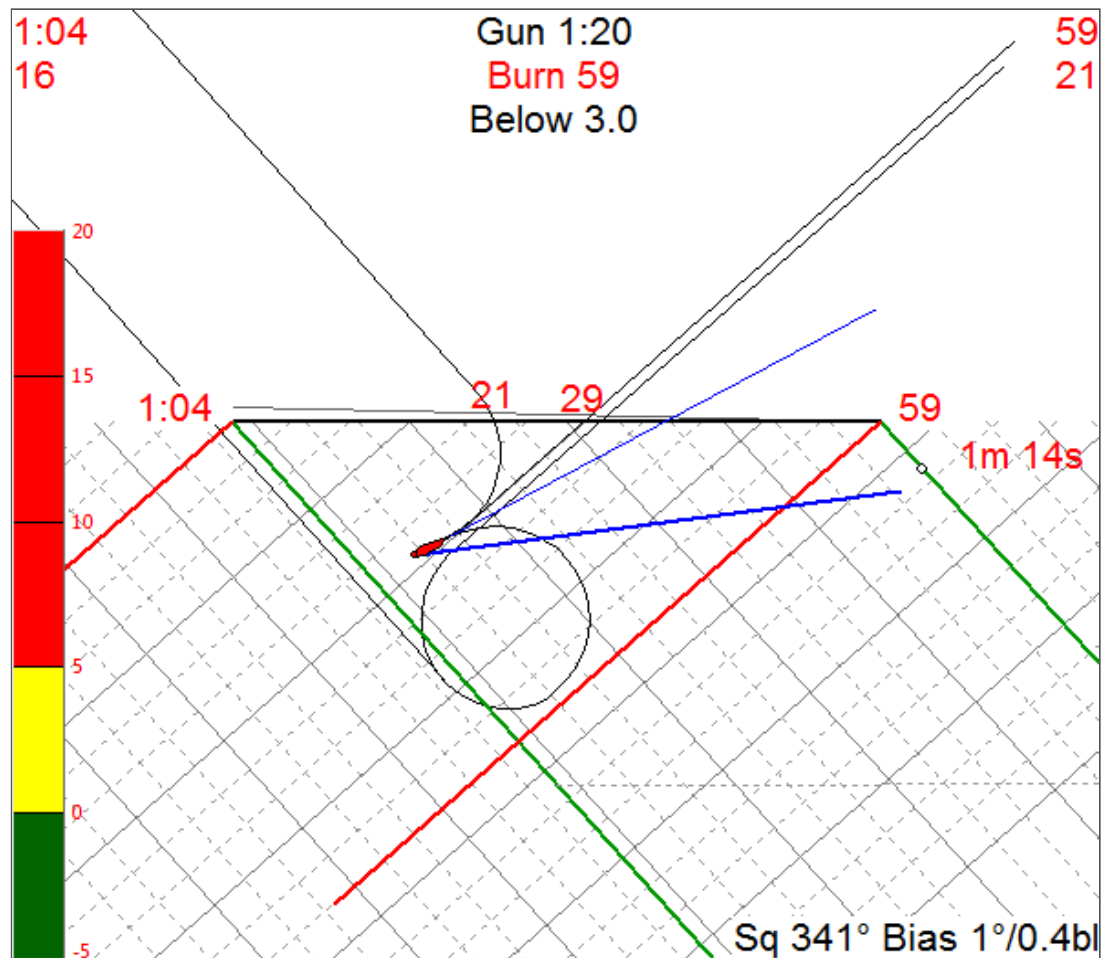
Selecting the *Display times* button on the [start](#) panel can reduce the number of these on screen numbers.

Time to gun, time to burn and distance below the line are displayed at the top centre of the screen. After the gun, the distance above or below

the line will also be displayed.

The bar on the left is a graphic indicator that shows time to burn.

The dot and time shown below is the time to burn after sailing to and turning onto the starboard end layline. The time below the start mark is defined on the [race](#) settings page.





Race and Start settings

Race options

Options used when racing (e.g. to a mark).

Use rate of turn in laylines

If this option is selected, Exp will include the time and distance to turn to the opposite tack or gybe when calculating laylines. This is based on the rate of turn [calibration](#) table.

Time and distance to each layline will be to the start of the tack or gybe.

Not used for start layline times.

Countdown past laylines

Exp can display negative numbers when past the layline. This can be useful to see how far the boat is past the layline the boat.

If unchecked, the times are set to zero when past the layline.

Race and start options

Options used in both the start and racing to a mark.

Ping marks at bow (else GPS)

The default option when pinging a start (or other) mark is to set the mark at the bow. This assumes the distance between the GPS and the bow is entered correctly as below. If this is de-selected, the mark is assumed to be at the GPS.

Start display

Options to determine items displayed on the start screen.

Grid

Draws a grid with increments of the boat length on the start display.

Boat length circles

Draws range circles in increments of the boat length on the start display.

Track

For clarity, there is an option to not draw the track in the start display.

Time to burn graphic

See the start [help](#).

Magnify line

If close to one end of the line, will magnify the start line to show the boat and that end of the line.

Show gybe

Shows more circle options in the start.

It shows turns great than 180 degrees.

This isn't really gybes, but for want of a better term ...

Predictor lines

Display heading, course and cog predictor lines.

Start settings

Mode options.

Switch to race state and chart at gun + 10s

Instructs Expedition to switch to the race display [state](#) and the last chart display at 10s after the gun.

Switch to race state and display at gun + 10s

Instructs Expedition to switch to the race display [state](#) and race chart at 10s after the gun.

Release held wind and current at gun

Deselects [held](#) wind and current values.

Switch to start display when set timer

...

Kill instrument start timer at gun

Kills instrument start/race timers where applicable.

Start time to line options

Select which numbers to use for time to the line and thus time to burn.

Expedition will use the minimum of the selected time values to determine the time to the line.

In a windward/leeward racing scenario, possibly just the starboard layline times to the line are desired. Add port layline approaches, reaching times, times to the ends as needed.

Time to line ends

Include times to port and starboard ends of the line.

GPS time to line

Include time to the line based on GPS COG and SOG.

Reaching time to line

Include the reaching time to the line.

Times on port

Times on starboard

Include times on upwind or downwind laylines to the line, including any tacks or gybes required.

Boat

To obtain maximum benefit from Expedition's starting tools, there are several things that are needed to describe the boat.

Boat length

The overall length (in metres) of your boat for proper scaling of Expedition's distance measurements.

Expedition uses the boat length to present various numbers in boat lengths. The desired units are selected on the [user settings](#) page. Note that if you have nautical miles as the default units, most start distances will still be displayed in boat lengths.

Bow to GPS

The distance (in metres) from the bow to the GPS antenna. The distance from the GPS to the bow is required for precise calculation of time and distance to the line. This will fine tune the distance and time to start line calculations.

Boat width

Boat width in metres.

GPS offset

For multi-hulls, not actually used anywhere in Exp.

Start

Roll

Start timer roll. Defaults to 5 minutes. See the [start](#) panel.

Twa & tws damping

Set & drift damping

Allows longer system damping of these instrument inputs during the prestart. Expedition will use the larger of these values or the normal [system damping](#) values.

Reverts to the normal system damping values at the gun.

Start calibration values

Shortcut to [calibration](#) values used by Expedition.

Also available from the [Application](#) menu.

Rate of turn and acceleration are always on, else the time to the line functions can not work.

Rate of turn

Set in the [calibration](#) section. Units are degrees per second and are a function of boat speed.

Acceleration

Set in the [calibration](#) section. Units are knots per minute and are a function of true wind speed and true wind angle. If you only have one wind angle in the cal file, then it will just be a function of tws.

Often the boat's speed will be less than target speed at any point in the pre-start. Expedition allows for acceleration when calculating time and distance to the line.

Braking

Set in the [calibration](#) section. Units are seconds to stop as a function of rate of turn.

This is optional and the default is off.



Setting the start line

There are several ways to create and modify a start line.

See the help on the [Start](#) panel.

See the help on the right-click [pop-up](#) menu.

Start marks are just like any other mark. They can drag them with the pointer after clicking on them (if they are unlocked). They can be manually edited. See the [marks](#) help for more information.

The right-click [pop-up](#) menu can be used to create start marks or to select an existing mark as a start mark. Note that to select an existing mark, it must be displayed on the screen - it must be part of the active route or be set to always drawn.



Start numbers

See [Appendix A](#).

Note that start distance units are set on the [system settings](#) page.



Calibration

Expedition's calibration functions are accessed from the [Application](#) menu or the relevant [instrument](#) page.

In most cases, the calibration functions will not be used. The common exceptions to this are the rate of turn and acceleration values that are necessary for [start](#) calculations. In higher end systems, the instrument system will perform calibration functions.

Calibration types can either be single value or tables of calibration values. See below for details for each calibration parameter.

Load and save calibration data using the Load and Save buttons to as text files in Expedition's config folder (in the [data](#) folder). The calibration values can either be edited from this dialog window or the calibration file may be edited directly, using any text editor.

Note that Expedition will store raw instrument values of Bsp, Twa and Tws in their own channels, which are logged and can be useful for later analysis.

Calibrations

The checkboxes with each value control whether that value will be calibrated by Expedition.

Some of these calibration tables are single values, others can be tables of numbers.

Interpolation of values uses a smooth spline technique by default. Linear interpolation is also an option.

Start calibrations

Used in start calculations to fine tune the time to the starting line.

Rate of turn and acceleration are always on, else the time to the line functions can not work.

Rate of turn

Table of values. Bsp across the table. Units are degrees per second.

One trick I do is adjust the cal values so the circles on the start display are about the same size as circles on the track.

Rate of turn can also be used to more accurately calculate time and distance to [laylines](#).

The Rate of Turn calibration is the most important calibration for starting numbers.

Acceleration

Table of values. Tws across and twa down the table. Units are knots per minute.

Only one row is needed - most users will just have one row in the table and ignore any TWA dependence.

Expedition has sample values, but it is easy to measure values using Stripchart. Slow to half speed or stop the boat, sheet on and measure how long it takes to accelerate to full speed and calculate this real acceleration value in knots per minute.

Once the boat reaches its polar speed (say 7.5 knots), Expedition will not assume further acceleration.

Braking

Table of values of time to stop in seconds. Rate of turn across.

Data calibrations

Heel (roll)

Table of added value corrections, heel across the table. Roll is positive to starboard (sailing on port).

Trim (pitch)

Table of added value corrections, trim across the table. Bow up pitch is positive.

Mast angle

Table of added value corrections, mast angle across the table.

Rudder angle

Table of added value corrections, rudder angle across the table.

Bsp

Table of added values. Bsp across and heel down the table.

Most users will just have one row in the table and ignore any heel dependence.

Aws

Table of added value corrections to correct for aws errors. Awa across and aws down the table.

For advanced users only.

The AWS cal is added to aws ($aws = cal + aws_{raw}$).

Twa

Table of added value corrections. Twa across and tws down the table.

Twa and Tws calibration tables are different from other calibration tables in that the default format is similar to the Expedition polar format.

Tws

Table of added value corrections. Twa across and tws down the table. Generally negative numbers as want to subtract from tws.

Heading

Normally a single calibration value to correct for compass installation errors, but could be a table dependent on heading.

Leeway

Often a single calibration value, but could be a table dependent on Twa across and tws down the table.

Expedition uses the same equation as used by systems such as Ockam and B&G to estimate leeway. The calibration values are leeway coefficient(s), k, such that

$$\text{Leeway} = k \cdot \text{heel} / \text{bsp}^2$$

The hard part is estimating the value of the coefficient, k . In practice we bias k to upwind sailing as the leeway is larger and more important upwind. For many purposes a value of 10 to 12 will suffice.

Because the ideal value of k varies at different angles, Expedition also allows the use of a table of leeway calibration values. These are just the coefficients at different wind angles and speeds.

So, given a VPP from the yacht designer with bsp, heel and leeway, it is relatively easy to generate a corresponding mean value or table of leeway coefficients.

Note that heel and leeway should be +ve on port, see [calculations](#).

Wind weight

Multiplier for polars, default 1.0.

Calibration and your instruments

Depending on the instrument system being used, Expedition may be able to exchange calibration values with it. In these cases, there will be an extra calibration page available dedicated to exchanging information with the instrument system.



Available from the Instruments button on the [Applications menu](#) or the Ctrl-i keyboard shortcut.

Expedition can be configured to talk to all the main types of instrument systems, either via a [network](#) (ethernet or wireless) or a [serial](#) connection. Expedition will automatically detect available serial ports on your computer and they will be listed on the left side the window. Select the page for the appropriate serial port or network connection and proceed with the configuration described for your individual instrument system.

Non-WGS84 [charts](#) are automatically corrected by Expedition to the WGS84 datum where possible. For this reason, it is generally preferable to use the WGS84 datum in your GPS or other position fixing device.

The common functions are described below. Depending on the instrument system selected, different options will be available on this page and from the *System* button.

GPS Source

Boats can have GPS data available from multiple sources - Instruments, AIS, Backup, NMEA 2000 etc.

Generally only one can be used at a time.

The [GPS](#) tab on the instruments window allows the desired source to be selected.

Also available from the Instruments menu on Expedition's [System](#) menu or Ctrl-G

User command

This box, although not commonly used, allows you to send a command to the instrument system. Beware of what you send!

Reconnect

Can be useful to pick up a dropped network connection, especially wireless.

Wake

Some instrument systems, need to be instructed what data to send by Expedition. Normally, this happens automatically but, if desired, this button forces all the wake-up commands to be immediately re-sent to the instruments.

Raw data

Displays data being received from the instruments. This can be very useful for tracing instrument connectivity problems. Use the *Export incoming data* button to save the incoming data to a text file.

System

Opens an interface specific for the instrument system selected on this page.

Broadcast

Forwards received data as a UDP broadcast to a port.

This is one method of sharing data from one source between different programs on the same or different computers.

For example, Expedition could receive AIS data and forward the same data stream to B&G Deckman software.



GPS Source

Boats can have GPS data available from multiple sources - Instruments, AIS, Backup, NMEA 2000 etc.

Generally only one should be used at a time.

Interface

Available from the Instruments menu on Expedition's [System](#) menu.

Also available on from the [Instrument](#) connection interface.

Ctrl-G keyboard short-cut.

Notes

This only feature only applies to Boat zero.

GPS data is available from most instrument systems. However, there are certain advantages (precision, timeliness and range of data available) to receiving the GPS position fix directly from its [NMEA](#) output rather than from your boat's instrument system.

It is recommended that any NMEA 0183 GPS data is sent to Expedition as NMEA 0183 v2.0 or higher rather than NMEA 0183 v1.5.

Example

GPS data may be available from each of the AIS, NMEA and B&G H5000 connections.

In this example, Expedition is using the NMEA feed on Com 4 for GPS data.

GPS

GPS Source to use

Currently selected: Com 4, USB Serial Port, NMEA 0183

→ Com 2

NMEA 0183 AIS

→ Com 4

USB Serial Port, NMEA 0183

→ Network 0

B&G H5000 WebSocket

→ Cosworth

→ Dfw

→ Expedition DLL

→ Garmin

→ Windows sensors

→ Cancel



Calculations

Expedition's calculation functions are accessed from the [System](#) settings page.

In most cases, the calculation functions will not be used.

Calculate

The checkboxes control the calculation of various parameters.

For example, if Twa is selected, it will be calculated from Aws, Awa, Bsp etc. If not selected, Expedition will use the value received from the instruments. This is useful if your instruments don't send some values.

Mostly, these options should not be checked, but it may be desired in cases where the instrument systems doesn't send all data.

If using [LogPlayer](#), these should all be de-selected.

Twa & Tws

Twc

Current set and drift

The calculations of set and drift use the Expedition system damped values of cog, sog, bsp, hdg and leeway.

The ROT limit for current calculations sets an upper limit at which the current set and drift will be calculated (you don't want to calculate current during a tack).

Leeway from heel and bsp

Leeway from bsp components

[Leeway](#) from bsp components is not often used and is currently only available if NMEA 2000 PGN 130578 data is available.

Pitch and roll rates

Yaw rate (ROT)

Yaw rate calculation is not recommended if ROT is available from the instruments.

Notes

[NMEA 2000](#)

Many NMEA 2000 systems do not send TWA, TWS or TWD, so these need to be calculated by Expedition.

[Nexus](#)

If Expedition is connected to the Silva NX2 Sail Performance program rather than directly to the instruments, TWA, TWS and TWD may need to be calculated by Expedition.

Nexus [FDX](#) apparent wind speed and direction are already corrected for heel if available, so do not select this option with Nexus instruments.

[Tactick](#)

TWD may need to be calculated by Expedition.

[KVH Quadro](#)

TWA and TWS may need to be calculated by Expedition.

Options

Correct for heel

Heel affects the apparent wind measurement.

Nexus [FDX](#) apparent wind speed and direction are already corrected for heel if available, so do not select this option with Nexus instruments.

Use pitch and roll rates

Corrects AWA and AWS for the motion of the masthead.

Use SOG as Bsp

Use SOG instead of Bsp in wind calculations, polar bsp%, target bsp% and start calculations.

Use COG as Hdg

Exp can use COG in wind calculations, polar bsp%, target bsp% and start calculations.

Notes on sign of data

Expedition uses the standard sign conventions, as below.

Twa & awa

Positive on starboard, negative on port.

Trim (pitch)

Bow up pitch is positive.

Heel (roll)

Roll is positive to starboard (sailing on port).

Leeway

Leeway is normally signed so clock-wise leeway is positive, thus leeway is > 0 on port.

$$\text{course} = \text{heading} + \text{leeway}$$

Thus

$$\text{leeway} \sim k \cdot \text{heel} / \text{bsp}^2$$



Serial ports

Available from the Instruments item on the [Application](#) button menu or the ctrl-i key stroke.

RS-422 and RS-232

Most PC serial ports use RS-232 whereas [NMEA 0183](#), [B&G 5000](#) etc. use the RS-422 electrical standard.

It is normally best to use a USB to RS422 adaptor rather than a low cost RS232 to USB adaptor. An RS422-RS232 convertor is an option for existing RS232 ports.

See the Expedition [instruments web page](#) for USB-RS422 adaptors. A common solution is the [Actisense](#) USG-1.

This overcomes issues associated with ground loops and the different electrical standards.

Different electrical standards

The RS232 standard as used by the PC uses ground and receive to get data from a connected system. Data is sent as voltages referenced to ground on a single transmit line. This type of drive is known as single-ended.

NMEA 0183 uses a differential system, with positive and negative data connections.

Ground loops

These occur when using a standard USB to serial converter or a RS232 port as the ground of the PC is then shared with the NMEA 0183 system ground, which may be different.

When the PC and the NMEA 0183 system have different grounds, data can be corrupted and damage can occur to the PC and/or NMEA device in extreme cases.

Instruments, baud rate etc. settings

Select the appropriate instrument type, serial port settings (baud rate etc.) and the boat to which the incoming data will be mapped. Most users will only want to map data to boat zero and throughout this help file we assume the system is set at Boat 0.

When you select an instrument system, Expedition sets the default settings that usually are best for that system.

DB9 pins

1 Received Line Signal Detector (Data Carrier Detect)

2 Received Data

3 Transmit Data

4 Data Terminal Ready

5 Signal Ground

6 Data Set Ready

7 Request To Send

8 Clear To Send

9 Ring Indicator



Networking

Available from the Instruments item on the [Application](#) menu or Ctrl-I.

Expedition can exchange data with instances Expedition on other computers or to any supported instrument system over a network using TCP, UDP or Websocket network protocols.

For data streaming, UDP is generally preferred. Websocket is used for connections to B&G [H5000](#).

By default, Expedition has no network connections available. If one or more network connections are required, this can be set with the Number of network connections item from the Instruments item on the [Application](#) button menu.

Receiving instrument data

Notes

It may be necessary to set Windows' firewall to enable both Public and Private networks for Expedition, especially for UDP connections.

NMEA has registered port 10110 for NMEA 0183 over UDP and TCP.

On a Network page

Select the instrument system to exchange data with and the port to connect to.

Expedition will automatically select default settings for each instrument system.

The port should be between 1024 and 65535. Ports below this are assigned for common services, so shouldn't be used.

Tx port

Normally the same as the (Rx) port, but some systems (Ockam and Onboard assistant) may require data to be sent to a port one greater than the Rx port.

Enter the port number to send to.

TCP client

Enter the address and port number of the server computer (eg. 192.168.1.128 port 5242).

UDP

UDP Broadcast address

Some default to specific addresses (eg Koden radar).

If just sending to a specific address, such as your instrument processor, the address of it should be used.

Expedition can also broadcast using a mask if you wish to send to more than one address.

The default Microsoft solution broadcasts to 255.255.255.255, but this may not be the best option for sending UDP broadcasts as some routers/firewalls block it unless configured otherwise.

A better option is to use the NIC's actual subnet broadcast IP instead.

For example, if a UDP socket is bound to local IP 192.168.0.1 with a subnet mask of 255.255.255.0, then the broadcast IP for that subnet is 192.168.0.255.

Note that the default x.x.x.255 is just the first address Exp finds - it may not be what you want.

Common connection scenarios

1. Ethernet, connecting the two computers through a router.
2. Ethernet, connecting the two computers directly using a cross-over cable.
3. Wireless, using a wireless DHCP access point.
4. Wireless, using an ad-hoc network.

In cases 2 and 4, you need to give each of the two computers their unique IP address. For example, set the IP addresses to 192.169.0.1 and 192.169.0.2 and the subnet mask on each computer to 255.255.255.0.

For cases 3 and 4, you also need to set up the wireless connection. Assuming a wireless system, you need to set the network wireless mode to infrastructure for case 3 or to Ad-hoc for case 4. You also need to make the SSID (e.g. default), the channel (e.g. 6) and the transmit rate (e.g. 11 Mbps) the same on all devices.

This is very brief and it is a good idea to read your manuals. The main disadvantage with an Ad-hoc network is that it is point-to-point - if you want more computers listening to the host computer, you need to use an Access point.

Sharing data between computers

Note that you should not select the same item to be both sent and received on the same PC.

Broadcasting data to Expedition on other computers

Expedition can broadcast selected data to other computers on your local network using UDP.

On a Network page

1. Select Expedition from the instrument drop-list.
2. Select UDP or UDP to IP address as the connection type.
3. Select a port between 1024 and 65535 to broadcast to. Ports below this are assigned for common services, so you shouldn't use these.
4. From the System button, select the desired Expedition channels to broadcast on the Expedition output filter page.

See also [Slave mode](#).

Receiving broadcast data

To receive Expedition data broadcast by another PC.

On a Network page

1. Select Expedition from the instrument drop-list.
2. Select UDP as the connection type.
3. Select a port to receive data on. This is the port number as set above to broadcast the data on.
4. From the System button, select the desired Expedition channels to receive on the Expedition input filter page.

Expedition data exchange between specific computers

Expedition data can also be exchanged between any two computers rather than the whole network.

On a Network page

1. Select Expedition from the instrument drop-list.
2. If using UDP (recommended), the sending computer can be set to send data to the IP address of the receiving computer.
3. If using TCP, one computer should be set as the TCP server and the other as a TCP client. The server listens for a connection request from the client. The IP address of the server needs to be entered on the client computer network page.
4. Select a port to receive and send data on.

5. From the System button, select the desired Expedition channels to send and receive.



AIS is a shipboard broadcast system operating in the VHF maritime band that is capable of handling well over 4,500 reports per minute and updates as often as every two seconds. The system coverage range is similar to other VHF applications.

Expedition can read and decode AIS messages from most standard AIS receivers.

AIS targets may be drawn on the chart significantly more accurately than with ARPA/MARPA radar target plotting. Expedition can also calculate the distance and time to the point of intersection with each AIS target.

Expedition can also display AIS-SART messages.

Instruments, baud rate etc. settings

The default AIS serial port settings are 38400, 8, N, 1.

Baud rate	38400
-----------	-------

Data bits	8
-----------	---

Parity None

Stop bits 1

Connecting an AIS device to the PC's RS232 serial port.

Most computers have a 9 pin serial port or can use a USB-serial port adaptor if lacking a serial port.

DB9 pin	PC	AIS device
2	Receive data +ve	AIS device output +ve
3	Transmit data +ve	AIS device input +ve
5	Signal ground	Signal ground

NMEA 2000

Exp can also receive AIS data from a NMEA 2000 source.

AIS Settings

More advanced interface options are available by clicking on this button, also available from the AIS target list [window](#), where double-clicking on the target list toggles the toolbar on and off.

AIS Pane

AIS parameters to display on the AIS pane. The list of attributes of each AIS target that may be listed include:

MMSI number

Vessel Name

Call sign

Vessel type

Destination

Navigation status

Safety message

Time since receiving the AIS signal

Range to target

Bearing to target

CoG

SoG

Hdg

Distance from target at closest point of approach (CPA)

Time to closest point of approach

True wind direction

True wind speed

Rate of turn

Position

Vessel dimensions

AIS SART, MOB and EPRIB

Expedition will display the icon for these with the special SART symbol (a red circle with a red cross in it).

In addition, non test messages will be shown as a pop-up notification. In order to save screen clutter, new messages are not displayed until the existing message is cleared.

AIS target to boat mapping

Map a target to an Expedition boat (1 to 7) by MMSI number. This can be useful if tracking a competitor. The cog, sog, range and bearing can then be displayed in number boxes, sent to instrument displays etc.

An AIS target may also be mapped to a boat by right clicking on it and selecting Map AIS target to boat from the [pop-up](#) menu.

AIS target filter

Log

Log raw AIS messages in the AisLog folder in the Expedition [data folder](#).

Warn if AIS target enters [race note](#) area

Displays a pop-up window if an AIS target enters a race note area with the 'Avoid' flag set. Also writes an entry to the [event list](#) in the [database](#).

This also makes the Windows Exclamation sound. To change the default sound, go to the Control panel, Sound, Change system sounds.

MMSI

Enter your boat's MMSI number if sending AIS data so you don't display your own boat.

This is also used when sending MOB PGN 127233.

Filter

Instructs Expedition to not decode data from targets not in \config\AisFilter.csv.

Just a list of MMSI numbers.

Only for specialised use.

CPA filter

Colours the target red (see below) and draw closest point of approach (CPA) for vessels where the CPA is less than the specified time and distance.

Distance between at CPA

The minimum CPA distance.

Time to CPA

Ignore CPA if the time to the CPA is more than this value.

Warnings

If Warnings is selected, a popup warning should show if a target's CPA is less than the filters.

Using AIS with Expedition

Displaying AIS targets on the chart

Received AIS targets as well as their ship name and/or Cog & Sog may be displayed on the chart. See the [display](#) settings help for more details.

Colours:

Red	CPA less than as defined in AIS settings, defaults 30 minutes and 0.5nm
White	no recent position fix received
Green	sog $\geq$ 1kt
Yellow	stationary or sog $<$ 1kt

List of AIS targets

Expedition can list the AIS targets in a customisable list, along with time and distance to point of intercept.

This list may be displayed at all times and may be toggled on and off using the AIS from the [Window](#) menu.

Individual elements listed may be turned on and off using from the AIS settings page or by clicking on the toolbar on the AIS target list [window](#).



Expedition supports aisstream.io AIS data from the web.

Web

aisstream.io

Configuration

Create an API key on the aisstream.io web site.

Then select aisstream.io as an instrument type and add the API key to into the settings.

Notes

This is not designed to replace AIS on board, but may be of use for shore-based interest.

See [AIS](#) for more details on AIS in Expedition.



H1000, 690, 790, H2000, H3000 and H5000 processors.

Connection settings

Hydra (690, 790, H1000, H2000 & H3000) instruments communicate using the NMEA 0183 protocol, so the default serial port settings are 4800,8,N,1. For H3000 using the USB connection or the RS-232 port, the connection speed default is 115k.

See the [NMEA](#) help for more information on the NMEA standard and interface. In addition, Hydra supports custom NMEA strings.

Wiring to connect a H1000 to a PC's serial port:

H1000		Colour	DB9
1	NMEA Tx (+)	Blue	2
2	NMEA TX (-)	Green	3
5	NMEA RX (+)	Red	5

Output

Select desired NMEA sentences to send back to your instrument system.

Hydra custom NMEA data

Expedition automatically sends some basic performance data to the B&G Hydra processor. However, you need a v9.0 NMEA FFD for this sentence support in H2000.

These may not yet be available on H3000 or H5000.

The following are available on the Hydra processor:

PERFORM

POL SPD (kt)
POL PERF (%)
LAYLINE distance (nm)
LAYLINE time hh/mm, Exp sends *Time to Gun* in the pre-start

WIND

TARG AWA (Expedition outputs Target Twa to this channel instead of Target Awa)

NAVIGATE

TARG HDG NL (near layline)
TARG HDG FL (far layline)

Not yet implemented

WAYPOINT - DIST TSL

WAYPOINT - TIME TSL HR/MS

Settings

More advanced interface options are available by clicking on this button.



B&G H2000 / H790 / H690



Including 690, 790 and 2000 performance processors with newer B&G software.

Connection settings

Expedition defaults to 9600, 7, E, 1 when the H2000/790/690 instrument system is selected (6.2 on the FFD).

Baud rate	9600
-----------	------

Data bits	7
-----------	---

Parity	Even
--------	------

Stop bits	1
-----------	---

Note: Occasionally, you may find 2 stop bits works best with B&G.

Wake

Each time a Hercules system is turned on, it has to be instructed what data to send by Expedition. Normally, this happens automatically. This button forces all the wake-up commands to be immediately resent to the instruments.

Instrument system wiring and set up

Recommended serial port wiring

	B&G 390	B&G 690+	DB9	DB25
RS232 gnd	7	11	5	7
RS232 Tx	2	10	2	3
RS232 Rx	3	9	3	2
Jumper			4 to 6	4 to 5
Jumper			7 to 8	6 to 20

While the normal PC hardware might well run with just Tx, Rx and Ground connected, some driver software will wait forever for one of the handshaking lines to go to the correct level. Depending on the signal state it might sometimes work, other times it might not. The reliable solution is to loop back the handshake lines if they are not used.

When the lines are handshake looped, the RTS output from the PC immediately activates the CTS input - so the PC effectively controls its own handshaking.

A more complete solution for B&G 690 onwards systems is

	B&G 690+	DB9
RS232 CTS	7 Green	4*

RS232 RTS	8 Violet	8
RS232 Rx	9 Red	3
RS232 Tx	10 Blue	2
RS232 Gnd	11 Black	5

* link pins 4 and 6.

B&G instrument setup

On an FFD, set the B&G to 9600 baud, even parity, 7 data bits and 1 stop bit. To do so, select *Waypoint / cross tr* on one section of the display and *Calibrate / cal val1* on the other. Set the value to 0.

Select *Waypoint / cross tr* on one section of the display and *Calibrate / cal val2* on the other. Set the value to 6.2.

GPS Position

Most Expedition users prefer to use a GPS position directly from [NMEA 0183](#) as the position received from the H2000/790/690 can have noticeably reduced precision.

B&G H2000 settings

More advanced interface options are available by clicking on this button.

Accesses the interface to instrument polars, calibration values and tables.

Expedition can exchange polars and calibration values with the instrument system. This is a very useful function for backing up calibration values.

Depending on the version of the instrument system, it occasionally takes several 'uploads or 'downloads' for Expedition to accurately send or retrieve the full polar.

B&G nodes used are those defined in custom configuration below.

Settings

Magnetic mode

Used if the B&G processor is in magnetic mode.

Send mark information

Tells Expedition to send mark bearing, range, time etc. to the H2000/790/690.

Polars

Expedition can update or retrieve performance [polars](#) stored on the instrument system.

Custom

External channels

Expedition can exchange data with the B&G processor's external channels. Simply choose the desired channel and select the number/function from the drop down menu. Note that you can also select one of Expedition's [alternating numbers](#).

Selecting the damped check-box next to any external channel will cause damped data to be sent.

Selecting refresh tags sends tag text every 10s even if not changing.

Selecting Send alternating tag names means tags on the external channels will be continually updated to change with the alternating channel.

Important note

It is recommended the [alternating numbers](#) period is not set to less than 3 seconds.

Linear channels

Hercules linear channels 1 to 10 (such as Forestay load) can be mapped to Expedition system channels.

Custom configuration

Advanced users only.

Expedition uses a configuration file to determine the functions Expedition instructs the B&G H2000/790/690 performance processor to send.

Expedition creates the file, H2000_config.txt, in the Instrument folder in the [Expedition data](#) folder.

This file can be edited - any line can be disabled or comments added by putting an exclamation mark (!) at the start of the line.

There are several options with H690, H790 and H2000 processors, but for more details on function and node numbers, please see your B&G documentation.

B&G #OV command

This command requires the B&G node and function number

```
! node, function  
5, 11 rudder
```

For example, you may have a compass on different nodes, so it is possible to specify which node the compass to be used is on.

The text is optional in the Expedition configuration file, but is useful for editing purposes.

B&G #OE command

This command just requires the Hercules or Remote channel number and a tag to say whether it is a Hercules or Remote channel being requested. These tags are just H or R characters.

```
!690/2000 Hercules functions  
H, 10 aws
```

```
!690/2000 Remote functions  
R, 17 cog True
```

The text is optional in the Expedition configuration file, but is useful for editing purposes.

Example of an *H2000_config.txt* configuration file

```
! use an exclamation mark to comment out functions  
!  
! H690,790,2000,3000 functions  
! node, function  
5, 11 rudder  
13, 16 Linear 5
```


13, 17 Linear 6
13, 18 Linear 7
13, 19 Linear 8
5, 28 air temp F
! 5, 29 air temp C
! 1, 30 sea temp F
! 1, 31 sea temp C
! 5, 52 heel
! 5, 56 Linear 1
! 5, 57 Linear 2
! 5, 58 Linear 3
! 5, 59 Linear 4
112, 64 Digital load cell
! 1, 65 bsp
! 16, 73 heading : Halcyon 2000 on Fastnet Bus
! 5, 73 heading : Super Halcyon 3 on CPU
! 15, 73 heading : Halcyon Gyro on Gyro Processor
! 18, 73 heading : Halcyon Gyro on ACP Pilot
! 96, 73 heading : NMEA compass on FFD
! 5, 77 aws
! 5, 81 awa
! 5, 85 tws
! 5, 89 twa
5, 102 keel angle
5, 103 canard / forward rudder, (node correct?)
5, 104 trim tab, (node correct?)
! 5, 109 twd
5, 130 leeway
! 9, 131 drift
! 9, 132 set
5, 135 barometer
! 5, 141 battery volts
5, 155 trim
! 15, 155 trim on Gyro compass

! 18, 155 trim on ACP pilot

5, 156 mast angle

5, 163 dagger board

5, 164 boom position

! 1, 194 depth ft

! 1, 233 cog True (may as well ask for it in its original reference frame, not magnetic)

! 96, 233 cog True (on first NMEA FFD)

! 1, 235 sog

! 96, 235 sog (on first NMEA FFD)

!

!690/2000 Hercules functions

H, 10 aws

H, 13 awa

H, 12 tws

H, 14 twa

H, 15 twd

H, 27 sea temp C

H, 0, heel

H, 1, bsp

H, 5, hdg

H, 23 leeway

H, 7, volts

H, 8, depth ft

H, 28 Linear 1

H, 20 Linear 2

H, 26 Linear 3

H, 18 Linear 4

!

!690/2000 Remote functions

R, 17 cog True (may as well ask for it in its original reference frame)

R, 19 sog

R, 28 set

R, 29 drift

B&G H3000 and H5000 processors

Connection settings

Expedition defaults to 115,200, 8, N 1 when B&G H3000 or H5000 instrument systems are selected.

Baud rate 115,200

Data bits 8

Parity None

Stop bits 1

H5000 RS-422 and RS-232

The H5000 port is a RS422 port. See the serial port connection [help](#).

GoFree alternatives

Expedition can use the H5000 [Web Server](#) interface.

Expedition can also use B&G's GoFree wireless interface to receive basic NMEA 0183 and AIS data. Note that this does not give the full HLink functionality.

Log on to the B&G on board GoFree Wi-Fi from the laptop

In Expedition instruments

1. Select Network 0
2. Select NMEA 0183 as the instrument type
3. Select TCP Client
4. Input your B&G Zeus 2 plotter IP address (as this unit supplies the NMEA data which it gets from the CPU). The IP address will be different for each plotter, but you can find it in the Network Settings pages.
5. Select Port 10110
6. Select "Reconnect"

Wake

Each time a H3000 or H5000 system is turned on, it has to be instructed what data to send by Expedition. Normally, this happens automatically, but if desired, this button forces all the wake-up commands to be immediately re-sent to the instruments.

B&G instrument setup

See the H3000 or H5000 manual.

It is recommended that awa heel correction (awa cal 2, heel_crn) is set to 1 (on).

It is recommended that the twa and twd calculations are set to use leeway (leeway cal 2, use_lwy).

B&G H3000 & H5000 settings

More advanced interface options are available by clicking on this button.

Accesses the interface to instrument polars, calibration values and tables.

Expedition can exchange polars and calibration values with the instrument system. This is a very useful function for backing up calibration values.

B&G nodes used are those defined in custom configuration below.

Settings

Magnetic mode

Used if the B&G processor is in magnetic mode. This should be set automatically for H3000 and H5000 systems.

Send mark information

Tells Expedition to send mark bearing, range, time etc. to the H3000 or H5000 .

Send COG Magnetic to H3000

B&G H3000 needs the [NMEA](#) VTG sentence for magnetic COG. If the H3000 is only receiving COG from RMC, it will only have true COG.

The solution is to always send VTG from the GPS to H3000. If this is not possible, it is possible to send magnetic COG to the H3000 from Exp.

This option should not be selected in the majority of cases.

Polars

Expedition can update or retrieve performance [polars](#) stored on the instrument system.

Custom

External channels

Expedition can exchange data with B&G's external channels. This is limited to Performance and possibly Hercules level H5000 CPUs.

Simply choose the desired channel and select the number/function from the drop down menu. Note that you can also select one of Expedition's [alternating numbers](#).

Selecting the damped check-box next to any external channel will cause damped data to be sent.

Selecting refresh tags sends tag text every 10s even if not changing.

Selecting Send alternating tag names means tags on the external channels will be continually updated to change with the alternating channel.

Important note

It is recommended the [alternating numbers](#) period is not set to less than 3 seconds.

Linear channels

Hercules linear channels 1 to 10 (such as Forestay load) can be mapped to Expedition system channels

Custom configuration

Advanced users only.

Expedition uses a configuration file to determine the functions Expedition instructs the B&G H3000 or H5000 performance processor to send.

Expedition creates the file, *H3000_config.txt*, in the Instrument folder in the [Expedition data](#) folder.

This file can be edited - any line can be disabled or comments added by putting an exclamation mark (!) at the start of the line.

For more details on function and node numbers, please see your B&G documentation.

B&G #OV command

The B&G #OV command requires the B&G node and function number

! node, function
5, 11 rudder

An example of an *H3000_config.txt* configuration file

! use an exclamation mark to comment out functions
!
! H3000 functions
! node, function
5, 11 rudder
13, 16 Linear 5
13, 17 Linear 6
13, 18 Linear 7
13, 19 Linear 8
5, 28 air temp F
!5, 29 air temp C
1, 30 sea temp F
!1, 31 sea temp C
5, 52 heel
5, 56 Linear 1
5, 57 Linear 2
5, 58 Linear 3
5, 59 Linear 4
112, 64 Digital load cell
1, 65 bsp
16, 73 heading : Halcyon 2000 on Fastnet Bus
!5, 73 heading : Super Halcyon 3 on CPU
!15, 73 heading : Halcyon Gyro on Gyro Processor

!18, 73 heading : Halcyon Gyro on ACP Pilot
!96, 73 heading : NMEA compass on FFD
5, 77 aws
5, 81 awa
5, 85 tws
5, 89 twa
5, 102 keel angle
5, 103 canard / forward rudder, (node correct?)
5, 104 trim tab, (node correct?)
5, 109 twd
5, 130 leeway
9, 131 tide drift
9, 132 tide set
5, 135 barometer
5, 141 battery volts
5, 155 trim
!15, 155 trim on Gryo compass
!18, 155 trim on ACP pilot
5, 156 mast angle
5, 163 dagger board
5, 164 boom position
1, 194 depth ft
1, 233 cog True (may as well ask for it in its original
reference frame, not magnetic)
!96, 233 cog True (on first NMEA FFD)
1, 235 sog
!96, 235 sog (on first NMEA FFD)



B&G Websocket



Expedition can connect to B&G

H5000, Edge, Hercules and Hercules WTP processors.

Connection settings

Either direct via ethernet or via a Wi-Fi router, see the B&G installation manual.

In the Expedition instruments interface

1. Select network 0
2. Select B&G Websocket as the instrument type
3. Websocket will be automatically set as the connection type
4. 2053 will be automatically selected as the port
5. Set the appropriate IP address

H5000 will default to 192.168.0.2

if it does not see a DHCP server.

Edge Wifi may default to 192.168.76.1.

B&G Webserver settings

More advanced interface options are available by clicking on this button.

Accesses the interface to instrument polars, linear and remote channels.

Calibration and other system settings are not included as they can be accessed with a web browser.

Race timer changes and setting line ends are automatically sent from Exp.

AIS Traffic

Accepts AIS data from GoFree. Not implemented in B&G processors.

AIS from [NMEA 0183](#) or [NMEA 2000](#)

is recommended over this option.

Send navigation data

Sends mark bearing, range, time, xte and layline range & time to H5. May not be implemented in the processor.

Send waypoint

Sends the active waypoint to the H5. May not be implemented in the processor.

Send start events

Send start timer and line ends pings to H5000. If disabled, Exp can receive the start timer from the processor.

Start times may only be sent in whole minutes.

Start line ends set from a GFD

or MFD may not be sent by the processor.

Polars

Expedition can exchange performance [polar](#) values with the instrument system.

H5000 only. Edge, Hercules and Hercules WTP can upload Expedition format polars from the web interface.

Custom

External channels

Expedition can exchange data with B&G's external channels. This may not be supported on Hydra and Edge CPUs.

Simply choose the desired channel and select the number/function from the drop down menu. Note that you can also select one of Expedition's [alternating numbers](#).

Selecting the damped check-box next to any external channel will cause damped data to be sent.

Selecting refresh tags sends tag text every 10s even if not changing.

Selecting Send alternating tag names means tags on the external channels will be continually updated to change with the alternating channel.

Caption names / tags

The Custom name fields lets the user over-ride the default external channel caption name.

Selecting refresh tags sends tag text every 10s even if not changing.

Selecting Send alternating tag names means tags on the external channels will be continually updated to change with the alternating channel.

Linear channels

Linear channels 1 to 10 (such as Forestay load and analogue channels) can be mapped to Expedition system channels.

Custom configuration

Expedition uses a configuration file to determine the functions Expedition instructs the H5000 performance processor to send over the Websocket.

The H5000_config.xml file is installed in the Expedition \instruments folder in Program Files.

This file may be copied to the instrument folder in the [Expedition data](#) folder and edited. If available, Expedition will use this file in preference to the installed configuration.

The file maps B&G Websocket vars and Expedition channels.

id_bg	the B&G Websocket var id
id_exp	the Expedition channel id, from usr_channels.h
sysval	true to use the B&G system value, false to use the user val
linear	id if a linear channel (1-16)
name	Expedition channel name, just for readability

sysval should be true for the fundamental channels so Expedition knows the units.

B&G channel (id_bg) 125, magnetic variation should not be deleted.

Rudders

If port and starboard rudders, enable Rudder toe in the configuration file.

Dfw2Exp

B&G does not allow full direct access to the WTP processor.

However, Expedition has a small utility ([Dfw2Exp](#)) that allows Expedition to exchange data with Deckman For Windows (Dfw) software.

Dfw2Exp is a 32 bit application, so is only available with the legacy 32 bit Expedition installations.

However, it can be used with Expedition x64 if the versions are identical.

WTP Fastout

Expedition can use the Fastout data from the WTP processor.

The data to be sent is specified in the WTP fastout.d file. Please see the WTP user manual for details.

There is a file (wtp_fastout_config.txt) in the config folder in the Expedition data folder that may be used to map WTP Fastout tags to Expedition channels.



Bravo Systems

Bravo Systems from Spain is universally acknowledged as being one of the highest performing instrument systems available. For example, it was used by 6 of the teams challenging for the 2007 America's Cup and Oracle in the 2010 Americas Cup.

Expedition provides a simple, intuitive and seamless interface to Bravo Systems .

Updates of settings and values are generally transferred automatically between Bravo Systems and Expedition. However, most pages in the interface allow the user to manually exchange data (such as calibration tables or waypoints) with Bravo Systems. In most cases, this will not be necessary.

Yes, the documentation below is short and simple, but it really is that simple to use Expedition with Bravo Systems.

Web

www.bravosystems.es

Connection settings

Expedition can connect to Bravo Systems via a [network](#) or a serial connection. The network connection is preferred for speed. A

serial connection could be a physical wire or Bluetooth or Wi-Fi.

Network

The default address the Bravo Systems server is 192.168.25.225. However, this can be changed by the user. The default port is 5242. Exp should be set up as a TCP client.

Serial

The default connection settings are 38400, 8, N, 1, but other settings are possible.

Bravo Systems settings

More advanced interface options are available by clicking on this button.

System

A summary of the Bravo Systems processor version. Please quote this with any queries - the text in this can be copied.

Events

The Bravo Systems processor supports a large number of *events*, such as a start timer, mark rounding etc. Expedition

does not yet support all events, but is adding new functionality as requested.

Vars

The Bravo Systems processor can support a vast number of variables in addition to the ones we are already familiar with.

This page provides a simple, graphical interface to map Bravo Systems vars to Expedition [variables](#) and boats and there are some default mappings to simplify the task. Similarly, Expedition variables can be mapped to Bravo Systems vars for display on instruments etc.

The variable mappings can be saved to a file (RB_Map2.dat) for archive purposes or for transfer to another computer.

The right hand column in the list contains the current values being received from Bravo Systems.

Settings

This is the list of Bravo Systems server settings/customised server parameters. For example, bsp, awa and aws calibration values.

Curves

These are just one dimensional tables. Often used for calibration, but can be used for other purposes.

Tables

Often used for calibration, but can be used for other purposes. There are two types - standard and advanced. For an example of the difference see the section on polars below.

Interpolation of values uses a smooth spline technique by default.

Polars

Polars in Bravo Systems can be in two forms - standard and advanced.

Examples of each are -

Standard

A two dimensional table in twa and tws. There will always be 10 rows for tws and 17 columns for twa from 20 to 180 and 4 extra columns for target speeds and wind angles.

[illegible]

6													
10													
12													
14													
16													
20													
23													
26													
30													

Advanced

The advanced form of tables and polars is similar to Expedition's [polar](#) format. The main difference is that the advanced polars in RB require a column for upwind targets and another for downwind targets whereas the Expedition polars can have the targets spread over several columns.

	v_0	a_0	v_1	a_1	v_2	a_2	$\dots$	v_n	a_n
2									
4									
6									
10									
12									
14									
16									
20									
23									
26									
30									

Caveats for exchanging Bravo Systems advanced polars with Expedition polars:

Requires size of polars to be the same.

Bravo Systems requires specific columns for upwind and downwind targets. Targets can be across different columns in Expedition, so for the moment we have restricted it to the second and second to last columns.

Dampings

Damping values in Bravo Systems. There are two types of Bravo Systems damping.

NODYN - a single parameter for the damping value in seconds.

DYN - dynamic damping.

Displays

Controls the behaviour of the Bravo Systems displays.

Waypoints

Bravo Systems can contain waypoints or marks. Expedition can retrieve these from the instrument system or send the active course to the Bravo Systems server.



Expedition can connect to Cosworth systems using a serial connection or directly to the Cosworth CAN bus.

Web

www.cosworth.com

CAN connection

[Cosworth2Exp](#) uses a [Kvaser](#) CAN adaptor to connect directly to a Cosworth CAN bus. The direct CAN interface is faster and generally superior to a serial connection.

Serial port connection

Expedition defaults to 115,200, 8, N 1 when Cosworth instrument systems are selected.

Baud rate 115,200

Data bits 8

Parity Even

Stop bits 1

Cosworth settings

Accesses the interface to instrument polars, calibration values and tables.

Cosworth uses a protocol very similar to B&G's [HLink](#) for compatibility with existing systems. In general, the function numbers and nodes are identical.

Expedition can exchange polars and calibration values with the instrument system.

Polars

If the performance [polar](#) contains more than 10 wind speed rows, only the first ten rows will be sent.

External channels

Expedition can exchange data with B&G's external channels. Simply choose the desired channel and select the number/function from the drop down menu. Note that you can also select one of Expedition's [alternating numbers](#).

Selecting the damped check-box next to any external channel will cause damped data to be sent.

Selecting refresh tags sends tag text every 10s even if not changing.

Selecting Send alternating tag names means tags on the external channels will be continually updated to change with the alternating channel.

Linear channels

Linear channels 1 to 10 (such as Forestay load) can be mapped to Expedition system channels.

Custom configuration

Advanced users only.

Expedition uses a configuration file to determine the functions Expedition will decode from the Cosworth processor data stream.

Expedition creates the file, *Cosworth_config.txt*, in the Instrument folder in the [Expedition data](#) folder.

This file can be edited - any line can be disabled or comments added by putting an exclamation mark (!) at the start of the line.

For more details on function and node numbers, please see your Cosworth documentation.

Each line contains the function number, node and the Expedition var to map to.

For example:

```
! function number, function, Exp var  
52, 5, 18 Heel Angle  
65, 1, 1 Boatspeed  
73, 1, 13 Heading
```



Expedition supports Digital Selective Calling (DSC) messages from [NMEA 0183](#) and [NMEA 2000](#) sources.

DSC targets are displayed in the list in the AIS and DSC [window](#) and on the chart. See the [display](#) settings help for more details.

Expedition can also display AIS-SART messages.



Web

www.pixelsurmer.com

Connection settings

Connecting to Exocet requires Tx and/or Rx network connections to each node you wish to connect to.

Select Exocet Tx or Exocet Rx as the instrument type.

Select UDP to IP address as the connection type, the address and the Rx port. The Tx port should be Rx plus one.

Exocet can also be configured to send Expedition or NMEA 0183 data, in which case, select Expedition or NMEA 0183 as the instrument type.

Sending data

Set the instrument type as Exocet Tx, then select the vars you wish to send.

Receiving data

Set the instrument type as Exocet Rx.

Expedition will ask the Exocet box for the vars are available.

These can then be mapped to Expedition vars.

Note that the mappings are stored in the Exp data folder - it may be a good idea to back them up.



Expedition simple protocol

Background

A simple protocol to exchange data over networks or via serial connections with Expedition on other computers or third party products such as A+T, Exocet, Sailmon, Ventus or Campbell Field's excellent apps. It is also useful for weather boat fleets etc.

Output (Tx) filter

Select desired parameters to send.

Send damped

Option to send **damped** vars instead of raw.

Default

Default data selection to send.

Tx Extra

Expanded default selections.

Input (Rx) filter

Select desired parameters to receive.

Developer information

The Expedition communication protocol is a simple text based protocol.

Each sentence starts with a # and ends with *XX\r\n, where the XX is a checksum.

The streaming data is of the format: #boat
id,var1,val1,var2,val2,var3,val3,...*XX\r\n External channel data
is sent in the format: E,var1,val1,var2,val2,...*XX\r\n

System channel data is sent in the format:
X,var1,val1,var2,val2,...*XX\r\n

This is currently used for held wind and current data.

Commands

There are various commands a third party can send to Exp.
This may not be complete and can be added to on request.

For alternating channels, the id of the mapped channel is
sent as the value. In the case of Sailmon, the value is sent
instead of the id.

Aknowledgement

Most commands will respond with ACK if successful. The payload is the checksum of the command being responded to.

#ACK,xx*xx\r\n

Gun

#G,SYNC*XX

Synchronises the [start](#) countdown timer.

#G,val*XX

Sets the [start](#) countdown timer. Val is in minutes. A negative number will kill the timer.

For example, #G,12.34*62\r\n

#G,KILL*XX

Kills the [start](#) countdown timer.

Line

#L,P*XX

Pings the [port](#) end of the line

#L,S*XX

Pings the [starboard](#) end of the line

#L,P,lat,lon*XX

Sets the [port](#) end of the line

#L,S,lat,lon*XX

Sets the [starboard](#) end of the line

Mark

#M,id,name,latitude,longitude,flag*XX\r\n Updates existing mark if id exists and is a number, else creates a new mark.

Mark id is normally 0,1,2,3,4 etc., but may be ACTIVE for the active mark. For example:

#M,ACTIVE,name,latitude,longitude,flag*XX\r\n A MOB position may appear as:

#M,MOB,MOB,latitude,longitude,X*XX\r\n
#M,ACTIVE,name,latitude,longitude,X*XX\r\n
Valid flags:

X	MOB
P	Leave to port
S	Leave to starboard
M	Sail direct to mark

#M,PING*XX

[Pings](#) a mark at the current location ([bow](#) or [GPS](#)).

#M,PINGACTIVE*

Sets the active mark to the current position.

#M,PREV*XX

Sets the previous [mark](#) in Exp's active route as active. Exp will reply with the new, active mark.

#M,NEXT*XX

Sets the next [mark](#) as active. Exp will reply with the new, active mark.

#M,QUERY*XX

Queries the active mark. Responds with the mark details, see [above](#).

#M,QUERYNEXT*XX

Queries the next mark.

#M,QUERYALL*XX

Queries all marks of the active route.

#M,MOB*XX

Sets Exp's [MOB](#). Exp will reply with the new, active mark, see [above](#).

#M,AT,N,rng,brg*XX

Creates a new active mark and route at the range and bearing.

#M,AT,A,rng,brg*XX

Adds a mark after the active mark at the range and bearing.

Route

#C,GETCOURSENUM*xx\r\n

Requests the number of routes. Returns:
#C,COURSENUM,n*XX\r\n

where n is the zero indexed number of routes.

#C,GETCOURSENAME,n*xx\r\n

Requests a route name for index n. Returns:
#C,COURSENAME,n,s*XX\r\n

where s is the route name.

#C,SETCOURSE,n*xx\r\n

Sets route n to be the active route.

#C,CREATECOURSE,s*xx\r\n

Creates an empty course 's' and makes active.

#C,WL,...*xx\r\n

Creates a W/L course.

#C,WL,axis,start to WM,start to LM,gate
width,finish width,Wi,laps,leeward/Gate above
L/C,marks to P/S*XX\r\n Notes

Axis is in degrees magnetic.

Start to WM

Start to LM

Distances to windward and leeward
marks Gate width
Finish width

Optional.

Wi

Distance to Wi, below W. Optional.

Laps

Optional.

Leeward/Gate above L/C

L Leeward mark above boat. If a gate, L will centre the gate above the line.

C Leeward above the committee boat.

G Gate centered above line

Marks to P/S

If not specified, marks are to port.

Event

Tells Expedition to create an event in the database.

#E,*xx\r\n

#E,comment*xx\r\n

Also supports Sailing Performance style phase events.

#E,PHASESTART,name*xx\r\n

#E,PHASESTOP,name*xx\r\n

#E,PHASECANCEL,name*xx\r\n

```
#E,PHASECOMMENT,name,comment*xx\r\n
#E,PHASESAILSUP,Mainsail,NumReefs,Headsail,
Staysail,Spinnaker,Comments*xx\r\n #E,*xx\r\n
```

Sails

Sends the sail name if set to be in Send (Tx) filter. See above.

```
#S,c,sailname*xx\r\n
```

c can be one of:

- A Current sail
- M Sail to mark
- N Sail for next leg
- E Sail set by events etc

System

```
#S,HEARTBEAT*xx
```

Heartbeat.

```
#S,RECONNECT*xx
```

Tells the receiving computer to reconnect all network connections.

```
#S,GETCHANNELNAME,channel*xx
```

```
#S,GETUSERCHANNELNAMES*xx
```

Gets the channel name. Exp replies with

```
#S,CHANNELNAME,channel,name,shortname*xx
```

```
#S,SETCHANNELNAME,channel,name,shortname*xx
```

Set the channel name. Name should be less than 28 characters and short name less than 10.

For example (for User0)

```
#S,SETCHANNELNAME,172,My new  
name,MyName,*xx\r\n #S,LICENCE*XX
```

Request the Exp site code. Exp replies with (e.g.)
#S,LICENCE,9ac6c16a*xx\r\n



Expedition event

Background

Uses a serial port to send events to Expedition.

MOB

While this can be used as an MOB button, we recommend using a fixed button onto a stand-alone GPS as the computer running Exp is more likely to experience a problem at a crucial time than a dedicated GPS display.

Events

Each serial port may have up to 4 event functions from this list.

- MOB
- Sync gun
- Kill gun
- Gun plus 1 min
- Gun 5 min
- Gun 4 min
- Gun 1 min
- Set port end
- Set starboard end
- Previous mark
- Next mark
- Generic event

This list can be added to as needed.

Serial port wiring

Using a DB9 connector, an event is triggered by briefly connecting

Pin 4 to

Pin 8 : Event 1

Pin 6 : Event 2

Pin 1 : Event 3

Pin 9 : Event 4



Furuno NAVnet

Furuno systems can send instrument data over ethernet. This also contains [NMEA 0183](#) and AIS data, so can be used by Expedition.

Connection settings

A network [connection](#) is needed in Expedition.

Set NMEA 0183 as the instrument type.

Set UDP Tx & Rx as the connection type.

Set the port to 10021.



Garmin GND10 and NX2 FDX

Expedition can also communicate with the GND10 and Nexus NX2/NXR servers using the FDX communication protocol. This allows faster transfer of data to and from the server and allows the upload of calibration values to the server.

If are using the Nexus NX2 Race software, it creates a virtual serial port that Expedition can talk to, so both software packages can talk and listen to the instruments at the same time.

FDX data received from the NX2 server is not damped, so may need to be [damped](#) within Expedition.

The heading is always sent from the FDX server in degrees true, so you may need to set variation in the server.

Web

www.garmin.com

Connection settings

Expedition will set the serial port settings below. Do not change these.

Baud rate	19200
-----------	-------

Data bits	8
-----------	---

Parity	None
--------	------

Stop bits	2
-----------	---

On start up, the FDX server will be in NMEA mode. Expedition switches the server to FDX mode. To revert to NMEA, the Nexus server has to be powered off and on.

Also see the [NMEA 0183](#) and [serial port](#) help

FDX Settings

More advanced interface options are available by clicking on this button.

Custom

Damp | Output

Allows Expedition data to be sent to the 10 custom channels available in the FDX processor. The data sent can also be damped.

Note that the alternating channels can be sent to these channels, effectively doubling the user channels available for display on the Nexus NX2 instruments. For example, Target Bsp and Target Twa could be mapped to Alternating 0, which could then be sent to CFD.

The Custom name fields lets the user over-ride the default external channel tag.

Input

Allows any of the 6 Nexus AdBox custom channels to be mapped to Expedition channels.

Nexus

Allows any Expedition channel to be sent to the three user channels available on the NX2 processor. The data sent can also be damped.

TBS

Target boat speed, but you could send other data such as polar boat speed.

CAD

Custom angle data - normally for angle or bearing data such as leeway, target TWA etc.

CFD

Custom fixed data - normally for other data such as Bsp percentage, distance to line etc.

Note that the alternating channels can be sent to these channels, effectively doubling the user channels available. For example, Target Bsp and Target Twa could be mapped to Alternating 0, which could then be sent to CFD.

Send mark information

NOTE: This must only be used with GND10 and not for connections direct to the FDX server.



Garmin NMEA 2000

Expedition can send data to the Garmin custom channels over [NMEA 2000](#).

At 20 September 2020, this applies to GNX120, GNX130 and GMI20 displays and GPSMAP plotters.

Also, Quatix 5, Quatix 6, MARQ Captain, MARQ American Magic watches and GPSmap 86 series hand held via an ANT connection from a GPSmap plotter.

Web

www.Garmin.com

Custom channels

Expedition can send data to Garmin displays. Simply choose the desired channel and select the number/function from the drop down menu. Note that you can also select one of Expedition's [alternating numbers](#) here.

Selecting the damped check-box next to any external channel will cause damped data to be output.

Timeout

This can be disabled if irregular data is being sent to the displays.

In this case, data displayed on the Garmin displays will not time out.

Tags

The Custom name fields lets the user over-ride the default external channel tag.

Selecting refresh tags sends tag text every 10s even if not changing.

Selecting Send alternating tag names means tags on the external channels will be continually updated to change with the alternating channel.

This may add significant load to the NMEA network, so may not be advisable and using a Custom name may be better.

Update rates

Each Garmin display updates at 5Hz at most.

These are large data packets, so sending more data will reduce the refresh rate.

For example, sending data to 10 displays will update about 1Hz.

CAN Load

If the network load is less than 40%, Expedition may send data at faster rates.

This is only allowed where Expedition can determine the CAN load percentage. In practice this is currently limited to Actisense NGT-1, NGX-1 and Digital Yacht iKonvert gateways.



Garmin NX2

Web

www.garmin.com

Connection settings

NX2 instruments communicate using the NMEA 0183 protocol, so the default serial port settings are 4800,8,N,1. See the [NMEA 0183](#) and [serial port](#) help for more information on the NMEA standard and interface. However, there are also several custom NMEA strings.

Baud rate 4800

Data bits 8

Parity None

Stop bits 1

NX2 Settings

By clicking on the Settings button, further Nexus and [NMEA](#) functions are available.

Nexus

Allows any Expedition channel to be sent to the three user channels available in the NX2 processor. The data sent can also be damped.

TBS

Target boat speed, but you could send other data such as polar boat speed.

CAD

Custom angle data - normally for angle or bearing data such as leeway, target TWA etc.

CFD

Custom fixed data - normally for other data such as Bsp percentage, distance to line etc.

Note that the alternating channels can be sent to these channels, effectively doubling the user channels available.

For example, Target Bsp and Target Twa could be mapped to Alternating 0, which could then be sent to CFD.

NMEA output

Select NMEA sentences to send back to your instrument system.



Garmin2Exp - Garmin USB interface

Expedition can receive position, cog, sog and satellite data from Garmin devices with a USB interface. This is quite useful as a backup GPS solution.

This is designed for devices such as the 18x USB, but can also be used with hand-held Garmin GPS devices that have a USB port.

Many handheld GPSs can send [NMEA](#) over the USB port instead of Garmin data, so it is arguably better to bypass Garmin2Exp and read the NMEA data directly into Expedition.

Another option is to use [GPSTGate](#).

Web

www.Garmin.com

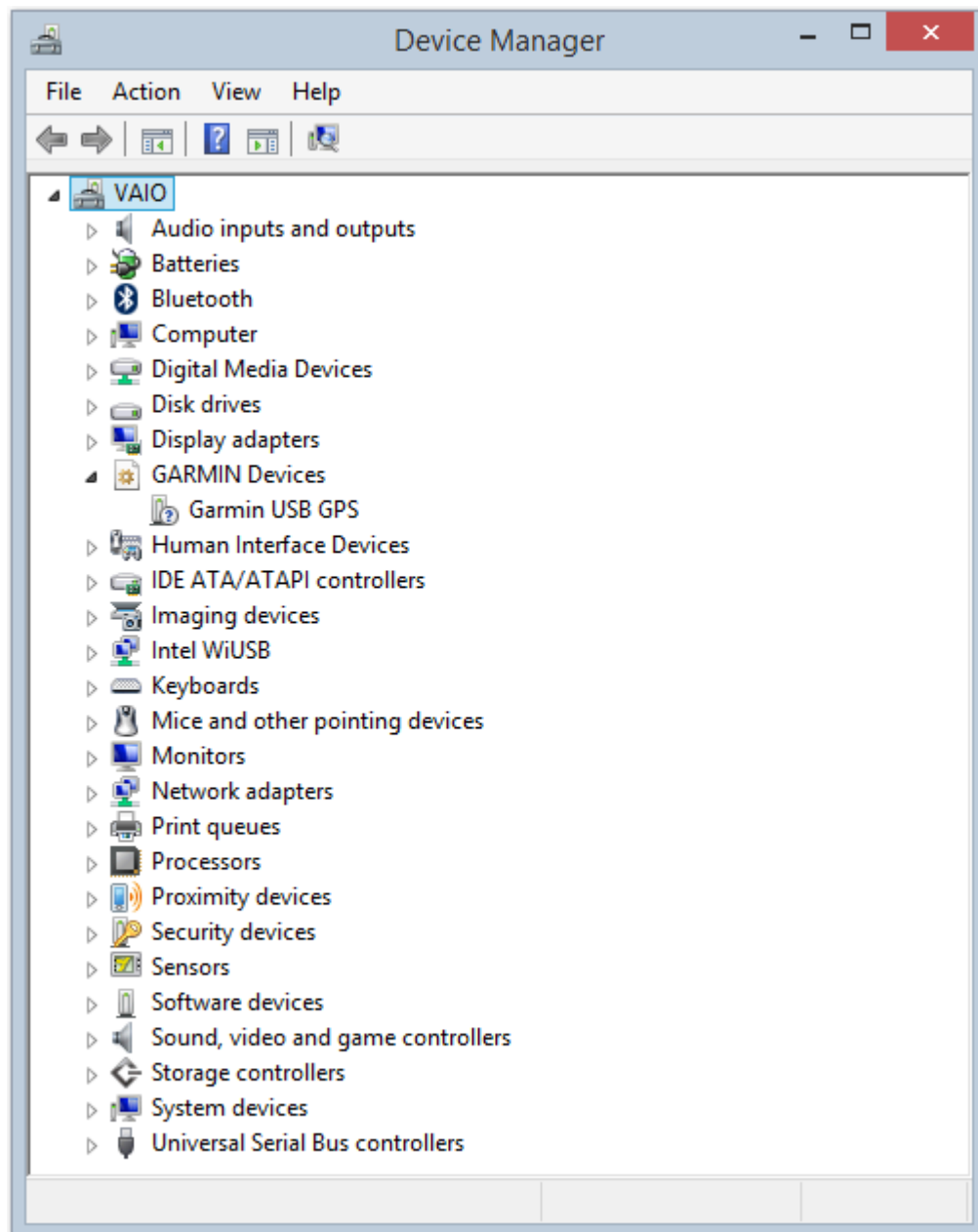
Driver

The Garmin USB driver needs to be installed.

As of 28 April 2012, this is available at

http://www8.garmin.com/support/download_details.jsp?id=591

If the driver is installed correctly and the device connected, it should be visible in Windows' device manager:





Sailing Performance

Supports Onboard Assistant.

Instructs Exp to save Onboard Assistant events in the events database and save tests.

Web

www.sailingperformance.com

Onboard assistant

For Onboard Assistant, it may be required that data to be sent to a port one greater than the Rx port.



Koden RADAR_{pc}

Expedition supports Koden radars - with the radar image displayed on the chart or in a dedicated radar window.

Web

[Koden](#)

Also sold by [Si-Tex](#)

Radar systems

The RB805 to RB809 require an MDS-11R interface box between the computer and the radar antenna.

Legacy MDS-1R/8R/9R/10R radars have an ethernet cable from the radar.

Legacy MDS-50 to 63R radars need an MDS-5R or MDS-6R interface box between the computer and the radar antenna.

There are advantages to both approaches. The MDS-1R to 10R solutions are lighter and cheaper, but cannot connect to a Koden display as is possible with the MDS-50 to 63R systems.

There are other differences. For example, the MDS-9R antenna is slightly smaller than the MDS-51R antenna so while both are rated at 4kW peak power, the larger 51R antenna has a smaller beam width. The 9R operates at 24rpm, whereas the 51R also has a 48rpm option.

Radome antennas



System	Interface	Antenna	Power		
MDS-1R		MRT-152R	2kW	24nm	0.9'
MDS-8R		MRT-158R	2kW	24nm	1.5'
MDS-9R		MRT-150R	4kW	36nm	1.8'
MDS-50R	MDS-5R	RB714A	2kW	24nm	1.2'
MDS-51R	MDS-5R	RB715A	4kW	36nm	2'
MDS-1105R	MDS-11R	RB805	4kW	36nm	2'

Open antennas



System	Interface	Antenna	Power		
MDS-10R		MRT-147R	4kW	48nm	3.5' or 4.5'
MDS-52R	MDS-5R	RB716A	4kW	48nm	3 or 4'
MDS-61R	MDS-6R	RB717A	6kW	64nm	4 or 6'
MDS-62R	MDS-6R	RB718A	12kW	72nm	4 or 6'
MDS-63R	MDS-6R	RB719A	25kW	96nm	4, 6 or 9'
MDS-1104R	MDS-11R	RB806	4kW	48nm	3' or 4'

MDS-1106R	MDS-11R	RB807	6kW	64nm	4 or 6'
MDS-1112R	MDS-11R	RB808	12kW	72nm	4 or 6'
MDS-1125R	MDS-11R	RB809	25kW	96nm	6 or 9'

Connection settings

Expedition connects to the Koden radar using an ethernet [network](#) connection.

For the integrated radars (MDS-1R,8R,9R and MDS-10R), select Koden MDS-1/8/9/10R from the Expedition drop-list.

The default radar UDP address is 192.168.0.1 and the default UDP port number is pre-assigned to 10001. The address should be disabled as Exp should be set to UDP broadcast as default.

For other Koden radars with the MDS-5R or MDS-6R control boxes, select Koden MDS-5/6R.

The Expedition radar interface is available by

- Clicking on the System button on the appropriate [network](#), settings page.

- Selecting the Radar item from Instruments on the [Application](#) button menu.

- The optional Radar toolbar button.

- The Ctrl + K key stroke.

It may be necessary to configure the fire-wall at the host PC.

Display

The radar image may be [displayed](#) on the chart.

It may also be drawn in its own, dedicated radar [window](#).

Warming up time

In order to protect the magnetron in the radar transmitter, it is required to warm the radar for 1.5 minutes (MDS-1R/8R), 2 minutes (MDS-9R/10R, MDS-50R/51R/52R/61R/62R) or 3 minutes (MDS-63R (25kW)) before use.

Transmission on & off

Stop or start radar transmission by using the Start Radar and Stop Radar items on the dedicated radar window or the [Application](#) menu.

If the radar is an open antenna system, it may be desirable to park the antenna when stopping transmission. This is achieved by selecting the Park antenna checkbox and entering an appropriate angle (in the adjacent edit box) to park the antenna at.

Koden settings

More advanced interface options are available by clicking on this button.

Range

The maximum range varies depending on the output power of the radar transmitter.

MDS-1R/8R	2kW Dome antenna	24nm
MDS-9R	4kW Dome antenna	36nm
MDS-10R	4kW Open antenna	48nm

Range can be selected on the Koden instrument settings page.

Auto-range

Selecting auto-range instructs Expedition to select the best range to fit the displayed chart scale.

Heading offset

Allows the radar to be aligned with the boat.

Resolution

Selects the Koden radar echo data resolution sent to Exp.

Number of pixels in each sweep range from 256 to 1024 and the number of sweeps in one rotation of the radar antenna range can be either 2048 or 4096.

256 x 2048

256 x 4096

512 x 2048

512 x 4096

1024 x 2048

1024 x 4096

Colour

Colour to paint radar echoes. Target intensity of shown in shades of this colour.

Other options are to fade the echoes by intensity or to use a colour spectrum.

Interference rejection

Used to reduce or eliminate interference from other radars.

Tuning

Gain, Tuning, STC and FTC is automatic on selected models.

However, manual tuning of the Koden radar may be accomplished on the System settings page above by selecting the relevant check-box and changing the value.

The Default button returns these values to their defaults.

Gain

Manually adjust the radar gain.

STC

Sensitivity Time Control. Also known as Anti Sea Clutter.

FTC

Fast Time Constant. Also known as Anti Rain Clutter.

Transmission trigger delay

This allows the user to make accurate distance measurement with the radar sensor by adjusting the transmission time delay. In practice, you need to align the start of the sweep with the leading edge of the transmission pulse. Use the following procedure.

1. Set up the radar controls as follows before doing the following set ups.

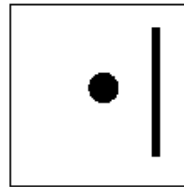
The range must be 0.25 nm, FTC minimum and the GAIN is set to the best picture. Adjust the STC until the TX pulse can be seen as a round point in the screen centre.

2. Adjustment of distance

Transmission trigger delay is adjusted to a point that the centre looks as shown in the diagram (b) below. Transmission trigger delay must be tuned so that a target such as a bridge or a breakwater displays as a straight line on the screen.



a) Too long



b) Correct



c) Too short



Web

KVH no longer manufactures or supports their Quadro range of instruments, however a few documents may be found at www.kvh.com.

Connection settings

KVH Quadro instruments communicate using the NMEA 0183 protocol, so the default serial port settings are 4800,8,N,1. See the [NMEA 0183](#) and [serial port](#) help for more information on the NMEA standard and interface. However, there are also several custom NMEA strings.

KVH Quadro does not send TWA or TWS, so Expedition needs to calculate them. Select these on the [calibration](#) page.

Output

Select desired NMEA sentences to send back to your instrument system.

KVH Quadro custom NMEA data

Expedition automatically sends some basic performance data to KVH Quadro instruments.

The following are available:

- Polar bsp %,
- Twa,
- Target twa,
- Target bsp,
- Vmg,
- Layline time,
- Layline distance,
- Vmc,
- Twa for optimum vmc,
- Mark range,
- Next mark twa,
- Sog,
- Twc,
- Mark bearing,
- Polar bsp.

KVH Quadro settings

More advanced interface options are available by clicking on this button.



Connection settings

There are no standard NMEA sentences that support output from laser range finders. Expedition supports several models, including Laser Atlanta, Lasercraft XLRic and the Trimble Pro-XL/XR

	Laser Atlanta	Lasercraft XLRic
Baud rate	4800	19200
Data bits	7	8
Parity	Even	None
Stop bits	1	1

The laser may be connected to the PC's serial port or, if using a Bluetooth equipped laser, to a virtual Bluetooth serial port.

If the laser is being used to 'ping' another boat, it is recommended to set the connection to say 'boat 1' as all derived numbers are relative to boat 0.



Mer Agitée Trim Control

Expedition can connect to Mer Agitée electronic telltales.

Web

www.meragitee.com

Connection settings

Either direct via ethernet or wireless, serial or UDP (virtual serial port), see the TrimContol installation manual.

Settings

Mer Agitée Trim Control settings on the instrument connection in Expedition.

Each tell-tails may be given a user-friendly name and mapped to an Expedition user channel.

The damping period sets the period the standard deviation of the angle is calculated over. This is used to determine whether the tell-tails is furled, attached or stalled.



navico

Navico radar

Expedition supports Navico radars, with the radar image displayed on the chart or in a dedicated radar PPI window.

Expedition does not currently support more than one radar or target tracking.

Web

www.navico.com

Navico distributes radars under the Simrad, Northstar and Lowrance brands.



Navico radars supported

BR24, 3G and 4G radar domes
Halo 20/20+ Halo 24 radar domes
Legacy Halo 3/4/6 ft open arrays
Halo 2000/3000 3/4/6 ft open arrays

Halo 50000 IMO radar
TXL-10/25 kW Magnetron Radars
SRT LAN 12/25 kW X Band & 30 kW S Band Magnetron IMO
Radars (R5000)

Navico licence

Note that the connection requires an additional Navico licence per radar.

When purchasing from your Navico dealer, as well as your name etc, please supply these details:

- The radar [Lock ID](#) , obtained from the Lock/Unlock button in Expedition's radar interface
- Radar serial number
- Radar type (4G, Halo20, Halo24 etc)
- Key type: LL1 or LL2
- Part Number: 000-10173-001 BRPC UNLOCK KEY - LL1
(or 000-10174-001 BRPC UNLOCK KEY - LL2)
- Customer: Expedition (BRPC104)

These can be copied from the Locked/Unlocked button.

Minimum radar firmware requirements

BR24 radars must be running RTM2 software (2.9.271 and above).

3G radars must have software version 3.0.38 or later.

4G radars must have software version 4.1.57 or later.

Older radars may require a firmware upgrade, available from your Navico dealer.

Connection settings

Expedition includes a separate program (BR24toExp.exe) to connect to the Navico radars using an ethernet network connection. At present, only one radar connection is supported. If multiple radars are required, we can extend this.

The radar interface can be started from Expedition's [application button](#), Windows' Start menu, by clicking on the Radar pane setting button or be [automatically](#) started with Expedition.

Notes

Older radars (BR24, 3G and 4G)

It is recommended that the radar is connected to the computer via an RI10 or RI11 interface box. This allows the radar to receive heading from an external compass, improving the radar image.

Navico has an adaptor cable to connect a standard ethernet RJ45 to the RI10 or RI11 interface. The part number is:

000-0127-56 ETHRNT YELW CBL 5 PIN
MALE:RJ45 FEMALE 2M

Halo 20, 20+ and 24 radars do not need RI10 or RI11 interface boxes.

Connecting via wireless is not recommended.

If connecting directly to the PC, a cross-over adaptor or cable may be required. Depending on the radar model, there may be issues connecting to the radar via a DHCP server.

Note that the BR24toExp.exe app needs to have firewall access granted for both public and private networks.

Symptoms associated with both firewall and DHCP issues include being able to connect to the radar, but data and commands not be exchanged.

Display

The radar image may be [displayed](#) on the chart. Expedition will need to be supplied with a GPS position. Exp also needs a heading input if the radar does not have a compass connected.

It may also be drawn in its own, dedicated radar [window](#).

Warming up time

Unlike magnetron based radars, warm-up time is negligible. However there may be a 15-20 second period between turning the radar on and it being available.

Transmission on & off

Stop or start radar transmission by using the Start Radar and Stop Radar items on the [Application](#) button menu, on the radar window toolbar or on the Navico radar to Exp application.

Expedition display settings

To draw radar echoes on the chart, select [Radar overlay](#) in [display](#) settings. Echo colours may be selected in the radar settings, as described below.

Colour

Colour to paint radar echoes. Target intensity of shown in shades of this colour. Other options are to fade the echoes by intensity or to use a spectrum of colours.

Settings

Scanner

As well as licensing and operation, this page gives the radar details and controls.

Lock / Unlock

Unlock the radar. This interface also gives the LockID needed when requesting a Lock key.

Control

Scanner power

Sets the power state of the radar to on or standby.

Transmit

Starts or stops transmission. Also available from the Start Radar and Stop Radar items on Exp's [Application](#) button menu, on the radar window toolbar.

Transmit period

The number of seconds to transmit for.

Standby period

The number of seconds remain in standby for.

Use mode

Select the use mode. The modes available are radar dependent, but include:

- manual
- harbour
- offshore
- weather
- bird

LEDs

Sets LED lighting level if available.

Doppler mode

Only available for LL2.

However, if there is a Navico display in the network that sets the radar into a Doppler mode, Expedition will display the Doppler data.

Select normal radar mode, show both approaching and receding targets or show approaching targets only

(receding targets will be coloured as for targets with no Doppler).

Doppler threshold

Only available for LL2.

Sets the lowest radial speed relative to the radar that will result in Doppler data.

Note these values are in m/s and normally accepted values are the range 0 to 16m/s.

Knot values are approximately twice m/s.

For example, 1kt threshold is about 2m/s.

1kt ~ 1/0.5144 ~1.94m/s

Image

It can be useful to have the radar PPI window showing when adjusting these values for feedback.

Factory defaults

Restores default radar settings.

Minimal

Default settings for minimal image processing.

Fast scan mode

Selects between normal and fast scanner rotation speeds. If available, options are

24 RPM (normal / off)

36 RPM

48 RPM

60 RPM

Note that other controls, such as range and noise-reject, can restrict the maximum RPM available to lower than this setting.

Also not supported when in use modes other than manual.

Not supported for BR24 and older 3G hardware.

IR Level

Sets the interference reject level in the radar. 0 means IR is disabled, 3 is maximum interference rejection.

Local IR

Can reduce interference from other on-board devices.
0 is disabled, 2 is high.

Beam sharpening

Improves the azimuth resolution of the radar.

Target boost

Used to expand targets. 0 (disabled) gives better target resolution and 2 gives maximum target emphasis.

Gain

Automatic or manual radar gain adjustment.

Rain

Manually set rain clutter rejection level. 0 is minimum clutter rejection, 255 is maximum.

FTC

Manually set the Fast Time Constant. 0 to 255.

Sea clutter

Automatic or manual control of the sea clutter sensitivity rejection level. 0 is minimum clutter rejection.

Side lobe

Automatic or manual control of the radar side-lobe mode and level. 0 to 255.

Noise rejection

Increases sensitivity or radar improving target detection, but reducing sharpness of the image. 0 is disabled, 2 is high.

Installation

Antenna height

Sets the antenna height for better automatic tuning.

Heading offset

This can be used to correct for errors due to misalignment of the radar installation in the boat.

Antenna height

Sets the antenna height. This improves the automatic radar tuning.

Radar to GPS

Distance radar is in front of the GPS in metres.

Range

Select the range of the radar image.

Auto-range to fit chart

Selecting auto-range instructs Expedition to select the best range to fit the displayed chart scale.

Allow Exp heading

Normally checked on. If the radar has a heading input, heading data for each radar spoke should be received from the radar. If this is selected, Exp will attempt to use its own heading for the radar data. If the heading into Exp and the heading from the radar do not agree, it is possible to turn the heading from Exp off.

Allow User heading

If no heading from the radar or Expedition, allows the user to define a heading value for the radar. This is useful in land based applications.

Allow User position

If no position is available from Expedition, allows the user to define a position for the radar. This is useful in land based applications. The format is (eg)

12 34.567s 123 45.678e

Send navigation data

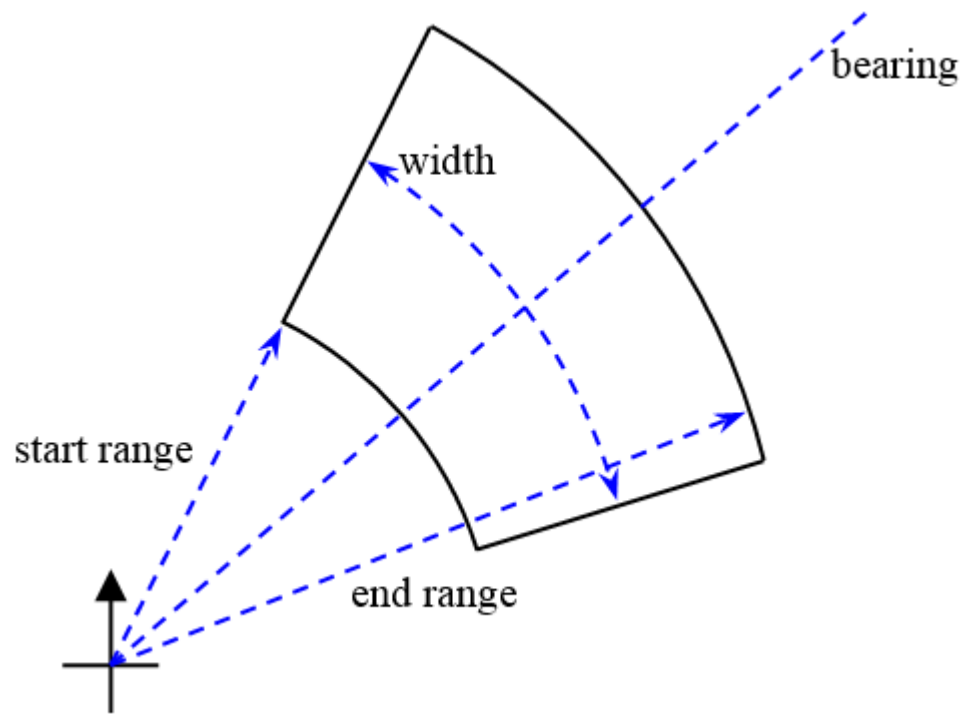
If COG, HDG, SOG or BSP are not available on the Navico network, this allows Expedition to send them to the radar.

The default is to have a lower priority than the Navico network, but the radar can be instructed to prefer the Expedition data.

Guard zones

Will show a warning for targets entering the guard zone.

Will also instruct Expedition to send a NMEA 0183 TTM sentence for ports with TTM Tx enabled.





Web

www.nke.fr

Connection settings

NKE instruments communicate using the NMEA 0183 protocol, so the default serial port settings are 4800,8,N,1. See the [NMEA 0183](#) and [serial port](#) help for more information on the NMEA standard. However, there are also several custom NMEA strings.

Custom NKE output

Expedition will send the following data to the NKE instruments:

- Target Bsp
- Opposite track
- Target Twa
- VMG%
- Polar Bsp %
- Set
- Drift

NKE Settings

More advanced interface options are available by clicking on this button.

NMEA 0183

Select standard NMEA sentences to send back to your instrument system.

Custom

Send Exp data to NKE Dynamic displays.

Web

www.nmea.org

Connection settings

The default NMEA 0183 serial port settings are 4800,8,N,1. Occasionally will you find NMEA devices that operate at different settings.

Baud rate	4800
Data bits	8
Parity	None
Stop bits	1

Tacktick, Silva, NKE and B&G Hydra instruments communicate using the NMEA 0183 protocol, but have extra custom sentences.

It is recommended that you configure your GPS to output NMEA 0183 v2.0 rather than NMEA 0183 v1.5. For example, the precision in latitude is just under 1m with 2.0 whereas it is just under 10m with 1.5.

Connecting a NMEA device to the PC's RS232 serial port

NMEA 0183 uses the RS422 electrical standard, so an optically isolated or RS-422 to RS-232 adaptor is recommended. See the serial port connection [help](#).

DB9 pin	PC	NMEA device
2	Receive data +ve	NMEA device output +ve
3	Transmit data +ve	NMEA device input +ve
5	Signal ground	Signal ground

So, for example, you connect the NMEA out from the instruments or GPS to Pin 2 of the DB9 adaptor of the isolated adaptor.

Network connection

If connecting to a NMEA 0183 [network](#) service over UDP or TCP, the default port is 10110.

NMEA 0183 settings

More advanced interface options are available by clicking on this button.

Send RTE and WPL

Sends mark and route information from the active route in RET and WPL sentences.

NMEA sentences decoded by Expedition

Expedition can send receive NMEA sentences to your instrument system.

Notes

HDT is normally turned off as Exp uses magnetic heading wherever possible.

DSC message data is viewed in the [AIS pane](#) and on the chart [window](#).

If BWC is selected and Expedition receives a valid BWC sentence, the mark received will replace the active mark being used by Exp.

In addition, XDR fields to be decoded are selected on this page.

BWC	Bearing and distance to waypoint
DBT	Depth below transducer
DPT	Depth below keel
DSC	Digital Selective Calling information
DSE	Expanded digital selective calling
FRPLC	Cariboni hydraulics, requires cariboni_config.txt
GGA	GPS fix

GLL	Position
GSA	GPS DOP
GSV	Satellites in view
GXA	Position (obsolete)
HDG	Heading
HDM	Heading - magnetic
HDT	Heading - true
LA1KA	Laser
LA1KD	Laser
PLCI	Laser
MDA	
MOB	Man Over-Board
MMB	Barometer
MWD	TWD and TWS
MTA	Air temperature
MTW	Water temperature
MWD	Wind direction and speed
MWV	TWA and TWS or AWA and AWS
OSD	Own ship data
OWD	Heading
PASHR	RT300 roll and pitch
PENON	Mer Agitee tell tails
PRDID	NKE attitude
RMA	
RMC	
ROT	Rate of turn

RPM	Revolutions
RSA	Rudder sensor angle
RSD	Radar System Data
TLL	Target latitude and longitude
TTM	Tracked targets
VBW	
VDR	Set and Drift
VDM	AIS
VDO	AIS Own ship
VHW	BSP and HDG
VLW	Trip log
VTG	COG and SOG
VWR	AWA and AWS
VWT	TWA and TWS
XDR	* see below
WIMDA	Airmar meteorological message
ZDA	

XDR

XDR sentences can be used for data not specified in other NMEA 0183 sentences.

Expedition can also send custom data in XDR type G sentences.

Currently (September 2020), Expedition options include:

	NMEA ID	Alternative IDs	Type	Expedition channel
Barometer	BARO	BAROMETER, BG, PRESSURE	P	Barometer
Humidity	AIR		H	Humidity

Leeway	LEEWAY	LWY, LWAY, LEEW	A	Leeway
Pitch / trim	PITCH	TRIM, PIT, PTCH, RDG10.IMU.PITCH	A	Pitch / trim
Roll / heel	ROLL	ROL, HEEL, RDG10.IMU.ROLL	A	Roll / heel
Heave	HEAVE		D	Heave
RPM	ENGINE1, ENGINE2		T	RPM1, RPM2
Rudder	RUDDER, RUDDER1, RUDDER2		A	Rudder, Rudder P, Rudder S
Temperature	AIR	TEMPAIR, ENV_OUTAIR_T	T	Air temperature
		ENV_WATER_T	T	Sea temperature
Voltage	BATTERY	BAT, HOUSE	U	Volts
Foils	BOARD		D	Board
	FOILP		A	Foil P
	FOILS		A	Foil S
	KEEL		A	Keel angle
Rig	MAST		A	Mast angle
	FORESTAY		D	Forestay length
	FORESTAY		N	Forestay load
	BOBSTAY		N	Bobstay
	RUNNERP		N	Runner P
	RUNNERS		N	Runner S
	D0P		N	D0 P
	D0S		N	D0 S
	D1P		N	D1 P
	D1S		N	D1 S
Sails	MAIN		N	Main sheet
	TRAV		D	Traveller
	J1		N	J1
	J2		N	J2
	J3		N	J3
	J4		N	J4
Stowe load cells			N	custom

These can be turned on or off as above. The alternative NMEA IDs are for non-standard or custom terminology.

Some mappings may depend on the data type. For example, forestay length or load.

Output

Expedition can send various NMEA sentences to your instrument system. Note that in the case of sentences containing Twd, Twa or Tws, the source (whether instrument, what-if? or calculated by Expedition) will be used. See the [What-if?](#) help for more information.

APB	Autopilot sentence 'B'
BWC	Bearing and distance to waypoint
DPT	Depth
GLL	Geographic position - latitude and longitude
HDG	Heading - deviation & variation
HDM	Heading - magnetic
MOB	
MTW	Water temperature
MWD	TWD and TWS
MWV	TWA, TWS, AWA and AWS
RMB	Recommended minimum navigation information
RMC	Recommended minimum navigation information
RSA	Rudder sensor angle
VBW	
VDR	Set and drift
VHW	BSP and HDG
VTG	CoG and SoG
VWR	AWA and AWS
VWT	TWA and TWS
WPL	Sends the active waypoint/mark.
XDR	Custom data
XTE	Cross track error
ZCD	

ZDA Time & date

ZTG UTC & time to destination waypoint



NMEA 2000



Web

www.nmea.org

Background

NMEA 2000 is a combined electrical and data specification defined and maintained by the National Marine Electronics Association ([NMEA](http://www.nmea.org)).

NMEA 2000 is much faster than NMEA 0183 (250k bits per second vs. 4.8k bits per second) and, unlike 0183, it supports a disciplined multiple talker and listener data network.

As well as the electrical and data specification, NMEA 2000 uses the DeviceNet cabling standard.

NMEA 2000 and Expedition

Exp supports several NMEA 2000 interfaces.

For network connections, it may be necessary to set Windows' firewall to enable both Public and Private networks for Expedition, especially for UDP connections.

Depending on the devices connected to the NMEA 2000 network, it may be necessary to [calculate](#) leeway, twd, tws, twa, set and drift in Exp.

Actisense

NGX-1 and NGT-1 USB and ISO

Select Actisense NGX/NGT-1 as the instrument type.

The NGT-1 is limited to 30 Tx slots. Exp uses some of these for system purposes, so the PGN Tx list is limited to 29.

Expedition sets all data to Rx.

Baud rate

115200 and 230400 are the baud rates used by Expedition.

230400 may be the default rate and is probably best for - USB devices.

NGT-1 devices with firmware earlier than v2.600 were limited to 115200 baud, but can have their firmware updated.

The NGT-1-ISO is limited to 115200 baud.

The baud rate stored in the device's eeprom can be set or changed with the Actisense NMEA reader in the Hardware config window or with the Actisense Toolkit.

-ISO note

The NGT-1-ISO and NGX-1-ISO ship with a default baud rate of 4800, for NMEA 0183 conversion.

Exp can use this as a NMEA 0183 connection.

Expedition requires the -iso baud rate to be 115k for a NMEA 2000 connection (Actisense NGX/NGT-1 instrument type option).

To change the device baud rate

1. Connect with Actisense NMEA Reader,
2. Select the Hardware config tab,
3. Change the Serial baud rate to 115200 or 230000 as appropriate
4. Click the Apply button
5. Power cycle the device.

[Actisense W2K-1](#) wireless

Select Actisense ASCII from the Expedition instrument type drop-list.

These notes apply for W2K-1 firmware 1.213 and later.

The W2K-1 data server can operate in several modes, see the W2K-1 user manual. Expedition supports:

N2K ASCII
RAW ASCII
NMEA 0183

Actisense N2K ASCII is recommended, especially as it can receive data from Expedition. We can't imagine a reason for using NMEA 0183 or RAW ASCII.

Actisense recommends TCP as the connection protocol, but UDP is often preferred. However, the W2K-1 does not currently support receive when using the UDP protocol.

Digital Yacht

[iKonvert](#) NMEA 2000 to USB Converter. Note that the ISO option may not be reliable in NMEA 2000 RAW mode.

[NavLink2](#) NMEA 2000 to Wi-Fi

Select RAW mode using the DIP switches (see the iKonvert manual), then select Digital Yacht RAW as the instrument type in Exp.

Yacht Devices

Select NMEA 2000 (RAW) in the YD interface, then select Yacht Devices RAW as the instrument type in Expedition.

USB Gateway [YDNU-02](#)

Ethernet Gateway [YDEN-02](#)

Wi-Fi Gateway [YDWG-02](#)

Throttling

Note that if Expedition can determine the CAN bus load, it will throttle sending data if the load exceeds 30 or 40%, depending on the data type.

This is currently only possible with Actisense NGT-1, NGX-1 and Digital Yacht iKonvert gateways.

NMEA 2000 settings

N2K PGNs

Controls receiving and sending of NMEA 2000 PGNs. If Exp is in [true](#) mode, these will be sent in degrees true where applicable.

See below for NMEA 2000 PGNs Expedition can receive and send.

N2K System

Data from any node can be set to be used or ignored.

N2K Sources

Allows the user to specify NMEA 2000 nodes to be used for different data.

For example, there may be multiple GPS sources in the system, but only the position from one may be wanted.

Custom

Expedition can send almost any data to [Garmin custom channels](#).

PGNs receivable by Exp

Note that 127245 may be used for a mast rotation sensor.

65280	Diverse
65288	Tinley load cell
65293	Diverse HLA load cell
126992	System time
126996	Product information
127233	Man Overboard Notification (MOB)
127245	Rudder
127250	Heading (magnetic, true)
127251	Rate of turn
127252	Heave
127257	Attitude (heel/roll and trim/pitch)
127488	Engine parameters, rapid update
127489	Engine parameters, dynamic
127505	Fluid level
127506	DC detailed status
127508	Battery status
127270	Manufacturer proprietary
128000	Nautical leeway angle
128259	Speed, water referenced
128267	Water depth
128275	Distance log
129025	Position, rapid update

129026 COG & SOG, rapid update

129029 GNSS position data

129038 AIS Class A position report

129039 AIS Class B position report

129040 AIS Class B extended position report

129041 AIS Aids to Navigation (AtoN) report

129285 Current set and drift

129539 DOPs

129540 Sats in view

129793 AIS UTC and date report

129794 AIS Class A static and voyage related data

129797 AIS Binary broadcast message

129798 AIS SAR Aircraft position report

129802 AIS safety related broadcast message

129809 AIS Class B static data

129810 AIS Class B static data

130306 Wind data (twc, twc, awa, awc, twa, twc)

130310 Environmental parameters (barometer, air and sea temperature)

130311 Environmental parameters (obsolete)

130312 Temperature

130313 Humidity

- 130314 Pressure
- 130316 Temperature, extended range
- 130323 Meteorological station data - these get filed as weather boats
- 130576 Trim tab
- 130577 Direction data
- 130578 Vessel speed components

PGNs sendable by Exp

Note the option for 127488 to send Polar Bsp % as RPM.

- 126720 [Garmin](#) custom data
- 127233 MOB. Note that the vessel's [MMSI](#) number should be set in AIS settings.
- 127245 Rudder
- 127250 Heading (magnetic, true) and variation
- 127251 Rate of turn
- 127257 Attitude (heel/roll and trim/pitch)
- 127258 Magnetic variation
- 127488 Engine parameters, rapid update
- 127488 Engine parameters, rapid update - Exp fudge to send bsp% as RPM
- 127505 Fluid level

127508	Battery status
128000	Leeway
128259	Speed, water referenced
128267	Depth
129025	Position, rapid update
129026	COG & SOG, rapid update
129029	GNSS position data
129283	Cross track error
129284	Navigation data (mark range bearing, latitude, longitude)
129285	Navigation - Route/WP information
129291	Set and drift
129301	Time to mark or layline
130067	Route and WP Service - WP Name and position
130074	Route and waypoint list
130306	Wind (twc, twc, awa, aws, twa, twc). The B&G twa/twc option may be required for B&G displays.
130310	Environmental parameters (barometer, air and sea temperature)



Web

www.ockam.com

Connection settings

The default Ockam serial port settings are 9600, 8, N, 1.

Baud rate 9600

Data bits 8

Parity None

Stop bits 1

Instrument system wiring and set up

Ockam recommended serial port wiring

	Ockam	DB9	DB25
RS232 gnd	7 black	5	7
RS232 Tx	3 green	2	3
RS232 Rx	2 blue	3	2
Jumper	7 to 8	4 to 6	4 to 5
Jumper		7 to 8	6 to 20

While the normal PC hardware might well run with just Tx, Rx and Ground connected, some driver software will wait forever for one of the handshaking lines to go to the correct level. Depending on the signal state it might sometimes work, other times it might not. The reliable solution is to loop back the handshake lines if they are not used.

When the lines are handshake looped, the RTS output from the PC immediately activates the CTS input - so the PC effectively controls its own handshaking.

Also see the [NMEA 0183](#) and [serial port](#) help.

DB9 pins

- 1 Received Line Signal Detector (Data Carrier Detect)
- 2 Received Data
- 3 Transmit Data
- 4 Data Terminal Ready
- 5 Signal Ground
- 6 Data Set Ready
- 7 Request To Send
- 8 Clear To Send
- 9 Ring Indicator

Network

For local UDP network connections to OckamSoft, it is required that data to be sent to a port one greater than the Rx port.

Ockam instrument set up

The default Expedition settings for Ockam are 9600, 8, N, 1 to suit the T1.

Older Ockam systems are often set to 4800,8,N,1.

To set the Ockam RS232 interface to 9600, 8, N, 1, set switch A to 4 and switch B to D.

To set the Ockam RS232 interface to 4800,8,N,1, set both switches A and B to 9.

NMEA 0183

Exp can also use the embedded NMEA 183 data from Ockam. If using DBT or DPT, Exp will ignore the Ockam 'w' depth tag. This is the preferred method of getting depth from Ockam systems.

If not using embedded NMEA 0183 data, de-select DBT and DPT from the Ockam NMEA options to enable Exp to receive depth data.

Settings

More advanced interface options are available by clicking on this button.

User channels

This tool is highly useful if you want to send some of Expedition's numbers/data to your Ockam instrument system for display on deck. For example you might want to display target speed when sailing a windward leeward course, or polar percentage when sailing a reaching course. Ockam has ten available external channels available for this purpose. Simply choose the desired channel and select the number/function from the drop down menu. Note that you can also select one of Expedition's [Alternating channels](#) here.

Selecting the damped check-box next to any external channel will cause damped data to be output instead of raw data.

The Custom name fields lets the user over-ride the default external channel tag.

Occasionally, your Ockam system will be receiving user data on a user channel (for example, a keel string-pot). In this case you don't want to send data to the particular Ockam user channel, but map the data Expedition receives from the Ockam system for that user channel to an Expedition channel. This is accomplished by checking the relevant send checkbox and selecting the appropriate Expedition channel to map the data to from the drop list. Note that the Ockam user data could be mapped to the Expedition User n channel.

Expedition uses Ockam's user tag 9 for its own purposes, so this is not available for use.

Auxiliary channels

Here you can map Ockam auxiliary channels 0, 1, 2 and 3 (such as Forestay load) to Expedition system channels.

Ockam user tag data

Ockam User tag data may also be stored in User0-8. Ockam user tag 9 is used by Expedition internally and is not available for other use.

Ockam settings

Heading and Twd in true

In some cases with a compass that sends heading in true, Ockam will not correct for this.

Update magnums at start-up

Updates the Ockam Magnums when Expedition is started.

Only one channel for Magnum displays, Matryx displays can have 1 to 4.

Damping

Allows upload of damping variables to the Ockam system. Note that these values are independent of Expedition's internal damping values.



Raymarine Seataalk

There are two Seataalk networks - Seataalk and SeataalkNG.

The original SeaTalk is a proprietary protocol used by Raymarine and cannot be connected to Expedition directly.

Seataalk NG

Seataalk NG is essentially NMEA 2000 with different connectors, so Exp can connect to the Raymarine SeataalkNG as a NMEA 2000 connection.

Actisense has an NGT-1 with a Seataalk connector.

Another option is to use a Raymarine SeataalkNG to NMEA 2000 [adaptor cable](#) and the Actisense [NMEA 2000](#) adaptor.

Seataalk

The original Seataalk network was originally developed by Autohelm in the early 1990s.

Raymarine manufactured a E85001 Seataalk to NMEA 0183 converter, but no longer offers this product.

Raymarine now has a [Raymarine SeaTalk1 to SeaTalk NG Converter Kit](#) (part number E22158). Expedition can then

connect to the SeataalkNG network as above.

There are several third party Seataalk to NMEA 0183 manufacturers. We have used or tested these, but note that they exist.

Digital Yacht <https://digitalyacht.co.uk/product/st-nmea-usb>

[ShipModule](#)



Raymarine Wireless (TackTick)

Raymarine Wireless instruments, previous Tacktick instruments.

Web

www.raymarine.com (previously www.tacktick.com)

Connection settings

Raymarine wireless / Tacktick instruments (via the T122 Wireless Interface) communicate using the NMEA 0183 protocol, so the default serial port settings are 4800,8,N,1. See the [NMEA 0183](#) and [serial port](#) help for more information on the NMEA standard and interface, however, there are also several custom NMEA strings available with Tacktick.

Baud rate	4800
Data bits	8
Parity	None
Stop bits	1

Note: Tacktick does not send TWD, so Expedition needs to calculate it. Select this on the [calculations](#) page.

Tacktick settings

More advanced interface options are available by clicking on this button.

Output

Any Expedition channel, including [alternating](#) channels, can be sent to one of four or six Tacktick proprietary NMEA sentences. See your Tacktick manual for further information.

Selecting the *damped* check-box next to any output channel will cause damped data to be output.

The *Custom name* fields lets the user over-ride the default external channel tag.

Some displays cannot display the proprietary data, please check your Raymarine/Tacktick product documentation. The T210 Maxi, T215 Maxi and T110 displays can, but the T111 dual line and 215 dual maxi displays can't.

Because of the limited display space, values below -100 may not show the - sign.

NMEA 0183

Allows [NMEA](#) messages to be selected to receive and/or sent.

Alternating

[Alternating](#) numbers.

Exp calibration

Advanced [calibration](#) and calculations.

Exp calculations

Tacktick does not send TWD, so Expedition needs to calculate it. Select this on the [calculations](#) page.



VSPARS

Web

www.vspars.com

Connection settings

The default VSPARS connection is a [UDP](#) connection on port 5685.

VSPARS settings

More advanced interface options are available by clicking on this button.



Communications

Expedition supports native Windows internet connections, Optimizer and Iridium GO!

See also

Expedition internet [settings](#).

Iridium GO! [help](#).



Internet settings

Connection

Expedition supports native Windows internet connections, Redport Optimizer and Iridium GO!

Connection type

Windows internet

Select Network from the connection drop-list.

Iridium GO!

See the Iridium GO! [help](#).

Optimizer

RedPort [Optimizer](#) is a satellite Wi-Fi router that blocks unwanted services to reduce satellite phone air time bills.

Newer versions of Optimizer let you easily set white lists.

Then select one of the [Optimizer](#) connection options from the drop-list.

- Optimizer Network
- Optimizer Iridium handset
- Optimizer Iridium Pilot
- Optimizer IsatPhone
- Optimizer Globalstar
- Optimizer Thuraya
- Optimizer HNS BGAN
- Optimizer GSM
- Optimizer Generic

You may need the Optimizer IP address and password.

Internet access type

Use Windows settings

The default access type is using Windows connection settings.

Direct to internet

Windows settings may be over-ridden by specifying direct access.

Time-out

Over-ride the default Windows connection and receive time-out values.

Mostly for simple, blocking connections.

Useful for slow internet connections or servers.

Any downloads that use the download window at the top left (grib downloads, Rainviewer, Sailflow etc.) have infinite timeouts. If they take too long, it is up to the user to cancel them.

Prefer DNS

Hosts

For services that do not allow DNS services, mainly Iridium. [Iridium GO](#) requires this.

Also, the hosts_map.xml file in the Expedition data folder may be manually edited.

See also the [IP addresses](#) help topic for more details.

Note this can also be achieved in Windows' hosts file.

Prefer secure http

Higher data transmission communication method like HTTPS/SSL can be difficult to operate on high latency networks like Iridium.

For Iridium, it is recommended to use HTTP webpages instead of HTTPS pages to reduce the load.

Note that some communication systems will only accept secure http (https) requests.

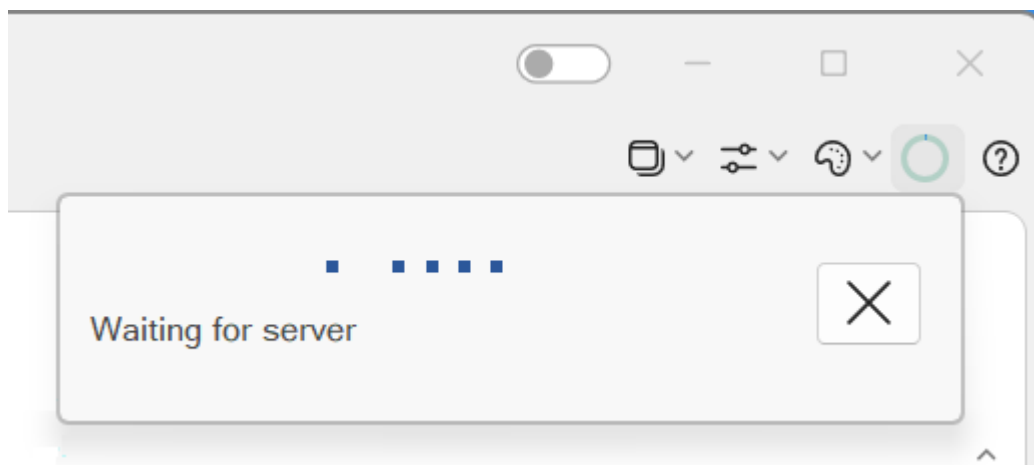
Currently this option is only supported by Expedition, Saildocs, SailFlow and YB Tracking.



Download manager

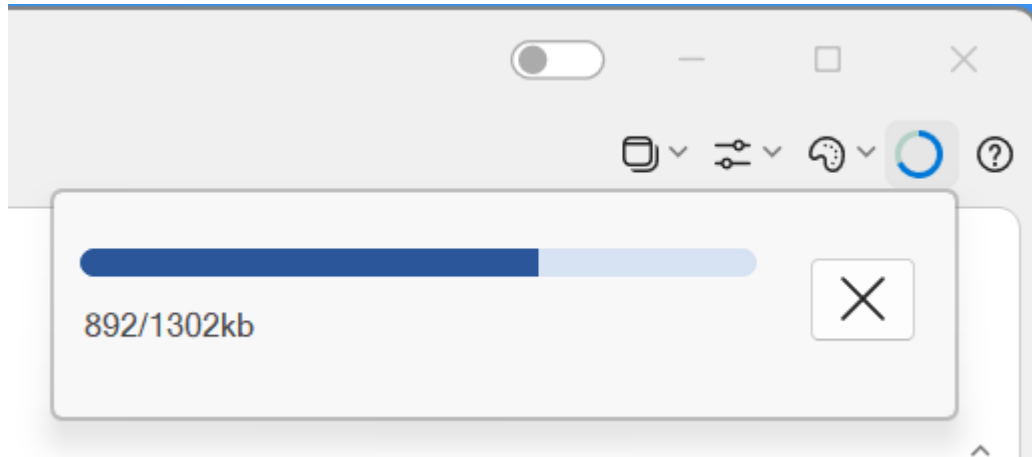
Most data downloads in Expedition are managed in a separate thread, controlled by the [Download manager](#) button near the top right of Expedition.

The button toggles the download window. Data downloads can be cancelled or resumed from the window.



Cancelling downloads

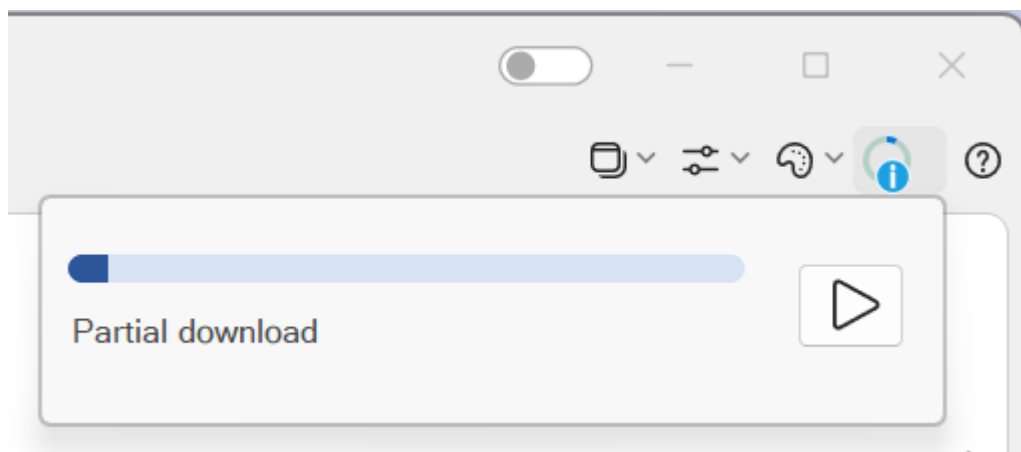
To cancel a download, just click the X button.



Resuming broken downloads

Failed downloads can be resumed for some services, including data from the Expedition grib server.

To resume, just click the resume download button.



Starting a new data download will clear the resume data state.

Unfortunately failed downloads from most third party services can not be resumed. This includes [Saildocs](#), [Squid](#), [Sailflow](#) and [Tidetech](#).



Iridium GO!

Iridium GO! is a very nice, small and portable device that provides global voice calling, text messaging and data solutions at about 2400bps via a Wi-Fi connection.

Expedition is a certified application for Iridium GO!

Note that Iridium GO! Exec is a different product, see note [below](#).

Web

www.iridium.com/products/iridium-go

Use with Expedition

There are several ways to use Iridium GO! to obtain data for Expedition.

Expedition can connect directly or a third party application may be used.

While a third party application such as Airmail or SquidX may sound inconvenient, there may be advantages in delegating what can be a lengthy download to another app.

Note that Iridium GO no longer seems to support DNS, so host IP addresses need to be specified in [settings](#).

Expedition direct connection

Expedition can connect directly to the Iridium GO! and automatically handle connections.

Airmail

Airmail from [Sailmail](#) is a nice way to use an Iridium GO!

Airmail is an email app that can be used to request and download data. The [Saildocs](#) grib request system integrated into Expedition can be used to request grib data, but it can be used to download almost anything.

SquidX

[SquidX](#) from Great Circle is an app that retrieves grib and other data from their service. The grib and satellite data can be then used in Expedition.

Direct Connection in Expedition

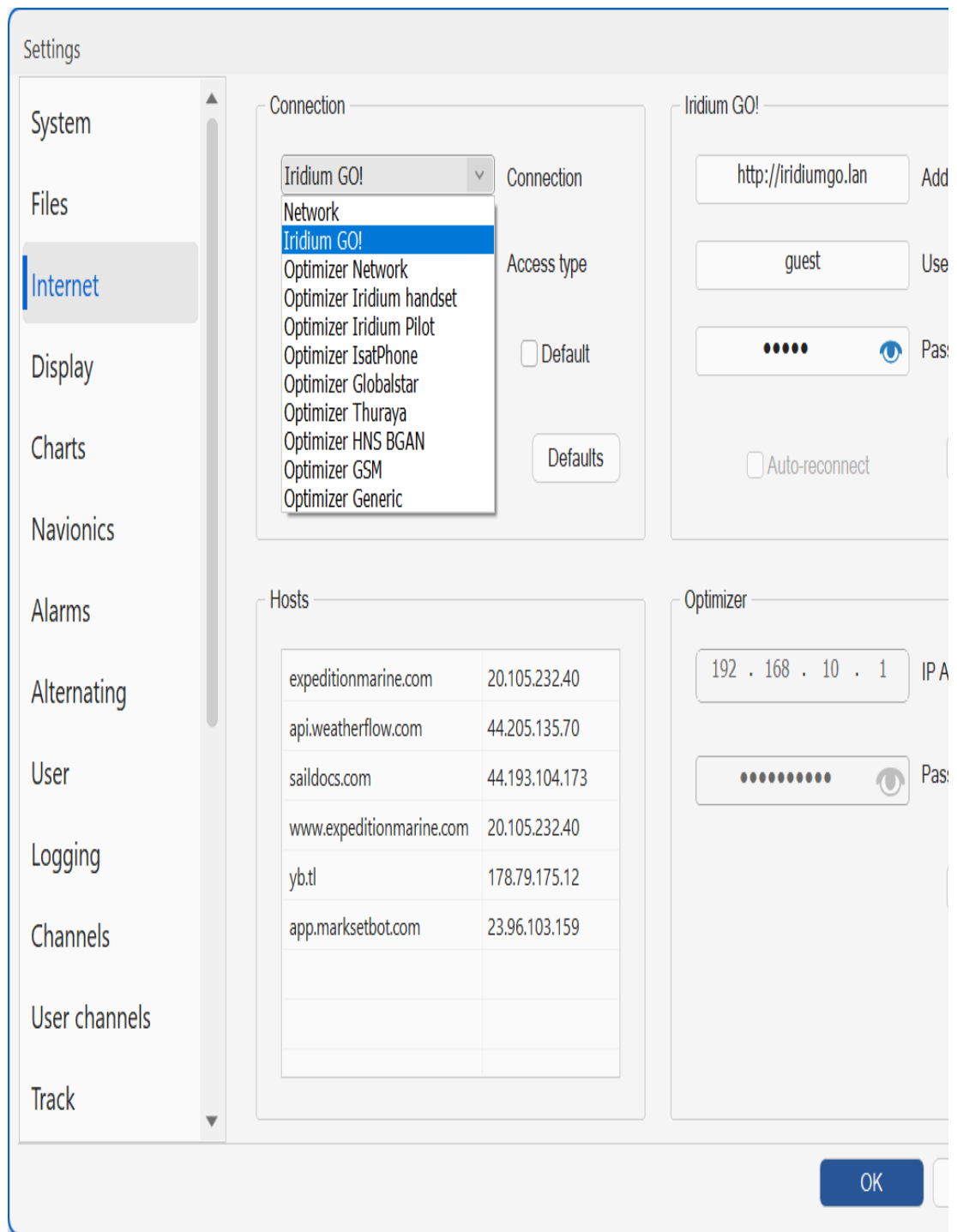
Open Expedition's [internet](#) settings page.

Select Iridium GO! as the connection type. This should automatically set Access Type to 'Direct to internet'.

If the address of the device, user-name or password are different from default, these can also be set here.

The Prefer DNS option should be selected.

With multiple networks, it may be required to specify the address of the device instead of the default `http://iridiumgo.lan` (normally `http://192.168.0.1`).



Usage notes

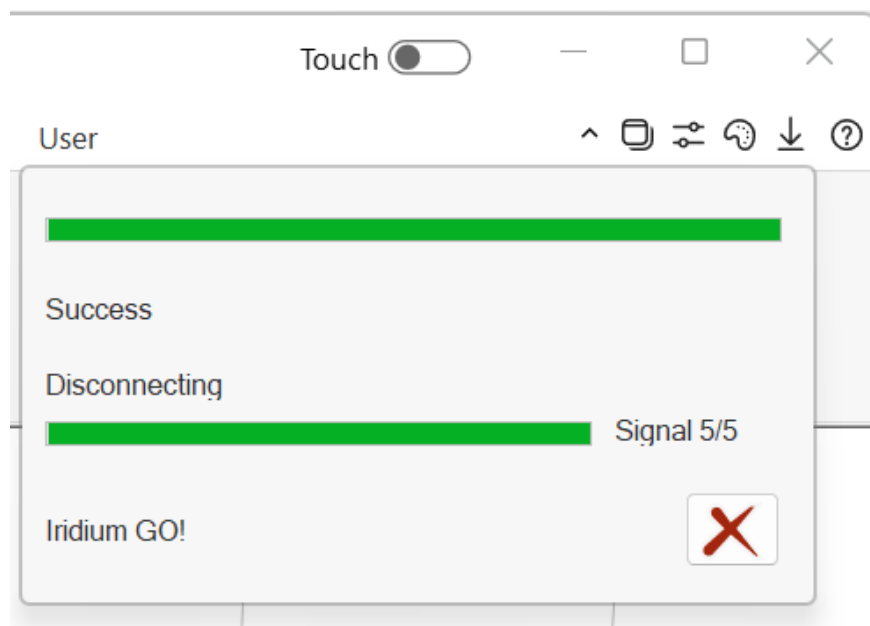
Always use an external antenna with GO! Connections will be at the same nominal speed, but reliability and effective speeds will be much higher

than using the pop-up antenna because of reduced loss of signal and re-connections.

Make sure the Iridium GO! is powered on and has 5/5 signal strength before use. It may work with a lower signal strength, but not as well. Expedition intermittently checks the signal strength, but the display on the device itself will give a more frequently updated indication.

When downloading data, Expedition will automatically instruct the Iridium GO! to make an internet connection. This may take 30-60 seconds and signal strength, status updates etc. will be displayed in the [download manager](#) at the top right of Expedition.

It may be necessary to increase the connection time-out value from Windows' default value (normally 30s). See Expedition's [Internet settings](#) page.



After the download is complete, Expedition will disconnect the internet call, but it is good practice to confirm the Iridium GO! internet connection has been disconnected.

Multiple devices can be connected to the Iridium GO! However, only one active voice or data connection can be active at a time. Expedition will not attempt to connect if another device is using the Iridium GO!

It may be worth changing the default login name and/or password if the Iridium GO! is powered on in an environment such as a marina where other users may be within Wi-Fi range. Don't forget to set the new user-name and password in Expedition's [internet](#) settings.

We recommend being careful with data requests as it is only a 2400bps connection. For this reason, consider restricting the area, number of time steps and resolution for grib data. As an example, downloading GFS data from Saildocs at 0.5 degree resolution will be about 1/4 of the size of 0.25 degree data.

A direct Wi-Fi connection to the Iridium GO! is recommended, unless it doesn't work. In which case, a router may be useful - see trouble-shooting below.

These comments also apply in most part to Airmail and SquidX.

Use http instead of https

Higher data transmission communication methods like HTTPS/SSL can be difficult to operate on a high latency network like Iridium. It is recommended to use HTTP instead of HTTPS pages to reduce the load.

Trouble-shooting

Because Windows' users commonly have multiple network connections, there can be more considerations than with an Android phone with a single wifi connection for example.

Sailmail has a useful Iridium GO! [trouble shooting](#) page.

Iridium GO

Ensure you have a recent firmware - at least 2.1.22 or 2.1.24.

Wi-Fi

It is important that the only the Iridium GO! connection is the only Wi-Fi network with automatic connection enabled.

Make sure the Iridium GO! Wi-Fi connection is set to 'Connect automatically' and disabled for other Wi-Fi networks.

Networks

Windows' automatic routing of internet connections tries to use what it thinks is the best internet connection. While this is desirable for most users, it can be can be sub-optimal with Iridium GO!, which mostly doesn't have an internet connection.

Waiting for server ...

If Expedition connects nicely to the Iridium GO! and it successfully makes an internet connection, but all you see is a 'Waiting for server ...' message and the request times out, the most likely cause is the network.

This is the most common issue people have.

Solutions and things to check

Firewall

Make sure Expedition is allowed through the firewall. We exclusively use the default Microsoft Defender.

IP addresses

Make sure you don't have another device (eg. router) at the same fixed IP address as the Iridium GO!

Networks

As mentioned above, mostly Windows will think the Wi-Fi connection to the Iridium GO! will lack an internet connection.

So if you have an ethernet connection to the instruments and a Wi-Fi connection to the Iridium GO!, Windows is likely try to connect to the internet via the instruments.

NCIS

It may help to turn off Windows NCSI, which will stop Windows testing for internet connectivity.

Metric

If the metric settings for the Iridium GO! Wi-Fi is set low and is set to a high number for other connections. See the link below for background and default values.

<https://learn.microsoft.com/en-us/troubleshoot/windows-server/networking/automatic-metric-for-ipv4-routes>

This may vary by Windows edition

1. Windows network and internet settings
2. Status
3. Change adaptor options
4. Select properties for the adaptor
5. Select the Internet protocol version 4 (TCP/IPv4)
6. Select Properties
7. Select the Advanced button on the General tab
8. Clear the Automatic metric check box
9. Assign a priority value for the Interface metric

Router

Pepwave routers are very popular with some Expedition users and let the computer connect to multiple wired, Wi-Fi and cellular connections.

<https://www.peplink.com/products/max-br-series>

Iridium GO! Exec

Iridium GO! Exec is a completely different device to the original Iridium GO!

It is much faster and the data plans are different.

- Select the default network internet connection type in Expedition.

Do not use the Iridium GO! connection option in Exp - it will not work.

- Set up a custom web/app profile

Call it (egg) Expedition.

DNS Blocking

Enable DNS blocking. It isn't needed.

Rule Policy

Select 'Block all except filters = Whitelist' - Block all traffic except the traffic defined in the rules.

Profile rules

Add rules, using the IP addresses and ports at [IP Addresses and ports](#). These will generally be TCP, not UDP.

- To connect

Select the profile created above.

Connect the Iridium GO! Exec to the internet.

Reference Iridium User manual

<https://www.iridium.com/idr-file/386680>



IP Addresses and ports

Some services may require knowing host names and/or IP addresses and ports for services used.

For example, Optimizer, Oceans Sidekick and [Iridium GO!](#)

Common host names used by Expedition

<https://www.expeditionmarine.com>

<https://saildocs.com>

<https://grib-server.greatcircle.be>

<https://yb.tl> See MyImage.xml and MyGrib.xml for URLs for other data.

IP Addresses

Currently (February 2025)

expeditionmarine.com	20.105.232.40
saildocs.com	44.193.104.173
grib-server.greatcircle.be	51.255.36.105
yb.tl	178.79.175.12

Note that these may change.

Use ping or nslookup at a command prompt to check IP addresses. For example:

ping saildocs.com

nslookup expeditionmarine.com Note that it may not be possible to ping expeditionmarine.com.

There are also web services, for example
<https://whatismyipaddress.com/hostname-ip>

Ports

Default ports include

http	80
https	443
ftp	21
ftps	990



Settings

The Expedition settings pages may be accessed from the [Application](#) button or the [charts](#) or [weather](#) panels.



System settings

Controls various aspects of the Expedition system.

System modes

Local time

Display all times (displayed or user input) in local time. If this box is unchecked, UTC (Universal or Greenwich Meridian Time) will be displayed. Normally you select local time for buoy racing and short offshore racing. It is often desirable to operate in UTC time mode in longer, offshore races.

Magnetic

Display all bearings and directions in degrees magnetic. Normally you will select this feature, but in certain cases it can be useful to operate in True magnetic mode, especially when you are in the Southern Ocean and the variation is 60°.

Fahrenheit

Display in temperatures in Fahrenheit. Default is degrees Celsius.

Keypad

Display pop-up keyboard when using edit controls for tablet use.

Tablet

More touch friendly ribbon with larger buttons for tablet use.

Also available with a switch at the top right of Expedition.

Auto-palette

Automatically select day, dusk or night [mode](#).

Auto-pan

Automatically pans the chart when using [visual marks](#), dividers etc. Only applicable in [drag](#) chart mode.

Simple ribbon

Simple and small ribbon, primarily for tablet use.

Also available with a switch at the top right of Expedition.

Slave

Puts Expedition in a slave mode. In this mode, Exp will not calculate most derived values.

This is useful if you want to send all the data from one computer to another using the Expedition instrument protocol, for example to a coach boat.

Port

For Port & Vesper ICAS customers. ICAS is a custom build.

ICAS users must specify the SQL database.

System options

Check for updates at start-up

Checks if there is an update available when Expedition is started. Also see the [system menu](#).

XTE calculated between marks

The default is for XTE to be the distance from the great circle path to the active mark from the preceding mark.

Deselect this to over ride the default behaviour. In this case, the preceding mark is over-ridden to use the GPS position

when a mark is set active. The Reset XTE option on the [marks panel](#) also resets the preceding position for the XTE calculation to the boat position.

Units

Distance

Changes the units used when displaying distance in most of Expedition's numbers. Normally this is set to nautical miles for navigation purposes, but it may be helpful at times to use a different measurement unit such as boat lengths.

If this setting is in nautical miles, distances in the starting line calculations will be displayed in boat lengths (e.g. Start distance to line).

Depth and height units

Select depth (feet, metres or fathoms) and height (feet or metres). This applies globally (ENC charts, C-MAP charts, tides and instrument data).

Speed

Start

Metres or boat lengths for start distances.

Pressure

Engine, transmission oil pressures etc.



System files settings

Data files used by Expedition.

Data folder

Where Expedition stores most data - log, grib files etc.

This may be changed to use (e.g.) a Microsoft OneDrive synchronised folder.

Default is C:\ProgramData\Expedition.

Marks database

Contains all Expedition's marks and routes.

Race tracking

The [race tracking](#) database. Often a different database is used for different races. The .mdb button allows importing from legacy Expedition Access databases.

User database

Some Expedition data is saved in a database - tests, sails, events &tc. The database file can be copied to other computers or backed up.

By default, this file is called Expedition.db. The default name is sufficient for most purposes. However, it may be more convenient to have unique databases if sailing on different boats.

Any Expedition database can be selected by dragging it onto the Files settings page.

Editing

The database is an SQLite database.

If needed, [DB Browser for SQLite](#) can be useful to edit the database.

It is recommended to not alter or edit

- Column formats in the default tables
- Entries in the default ATTR_TYPE table
- Entries in the default EVENT_TYPE table

Compact database

This compacts and repairs the database file currently being used.

Legacy database import

Earlier versions of Expedition used a Microsoft Access database. Legacy Access (.mdb) databases can be converted to the new SQLite (.db) database.



Display settings

Various check boxes and controls on this page allow you to select whether or not to display various details on the chart.

Update frequency

Sets maximum screen refresh rate in Hz.

The default value is 8Hz. If using a remote desktop or display, it can be a good idea to reduce the refresh frequency to say 4Hz.

Boat real size

Select small, medium, large or to scale boat icon.

Boat colour

Select colour to paint boats.

Route colour

Select colour for marks and route.

Weather observation colour

Select colour to plot weather observations. Can also be drawn using the 10m wind colouring scheme, see below.

Font size

Size of the text font used.

Range circle radius

Controls the size of the range circle drawn around the boat. See range circle below.

Period for Cog and heading

Controls the size of the predictor lines and the polar range circle. See below.

Layline bounds period

Sets the period of time for the Layline bounds as described below.

Also used for the Twd minimum and maximum values.

Units are minutes.

Notifications

MOB and SART pop-up warnings

Display a pop up message on receipt of a [SART](#) or [MOB](#) message.

Marks and routes

Leg details

Display range and bearing of all legs in a route and Divider measurements.

The range will be displayed in user selected units. The bearing can be displayed in either degrees true or magnetic. On a long leg a straight line

appears curved because it is a great circle. In a great circle, the bearing displayed is the bearing at the starting point.

Current leg

Draw the leg of the route from your boat to the next mark.

Mark names

Display mark names. If you move the mouse over a mark, any comments associated with it (that you have entered when editing or creating a mark) will pop up.

Mark range circles

Draw range circles around marks for which you have set up a range circle (when editing the mark's details).

Racing

Laylines

Draw laylines that will emanate from the active (yellow or green dot) mark. By default, these laylines are a function of the active polar, the leeway, the damped true wind and the damped current.

Expedition allows you to override the instrument wind/current and to have the laylines drawn oriented to a contrived, user selected wind and current. This is especially helpful when the wind/current instruments are acting up, or when studying possible tactical scenarios. Learn how to force what if wind and/or currents using the [What if?](#) help.

You can change the active mark (displaying the yellow or green dot) to another mark in a route by clicking the previous or next buttons on the

[marks](#) panel or the right/left arrows on the keyboard.

Layline bounds

Similar to laylines, but are extreme laylines for a preceding period of time as defined in the Layline bounds (minutes) control. Often if displaying laylines, you may wish to also display these.

Laylines from boat

Similar to laylines, but emanating from the boat's position.

Laylines using predicted tides

As for laylines above, but uses tidal model data if available instead of using set and drift at the boat (from instruments, predicted or [what-if?](#) values).

This will result in laylines curved to take account of the tidal current model.

Note that these laylines use the instrument (or [what-if?](#)) wind values at the boat, which are even more likely to change than the current.

In addition to the graphical representation, there are 6 channels that can be displayed in number boxes or lists. These are the times to the port and starboard sides of the course:

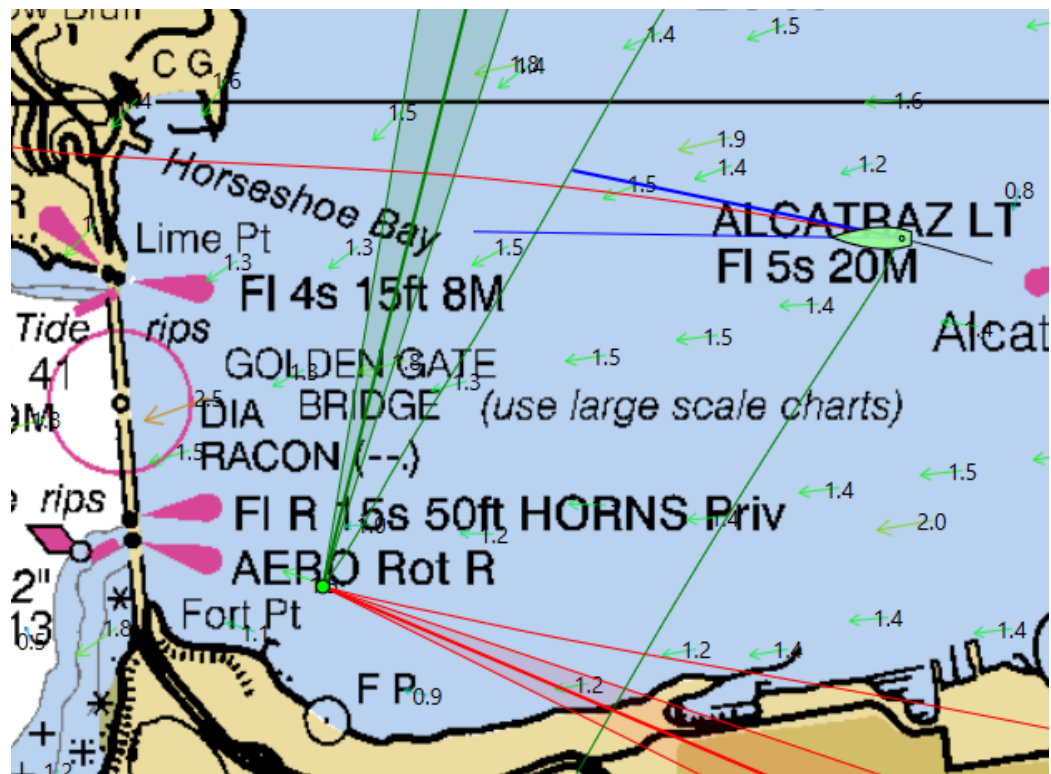
To the port side of the course

- Lay tide port side port time - Time on port from the boat
- Lay tide port side stbd time - Time on port to the mark
- Lay tide port side time - total time to the mark

To the starboard side of the course

- Lay tide stbd side port time - Time on starboard from the boat
- Lay tide stbd side stbd time - Time on starboard to the mark
- Lay tide stbd side time - total time to the mark

In this example, the solid lines are the laylines based on the measured current and the thin lines are laylines based on the tidal current model for San Francisco:



Opposite tack

Draws a predictor line for the opposite tack or gybe.

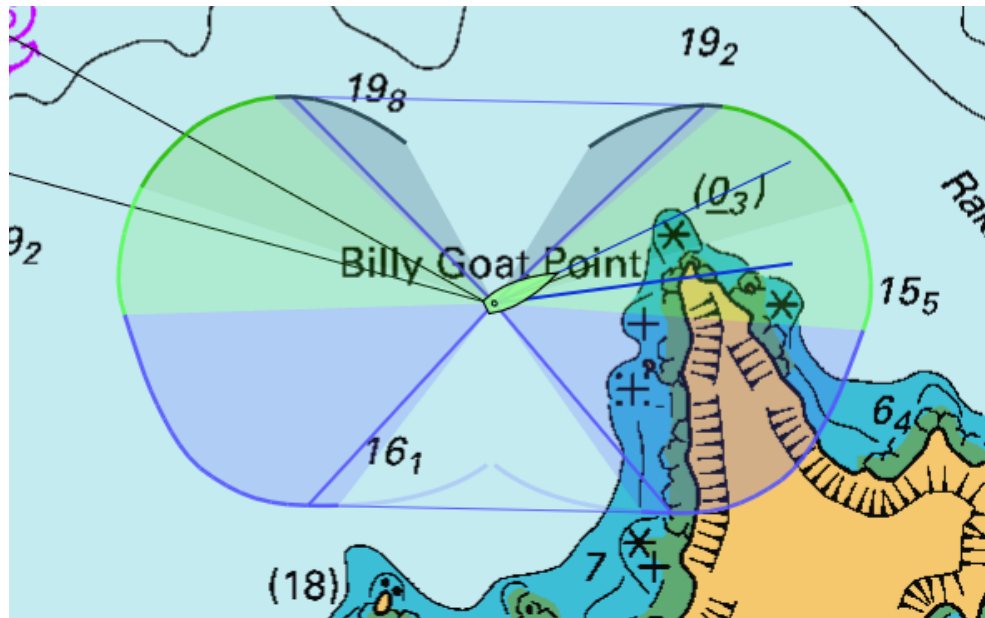
Race notes

Draws race notes on the chart.

Polar range circle

Draws a polar around the boat. The size of the polar dependent on the number of minutes entered.

The polar curve will be drawn in the sail colours.



Start line

Draw the start line and laylines for it.

Race tracking

Boats

Display other boat [Race tracking](#) positions. This is very useful in an ocean race when there are daily position reports that you enter into Expedition's [Race tracking](#).

Select which boats to display in [Race tracking](#) and on the chart from the [Edit boats](#) interface, available on the [sail](#) panel.

Boat positions will be interpolated from the position reports to the current display time.

Use Set display time in [Race tracking](#) or the Time button on the [weather display](#) panel to set the display time.

Tracks

If drawing [Race tracking](#) boats, this will add tracks.

The reference boat track will be drawn in bold.

Tracks window

Draws boat tracks only in the selected time window up to the display time.

User definable time window on this settings page. Default is 24 hours.

Useful for de-cluttering the display or when playing back a race.

Boat

Advantage line

Boat winds

Display a boat's damped wind speed and direction in a data box next to the boat and a wind barb emanating from the boat's position. This is normally turned off, but can be especially useful when there are several boats in the system (e.g. 2 boat testing or a multi boat weather program).

This was a very nice part of the Stars & Stripes weather program where we had up to a dozen instrumented boats connected to Expedition - both race boats, all the weather boats and the support vessels as well as several weather buoys and land stations.

Compass

Draw a compass rose (in magnetic degrees corrected for the local variation) on the chart. You can drag the compass around the screen.

Heading, course and CoG

Draw heading, course and/or CoG predictor lines from the bow of the boat. The user enters how long the line will be by entering a time value in minutes. For example, if you enter 5 minutes , the predictor line will show you where you will be in 5 minutes if you keep the present course and speed). This is very helpful for coastal navigation and when monitoring how long you have to go before a manoeuvre/course change. See the help on centre chart ahead of boat.

Radar overlay

Draws [Navico](#) or [Koden](#) radar echoes on the chart.

Estimated horizontal position error

Draws a series of shaded rings around the boat given information from the GPS. Each ring is approximately one standard deviation.

For example, this means any given GPS fix has about a 95% chance of being within the first two rings.

This feature is intended purely as a visual aid to monitor GPS performance and in no way reduces or replaces the requirement to use official printed charts or prudent use thereof. See the Expedition [licence](#) and also the disclaimer when starting Exp.

Name

Boat name.

AIS and DSC targets

Targets

Draws AIS and DSC targets on the screen.

De-clutter

Show less AIS targets at lower chart scales.

Name

Displays the name of each AIS target next to its symbol if AIS targets is also selected.

Call Sign

Displays the radio call sign of each AIS target next to its symbol if AIS targets is also selected. Note this also applies to both boat descriptors for boats > 0.

CoG & SoG

Adds CoG and SoG text to each AIS target next to its symbol if AIS targets is also selected.

Predictor lines

Draws a predictor line from each target, based on its Sog, Cog and the specified period.

Status

Target status.

Trails

Trails.

CPA

Displays a blue line at the closest point of approach (CPA).

Winds

Displays wind barbs and text for AIS meteorological stations.

Tooltips

General

Enables tooltips for marks, events, race notes, tide stations, weather boats etc.

AIS

Shows tooltips for individual AIS targets.

Chart

Shows ENC, Navionics and C-Map object information in a tooltip, similar to the right click [Query chart data](#).

Also toggled with the Tooltip button on the [chart panel](#).

Optimal

Also toggled with the Tooltip button on the [optimal panel](#).

Weather

Also toggled with the Tooltip button on the [weather panel](#).

Track

Track tooltips.

Weather observations

Observations

Displays [weather observation](#) data on the chart.

Wind observations only

Only draw weather observations that include wind data.

Observation wind text

Wind direction and speed text.

Observation all text

Mslp, temperature and dew point text.

Use 10m colour

Colour weather observations using the 10m wind colouring scheme, else specify a colour as above.

Other

WRF

Draws Expedition [WRF](#) forecast areas on the chart.

MyGrib

Draws [MyGrib](#) forecast areas on the chart.

Marine mammals strikes

Draws [MMAG](#) strikes on the chart.

Tidetech

Draws Tidetech forecast areas on the chart.

Predictwind

Draws Predictwind areas on the chart.

Status-bar

Controls the parameters listed on the [Status bar](#).



Alarms settings



Alternating number settings

Expedition has ten channels that can be set to alternately display different channels. These numbers can be displayed on the screen or output to the instrument system.

Selecting alternating numbers

For each alternating number, select desired numbers for each situation. Set any cell to blank if that option is not desired. If redundant options are selected, the precedence when selecting the channel is:

1. MOB

Man Over Board.

When the **MOB** attribute is set for the active mark.

2. Pre-start

Channel to use before the gun

3. Timer 1, Timer 2

These two numbers alternate on the display at the period as entered in the Alternating period (s) box

4. Reaching, Upwind & Downwind

The alternating number will be one of the numbers selected in these columns depending on the true wind angle

Twa from targets to display reaching number determines how far from the target angle to display the up-wind, down-wind or the reaching number

5. Port, Starboard

Channel to use when on port or starboard

Alternating numbers can be nested. In the example below, time to gun is displayed in the pre-start. Then the display shows target bsp when upwind, polar bsp% when reaching and alternates between target bsp and target twa when downwind.

Settings

- System
- Files
- Internet
- Display
- Charts
- C-MAP
- Alarms
- Alternating**
- User
- Channels
- User channels
- Racing / start
- Track

	Timer 1	Timer 2	Upwind	Reach	Downwind	Pre-start	Port	Starboard
0			Targ Bsp	PolBsp%	Alt 1	TmToGun		
1	Targ Bsp	Targ Twa						
2								
3								
4								
5								
6								
7								
8								
9								

Alternating period (s)
 Twa from targets to display reaching number

OK

Alternating period(s)

This determines the period at which the numbers alternate at. Please note that some instrument systems (e.g. B&G 20/20 or GFD displays) will have

problems and may lock-up if the alternating period is too short (e.g. less than 3 sec).

Twa from targets to display reaching number

This determines when to switch from display the up-wind, down-wind or reaching number.

For example, if the target wind angle is 40 and, then with 10° entered, Exp will display the upwind selection from zero to 50°, then the reaching number at greater angles.



User settings

Use this page to control many functions in the system.

Auto mark advance

Automatically advances the active mark to the next mark of the route.

The decision to advance the mark depends on the mark range and/or when pass the bisector between the legs.

Auto-start

Select Stripchart, Dfw2Exp etc. to start these applications when Expedition starts.

Boats

Edit boat names. Your boat is normally boat zero, other boats may be from telemetry or mapped from [AIS](#) targets.



Logging settings

Controls Expedition data logging.

Logging list

Select channels to log.

Log boats

To log instrument data, enable the Log boats check-box. The No Log warning will appear in the status bar if you are not logging data.

The frequency at which data is logged can also be changed. The default is to log data at 1Hz.

Notes

Data is saved in a simple comma delimited text file.

The time stamp is a Microsoft DATE type and is always in UTC for consistency.

All bearings, headings etc are saved in magnetic.

All data for a particular day is saved in a file that is named for the date it is logged. For consistency, the date is in UTC.

Logged data that is saved in a folder (usually named log) located in the Expedition app data folder. This folder can be easily accessed by clicking on the data folder button on the [Application](#) button.



Channels settings

Damping

Sets damping for each Expedition var in seconds.

This is similar to the damping in most instrument systems. Entering a non-zero number (seconds) next to a particular variable (channel) dampens that variable.

Some instrument systems will output damped data to Expedition and others will output raw data. Obviously, if the instrument system is sending damped data, it may be undesirable to further damp numbers in Expedition.

Targets, Polars, Layline and start line calculations use the damped values of Twd and Tws. For twd and tws, good starting damping values are 3-5 seconds for high end instrument systems and say 6-10s for simpler instrument systems.

Normally only damp Twd and Tws need to be damped - all other functions are left at 0 seconds damping.

Both raw and damped values in number bars in the main chart window. The damped values will be underlined.

The default Exp damping weights the value to the most recent measurements. The box-car option uses an average of all data samples over the period.

Note that some vars cannot be damped. For example it makes no sense to damp time to layline.

Precision

Precision is the number of digits displayed after the decimal point. Leave blank for defaults.



User channels settings

User channels

Allows Expedition's user channels to be renamed. Note that in some situations (with Racing Bravo for example), this happens automatically.

The settings are saved in a file in the Exp data folder -
`\config\UserChannels.csv`

Current options for var type are:

- Bearing
- Boat length
- Distance
- Latitude
- Longitude
- Percentage
- Sail
- Time
- Time span
- Vector



Track settings

Track colouring

Limit track

Enable this to specify the number of minutes of track to draw.

Load day track at startup

Loads the day's logged track if restart Exp.

Track width

In pixels.

Track

Draw your track/trail on the screen. This is normally selected and is very valuable for repositioning marks that you have rounded.

You can load a track from a log file by selecting the *Load track* button on the [Tools](#) panel.

Track currents

Draws current vectors along your track. The length of the vector can be controlled by changing the *maximum value* for current on the [weather](#) tab. For best resolution, set your maximum current about equal in magnitude to the maximum expected current.

This feature is very useful in studying current flow over a race track and to troubleshoot instrument (heading, leeway and speed) calibration errors - if your current changes from tack to tack, you may have a calibration error in your compass or boat speed.

Currents obtained from the instruments will be displayed unless using Expedition derived currents for calculations (see [what-if?](#)).

Track winds

Winds obtained from the instruments will be displayed unless using Expedition derived winds for calculations (see [what-if?](#)).



Tides settings

Using the Tides tab you are able to study tidal information for various areas and select it for display on the main chart window.

Disclaimer

The tides tools were developed to allow the addition of tidal stream information for the purposes of yacht racing. The ability to predict heights is just a bonus. As for the charting functionality in this system, do not rely on the heights computed by this software for anything that could in any way endanger you, your vessel, anyone else or anything else. Always use official tide height data and your own good judgment and seamanship when piloting shoal waters. See the [licence](#) document.

Displaying tide data for individual tide stations

Individual tidal stations can be accessed by right-clicking on the  tide icon on the chart.

For C-MAP tide stations, select 'Query C-MAP object' from the [pop-up](#) menu.

For Expedition tide stations, select 'Tides' from the [pop-up](#) menu.

The full list of Expedition tide stations loaded can be accessed by clicking on the Tide stations button on the [weather data](#) panel.

Display

Tidal stations

Displays primary tidal stations and their information on the chart. These locations are listed in Tides on the tree in the window to the right.

When selected, these a pop-up tool-tip will display their name when the mouse pointer is moved over the icon. Right-clicking on the icon will display tidal height information.

Secondary tidal stations

Displays similar tidal height information for even more locations (from the Secondary tides list).

Tidal streams

Displays tidal current stream information on the chart.

Make sure the maximum displayed current speed is set to a reasonable number on the [weather](#) settings tab. This is venue dependant, but usually a setting of about 2 knots will suffice.

Selecting the [Weather tooltip](#) will display the tidal set and drift as the mouse pointer is moved over an area with tidal stream data.

The time displayed in the upper left of the main chart window is the time the tidal streams are drawn at. See the help on weather to review Expedition's time controls.

Tidal stream labels

Draws the predicted drift as text label beside each current arrow.

Tidal stream offset

Allows the tidal streams predictions to be shifted in time if they are early.

For example, if you expect the tide to change 30 minutes before the models expect it to, enter -30.

So that changing this value is a conscious decision each time you run Expedition, this value is not saved on exiting. So, if you restart Expedition, it will be reset to zero.

Tide sources to use

Select or deselect tide sources to use and display in Expedition.

Expedition performs faster when unnecessary tides sources are not selected.

For example, when in the following locations:

Europe

Most data is now in grib format from the Expedition grib server, but Expedition also includes

[SHOM](#) tidal current models (Modèles de courant de marée en 2D) for France.

[Winning Tides](#) for the Solent. To use this dataset, it needs to be licensed (see Licences on the [Application](#) button). Contact [Winning Tides](#) for further information.

Messina Strait.

New Zealand

Select New Zealand.

USA

Select US West Coast currents and/or heights or US East coast and/or heights depending on location. These options have been deprecated in favour of XTide data.

[XTide](#) is a nice repository of tides and currents for the USA. These are mostly sourced from NOAA and are relatively up to date.

Expedition also includes a range of model datasets for the north-west Atlantic and north-east Pacific. Because of size, these are limited in extent, but we can add areas as they are requested. These are relatively low resolution models and are averaged across the water column, but may be sufficient in many cases.

XTide

Expedition can read XTide data in tcd or the legacy text file format and includes a copy of a recent tcd data file for convenience.

Other XTide data can be copied to the tides folder in the Expedition data folder and Expedition will automatically detect them.

Note that some windows users may experience issues extracting the tcd file from the tar if using Winzip.

Important note on model data

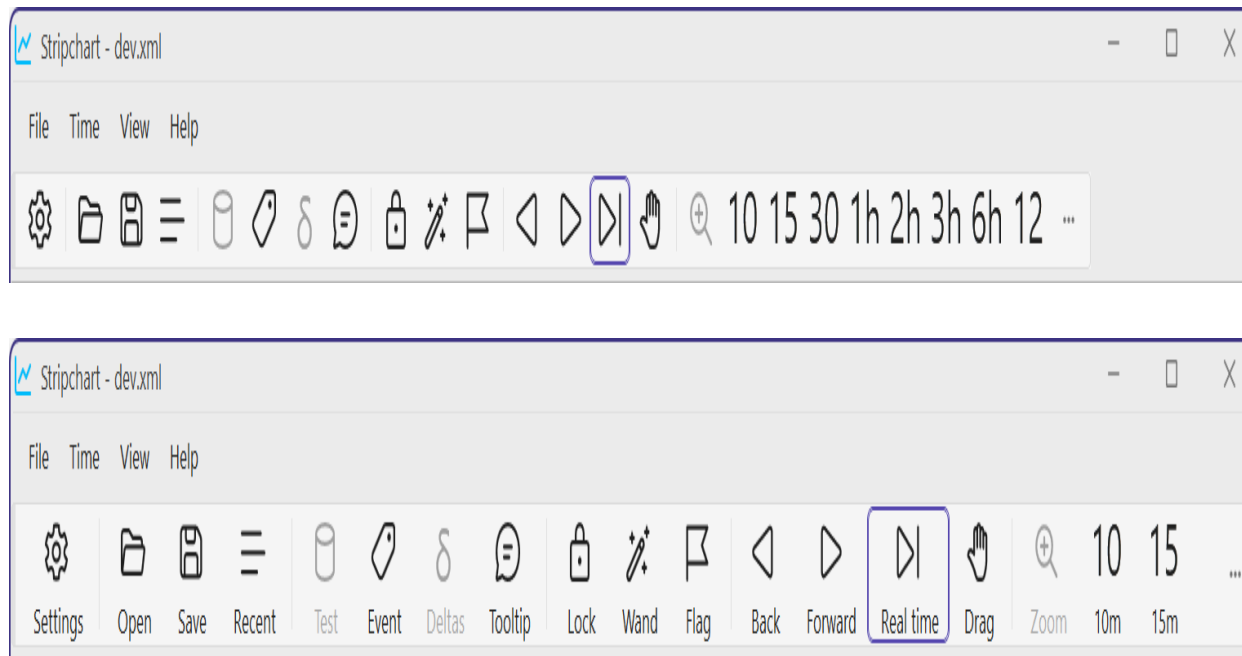
The tidal currents from model predictions (the US models, NZ and SHOM models) are just that - models. They may not show features in complicated situations or features of an inherently unpredictable nature.

For example, the San Francisco model doesn't show the tide changing first inshore along the city front. This is because the effect is mostly smaller than the resolution of the model. When racing there, we are interested in every extra metre we can get out of or into the tide. The model does however give a good picture of the general flow and when changes happen.



Other apps

Other Expedition apps.



Stripchart is a very powerful and easy to use program that graphs numbers, follows trends, computes averages and saves tests and events (notes). It is especially valuable for quickly and efficiently reviewing data from a race and calibrating instruments.

Using Stripchart

Stripchart can be started from Expedition's [application button](#) or Windows' Start menu. Stripchart can also be [automatically](#) started with Expedition.

Display and time

Use the Time menu or toolbar buttons to change the time scale displayed.

The Time drag mode, Back in time and Forward in time toolbar buttons are used to navigate in time.

The Draw in real time mode always shows the most recent data.

Display frequency

Select the display frequency from the Frequency menu. The display frequency is always displayed on the status bar.

The default update frequency is 1Hz. Lower frequencies use less memory as Stripchart has to store less data. Cruising boats may be content with 0.5Hz (one sample every two seconds), while in short course races 2Hz or even 5Hz may be useful. Obviously collecting data at 5Hz uses 5 times the memory as sampling at 1Hz.

Setting configurations

Set the number of channels to graph from the View menu.

The Stripchart panes can be dragged to rearrange their order.

Right clicking on a channel's horizontal pane displays a pop-up menu

Delete

Enables any individual strip-chart to be removed.

Edit

Allows all attributes for that strip-chart to be edited.

Move up

Shuffles a strip-chart up.

Move down

Shuffles a strip-chart down.

Set event

Sets an event, see below.

Notes

Take some time to experiment with these settings to get comfortable with what things like Shift Max (setting the maximum amount the top and bottom value of the graph will jump) and roll (keep the min and max span the same and simply roll the top and bottom values up and down to keep the new data on the screen) do when you change them.

Remember, unless you are configured in a multi boat program, your data is for Boat 0.

Saving and loading configurations

StripChart configurations can be saved as files (use the File menu or the toolbar buttons), so it is easy to change plots simply by loading pre-saved configurations. These files can be copied to other computers. The most recently used configurations are available from the File menu and the Recently used file list on the toolbar.

StripChart configuration files can also be opened by being dragged onto the Stripchart display.

Post race analysis

You can also load log files from the File menu for analysis later. This is especially useful to analyse data and save tests after a race or testing session. It can also be very helpful in instrument calibration.

Expedition log files can also be opened by being dragged onto the Stripchart display.

Averaging the data & setting wands

Use the left mouse button to set wands (up to 4 red vertical lines) and thereby select the start and finish of up to two discrete time

sections. Once you set these wands across a particular time span, Stripchart will display the average of the data for that time span in the wanded section of each pane. At the top of the screen, just below the toolbar, Stripchart will display the time of the left hand (oldest) wand and also the duration of the time span that is wanded.

Use the Show 4 wands item from the View menu to select two or four wand display modes.

You can also use the Set wand now toolbar button to lay a wand at the current time. You can insert a permanent marker at a particular point in time with the Set flag toolbar button. This is a very useful feature for temporarily flagging events such as a sail change.

Averages for the display Stripchart data can be turned on or off using the *Show means* item on the *View* menu.

Use the delta toolbar button to display differences between two wanded regions. This can be useful for calibration.

This process of setting wands is very valuable and powerful for instrument calibration and studying data. For example, you can quickly and accurately determine average rudder angles or wind angles for each tack this way.

Each panel can show a second data plot. This may be useful for comparison purposes or to save space. In this case, the mean and standard deviation for the second parameter will be in parentheses ().

For GPS parameters COG and SOG, a delta value is computed based on the position at each wand and the time difference. This value will be presented in square brackets [].

Saving tests

Once you have set the wands at desired points in time, you can save the averages of the time span in Expedition's test database. A typical time span for a test would be in the 2 to 5 minute range.

Simply use the Save test to database button to save the averages within the *wanded* time span as a discrete test. A window will open where you can enter or change other test details such as sails or sea state as well as comments before finally saving.

Items such as sails available in the test database can be added to the test (see [polars](#)).

Once a test has been saved, that section of time will be shaded grey in Stripchart so you know not to save another test during that time span.

These saved tests are then available for viewing in the [tests list](#) and [analysis](#). The tests will also appear as data points in the [polar edit window](#) and can be displayed on the [sail chart](#).

Saving events

Use the Event item on the Test menu to set an event at the current time. Events can be useful to record sail changes or items of interest. Events can also be set using the [set event](#) button in Expedition.

Events can also be set at point in time on the Stripchart using *Set event* from the right click pop-up menu.



System restore

System restore is an application installed with Expedition that allows the user to clear all settings they have customised for Expedition.

It can be found from Windows' start menu or from the Applications button on Expedition's [system menu](#).

Make sure you close other Expedition apps before use.

It can also be used to delete the marks and chart databases and to delete the licence keys from the computer.

Note that it is always a good idea to regularly back-up the contents of the data folder.



Exp DLL

The ExpDLL.dll file is supplied to allow third party software to query Expedition data as well as to send data to Expedition. It is supplied with a header file that may be used to link to the DLL.

As an aid to developers, the location of the Expedition program files folder is saved in the registry key

```
HKEY_CURRENT_USER\SOFTWARE\Expedition\Core\Location
```

Note that any directions returned will be in degrees magnetic.

The relevant lines in the header file include:

```
extern "C" __declspec( dllimport ) void __stdcall  
SetExpVar(short id, double value, short iBoat);  
extern "C" __declspec( dllimport ) void __stdcall  
GetExpVar(short id, double* pValue, short iBoat);  
etc.
```

id

Identifier of the parameter to pass. See `UsrChannels.h`.

value

The value to pass to Expedition.

iBoat

The boat id. For GetExpVar, SetExpVar and SetBoatName, the allowable values are 0 to 99 (as at 01 Sep 2020) inclusive.

name

The maximum size allowed for the boat name is 32 characters. Any more characters than this will be discarded.

Accept GPS data from DLL

Can disable GPS data obtained from the Exp dll. See the Exp system [settings](#).



Dfw2Exp32 is a utility that lets Expedition exchange data with Deckman for Windows.

User channel mappings are saved in the Dfw2Exp folder in the Exp data folder.

Expedition x64

Deckman is a very old 32 bit application, so Dfw2Exp32 is also 32 bit.

Dfw2Exp32 is rarely used, so the Microsoft components are not distributed with Expedition.

If not using Expedition MAX, it is recommended the latest [vc_redist.x86.exe](#) from Microsoft is installed. Link also found on the Expedition Resources web page.

Receiving data

Dfw2Exp32 allows data defined in Dfw's j_varsxx.d file to be passed to Expedition.

The mappings are set in Dfw2Exp.xml in the instruments folder in the Expedition data folder, see below.

Any Dfw channel defined in j_varsxx.d can be mapped to any Expedition User channel.

Exp looks first for J_vars06.d. If it can't find it, Exp will look for other files called j_varsxx.d.

Sending data

Any Expedition data can be sent to a WTP remote channel/function number.

The channels need to be defined in the \data\usermenu.d file on the WTP.

Dfw2Exp.xml

See the sample dfw2exp.xml file installed with Exp.

There is a var element to map each Dfw var with a corresponding Exp var.

For example:

```
<var exp="1" dfw="Boatspeed"></var>
```

Expedition can also send to remotes. To send Polar Bsp %:

```
<remote exp="58" remote="0" boat="0" damp="false">  
</remote>
```

Notes

Data received by Dfw2Exp32 is not affected by Expedition calibration functions.

Heel is opposite in Exp from WTP (and Dfw when connected to WTP). This is because WTP is opposite to convention where a roll to starboard is positive.

Other settings

Depth in feet from Dfw

If the expected depth units from Dfw are in feet instead of meters.

Send names to remote channels

When sending data to remote channels, also update the tags.

Heading, twd and cog in degrees true

If the data from Dfw is not in magnetic.

Use Dfw timer channel to set Exp timer.

Uses the TIMER channel to set the Exp timer.



Cosworth2Exp

Cosworth2Exp is a utility that lets Expedition exchange data directly with Cosworth systems.

Cosworth2Exp uses a [Kvaser](#) CAN adaptor to connect to a Cosworth CAN.

Changes to settings will be saved if Cosworth2Exp is closed using the OK button, but not if using the Cancel button.

Sending data

See the Mappings tab. Changes to mappings can be applied by clicking the Apply button.

Exp can send data to the Cosworth processor's ten remote channels. In addition, Exp can send predefined data to the Cosworth system:

- Layline time and distance,
- Mark position,
- Current set and drift,
- Magnetic variation,
- Cross track error (xte),
- Start line ends,
- Time to gun

Receiving data

Cosworth2Exp automatically sends data received from Cosworth to Expedition. This includes

- GPS Position,
- GPS SV number,
- GPS Quality,
- GPS CoG and SoG,
- Sea and temperature,
- AWA, MWA, TWA, TWD, AWA, MWS, TWS,
- BSP,
- Barometer,
- Heading,
- Voltage,
- Forestay,
- User 0 - 31

Calibration

Damping

Polars



Technical notes & help

For common problems and how to resolve them.



LogPlayer won't open today's log file

The log file currently being written to (today's log file) cannot be read by [LogPlayer](#) unless [logging](#) is turned off as Expedition would then be logging data from earlier in the log file.

One solution is to temporarily disable [logging](#).



Crazy mouse

NMEA input detected as mouse input by windows

This rarely occurs with modern systems.

Occasionally, Microsoft Windows will detect instrument input into a serial port as a mouse. This can lead to the mouse pointer jumping unpredictably around the screen. This is most common with NMEA input and USB-Serial port adaptors.

If you turn the instruments or GPS off, it will stop.

Solution

One option is to start the computer before turning the instruments or GPS on.

Microsoft has posted a fix for this as Knowledge Base Article Q131976.

Note that to regain control of your computer you need to disconnect or turn off the instruments or GPS causing the problem.

My personal quick solution

When the problem next occurs (it may not happen every time), I try and disable the mouse. If it is deleted, Windows may still detect it at some future time.

1. Disconnect or turn off the NMEA GPS or instruments causing the problem. You need to do this to regain control of your computer.
2. Open Windows' Device Manager by clicking on the Device Manager button on the Hardware page of System Properties. You can access System Properties by right-clicking on My Computer on the desktop and selecting Properties or from System from the Control Panel.
3. Expand the Mice and other pointing devices section. Then disable the mouse Windows thinks your GPS is. It will probably be called Serial Ballpoint Mouse or just Ballpoint mouse.
4. Right click on the offending mouse entry and select Properties from the pop-up menu. On the page that appears, select the Do not use this device (disable) option from the Device Usage list.



No start numbers

Expedition requires a start polar and estimates of the boat's acceleration and rate of turn characteristics in order to calculate the various [start numbers](#).

So, if you have a start line defined and the [display settings](#) set to draw the start line, make sure you have a start polar loaded and values set for acceleration and rate of turn (these can be linear values or tables in your [calibration](#) file).



Appendix A : Channels

Expedition has channels for each boat.

Ahead of

Distance boat zero is ahead of boat n vmg-wise.

Air density

The air density. Requires pressure, temperature and preferably dew point or relative humidity. Will be dry air density if dew point and relative humidity are not available.

Air temperature

The air temperature.

Alternating 0 - 9

See [alternating numbers](#).

Awa

Apparent wind angle. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Aws

Apparent wind speed. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Backstay

Barometer

Atmospheric pressure.

Battery current.

Amps.

Board

Board port

Board starboard

Boat bearing from boat 0

Boat range from boat 0

Range and bearing of boat n from boat 0.

Boom

Bsp

Boat speed. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Bsp transverse

CAN Load

CAN fast packet errors

From Actisense or B&G Websocket.

Charge state

Battery charge state %.

Cog

Course over the ground. Heading + leeway + current.

Course

Heading + leeway.

Course - COG

Cross track error

XTE.

Current drift

Current drift speed in knots. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Current set

Current set direction. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Current set predicted

Current drift predicted

Current set as predicted by diamonds, NOAA tides, Grib data, Winning tides etc.

Bsp - SoG

Deflector

Delta polar bsp

Delta target bsp

The difference between the current boat speed and the target boat speed at the current true wind speed.

Delta target heel (roll)

Delta target twa

The difference between the current true wind angle (see Twa) and the target wind angle at the current true wind speed.

Depth

Depth. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Dew Point

Dew point temperature.

Diff station

Differential reference station ID.

Distance to finish

Distance to current mark and on to last mark in the active route.

Downhaul load

Downhaul load 2

Load on the downhaul. Mapped on an instrument page from a linear channel in B&G &tc.

Engine temperature

Engine oil temperature

Engine oil pressure

Engine temperature and oil pressure may be flagged red in number boxes for NMEA 2000 over temperature and low oil pressure warnings.

Error code

Exp value.

Finish

Forestay load

Forestay inner load

Load on the forestay. Mapped on an instrument page from a linear channel in B&G &tc.

Forestay length

Length of the forestay.

Forestay plus tack load

May be useful for (e.g.) structured luff type sails.

Fuel level

Fuel level 2

NMEA 2000 instance 0 and 1.

Gate bearing

Gate range

Gate time

To gate mark.

Gate lay dist on port

Gate lay dist on starb

Gate lay time on port

Gate lay time on starb

Distances and times to the gate laylines.

Gate spot time on port

Gate spot time on starb

Distances and times to the gate spot - the intersection of the laylines to the two gate marks.

Gate square wind

Wind direction at 90° to the gate.

Gate bias length

Distance bias of the gate given the current wind direction.

Port/starboard sense is from the previous mark.

GPS age

Age of differential GPS data, time in seconds since last SC104 type 1 or 9 update, null field when DGPS is not used.

GPS altitude

Antenna altitude above geoid.

GPS estimated position error

GPS geoidal separation

Geoidal separation is the difference between the WGS-84 earth ellipsoid and mean sea level (geoid). Negative if mean sea level below ellipsoid.

GPS PDOP

GPS HDOP

GPS VDOP

Dilution of precision, horizontal and vertical dilution of precision in metres.

GPS mode

0 fix not available.

1 GPS fix.

2 Differential fix.

GPS number

Number of satellites in view.

GPS position fix

1 not available, 2 = 2D, 3 = 3D.

GPS time

GWD

Ground wind direction. Wind direction over the surface of the earth. Twd is wind direction with respect to the surface of the water without regard to current set/drift.

GWS

Ground wind speed. Wind speed over the surface of the earth. TwS is wind speed with respect to the surface of the water with regard to current set/drift.

Heading

Compass heading. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Heading to steer

Heading to steer polar

Heading to steer to the mark, allowing for current. At Bsp or polar bsp.

Heel (roll)

Boat heel. Heel/roll is positive in for a roll to starboard (sailing on port).

Heel (roll) rate

Instrument trip log

Keel Angle

Keel angle (for canting keels).

Keel Height

Keel height (for retracting keels).

Rake

RPM1

RPM2

Engine rpm.

Rudder

Rudder forward

Rudder port

Rudder starboard

Rudder toe

Latitude

Longitude

Layline bearing

Bearing to the mark of the layline boat n is heading towards.

Layline distance on port

Distance to the starboard layline.

Layline distance on starboard

Distance to the port layline.

Layline distance

Distance to the layline boat n is heading towards.

Layline port bearing

Bearing to the mark along the port layline.

Layline starboard bearing

Bearing to the mark along the starboard layline.

Layline time

Time to the layline boat n is heading towards at the target speed and angle.

Layline time GPS

Time to the layline boat n is heading towards based on cog and sog.

Layline time on port

Time to the starboard layline at the target speed and angle.

Layline time on starboard

Time to the port layline at the target speed and angle.

Layline tide port side port time

Layline tide port side stbd time

Layline tide stbd side port time

Layline tide stbd side stbd time

Layline tide port side time

Layline tide stbd side time

As above, but with predicted tidal stream at the mark.

Lay max bearing on port

Lay max bearing on strb

Lay min bearing on port

Lay min bearing on strb

Layline bearing on port

Layline bearing on strb

Layline dn bearing on port

Layline dn bearing on strb

Layline time on port ratio

Layline time on strb ratio

Layline up bearing on port

Layline up bearing on strb

Lead port

Lead starboard

Leeway

If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Load cell port

Load cell starboard

Log Bsp

Log of distance based on Bsp.

Log Sog

Log of distance based on Sog.

Magnetic variation

The variation between true north and magnetic north at boat n's current position.

MainSheet

Mark bearing

Mark range

Bearing and range to the current mark.

Mark current drift

Mark current set

Predicted current at mark at estimated time of rounding, based on Mark polar time.

Mark latitude

Mark longitude

Mark polar time

Time to the current mark based on the current wind direction and speed as well as the polar.

Mark time

Time to the current mark based on the current polars.

Mark GPS time

Time to the current mark based on the GPS CoG and Sog.

Mark twa

True wind angle if heading directly to the current mark given the current wind direction.

Mast angle

For rotating masts - the angle of the mast.

Mast butt

Mast butt position.

Mast rake

Mast rake.

Motion

MSLP predicted

MSLP predicted by loaded grib data.

MWA

MWS

Measured apparent wind angle and speed.

Next mark Awa

Expected apparent wind angle on the next leg.

Next mark Aws

Expected apparent wind speed on the next leg.

Next mark bearing

Next mark range

Bearing and range from the current mark to the next mark.

Next mark polar time

Expected time for the next leg based on the current wind and your polars.

Next mark time on port

Time to the next mark spent on port based on the current wind direction and speed as well as the polar target angles and speeds.

Next mark time on starboard

Time to the next mark spent on starboard based on the current wind direction and speed as well as the polar target angles and speeds.

Next mark twa

True wind angle if heading directly to the next mark from the current mark given the current wind direction.

Opposite track

What the Cog (heading + leeway + current) would be on the opposite tack or gybe.

Opposite track cog

$\text{Cog} \pm 2 \times \text{TargTwa}$. No current effects included.

Polar 5, 6, 7, 8, 5%, 6%, 7%, 8%

Custom polars.

Polar bsp

The polar boat speed at the current wind speed and angle.

Polar bsp%

The fraction of the current boat speed to the polar bsp, expressed as a percentage.

Polar heel

The polar heel angle at the current wind speed and angle.

Polar leeway

The polar leeway at the current wind speed and angle.

Polar Tws

Polar Tws delta

Polar Tws %

Uses the bsp nav polar and/or heel/roll and the heel polar to estimate tws.

Port gate lay dst on strb

Distance to the port layline to the port gate.

Port gate lay tm on strb

Time to the port layline to the port gate.

Port gate lay dst on pt

Distance to the starboard layline to the port gate.

Port gate lay tm on pt

Time to the starboard layline to the port gate.

Race note GPS reach distance to

Race note GPS reach time to

Radar bearing

Radar range

Rate of turn

ROT.

Relative humidity

Runner port

Runner starboard

Sail

Sail mark

Sail next mark

From the sail chart.

Sail AI

Lower Stripe Camber

Camber(at 25 % of P) expressed as a percentage of the local sail chord.

Lower Stripe Draft

Position of max camber(at 25 % of P) expressed as a percentage of the local sail chord.

Lower Stripe Twist

Twist in degrees(at 25 % of P) relative to the base chord.

Lower Stripe Entry

Entry angle(at 25 % of P).

Lower Stripe Exit

Exit angle(at 25 % of P).

Mid Stripe Camber

Camber (at 50 % of P) expressed as a percentage of the local sail chord.

Mid Stripe Draft

Position of max camber (at 50 % of P) expressed as a percentage of the local sail chord.

Mid Stripe Twist

Twist in degrees(at 50 % of P), relative to the base chord.

Mid Stripe Entry

Entry angle (at 50 % of P).

Mid Stripe Exit

Exit angle (at 50 % of P).

Upper Stripe Camber

Camber (at 75 % of P) expressed as a percentage of the local sail chord.

Upper Stripe Draft

Position of max camber (at 75 % of P) expressed as a percentage of the local sail chord.

Upper Stripe Twist

Twist in degrees(at 75 % of P), relative to the base chord.

Upper Stripe Entry

Entry angle(at 75 % of P).

Upper Stripe Exit

Exit angle(at 75 % of P).

Sea temperature

SST.

Shadow

The bearing of the centre of boat n's wind shadow.

Shadow opposite gybe

The bearing of the centre of boat n's wind shadow if on the opposite gybe or tack.

Slam

Volvo wave sensor parameter.

Sog

Speed over the ground.

Start bias angle

Bias of the start line in degrees given the current wind direction. Negative means the port end is favoured, positive means the starboard end is favoured. In the number bars, a port end favoured line will have a 'P' next to the number as in the display above.

Start bias length

Distance bias of the start line given the current wind direction at the favoured end. Negative means the port end is favoured, positive means the starboard end is favoured.

Start boat to pin time

Time to reach from the boat to pin end along the line.

Start distance below line

The minimum of the bow and GPS position below the line. This will be negative if over the line.

Essentially the XTE from the line.

Start distance to line

Minimum distance of the options selected (reach, starboard etc.), including turns. This will be negative if over the line.

Start distance to line reach

Distance to the line at current heading based on the start polar.

Start GPS time to burn

Start GPS time to line

Start numbers based on GPS COG and SOG only.

Start gun dist below line

Distance below/above line at the gun.

Start gun bsp target %

Start gun bsp polar %

Target and polar bsp percentages at the gun.

Start layline on port

Start layline on starboard

Start line square wind

Wind direction at 90° to the start line.

Start pinch time to port

Start pinch time to starboard

Time to line end sailing at polar bsp above (or below) targets.

Start port latitude

Start port longitude

Position of port end.

Start port with left turn

Start port with right turn

Start reach dist to line

Start reach speed to line

Start reach time to line

Time to the line at current heading based on the start polar.

Start speed to port

Start speed to starboard

Speed required to reach an end of the line at the gun.

Start speed on port

Start speed on starboard

Speed required to reach the line on port or starboard.

Start stbd latitude

Start stbd longitude

Position of starboard end.

Start stern below line

Start strb end time to burn, X secs

Start port end time to burn, X secs

Time to burn after tacking onto a point X seconds from the line on the starboard layline to the starboard end of the line.

Also for starboard layline to pin end.

These may not include current.

Start strb with left turn

Start strb with right turn

Times to the line on starboard after a turn to port or starboard.

Start target bsp

Start target twa

Start time to burn

Difference between time-to-line and time-to-start.

Start time to gun

Time to start gun.

Start time to layline P

Start time to layline S

Time to reach the starboard layline to port and starboard ends of the line, after turning and sailing parallel to the line.

Start time to line

Shortest time to the line at targets (includes tacking or gybing and acceleration) based on the start polar, rate of turn and acceleration settings. This will be negative if over the line.

Start time to port

Start time to starboard

Time to reach the port and starboard ends of the line based on the start polar.

Start time to port pinch

Start time to starboard pinch

Time to reach the port and starboard ends of the line, allowing to sail above targets (or below downwind targets).

Start time to port simple

Start time to starboard simple

As above, but without current, turning, acceleration or braking effects.

System - GPS time delta

Difference between system clock time and GPS.

Tab

Angle of the keel tab.

Tack/gybe loss metres

Tack/gybe loss time

Tacking angle

Tacking angle polar

Target awa

Target bsp

Target bsp nav

Target boat speed upwind or downwind based on the current performance polar.

Target bsp %

The fraction of the current boat speed to the target bsp, expressed as a percentage.

Target heel

Target heel upwind or downwind based on the current heel polar.

Target leeway

Target leeway upwind or downwind based on the leeway polar.

Target twa

Target twa nav

Target twa without leeway

Target true wind angle upwind or downwind based on the current performance polar.

Target Twd

Target Twd Delta

Estimated Twd based on course and target twa from the nav polar. The delta is the difference between this and the instrument Twd.

Target vmg

Target vmg upwind or downwind based on the current performance polar.

Test time

Time since test start.

Tide left port time

Tide left stbd time

Tide left time

Tide right port time

Tide right stbd time

Tide right time

Tide affected laylines.

Tide station

Nearest tide station.

Time

The current date & time and is stored internally in UTC.

Transmission oil temperature

Transmission oil pressure

Traveller

Trim (pitch)

Trim (pitch) rate

Fore-aft trim. Bow up pitch (trim) is positive.

Turn to mark

Twa

True wind angle includes leeway. If your instrument system calculates Twa to the centreline of the boat, then Expedition will add the leeway value (if any) to that to generate Expedition's

Twa. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Twd

True wind direction. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Twd -90

Twd +90

True wind direction +/- 90°.

Twd from targets

Twd Period

Period of the dominant wind shift

TWD predicted

TWS predicted

TWD and TWS predicted by loaded grib data.

Twd to lay mark

Twd to lay mark opposite tack

Twd to lay the mark - what the wind direction would have to shift to in order to lay mark.

Twd twist

Tws

True wind speed. If selected under [calibration](#), will be Expedition's internally calculated and calibrated value.

Tws Period

Period of the dominant tws cycle.

TWG

TWDG

Gust speed and direction.

User 0-31

Channels for custom use.

[Dfw2Exp](#) will use some of these and rename them.

[Ockam](#) calibration values are mapped to user channels 10 to 18. Ockam User tag data may also be stored in User 0 to User 8.

Vang

Vmc

Velocity made course - the component of sog towards the current mark.

Vmc%

The fraction of the current vmc to the optimum polar vmc, expressed as a percentage.

Vmc optimum

Given the current polar and wind, what the optimum Vmc would be to the current mark.

Vmc optimum heading

The heading to achieve optimum vmc to the current mark.

Vmc optimum twa

The wind angle that would be experience if sailing at the optimum vmc heading.

Vmc polar

The vmc to the mark at the current heading based on the polar and given the current true wind angle and speed.

Vmc to mark polar

The vmc to the mark if headed directly towards the mark based on the polar and given the current true wind angle and speed.

Vmg

Velocity made good - the component of bsp and leeway upwind or downwind.

Vmg%

The fraction of the vmg to the target vmg, expressed as a percentage.

Voltage

System voltage.

Water level

Water level 2

NMEA 2000 instance 0 and 1.

Wind weight

In the Ockam sense - as a fraction.

Wave max height

Wave max period

Wave sig height

Wave sig period

From Volvo wave sensor.